The Receiver Geometry of the Riemann Hypothesis

Parametric transport, effective tension, and the inductive support successor

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Abstract. This paper reconstructs the laboratory's scattered work on primes, zeta, theta/Mellin transport, explicit formulas, and semilocal operator geometry into one proof-auditable argument. It does not claim a proof of the Riemann Hypothesis. The classical spine is the Euler product, theta completion, functional equation, explicit formula, and Weil positivity criterion. The laboratory contributes a receiver-covariant synthesis of published semilocal Sonin intertwiners. A three-frame counting correspondence removes a further hidden absolute: arithmetic place/valuation, archimedean Fourier–Poisson–Mellin transport, and the receiving test/aperture frame have independent event lineages and meet only through actual incidences. A scalar parameter is a chart on one chosen section, not their master clock. Their first closed relational object is a triangular return, covariant up to conjugacy under independent recharting. A Weil-response bundle over that incidence replaces the misleading picture of one absolute scalar sign: test currents inhabit changing fibers, transition maps carry their Hermitian responses, and the critical line is one affine chart of the completed involution's fixed seam. Adjoining one prime becomes a whole-receiver Euler transport with an exact pulled-back metric and receiving aperture. Its arithmetic preimage is now explicit: primorial COPY/CUT/JOIN, arithmetic Möbius exclusion, the finite Euler difference, and the prime-power resolvent are finite, scalar, and operator faces of the same valuation-stratified event. The logarithmic normal derivative of that metric is the symmetric prime-power recurrence measure. An exact geometric-remainder squeeze now places that recurrence inside centered prime-character sheets: circular phase and reciprocal hyperbolic normal motion share one complex exponential carrier, while the local sheet metrics remain flat. This local flatness identifies the turned receiving aperture, rather than the geometric series itself, as the first nontrivial semilocal connection. The resulting Sonin trace changes by an aperture-connection term, and the completed explicit-formula defect obeys an exact one-prime recurrence. A further squeeze audit separates resolution from structure: finite-rank Sonin tails vanish inside one receiver, while a negative compact witness persists through every larger support aperture. The complete error becomes automatic only after the completed defect has an independently positive remainder carrier. A covariant coarea refinement now gives that carrier an exact mathematical type: a receiver-local field of squared amplitudes whose chart changes carry the square-root Jacobian. Its covariant fundamental theorem keeps every contemporary response nonnegative while allowing the interior connection current and discrete prime seams to have either sign. The logarithmic change of variables produces the Mellin half-density $e^{\frac{u}{2}}$ and the critical $\frac{1}{2}$ chart; the square root $G_p^{\frac{1}{2}}$ of the Euler metric supplies the corresponding operator half-Jacobian and an exact positive successor

complement. None of these identities yet proves that this complement is the completed defect. The published finite archimedean tail order supplies one exact bounded squeeze on each certified finite Galerkin band, but not the band or support completion. A final source audit corrects the amplitude space to the codimension-two Mellin-moment kernel. Its support successor is a graph: every new shell direction carries the unique old-region compensation preserving those moments. Consequently a newly admitted prime is purely cross-incidence only in the raw support split; conditioning returns part of that incidence into the shell diagonal. For the first $P(2) \to P(3)$ cell, positive definiteness and compact resolvent of the published base already carry the complete cross through the old energy space. The first unsupported statement is now one sign: the conditional shell energy left after that carried component is removed must be nonnegative. The complete prime-power threshold combinatorics is now derived as well. Every prime axis is an equally spaced logarithmic family $k \log p$ with geometrically weighted half-density; an integer support successor births one paired hinge cell exactly at a prime power. Those hinges are signed difference-energy/potential balances, and conditioning creates a complete Gram of cross-terms among their old energy carriers. The invariant object behind that coordinate expansion is now identified exactly: it is the shell's effective Dirichlet-to-Neumann tension after the old body relaxes. Published strict positivity at $R = 2$, together with continuity of the localized Weil ground value, proves positivity on a genuine but nonquantitative prime-active interval $2 < R \leq R_*$ for some $R_* > 2$. Existing Galerkin evidence sees continuation through $R = 3$, but no continuum bound reaching that endpoint is supplied. A uniform positive effective-tension law would let ordinary induction carry every compactly supported returned current and imply RH; no exhaustive computation is involved. The local induction is now placed inside one completed field. The endpoint, Gamma, and Euler metrics multiply to the completed $|\xi|^2$ return metric, and their responses add as its normal logarithmic derivative. RH is equivalent to this normal impedance being positive real on the centered right half-plane. Folding the opposed seam axes through $w = z^2$ turns the same statement into a Stieltjes impedance and an exact pair of connected Hankel Gram laws derived recursively from the positive theta moments. This supplies a coefficient-level proof construction which does not enumerate zeros or support cells: one uniform source-side Gram factorization would close the complete Weil sign. Knot monodromy sharpens the same boundary. The Euler product is written exactly as a primitive-orbit product with orbit lengths $\log p$; the completed involution becomes reciprocal-conjugate spectral symmetry; and the figure-eight monodromy gives an exact counterexample showing that reciprocal topology alone does not force unit modulus. A positive covariantly preserved monodromy metric would force every returned eigenmode onto the unitary seam. The archimedean remainder supplies one genuine base metric, while its finite-place continuation and a complete prime-orbit trace identity remain open. The fixed dyadic Euler polarization is correspondingly one exact edge face of the asynchronous return, not the whole joint support/place event. The positive half has now been factored as an oriented first-order boundary operator, realized geometrically as a convex half-overlap surface, and periodized on the dyadic cell. That periodization constructs the unique minimum local symmetric-character filler. The published Sonin maps already give invertible prime admission and its dual population return on every aperture. This disproves the former target of embedding the dyadic quotient as a restriction of the successor: the quotient has a nonzero kernel. Its correct role is a metric-defect face inside the still-open form-level support successor. Geometric figures are used as mathematical charts rather than illustrations: valuation simplices, parameter trajectories, transported response cones, reciprocal Gaussian scale, Euler chords, Mellin reflection, moving theta boundaries, support/place filtrations, metric-turned apertures, and the final defect balance all display the same addressed objects used in the equations.

Contents

I Status, claim discipline, and the central result

The present paper has one purpose: to say exactly how far the laboratory's Riemann-Hypothesis line has reached when its number-theoretic, geometric, and operator-theoretic parts are placed in one dependency order.

The result is not "a pattern in the primes." It is a sharper mathematical boundary:

Geometric construction

The current RH result in one sentence

Problem. Relate the prime-power current to the critical seam through the returned incidence of independently changing arithmetic, archimedean, and receiving frames, without freezing an old manifold, synchronizing them by an external clock, or assuming the desired sign.

Euler / theta completion

↓

Mellin adjoint seam

↓

Weil explicit formula

↓

OPEN: first conditioned residual, then its uniform successor

Construction and demonstration. The completed Mellin return is a norm square on the critical seam. A published semilocal Euler intertwiner supplies the one-prime transport. Pulling the successor geometry back gives an exact metric, aperture, and adjoint law. Differentiating that metric normally recovers every repetition of the prime. Comparing the positive Sonin trace with the completed Weil response yields an exact defect recurrence.

Algebraic face.

$$D_{S\cup\{p\}}(f) = D_{S(f)} - W_{p(f*f^\sharp)} - \kappa_{S,p}(f).$$

Therefore. The recurrence and independent-rechart covariance are proved. The space-level prime/support induction is also already carried by the published Sonin maps. Moment conditioning now gives the exact quotient lift, and the published base gap carries the first cross. The residual is exactly the effective tension after the old body relaxes. Continuity carries its positive sign through some nonzero $p = 2$-active interval beyond $R = 2$. Reaching $R = 3$ remains quantitative; the uniform support-successor analogue would prove every $P(n)$.

I.1 Four grades of statement

Every substantial statement is assigned one of four grades.

Grade	Meaning in this paper
Classical / published	A theorem or construction inherited under its published hypotheses.
Exact laboratory derivation	A calculation proved from declared objects. It may be a new synthesis without being a globally novel theorem.

Grade	Meaning in this paper
Bounded measurement	A finite exact experiment showing that a representation or lifecycle is physically available. It is not evidence for a universal analytic sign.
Open hypothesis	A statement for which neither the paper nor the cited literature supplies a proof. If it implies RH, that implication is stated explicitly.

I.2 What is actually distinctive to this laboratory

The following inventory answers the originality question directly.

Object	Status	Assessment
Mellin return fixed-seam factorization	Exact derivation	Classical Mellin identities, assembled in the laboratory's current/return language. Useful and exact; not claimed as new mathematics.
One-prime Euler successor geometry	Exact lab synthesis	Published semilocal intertwiners are combined with a pulled-back metric, metric-orthogonal aperture, and adjoint covariance as one receiver event. This package was not found stated verbatim in the audited sources.
Prime-power recurrence as normal metric deformation	Exact lab derivation	The derivative of the Euler chord metric is identified with the symmetric prime-power atomic measure. This is the most compact original formulation in the present line. Global novelty is not asserted.
Receiver-covariant completed-defect recurrence	Exact lab synthesis	The explicit prime-power change and the aperture-connection change are joined in one balance law. The identity is new to the laboratory's framework; its sign is open.
Fixed-seam conjugacy and signature transport	Exact lab refinement	The real-part line is replaced by the fixed locus of the completed involution, and positive congruence is shown to preserve the complete positive/null/negative cone. This is the parametric correction made in the present cut.
Three-frame counting correspondence	Project definition	Arithmetic, archimedean, and receiving lineages are no longer synchronized by one undeclared parameter. Their actual co-presence is an incidence object, and each numeral or coordinate is explicitly a receiver face of a counted construction.
Triangular return covariance	Exact lab theorem	The three-edge return changes by conjugacy under three independent recharts. Its spectrum, fixed subspace, and congruently transported Hermitian response are therefore relational invariants; its plotted edge coordinates are not.
Inductive Weil support successor	Exact reduction / open construction	The archimedean cell is the base case. The first source-derived cross and conditioned shell are constructed, and their Schur complement is the exact effective tension. Strict positivity con-

Object	Status	Assessment
		tinues nonquantitatively into the first prime-active interval. Its sign through $R = 3$, and the corresponding uniform law for every successor, remain open.
Conditioned effective tension	Exact lab derivation	Variational shorting is identified with the Dirichlet-to-Neumann response after the inherited interior relaxes. Congruence localizes every new negative or null direction to this one operator, and sequential shorting is associative. The application is exact; its required zeta-kernel sign is not assumed.
Covariant coarea carrier and signed-seam FTC	Exact lab synthesis	Classical change of variables, coarea, polar decomposition, and the metric derivative of a squared norm are assembled into a receiver-local carrier law. It supplies the exact type of a possible positive remainder, but not the arithmetic amplitude field which would identify it with the Weil defect.

The foundation paper's simplicial boundary identity and transported fundamental theorem are classical. The terms *holon*, *standing*, *current*, *FOUND*, *RIDE*, *receiver*, and *compression* are project vocabulary. Their definitions can organize mathematics, but definitions do not become new theorems merely because they are useful.

I.3 Lambda is not yet the ideal mathematical type

Definition 1 — Situated event as a typed correspondence Let

$$X_e^- = Q_e \sqcup_{F_e} A_e \sqcup_{F_e} L_e$$

denote the source boundary actually co-present at event e: an oriented question Q_e, standing atlas A_e, and live source lineages L_e in frame F_e. A **situated event** is a typed local correspondence

$$X_e^- \xrightarrow{\iota_e^-} B_e \xleftarrow{\iota_e^+} P_e, \quad q_e : B_e \to \Pi_e,$$

where B_e is the event body, P_e is its successor boundary, and q_e is a receiver testimony map. Boundary orientation and admitted transport carry the causal hand; the cospan arrows alone do not.

> Boundary. The amalgamated notation asserts only supplied co-presentation. A specialization must name the category, gluing maps, and any universal property it uses. The historical abbreviation $\Lambda_{F(Q\otimes A\otimes L)} \to (P, \pi_B)$ may name a deterministic law selecting such event bodies, but it is not the general definition and $\otimes$ is not an algebraic tensor product without additional structure.

The historical expression

$$\Lambda_{F(Q\otimes A\otimes L)} \to (P, \pi_B)$$

was retained because its semantics were ratified: the question, standing atlas, and live lineages participate co-presently in one event, and testimony does not reconstruct the successor. That semantic boundary remains valuable.

The notation itself was not independently derived as uniquely correct. $\otimes$ is too strong when no tensor category or universal multilinear law has been supplied, and an ordinary function arrow hides the event interior and its two boundary cuts. In this paper, Λ names a **law class** which admits or selects situated event correspondences. One event occurrence is the typed body B_e with source boundary, successor boundary, orientation, and testimony map. A deterministic specialization may collapse that correspondence to the compact historical arrow, but the compact arrow is not the foundation.

II Parametric receiver geometry and relational sign

II.1 Counting does not supply a master clock

Definition 2 — Three-frame counting correspondence For $i \in \{1, 2, 3\}$, a **local counting frame** is a tuple

$$\mathcal{F}_i = (B_i, \pi_i : \mathcal{H}_i \to B_i, \nabla_i, \rho_i).$$

Here B_i is the frame's oriented lineage of actual local events, $\mathcal{H}_{i,b}$ is the family of constructions participating at $b \in B_i$, ∇_i is the admitted partial transport along that lineage, and ρ_i is a family of receiver maps. A numeral or other glyph

$$n_{i,b}(v) = \rho_{i,b}(v)$$

is a face of the counted construction v in that receiver; it is not the identity of v or of its lineage.

Co-presence of the three frames is an incidence object E with structure maps

$$\tau_i : E \to B_i, \quad \tau = (\tau_1, \tau_2, \tau_3) : E \to B_1 \times B_2 \times B_3. \quad \text{(INCIDENCE)}$$

The Cartesian product only addresses the three local positions. E contains the incidences which actually occur and may distinguish multiple occurrences with the same addressed triple. It need not be the full product and does not impose a common order on the B_i.

At $e \in E$, abbreviate

$$\mathcal{H}_{i,e} = \mathcal{H}_{i,\tau_{i(e)}}.$$

An oriented triangular comparison consists of partial cross-frame transports

$$T_{12,e} : D_{12,e} \to \mathcal{H}_{2,e}, \quad T_{23,e} : D_{23,e} \to \mathcal{H}_{3,e}, \quad T_{31,e} : D_{31,e} \to \mathcal{H}_{1,e},$$

where

$$D_{12,e} \subset \mathcal{H}_{1,e}, \quad D_{23,e} \subset \mathcal{H}_{2,e}, \quad D_{31,e} \subset \mathcal{H}_{3,e}.$$

Its complete admissible return domain is

$$D_{123,e} = D_{12,e} \cap T_{12,e}^{-1}(D_{23,e}) \cap (T_{23,e}T_{12,e})^{-1}(D_{31,e}).$$

On this domain the first closed relational operator is

$$M_{123,e} = T_{31,e}T_{23,e}T_{12,e} : D_{123,e} \to \mathcal{H}_{1,e}. \quad \text{(RETURN)}$$

If the transports are connection isomorphisms around a closed comparison cell, $M_{123,e}$ is its triangular holonomy. If its domain is invariant, the returned defect

$$\Omega_{123,e} = M_{123,e} - I$$

records failure of that oriented triangle to commute.

When the incidence changes the participating frames, its successor must be supplied jointly:

$$(\mathcal{F}_1^+, \mathcal{F}_2^+, \mathcal{F}_3^+) = \Phi_{e(\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3; T_{12,e}, T_{23,e}, T_{31,e})}. \quad \text{(JOINT SUCCESSOR)}$$

No one edge, counter, or receiver owns that successor.

> Boundary. The three lineage bases may each be discrete, continuous, or mixed, but any calculus used on them must be declared locally. A parameter t may enumerate one chosen receiver section through E; it is then a chart on that section, not a clock shared by all three frames.
>
> The definition does not assert that arbitrary frames admit transports, that the transports are invertible, or that a returned defect is positive. A partial or noninvertible comparison remains a correspondence rather than a groupoid arrow. The triangular return is the least closed relational face; it does not reconstruct the complete three lineages or reduce higher incidences to triples.

Definition 3 — Parametric Weil-response bundle Let

$$\tau : E \to B_{\text{arith}} \times B_{\text{arch}} \times B_{\text{recv}}$$

be the incidence object of three local counting frames: arithmetic place/valuation, archimedean Fourier–Poisson–Mellin transport, and the receiving test/aperture/normalization frame. Let $\pi : \mathcal{H} \to E$ be a bundle whose fiber $\mathcal{H}_e$ consists of the test currents admitted at the actual incidence e. Each fiber carries a Hermitian response

$$W_e : \mathcal{H}_e \times \mathcal{H}_e \to \mathbb{C}.$$

Its evaluated quadratic response and nonnegative cone are

$$q_{e(f)} = W_{e(f,f)}, \quad \mathcal{C}_e = \left\{ f : q_{e(f)} \geq 0 \right\}.$$

An admitted incidence comparison $\alpha : e \to e'$ may carry a partial transport

$$T_\alpha : D_\alpha \to \mathcal{H}_{e'}, \quad D_\alpha \subset \mathcal{H}_e.$$

For $f, g \in D_\alpha$, its response defect is

$$\mathcal{K}_{\alpha(f,g)} = W_{e'}(T_\alpha f, T_\alpha g) - W_{e(f,g)}.$$

When the fibers are anchored by maps $\iota_e : \mathcal{H}_e \to \mathcal{H}_W$ into the classical admissible Weil test space, receiver exactness requires the transition to represent the same completed test occurrence:

$$\iota_{e'} T_\alpha = \iota_e.$$

A one-parameter chart

$$\gamma : J \to E$$

selects a receiver-readable section through the incidence network. It induces the familiar displayed family

$$q_{\gamma(t)} = W_{\gamma(t)}(f(t), f(t)),$$

but t orders only that chosen section. The three projections $\tau_i \gamma$ retain their own local lineages and need not be synchronized beyond the incidences selected by γ.

> Boundary. The inequality $q_{e(f)} \geq 0$ is an evaluation of a Hermitian form on one transported direction, not an observer-free scalar property. Positivity of the form means that every admitted direction in the fiber is nonnegative. A chart which merely selects some different positive direction does not establish that statement. The bundle, connection, anchors, and transition laws must be constructed; naming a parameter does not supply them.
>
> The product of the three lineage bases is an addressing space, not an enlarged simultaneous state field. Only incidences in E are co-present. A common scalar parameter may be useful for differentiation after a section is selected, but it may not silently become a master chronology for prime admission, Fourier/Mellin scale, support, and the receiving current.

The earlier draft corrected a frozen receiver by writing one moving configuration

$$\gamma(t) = (F(t), A(t), L(t), S(t), R(t), f(t), \ldots).$$

That was still too absolute. One scalar t silently synchronized prime-place admission, Fourier/Mellin scale, support growth, and the test receiver. It described a useful observed movie, but not the underlying relational body.

The corrected base has three independently ordered counting frames:

1. B_{arith} carries prime valuation, Euler admission, and repetition;
2. B_{arch} carries Fourier–Poisson–Mellin scale and completion; and
3. B_{recv} carries the admitted current, support aperture, and normalization.

Their co-presence is not the full Cartesian product. It is the actual incidence

$$\tau : E \to B_{\mathrm{arith}} \times B_{\mathrm{arch}} \times B_{\mathrm{recv}}.$$

The product addresses three local positions; E says which triples actually meet and can distinguish repeated meetings at the same addresses. A numeral, place label, complex coordinate, or support bound is a receiver glyph for a construction on one of these lineages. It is not the construction's observer-free identity.

The third frame is therefore not a passive container for a relation between the other two. At a consequential incidence all three successors may depend on the complete cyclic meeting:

$$(\mathcal{F}_1^+, \mathcal{F}_2^+, \mathcal{F}_3^+) = \Phi_{e(\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3; T_{12}, T_{23}, T_{31})}.$$

Solving three isolated pair problems and appending their answers would miss that mutual reorientation. This is the precise sense in which the construction has a three-body character.

The first closed comparison available at one incidence is the oriented triangular return

$$M_e = T_{\mathrm{recv},\mathrm{arith},e} T_{\mathrm{arch},\mathrm{recv},e} T_{\mathrm{arith},\mathrm{arch},e},$$

on its complete composability domain. The naming of the start vertex is a chart choice; the oriented cycle is the relational content.

Theorem 4 — The triangular counting return is covariant under independent recharting Fix an incidence e of a three-frame counting correspondence. Let

$$g_i : \mathcal{H}_{i,e} \to \mathcal{H}'_{i,e}$$

be independent invertible recharts, taken to be bounded isomorphisms whenever the fibers are infinite-dimensional normed spaces. For $(i, j) \in \{(1, 2), (2, 3), (3, 1)\}$, carry each domain by $D'_{(i,j),e} = g_{i\left(D_{(i,j),e}\right)}$, and define

$$T'_{(i,j),e} = g_j T_{(i,j),e} g_i^{-1}. \quad \text{(RECHART)}$$

Then

$$D'_{123,e} = g_1\left(D_{123,e}\right)$$

and the returned operators obey

$$M'_{123,e} = g_1 M_{123,e} g_1^{-1}. \quad \text{(COVARIANT RETURN)}$$

Whenever the returned defect is defined,

$$\Omega'_{123,e} = g_1 \Omega_{123,e} g_1^{-1}.$$

Consequently, when the returns are bounded endomorphisms and the recharts are bounded isomorphisms, their spectra and fixed subspaces agree up to the rechart. In finite dimension their characteristic polynomials, traces, and determinants agree exactly.

Let W_1 and W_1' be Hermitian responses related by

$$W_1'(g_1 v, g_1 w) = W_1(v, w). \quad \text{(RESPONSE RECHART)}$$

For every admissible anchored current v,

$$W_1'(M' g_1 v, M' g_1 v) - W_1'(g_1 v, g_1 v) = W_1(Mv, Mv) - W_1(v, v). \quad \text{(RETURNED RESPONSE)}$$

Thus both the conjugacy class of the closed return and the congruently transported Hermitian response are independent of the three chosen coordinate charts.

Demonstration. Substituting ("RECHART") into the three-edge composite cancels the two interior rechart pairs:

$$T_{31}' T_{23}' T_{12}' = g_1 T_{31} g_3^{-1} g_3 T_{23} g_2^{-1} g_2 T_{12} g_1^{-1} = g_1 M_{123} g_1^{-1}.$$

The domain identity follows by applying the same substitutions to the three composability conditions. Similar operators have the same spectrum; in finite dimension they have the same characteristic polynomial, trace, and determinant. Also

$$\ker(M' - I) = g_1 \ker(M - I).$$

Finally substitute $M' g_1 = g_1 M$ into ("RESPONSE RECHART") twice to obtain ("RETURNED RESPONSE").

Boundary. Covariance does not mean that an event leaves the system unchanged. Replacing the incidence, current, aperture, transport law, or Hermitian response is a new situated comparison, not a rechart of the old one. Such a change may alter a scalar response or its sign.

Conversely, a genuine negative Hermitian direction cannot be made positive by independently rotating the three charts. The sign belongs to the anchored current and response at the declared incidence; it is neither an observer-free label on one object nor disposable coordinate noise.

Edge matrices and plotted coordinates are not conjugacy invariants. The closed return also forgets some edge-wise lineage, so equal holonomy does not identify complete three-frame constructions. Noninvertible cross-frame maps require the wider span/correspondence calculus and do not acquire a spectrum merely by notation.

Under independent invertible recharts of all three frames, M_e changes by similarity. Its coordinate matrix changes, while its conjugacy class, spectrum, and fixed subspace do not. If the Hermitian response is carried congruently, the returned response of the same anchored current is also unchanged. Conversely, changing the incidence, current, aperture, transport law, or response form is a new event and may change the response. This is the precise division between a coordinate rotation and a real receiver-relative change.

The returned operator, its defect $M_e - I$, or a receiver face of its action may be the **emanated fourth** of this closed relating. That consequence must not be confused with a **supplied fourth contact** required to evaluate a projective cross-ratio. The former is produced by the event; the latter is additional input to a different measurement.

II.2 A parameterized trajectory is one receiver section

A chosen chart

$$\gamma : J \to E$$

may still order a readable section through the incidence network. Its three projections $\tau_i \gamma$ need not be clocks for each other. Along that declared section, the contemporary response is

$$q_{\gamma(t)} = W_{\gamma(t)}(f(t), f(t)),$$

not a list of unrelated numbers and not an absolute history of all three frames.

Where the pulled-back bundle is differentiable and carries a connection ∇, differentiation separates motion of the response geometry from motion of the current inside that selected section:

$$\frac{\mathrm{d}}{\mathrm{d}t} q_{\gamma(t)} = \left(\nabla_{\dot{\gamma}} W\right)(f, f) + 2\operatorname{Re} W\left(\nabla_{\dot{\gamma}} f, f\right).$$

For a compatible connection and a parallel current, the value is retained. For an indefinite response, an individual section can cross its null cone and change sign. That is a real oriented event. It is different from proving that the whole fiber has no negative direction. Reparameterizing the same section does not alter the event order or the integrated covariant response; choosing a different incidence path may expose genuine triangular holonomy.

II.3 Negative responses are possible unless RH removes them

The Weil condition concerns the specified adjoint-return family $h = f * f^{\sharp}$ and every admissible f. It does not assert that every arbitrary test family or every locally chosen pairing must be positive. Within the prescribed family,

$$\text{RH} \quad \leftrightarrow \quad Q_{W(f)} \geq 0 \quad \text{for every admissible } f.$$

Consequently, if RH is false, the criterion entails the existence of an admissible returned current with $Q_{W(f)} < 0$. If RH is true, no such direction exists. A restricted parameterized family can remain positive while simply missing a negative direction elsewhere.

This is where “orientation” needs a precise boundary. Reversing the current or multiplying it by a phase does not reverse a Hermitian square:

$$Q_{W(-f)} = Q_{W(f)}, \quad Q_{W(e^{i\theta} f)} = Q_{W(f)}.$$

Choosing a different vector in a positive sector is possible even in an indefinite geometry, but that vector is not the one whose negative response must be explained.

Theorem 5 — A genuine negative direction cannot be removed by recharting Let $T : \mathcal{H}_\lambda \to \mathcal{H}_\mu$ be invertible and suppose

$$W_{\mu(Tf,Tg)} = c W_{\lambda(f,g)}$$

for one real $c > 0$ and all f, g. Then

$$q_{\mu(Tf)} = c q_{\lambda(f)},$$

so T carries the positive, null, and negative cones bijectively. In particular, W_λ is positive semidefinite if and only if W_μ is, and their maximal negative-subspace dimensions agree.

> **Demonstration.** Set $g = f$ in the covariance identity. Because $c > 0$, the sign is unchanged. Invertibility gives the converse and maps every negative subspace isomorphically to a negative subspace. Applying the same argument to T^{-1} proves equality of the maximal dimensions.

> Boundary. An individual path in an indefinite fiber may cross its null cone and change sign. That does not eliminate the fiber’s negative directions. Multiplying a test current by -1 or by a unit complex phase also leaves a Hermitian quadratic response unchanged. A transformation which flips the sign of the form is not a harmless orientation change; it changes the positive structure being tested.

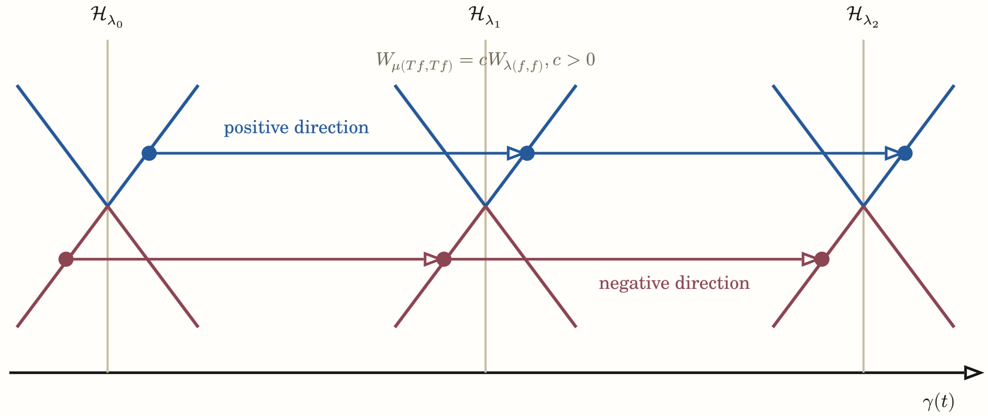

Figure 1: A parametric response is a field of Hermitian geometries, not a sequence of detached scalars. Positive-congruence transport carries both positive and negative directions. Finding another positive direction is therefore insufficient; a proof must show that the exhaustive admitted bundle has no transported negative direction.

Thus a genuine negative direction cannot be rotated away by an invertible coordinate change. It can only be excluded by the mathematics of the completed response, or carried to an independently positive chart through a transition which preserves the same anchored occurrence and its sign.

II.4 Real part is one chart of the completed fixed seam

Lemma 6 — The critical line is a chart of a fixed seam On the completed spectral chart define the antiholomorphic involution

$$J(s) = 1 - \overline{s}.$$

Then

$$\mathrm{Fix}(J) = \left\{ s : \mathrm{Re}(s) = \frac{1}{2} \right\}.$$

If φ is a biholomorphic rechart, the transported involution is

$$J_\varphi = \varphi \circ J \circ \varphi^{-1},$$

and

$$\mathrm{Fix}\left(J_\varphi\right) = \varphi(\mathrm{Fix}(J)).$$

Thus $\mathrm{Re}(s) = \frac{1}{2}$ is the affine-chart expression of the invariant fixed seam, not an absolute line visible outside the completed relation.

Demonstration. The equality $J(s) = s$ is equivalent to $s + \overline{s} = 1$, hence to $\mathrm{Re}(s) = \frac{1}{2}$. If $J(s) = s$, then $J_{\varphi(\varphi(s))} = \varphi(J(s)) = \varphi(s)$. Applying φ^{-1} proves the reverse inclusion.

Boundary. Conjugating the seam does not move a zero onto it. It only expresses the same fixed locus in another chart. A proof of RH still owes a mechanism which excludes zero orbits away from that locus.

In the standard affine chart the involution itself supplies a normal and a seam projection:

$$\varepsilon_{J(s)} = \frac{s - J(s)}{2} = \mathrm{Re}(s) - \frac{1}{2}, \quad c_{J(s)} = \frac{s + J(s)}{2} = \frac{1}{2} + i\,\mathrm{Im}(s).$$

Along one receiver-selected trajectory $s = s(t)$, $\varepsilon_{J(s(t))}$ is its signed normal displacement in that chart. The parameter orders that displayed section only. Under a general rechart one should retain the fixed locus and its normal bundle rather than insist that the image remain a straight vertical line.

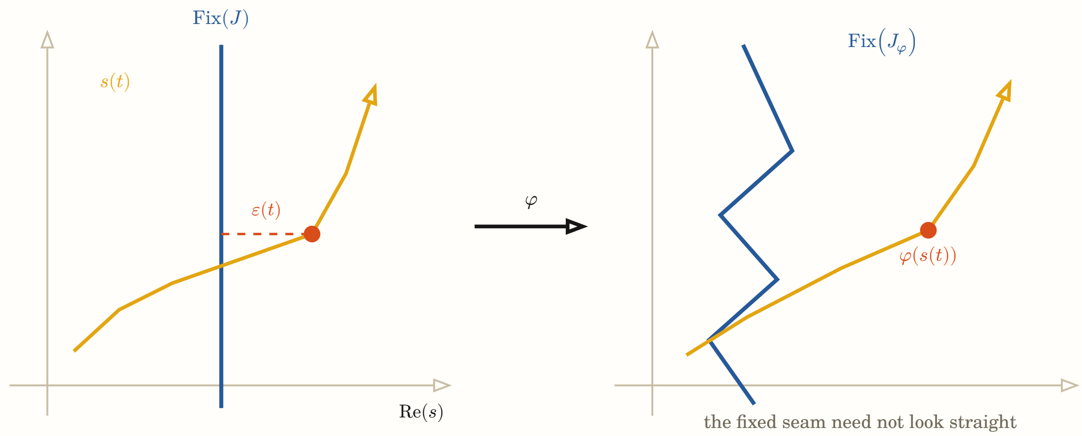

Figure 2: The real coordinate is one affine chart. The invariant object is the fixed seam of the completed antiholomorphic involution. A biholomorphic rechart conjugates both the trajectory and the seam; it does not move an off-seam point onto the seam.

The relativistic statement is therefore not that the real coordinate is illusory. It is that $\mathrm{Re}(s) = \frac{1}{2}$ has meaning because the completed relation selects an involution, and that coordinate expression must transform with the involution. Merely redrawing an off-seam zero in coordinates where its first component looks positive or vanishes does not change its incidence with the fixed seam. In the three-frame formulation, the intrinsic candidate is the self-dual or unitary stratum of the returned arithmetic–archimedean–receiver transport. The vertical line is its standard Mellin-coordinate face, not a plane standing outside that return.

II.5 The parametric π–e bridge is reciprocal Gaussian scale

Use the Fourier convention

$$\mathcal{F}f(\xi) = \int_{\mathbb{R}} e^{-2\pi i x\xi} f(x)\,\mathrm{d}x$$

and the half-density Gaussian family

$$g_{u(x)} = e^{\frac{u}{2}} \exp\left(-\pi e^{2u} x^2\right).$$

The Gaussian transform gives the exact parametric return

$$\mathcal{F}g_u = g_{-u}.$$

Thus $a = e^{2u}$ turns additive log-scale into multiplicative width, Fourier transport sends a to a^{-1}, and $u = 0$ is the self-dual fixed seam. The constant π fixes the conventional self-Fourier normalization; e is both the scale exponential and the Gaussian transport.

Sampling this family on the integer lattice and applying Poisson summation gives theta reciprocity. Mellin transport carries reciprocal scale into $s \to 1 - s$. On the arithmetic side,

$$p^{-s} = \exp(-s \log p)$$

uses the same additive-to-multiplicative lift. Its logarithm separates pure prime-axis repetitions, while formal exponentiation glues repeated and mixed axes into prime-power, semiprime, and higher multiplicative faces.

scale chart u $\xrightarrow{\mathcal{F}}$ **reciprocal chart** $\xrightarrow{\mathcal{M}}$ **spectral chart**

$$g_{u(x)} = e^{\frac{u}{2}} \exp\left(-\pi e^{2u} x^2\right) \qquad \mathcal{F} g_u = g_{-u} \qquad s \to 1 - s$$

$$a = e^{2u} \qquad a \to a^{-1} \qquad \mathrm{Fix} = \tfrac{1}{2} + i\mathbb{R}$$

$u = 0$ is the self-Fourier Gaussian seam; Mellin transport carries reciprocal scale into the completed zeta reflection.

Figure 3: A parametric π–e bridge. Exponentiation turns additive log-scale u into the multiplicative width a; π fixes the standard self-Fourier normalization. Reciprocal Gaussian scale is transported by Mellin analysis into the zeta reflection.

This is the topology the earlier formula-ecology experiments were intended to expose. The useful object is not a census of formulas for π and e. It is the groupoid generated by logarithmic lift, exponential return, Fourier–Poisson reciprocity, Mellin transport, Euler-place admission, analytic continuation, and completed reflection, together with their domains, seams, and composition laws.

The proposed RH route can now be stated without absolutizing positivity: every anchored admissible current must lie in an orbit which reaches an independently positive base cell, and every arc in that orbit must preserve the completed Hermitian response by positive congruence. The existence of some unrelated positive orientation is not enough.

III Arithmetic geometry before analytic continuation

III.1 Prime axes are primitive valuation directions

For a positive rational number x, let

$$\nu(x) = \left(v_{p(x)}\right)_{p \text{ prime}}$$

be its finitely supported valuation vector. Then

$$\nu(xy) = \nu(x) + \nu(y), \quad \nu\left(\frac{x}{y}\right) = \nu(x) - \nu(y).$$

Unique factorization identifies a prime p with the primitive basis direction e_p. A prime power p^m is repeated traversal me_p; a semiprime pq is a degree-two mixed face $e_p + e_q$, while p^2 is a degree-two diagonal face.

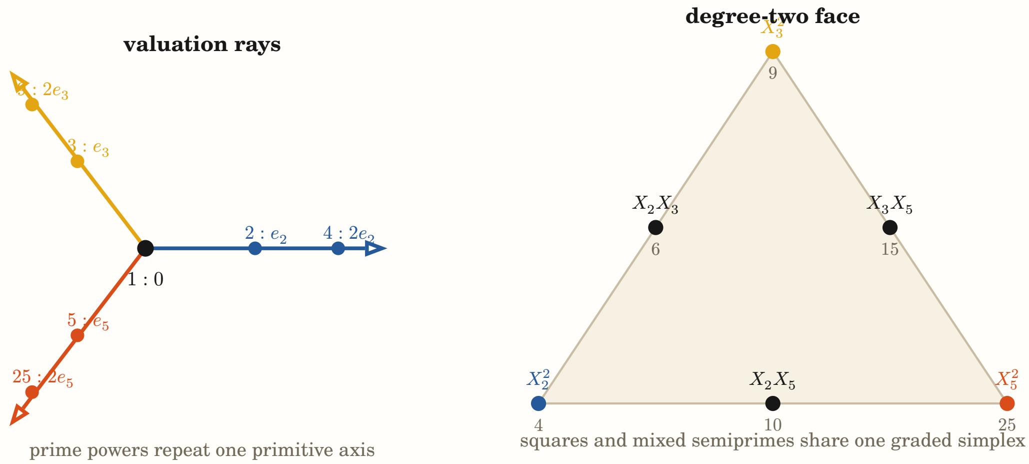

Figure 4: Prime axes are primitive valuation directions. Prime powers are repeated traversal of one ray; products of distinct primes occupy mixed faces. This geometry is exact in the exponent lattice. It does not assert a Euclidean metric on the integers.

This geometry does not place the integers in one privileged Euclidean embedding. It states an exact affine-additive law in the exponent lattice. Different receivers may project that lattice to residues, decimal glyphs, polynomial factor species, or logarithmic lengths. Those projections can co-locate or separate the same integer occurrences without changing the valuation relation.

III.2 Residue, winding, and Euclidean word are different faces

For a candidate n and a prior prime axis p,

$$n = q_p p + r_p, \quad 0 \le r_p < p.$$

The residue r_p is cyclic phase. The quotient q_p retains completed winding. The exact rational $\frac{n}{p}$ retains both, and its continued fraction retains the ordered Euclidean rebase word. A unimodular matrix product therefore gives an exact reversible path in a projective rational chart.

The bounded fraction study at $n = 17$ confirmed that these faces carry useful navigation, but it also supplied the necessary control: long continued fractions and affine offset closures occur for composites as well as primes. The defining prime event remains nonclosure against every already-founded axis $p \le \sqrt{n}$. The surrounding fraction atlas enriches navigation; it is not a prime classifier.

At a finite receiver horizon H and observation word O, the laboratory's first-person quotient is

$$\Pi_{\sigma,H}(\text{prime } \texttt{mid } O(x)) = \frac{\sum_{n \in F_{O(x)}, n \text{ prime}} n^{-\sigma}}{\sum_{n \in F_{O(x)}} n^{-\sigma}},$$

where $F_{O(x)}$ is the still-indistinguishable fiber. This is deterministic uncertainty relative to a receiver's current distinctions. Once x is completely specified, it is not ontic chance that x is prime.

III.3 Radix locality and character winding belong to one exact receiver

Definition 7 — Radix, residue, and character faces of one bounded counting occurrence
Fix integers $b, q \ge 2$ and a digit depth $K \ge 1$. The radix word space, value map, and residue receiver are

$$D_{b,K} = \{0, 1, ..., b-1\}^K, \quad V_{b,K}(d) = \sum_{k=0}^{K-1} d_k b^k,$$

$$\rho_{b,q,K}=\pi_q\circ V_{b,K}:D_{b,K}\to C_q,\quad C_q=\frac{\mathbb{Z}}{q}\mathbb{Z}.$$

The **place-incidence word**

$$w_{b,q}(k)=b^k \bmod q$$

gives the receiver directly:

$$\rho_{b,q,K}(d)=\sum_{k=0}^{K-1}d_k w_{b,q}(k)\quad\text{in } C_q.\quad\text{(PLACE)}$$

Let X denote the generator of the integral group algebra $\mathbb{Z}[C_q]=\frac{\mathbb{Z}[X]}{X^q-1}$. The complete bounded population is

$$G_{b,q,K}(X)=\prod_{k=0}^{K-1}\left(\sum_{d=0}^{b-1}X^{dw_{b,q}(k)}\right)=\sum_{a\in C_q}c_aX^a.\quad\text{(POPULATION)}$$

The coefficient c_a is exactly the cardinality of the complete fiber $\rho_{b,q,K}^{-1}(a)$.

After extending scalars to a field containing a primitive q-th root ω_q, the character face and its inverse are

$$\hat{G}(j)=G_{b,q,K}(\omega_q^j)=\sum_{a\in C_q}c_a\omega_q^{ja},$$

$$c_a=\frac{1}{q}\sum_{j\in C_q}\omega_q^{-ja}\hat{G}(j).\quad\text{(CHARACTER)}$$

For an actual bounded integer occurrence set O, a radix presentation is an injective map

$$r_b:O\to D_{b,K}$$

satisfying $V_{b,K}r_{b(n)}=n$. Two such radix maps $r_b, r_{b'}$ are invertible recharts of the same occurrences through O. The residue map $\pi_q:O\to C_q$ is instead a quotient receiver with complete fibers. The coefficient-to-character transform in ("CHARACTER") is again invertible after the declared scalar extension.

> Boundary. A radix presentation, a residue quotient, and a linear character basis are three different operations. Calling all three a "basis change" erases whether the passage is invertible. A change of radix is a rechart only on a common occurrence set and with enough digits to represent it. The factor $\frac{1}{q}$ is the inverse finite-character normalization; it is not caused by the radix and it does not identify all other occurrences of the same rational glyph. For composite q, a cyclotomic scalar extension and its chosen basis must still be declared.

Theorem 8 — A radix receiver splits exactly into finite locality and cyclic winding In a radix-residue-character cell, factor

$$q=q_\parallel q_\perp,\quad q_\parallel=\prod_{p\mid b}p^{v_{p(q)}},\quad \gcd(b,q_\perp)=1.\quad\text{(SPLIT)}$$

Thus $q_\parallel$ is the largest divisor of q supported on prime axes already present in b, and CRT gives

$$C_q\simeq C_{q_\parallel}\times C_{q_\perp}.$$

Put

$$K_\parallel=\max_{p\mid q_\parallel}\left\lceil\frac{v_{p(q)}}{v_{p(b)}}\right\rceil,$$

with $K_\parallel=0$ when $q_\parallel=1$. Then

$$b^k = 0 \quad \text{in } C_{q_\parallel} \quad \text{for every } k \geq K_\parallel. \quad \text{(LOCAL)}$$

Hence the $q_\parallel$ receiver depends only on the finite suffix $\left(d_0, ..., d_{K_\parallel - 1}\right)$.

If $q_\perp > 1$, let

$$r = \operatorname{ord}_{q_\perp}(b).$$

Multiplication by b is invertible on $C_{q_\perp}$ and

$$b^{k+r} = b^k \quad \text{in } C_{q_\perp} \quad \text{for every } k \geq 0. \quad \text{(WINDING)}$$

The coprime receiver can therefore be written

$$\rho_{\perp(d)} = \sum_{c=0}^{r-1} b^c \left(\sum_{0 \leq k < K, k = c \bmod r} d_k \right) \quad \text{in } C_{q_\perp}. \quad \text{(PHASE CLASSES)}$$

The complete place-incidence word $b^k \bmod q$ is consequently periodic after the finite local cutoff. It is purely local when $q_\perp = 1$ and purely cyclic when $q_\parallel = 1$.

For a reduced rational number $\frac{a}{q}$, its radix-b expansion terminates exactly when $q_\perp = 1$, equivalently when q divides b^K for some K. In the character face, the complete population factors as

$$\hat{G}(j) = \prod_{k=0}^{K-1} \sum_{d=0}^{b-1} \omega_q^{j d b^k}, \quad \text{(DIAGONALIZED POPULATION)}$$

so the place winding changes the factors while character inversion returns the same residue fibers exactly.

Finally, let O^+ be a common positive integer occurrence set whose prime support is contained in a finite set S. The same occurrence has the radix face $r_{b(n)}$, residue face $\pi_{q(n)}$, valuation face $\nu_{S(n)}$, logarithmic length

$$\ell_{S(n)} = \sum_{p \in S} v_{p(n)} \log p,$$

and Mellin character face

$$\Phi_{s(n)} = \exp\left(-s \ell_{S(n)}\right) = \prod_{p \in S} p^{-s v_{p(n)}}. \quad \text{(VALUATION--MELLIN)}$$

Radix recharting changes the place-incidence word but leaves this common occurrence and its valuation–Mellin identity exact.

Demonstration. The two factors in (“SPLIT”) are coprime by construction, so CRT applies. For each p dividing $q_\parallel$ and $k \geq K_\parallel$,

$$v_{p(b^k)} = k v_{p(b)} \geq v_{p(q)},$$

which proves (“LOCAL”). Since b is a unit modulo $q_\perp$, its powers form a finite cyclic subgroup and have the stated multiplicative order. Grouping (“PLACE”) by congruence classes of k modulo r proves (“PHASE CLASSES”). A reduced denominator terminates in base b exactly when it divides a power of b; prime valuations make this equivalent to $q_\perp = 1$. Evaluating each group-algebra factor at ω_q^j proves the diagonalized product, and finite-character orthogonality gives its exact inverse. The final identity follows from unique factorization and $n = \prod_p p^{v_{p(n)}}$.

Boundary. Radix locality does not make the shared prime factors of b intrinsically simpler; it makes their residue face accessible from a finite suffix in that radix. Cyclic winding is not randomness. Equal residue or character faces do not reconstruct the digit word, integer occurrence, valuation address, or path. The finite Fourier transform and the Mellin character are both character receivers of abelian transport, but they are not the same transform and no direct isomorphism between their domains is asserted.

For radix b and modulus q, the complete place word is

$$(1, b, b^2, \ldots) \bmod q.$$

Its CRT decomposition distinguishes two actual mechanisms. Prime powers in q which also divide b disappear after finitely many positions, so that receiver becomes suffix-local. The coprime component cannot disappear: multiplication by b is a permutation there, and the place word winds with period $\mathrm{ord}_{q_\perp}(b)$.

Thus the familiar decimal rules are local faces of a more general transport. Modulo 5, base ten has place word $(1, 0, 0, \ldots)$; modulo 5, base three has the four-cycle $(1, 3, 4, 2, \ldots)$; modulo 11, base ten alternates $(1, -1, 1, -1, \ldots)$. The residue group has not changed. The radix has changed which digit incidences conduct to it.

The group-algebra population

$$G_{b,q,K}(X) = \prod_k \sum_{d=0}^{b-1} X^{db^k} \quad \text{in } \frac{\mathbb{Z}[X]}{X^q - 1}$$

retains every residue fiber. Evaluation at the q finite characters diagonalizes cyclic translation, and the inverse character transform returns those fibers with the factor $\frac{1}{q}$. That factor belongs to averaging over C_q; it is not a decimal feature.

On a common integer occurrence set, radix charts are invertibly related through the integer itself. Projection modulo q is not invertible unless its complete fiber is retained. The same positive occurrence also has its valuation address and exact logarithmic character

$$n \to \left(v_{p(n)}\right)_p \to \sum_p v_{p(n)} \log p \to \exp\left(-s \sum_p v_{p(n)} \log p\right) = n^{-s}.$$

This is the precise basis transport needed here: digit locality, finite cyclic phase, prime valuation, and Mellin phase are different receiver faces joined by the same occurrence, not interchangeable coordinate matrices. The bounded exact calibration preserves every member under the decimal–ternary rechart: it gives identical modulo-5 fibers while the decimal path is suffix-local and the ternary path winds through all four place phases. Its character face preserves the common residue population but cannot reconstruct either radix path.

III.4 The wheel is the arithmetic preimage of Euler admission

Theorem 9 — The prime wheel, Möbius cut, and Euler return are one transported incidence Let S be a finite set of rational primes, put

$$P_S = \prod_{q \in S} q, \quad \mathcal{C}_S = \{n \in \mathbb{N}^+ : \gcd(n, P_S) = 1\},$$

and let $p \notin S$. With $S' = S \cup \{p\}$, unique factorization gives the disjoint valuation stratification

$$\mathcal{C}_S = \{p^k m : k \geq 0, m \in \mathcal{C}_{S'}\}, \quad \text{uniquely.} \quad \text{(VALUATION STRATA)}$$

Its first difference is

$$\mathcal{C}_{S'} = \mathcal{C}_S \setminus p\mathcal{C}_S. \quad \text{(CUT)}$$

On the finite wheel

$$\mathcal{U}_S = \left(\frac{\mathbb{Z}}{P_S}\mathbb{Z}\right)^{\times},$$

each residue $r \in \mathcal{U}_S$ has the p lifts

$$r + jP_S \bmod pP_S, \quad 0 \le j < p.$$

Exactly one lift is divisible by p. Removing that lift leaves $p-1$ successors over every old residue, so

$$|\mathcal{U}_{S'}| = (p-1)|\mathcal{U}_S|. \quad \text{(COPY--CUT)}$$

Reading the surviving residues in cyclic order joins the two gaps incident to every removed lift. The primorial COPY/CUT/JOIN wheel is therefore the finite quotient of ("CUT"), not a separate prime-generating law.

If μ denotes the arithmetic Möbius function, the complete finite cut is the inclusion–exclusion identity

$$\mathbf{1}_{\mathcal{C}_S}(n) = \sum_{d|P_S} \mu(d)\mathbf{1}_{d|n}. \quad \text{(MOBIUS CUT)}$$

For $\mathrm{Re}(s) > 1$, its Dirichlet/Mellin character is

$$D_{S(s)} = \sum_{n \in \mathcal{C}_S} n^{-s} = \zeta(s) \prod_{q \in S} (1 - q^{-s}),$$

and one prime admission obeys

$$D_{S'}(s) = (1 - p^{-s}) D_{S(s)}, \quad D_{S(s)} = (1 - p^{-s})^{-1} D_{S'}(s). \quad \text{(EULER ADMISSION--RETURN)}$$

Let U_t be logarithmic translation on $L^2(\mathbb{R}, \mathrm{d}u)$ and define

$$J_q = I - q^{-\frac{1}{2}} U_{\log q}, \quad J_S = \prod_{q \in S} J_q.$$

Since the translations commute, ("MOBIUS CUT") lifts exactly to

$$J_S = \sum_{d|P_S} \mu(d) d^{-\frac{1}{2}} U_{\log d}. \quad \text{(OPERATOR CUT)}$$

Its inverse converges in operator norm and returns every valuation stratum:

$$J_S^{-1} = \sum_{m \in \mathcal{M}_S} m^{-\frac{1}{2}} U_{\log m}, \quad \mathcal{M}_S = \{m \in \mathbb{N}^+ : \text{every prime divisor of } m \text{ lies in } S\}. \quad \text{(VALUATION RETURN)}$$

The same strata have an exact projective face. For $x_0, x_1, x_2 \in \mathcal{C}_{S'}$ and every $k \ge 0$,

$$\frac{p^k x_2 - p^k x_1}{p^k x_1 - p^k x_0} = \frac{x_2 - x_1}{x_1 - x_0} \quad \text{(when the denominators are nonzero)},$$

and dilation by p^k likewise preserves every ordered cross-ratio. In the logarithmic chart this common dilation is the translation

$$\log x \to \log x + k \log p. \quad \text{(SIMILARITY--TRANSLATION)}$$

Finally, for $0 < a < 1$ put

$$J_{p(a)} = I - aU_{\log p}, \quad \mathcal{K}_{p(a)} = \left(J_{p(a)}^*\right)^{-1} J_{p(a)}^{-1}, \quad \mathcal{R}_{p(a)} = \sqrt{1 - a^2}\, J_{p(a)}^{-1}.$$

Functional calculus gives the norm-convergent identity

$$a\partial_a \log \mathcal{K}_{p(a)} = \mathcal{R}_{p(a)}^* \mathcal{R}_{p(a)} - I = \sum_{k \ge 1} a^k \left(U_{k \log p} + U_{-k \log p}\right). \quad \text{(RETURN DERIVATIVE)}$$

At $a = p^{-\frac{1}{2}}$, for a returned current represented by x,

$$W_p = (\log p)\langle x, a\partial_a \log \mathcal{K}_{p(a)} x\rangle. \quad \text{(PRIME RESPONSE)}$$

Thus the symmetric prime-power response is the logarithmic degree derivative of the metric of the exact valuation return.

Demonstration. Every $n \in \mathcal{C}_S$ has one p-adic valuation $k = v_{p(n)}$, and $m = \frac{n}{p^k}$ belongs to $\mathcal{C}_{S'}$. This proves ("VALUATION STRATA") and ("CUT").

Because p does not divide P_S, multiplication by P_S permutes the residue classes modulo p. Hence exactly one of the p displayed lifts is zero modulo p. The cardinality and cyclic gap statements follow. Arithmetic Möbius inversion gives

$$\sum_{d \mid \gcd(n, P_S)} \mu(d) = \begin{cases} 1 & \gcd(n, P_S) = 1 \\ 0 & \text{otherwise} \end{cases},$$

which is ("MOBIUS CUT").

Absolute convergence for $\mathrm{Re}(s) > 1$ permits the Euler-factor calculation of D_S. Splitting ("VALUATION STRATA") into its p-adic layers gives the reciprocal relation in ("EULER ADMISSION–RETURN").

Expanding the finite commuting product J_S gives ("OPERATOR CUT"): squarefree subsets of S are precisely divisors of P_S, and their signs are $\mu(d)$. Expanding each inverse as a Neumann series gives ("VALUATION RETURN"). Its operator norm convergence follows from

$$\sum_{m \in \mathcal{M}_S} m^{-\frac{1}{2}} = \prod_{q \in S} \left(1 - q^{-\frac{1}{2}}\right)^{-1} < \infty.$$

Common dilation cancels from the displayed ratio and from all four factors of an ordered cross-ratio. Taking logarithms turns that dilation into translation, proving ("SIMILARITY–TRANSLATION").

Since $J_{p(a)}$ is an invertible normal function of one unitary,

$$\log \mathcal{K}_{p(a)} = \sum_{k \geq 1} \frac{a^k}{k} \left(U_{k \log p} + U_{-k \log p}\right)$$

in operator norm on compact subintervals of $0 < a < 1$. Applying $a\partial_a$ gives the series in ("RETURN DERIVATIVE"). The normalized Euler-resolvent identity identifies the same series with $\mathcal{R}^*_{p(a)} \mathcal{R}_{p(a)} - I$. The established prime-power aperture-incidence formula then gives ("PRIME RESPONSE").

Boundary. This theorem identifies the exact arithmetic preimage of the semilocal Euler successor. Candidate succession, growth of the admitted place set S, and growth of the analytic support aperture are three distinct orderings; the theorem does not collapse them into one stage. Dilation similarity explains a lawful source of recurring ratio and cross-ratio faces, but those faces also occur among composites and are not primality tests. Positivity of the return metric does not determine the sign of its degree derivative. The theorem therefore does not sign the completed Weil form or prove RH; that obligation begins only after arithmetic admission, archimedean completion, moment conditioning, and support shorting meet in one receiver.

This closes a gap in the argument's direction of travel. The semilocal factor

$$\Delta_p = I - p^{-\frac{1}{2}} U_{\log p}$$

does not first appear as an analytic black box. Before Mellin transport, the same event is the exact cut

$$\mathcal{C}_{S \cup \{p\}} = \mathcal{C}_S \setminus p\mathcal{C}_S$$

of integers not divisible by the already admitted prime set. Modulo the primorial this is the measured COPY/CUT/JOIN wheel. Arithmetic Möbius inclusion–exclusion gives its finite signed difference; the Dirichlet character gives the Euler factor; and the inverse Euler operator restores the complete p-adic tower.

The source also explains one precise kind of recurring geometry. Every stratum $p^k \mathcal{C}_{S \cup \{p\}}$ is a common dilation of the same coprime body. Gap ratios and ordered cross-ratios survive that dilation, while the logarithmic chart turns it into translation by $k \log p$. This is an exact route from valuation recurrence to similar-triangle and projective faces. It is not the claim that consecutive-prime triangles classify primes: composites inhabit the same projective ratio classes, and primality remains nonclosure against the bounded factor horizon.

Three orderings must consequently remain visible:

- candidate succession determines when an irreducible axis is founded;
- the finite place set determines which divisibility cuts are active; and
- the support aperture determines which translated prime-power incidences a returned test current can meet.

Their incidences form the arithmetic–analytic bifiltration. Treating any one of these orderings as a master clock was the source of the earlier hyperfocus on the final support short.

III.5 The formal exponential generates the multiplicative strata

Let X_p be commuting formal variables and put

$$P_m = \sum_p X_p^m.$$

In the degree-completed formal power-series ring,

$$\exp\left(\sum_{m\geq 1} \frac{P_m}{m}\right) = \prod_p \left(1 - X_p\right)^{-1}.$$

Indeed,

$$-\log\left(1 - X_p\right) = \sum_{m\geq 1} \frac{X_p^m}{m},$$

and summing over p before exponentiating gives the product. This is a classical formal identity. Its geometric use here is exact:

- degree one is the prime-axis face;
- degree two consists of prime squares and products of two distinct primes;
- degree k is the k-fold symmetric composition of prime directions; and
- the term $\frac{P_m}{m}$ supplies the missing diagonal weight for m repeated traversals of one axis.

The formula-ecology experiment measured the degree-two and degree-three instances as causal finite growth. Withholding only $\frac{P_2}{2}$ left residual $-\frac{X_p^2}{2}$ on each square and no mixed residual. Withholding only $\frac{P_3}{3}$ left $-\frac{X_p^3}{3}$ on each cube and no mixed residual. Those runs establish exact embodiment of the recurrence; the identity itself is algebra, not an empirical discovery.

III.6 Zeta is one analytic specialization of the formal ecology

For $\mathrm{Re}(s) > 1$, substitute

$$X_p = p^{-s} = \exp(-s \log p).$$

Then

$$\zeta(s) = \prod_{p(1-p^{-s})}^{-1}$$

and

$$\log \zeta(s) = \sum_p \sum_{m\geq 1} \frac{p^{-ms}}{m},$$

while

$$-\zeta' \frac{s}{\zeta(s)} = \sum_p \sum_{m\geq 1} (\log p) p^{-ms}.$$

This is the first exact place where the laboratory's formula research meets RH. The exponential transports additive logarithmic scale into multiplicative amplitude. The prime axes are primitive translations by $\log p$, their powers are repetitions $m \log p$, and the logarithmic derivative gives each repetition the von-Mangoldt weight $\log p$.

The Euler product converges only in its stated half-plane. It does not by itself define a pointwise prime geometry throughout the critical strip. Analytic continuation and the archimedean place are not optional decorations; they form the completion which makes the reflection and zero spectrum available [1], [2].

IV Why e and pi appear in the completed geometry

IV.1 e is the additive-to-multiplicative transport law

The constant e is one value of the exponential, but the structural object is the map

$$u \to \exp(u), \quad \exp(u+v) = \exp(u)\exp(v).$$

Its inverse chart is $\log$. In zeta geometry this chart turns an arithmetic length $\log p$ into the character $p^{-s} = \exp(-s \log p)$. A formula for e and an Euler product for zeta are therefore not unrelated curiosities: both are finite presentations of exponential transport under different supplied generators.

The many formulas for e do not imply one hidden algebraic path among all constants. They form an atlas of nonidentical algorithms whose endpoint receiver may agree. What matters for RH is the particular exponential character of logarithmic scale and its completed return.

IV.2 pi enters through Fourier self-duality and circular normalization

Let

$$\theta(t) = \sum_{n \in \mathbb{Z}} \exp(-\pi n^2 t), \quad t > 0.$$

Poisson summation for the Gaussian gives

$$\theta(t) = t^{-\frac{1}{2}} \theta\left(\frac{1}{t}\right).$$

For $\mathrm{Re}(s) > 1$,

$$\int_0^\infty \frac{\theta(t) - 1}{2} \cdot t^{\frac{s}{2}} \, \mathrm{d}\frac{t}{t} = \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

After the endpoint terms are handled, the completed function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

is entire and obeys

$$\xi(s) = \xi(1-s).$$

Here π is not inserted because a perfect circle is physically prior. It is the exact Fourier/Gaussian normalization joining a lattice to its reciprocal lattice. The Gamma factor is the Mellin face of that Gaussian transport. Different formulas for π are different formulation paths to the same constant; the theta path is the one whose interior is relevant to zeta completion [3], [4].

IV.3 Finite theta faces owe their tails

A raw finite truncation

$$\Theta_N^d = d_0 + 2\sum_{n=1}^{N} d_n$$

is not equal to its reciprocal image

$$\Theta_N^r = r_0 + 2\sum_{n=1}^{N} r_n.$$

The exact completed charts are

$$C_d = P_{d,N} + T_{d,N}, \quad C_r = P_{r,N} + T_{r,N}.$$

When Poisson transport supplies $C_d = C_r$, the visible finite mismatch is exactly the complementary tail mismatch:

$$P_{d,N} - P_{r,N} = T_{r,N} - T_{d,N}.$$

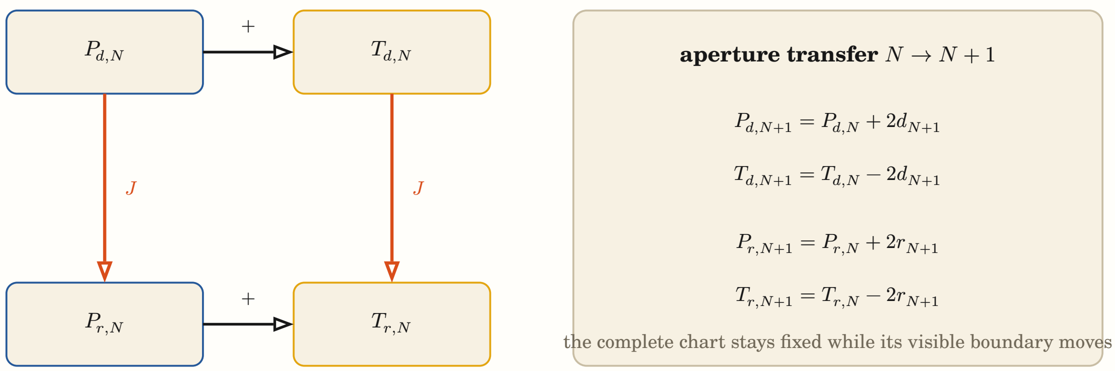

Figure 5: Finite theta truncations are not self-dual. Direct and reciprocal partials become equal only after their generative tails are included. Aperture growth transfers the same shell across the finite/remainder boundary in both charts.

The theta/Mellin and Poisson-tail experiments carried this boundary through one actual aperture growth $4 \to 5$. They established that a finite participating face and a generative remainder can move together without materializing an infinite population. They did not discover Poisson summation, prove analytic continuation, or supply Weil positivity.

V The critical line as a fixed and unitary seam

V.1 Center logarithmic scale at one half

Write

$$s = \frac{1}{2} + z, \quad z = \varepsilon + i\gamma.$$

The centered scale character is

$$\chi_{z(u)} = \exp(zu).$$

Under rebase $u \to u + a$,

$$\frac{\chi_{z(u+a)}}{\chi_{z(u)}} = \exp(\varepsilon a)\exp(i\gamma a).$$

Its magnitude is preserved for every real a exactly when $\varepsilon = 0$. Thus $\mathrm{Re}(s) = \frac{1}{2}$ is the unitary dual of scale translation in the centered chart.

That statement explains the geometric role of the critical line. It does not say that every zero must select a unitary character. The functional equation reflects the analytic body across the seam; RH asserts that all nontrivial zero singularities lie on it.

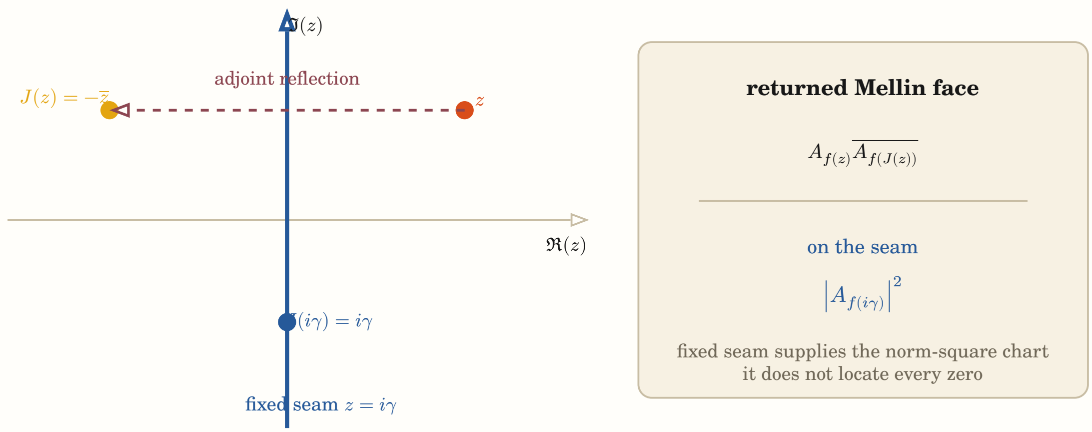

Figure 6: Centering at $s = \frac{1}{2}$ turns the functional reflection into $J(z) = -\overline{z}$. The critical line is its fixed locus, and only there does the returned Mellin pair become a norm square.

V.2 The adjoint return fixes the same seam

Lemma 10 — Mellin return and its fixed seam On $G = \mathbb{R}^{\times}_{>0}$ with $\mathrm{d}^{\times}x = \mathrm{d}\frac{x}{x}$, define

$$\mathcal{M}f(s) = \int_0^{\infty} f(x)x^s \mathrm{d}^{\times}x$$

and

$$f^{\sharp(x)} = x^{-1}\overline{f(x^{-1})}.$$

For $h = f * f^{\sharp}$, $z = s - \frac{1}{2}$, $A_{f(z)} = \mathcal{M}f\left(\frac{1}{2} + z\right)$, and $J(z) = -\overline{z}$,

$$\mathcal{M}h\left(\frac{1}{2} + z\right) = A_{f(z)}\overline{A_{f(J(z))}}.$$

The response is the norm square $\left|A_{f(z)}\right|^2$ exactly on the fixed locus $J(z) = z$, equivalently $\mathrm{Re}(s) = \frac{1}{2}$.

Demonstration. Multiplicative convolution gives $\mathcal{M}(f * g) = \mathcal{M}f\mathcal{M}g$. Substitution $y = x^{-1}$ gives

$$\mathcal{M}(f^{\sharp})(s) = \overline{\mathcal{M}f(1 - \overline{s})}.$$

Evaluating at $s = \frac{1}{2} + z$ yields the displayed factorization. Finally, $J(z) = z$ iff $z = -\overline{z}$, which is exactly $\mathrm{Re}(z) = 0$.

Boundary. This identity explains why the critical line is the adjoint fixed seam. It does not imply that every nontrivial zeta zero lies there; that implication requires positivity of the completed explicit-formula return on the full admissible test family.

This lemma is stronger than a visual symmetry claim because it names the operation which becomes positive: the returned current $h = f * f^{\sharp}$. Off the fixed seam, the two factors are evaluated at distinct reflected points. Their paired sum is real under the completed zero symmetry, but it is not sign-definite for arbitrary f.

V.3 Zeros are singularities; the line is not

Let

$$\Xi(z) = \xi\left(\frac{1}{2} + z\right).$$

Away from its zeros,

$$U(z) = \log|\Xi(z)|$$

is harmonic. Distributionally,

$$\Delta U = 2\pi \sum_{\Xi(\rho)=0} m_\rho \delta_\rho,$$

with the usual local convention for the planar Laplacian. A loop around a zero has argument-principle charge

$$\frac{1}{2\pi i}\int_\gamma \mathrm{d}\log\Xi = \text{enclosed zero multiplicity.}$$

This is the rigorous part of the laboratory's potential-field and vortex language. A zero is a source singularity of the log-amplitude and a phase vortex. The critical line is a reflection/fixed seam; it is not itself a source. Nor is there a one-prime/one-zero matching. The explicit formula relates complete populations.

VI Knot monodromy joins the transport-atlas and spectral routes

VI.1 Euler factors are primitive return factors

For $\mathrm{Re}(s) > 1$, put

$$\ell_p = \log p.$$

Then the Euler product and its logarithmic derivative are exactly

$$\zeta(s) = \prod_p \left(1 - e^{-s\ell_p}\right)^{-1},$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m\geq 1} \ell_p e^{-sm\ell_p}. \quad \text{(ORBIT)}$$

This is the formal dynamical-zeta grammar:

- p is one primitive closed return of length ℓ_p;
- p^m is its m-fold traversal; and
- the von Mangoldt weight is the primitive length carried through every repetition.

Deninger's prime–knot–orbit dictionary makes this relation geometric: closed prime points correspond to periodic orbits, their lengths are $\log p$, and the explicit formula is sought as a transversal trace relation [5]. A periodic orbit embedded in a three-dimensional ambient body is a knot up to isotopy. The knot face therefore refines the primitive-orbit face by retaining how the orbit sits inside the total return space.

Equation ("ORBIT") is exact without such an ambient body. What remains open is the natural completed flow whose primitive orbit trace is exactly the prime-power side, whose fixed-point contribution is the archimedean place, and whose generator has the complete nontrivial zero spectrum. Declaring a formal orbit for each prime does not construct that flow.

VI.2 One closed precession returns a monodromy

For a fibered knot, the complement is a mapping torus of a surface F under one return map

$$h : F \to F.$$

The induced homological monodromy h_* carries the return independently of any one braid diagram. Equivalently, for some $\varepsilon \in \{\pm 1\}$ and $k \in \mathbb{Z}$,

$$\Delta_K(t) = \varepsilon t^k \det\Big(tI - h_*|_{H_1(F)}\Big). \quad \text{(ALEX)}$$

Crossings in a braid word are therefore a presentation of a return map; they are not its complete identity [6].

Now center the completed spectral parameter,

$$z = s - \frac{1}{2},$$

and convert one positive return length ℓ into a multiplier

$$\lambda_\ell(z) = e^{-\ell z}.$$

The completed antiholomorphic involution sends z to $-\overline{z}$, hence

$$\lambda_\ell(-\overline{z}) = \frac{1}{\overline{\lambda_\ell(z)}}. \quad \text{(RECIP)}$$

Thus the completed zero orbit becomes reciprocal-conjugate monodromy data. The fixed seam is

$$\mathrm{Re}(z) = 0 \quad \leftrightarrow \quad |\lambda_\ell(z)| = 1.$$

Phase around the return is $-\ell\,\mathrm{Im}(z)$; normal growth or decay per traversal is $-\ell\,\mathrm{Re}(z)$. This is the algebraic content of the precessing polarized axis relevant to RH.

The full Euler weight is

$$e^{-s\ell} = e^{-\frac{\ell}{2}}\lambda_\ell(z).$$

The factor $e^{-\frac{\ell}{2}}$ is the centered half-density carried by the orbit; $\lambda_\ell(z)$ is its phase/normal return. Calling the full Euler weight unitary would therefore collapse two distinct faces.

VI.3 Reciprocal topology is not the critical-line theorem

Lemma 11 — Reciprocal monodromy does not force unit modulus Let

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then

$$M^T \Omega M = \Omega,$$

so M preserves the oriented symplectic pairing. Its characteristic polynomial and eigenvalues are

$$\det(tI - M) = t^2 - 3t + 1, \quad \lambda_\pm = \frac{3 \pm \sqrt{5}}{2}.$$

Hence

$$\lambda_+\lambda_- = 1, \quad \lambda_+ > 1, \quad 0 < \lambda_- < 1.$$

The return therefore has exact reciprocal symmetry while neither eigenvalue lies on the unit circle.

The matrix is a homological monodromy representative for the fibered figure-eight knot, and its characteristic polynomial is the knot's Alexander polynomial up to multiplication by a unit and the usual choice of sign.

Demonstration. Direct multiplication gives $M^T\Omega M = \Omega$. The determinant expansion gives $t^2 - 3t + 1$, whose quadratic roots are the displayed reciprocal pair. Since $\sqrt{5} > 1$, one root is greater than one and the other is its positive reciprocal.

Boundary. A symplectic or reciprocal return preserves oriented incidence, not a positive Hermitian norm. This is the exact finite-dimensional reason that the completed reflection $s \mapsto 1 - s$ cannot by itself imply RH. Forcing unit modulus requires an independently positive metric preserved by the return, not another reciprocal presentation.

The figure-eight return is an exact warning against deriving RH from reflection alone. Its homological monodromy preserves the oriented intersection form and its Alexander polynomial is reciprocal, yet its two eigenvalues lie on opposed inside/outside radial sheets. In the centered zeta chart this is the difference between

$$z \mapsto -\overline{z}$$

pairing the sheets and proving $\mathrm{Re}(z) = 0$.

The example also clarifies what kind of geometry is insufficient. A symplectic, alternating, or indefinite form can preserve reciprocal incidence while admitting hyperbolic expansion. The desired return must preserve an independently positive Hermitian metric. This is not an academic distinction: it is exactly the difference between the completed functional equation and Weil positivity.

VI.4 Positive covariant monodromy fixes the seam

Theorem 12 — A positive covariant monodromy law fixes the critical seam Let $\mathcal{E} \to X$ be a complex vector or Hilbert bundle. Suppose each fiber $\mathcal{E}_x$ carries an independently constructed positive-definite Hermitian form H_x, and every admitted path $\gamma : x \to y$ carries an invertible transport

$$T_\gamma : \mathcal{E}_x \to \mathcal{E}_y$$

satisfying

$$H_y\big(T_\gamma v, T_\gamma w\big) = H_x(v, w). \quad \text{(COV)}$$

Then the holonomy $M_\gamma = T_\gamma$ of every closed admitted path based at x is H_x-unitary and

$$\mathrm{spec}\big(M_\gamma\big) \subset \{\lambda \in \mathbb{C} : |\lambda| = 1\}. \quad \text{(UNIT)}$$

In particular, let $U_t = e^{tA}$ be a strongly continuous returned flow satisfying (COV) in one fiber for every real t. Every eigenvalue z of A is purely imaginary. If an independently constructed trace correspondence identifies the complete nontrivial zero spectrum of $\xi(s)$ with the point spectrum of

$$\Theta = \frac{1}{2}I + A,$$

including multiplicity and every arithmetic and archimedean trace term, then every nontrivial zero has real part $\frac{1}{2}$ and RH follows.

Demonstration. For a closed path, (COV) gives

$$M_\gamma^* H_x M_\gamma = H_x.$$

Conjugating by the positive square root $H_x^{\frac{1}{2}}$ turns M_γ into an ordinary unitary operator, whose spectrum lies on the unit circle.

If $Av = zv$ with $v \neq 0$, then $U_t v = e^{tz} v$. Norm preservation gives

$$H(v, v) = H(U_t v, U_t v) = e^{2t\,\mathrm{Re}(z)} H(v, v)$$

for every real t. Positivity of $H(v, v)$ forces $\mathrm{Re}(z) = 0$. Under the stated independent spectral correspondence, every nontrivial zero is $\rho = \frac{1}{2} + z$ with $\mathrm{Re}(z) = 0$.

Boundary. The theorem does not construct the bundle, positive metric, returned flow, generator, or trace correspondence. Defining Θ from the known zero set would be circular. In the current semilocal programme the archimedean Sonin/prolate amplitude supplies one positive-semidefinite base form. To use it as the displayed metric one must additionally prove either nondegeneracy or invariance of its radical and then work on the positive quotient, while showing that no zero mode is discarded

> there. Its receiver-covariant continuation through finite-place and support growth, together with a trace identity identifying the generator's spectrum with the completed zeros, remains the proof-bearing object.

The theorem joins two closure routes which were previously listed separately. The transport-atlas route must construct the positive metric and its covariant path law. The spectral route must construct the returned flow, its generator, and the complete trace correspondence. If both structures belong to the same body, unitary holonomy forces the centered generator onto the imaginary axis and the zero spectrum onto the critical seam.

The current archimedean theorem supplies a genuine initial cell. On its conditioned current space,

$$W_{\infty(g*g^*)} - \Sigma_{\infty(g)} = \left\|A_2\xi_g\right\|^2. \quad \text{(BASE)}$$

Consequently

$$H_\infty(g) = \Sigma_{\infty(g)} + \left\|A_2\xi_g\right\|^2$$

is independently nonnegative and equals the declared archimedean response there. It is not a form invented from the zero set. The published result does not by itself prove that this form is nondegenerate. To enter the positive-covariant-monodromy theorem it must either become a genuine metric, or its radical must be transported exactly and the complete zero-bearing trace must survive passage to the positive quotient.

For a general receiver, the required successor has the constructive form

$$H_{S,R}(f) = \Sigma_{S,R}(f) + \left\|A_{S,R}f\right\|^2, \quad \text{(METRIC)}$$

with $A_{S,R}$ built from the completed arithmetic and archimedean transport, not defined as a square root of a response whose positivity is being sought. The two missing identities are

$$H_{S,R}(f) = Q_{W(f)} \quad \text{on the complete admitted current space,} \quad \text{(TRACE)}$$

and, for every admitted incidence comparison $\alpha : e \to e'$,

$$H_{e'}(T_\alpha f, T_\alpha g) = H_e(f, g). \quad \text{(COVARIANCE)}$$

On a closed triangular return, ("COVARIANCE") makes the holonomy unitary. This does not require one clock to order all three edges; it requires actual composable incidences and the same anchored current. Along one finite-place successor, ("TRACE") requires the independently constructed remainder amplitude to obey the already-derived signed recurrence

$$\left\|A_{S\cup\{p\},R}f\right\|^2 - \left\|A_{S,R}f\right\|^2 = -W_p\left(f * f^\sharp\right) - \kappa_{S,p}(f). \quad \text{(SEAM)}$$

Equation ("SEAM") is not a demand that a prime contribute a positive piece. It is the signed change between two positive contemporary metrics.

VI.5 Connected sum explains why the seam is not additive

Brittenham and Hermiller prove that the unknotting number is not additive under connected sum. In particular, for the $(2, 7)$ torus knot K and its mirror $\overline{K}$,

$$u\left(K \# \overline{K}\right) \leq 5 < 6 = u(K) + u\left(\overline{K}\right).$$

Their crossing changes are separated by isotopies: one event changes the complete standing topology, after which another diagram exposes a newly available consequential crossing [7].

This is not identified numerically with the completed defect. Its exact structural consequence is narrower and important: prime decomposition of an object does not imply additivity of a path metric on transformations of the composite. The RH recurrence already has the corresponding non-product shape,

$$D_{S\cup\{p\}} = D_S - W_p - \kappa_{S,p}.$$

The connection current $\kappa_{S,p}$ records the reorientation of the whole receiving aperture. Removing it in order to obtain independent positive prime blocks would commit the same composition error that the knot theorem rules out for unknotting distance.

VI.6 Knot presentation gives a local grammar, not a favorable orientation

Definition 13 — Situated knot receiver and its crossing field Let $K : S^1 \to M^3$ be an oriented tame embedding in an oriented three-manifold. A **situated knot receiver**

$$\rho = (\Sigma_\rho, \pi_\rho, \nu_\rho)$$

consists of an oriented receiving surface, a generic projection $\pi_\rho : M^3 \to \Sigma_\rho$, and a coorientation or depth law ν_ρ sufficient to order the two preimages of every transverse double point.

Its crossing population is

$$\mathcal{C}_{\rho(K)} = \Big\{\{s,t\} : s \neq t, \pi_{\rho(K(s))} = \pi_{\rho(K(t))}, \text{ the projected tangents are transverse}\Big\}.$$

Together with the source orientation, depth order, and local crossing sign, this population forms the received diagram $D_{\rho(K)}$.

The embedded knot K, its ambient-isotopy class $[K]$, its received diagram $D_{\rho(K)}$, and the scalar crossing count $\left|\mathcal{C}_{\rho(K)}\right|$ are four different objects. A crossing is a receiver-relative incidence between two source parameters. It is not an intrinsic vertex of the embedded knot.

> Boundary. The receiver must be generic for the crossing population to consist of transverse double points. Tangencies, cusps, triple points, and failed depth order belong to the projection discriminant. Switching one over/under relation is a crossing change and may change knot type; it is not a Reidemeister re-presentation of the same knot.

For an embedded knot $K : S^1 \to M^3$, a generic receiver projection $\pi_\rho : M^3 \to \Sigma_\rho$ produces a diagram crossing when two distinct source parameters project transversely to the same received point. The crossing therefore belongs to the pair (K, ρ), not to K alone. Moving ρ through its projection discriminant changes the diagram by Reidemeister events while retaining the ambient-isotopy class.

Three further presentation laws must remain distinct. Alexander proves that every oriented link is the closure of a braid; Markov moves relate braid presentations with isotopic closures; and Schubert decomposition gives prime factors under connected sum [8], [9], [10], [11]. Thus a diagram, an axis-relative braid word, a closed knot type, and its prime factorization are not interchangeable “faces” with the same retained data.

This makes the perspective claim exact but limited. A receiver can change the visible crossing set. It cannot move a completed-zeta zero onto the critical line, change the inertia of the Weil form, or turn a nontrivial knot into an unknot by re-presentation alone.

VI.7 Tangles make contextual substitution exact

Definition 14 — Contextual tangle equivalence and receiver-exact substitution Fix an oriented marked boundary B. Let $\mathrm{Tan}(B)$ be the collection of tangles with boundary B, modulo ambient isotopy relative to B. A compatible exterior context C composes with an interior tangle T by gluing, written $C[T]$.

For a declared compositional receiver family $\mathcal{R}$, define **contextual receiver equivalence**

$$T \underset{\mathcal{R}}{\approx} T' \quad \text{iff} \quad R(C[T]) = R(C[T'])$$

for every $R \in \mathcal{R}$ and every admitted compatible context C. Because contexts themselves compose, $\underset{\mathcal{R}}{\approx}$ is a congruence: an equivalent interior may be substituted inside every larger admitted exterior.

A receiver-exact tangle compression is a factorization through the quotient

$$q_{\mathcal{R}} : \mathrm{Tan}(B) \to \mathrm{Tan}\,\frac{B}{\underset{\mathcal{R}}{\approx}}.$$

It may discard distinctions between interiors only at the stated receiver and context boundary.

After linearizing over a ring R_0, a skein relation is a local module relation such as

$$\alpha[T_+] + \beta[T_-] + \gamma[T_0] = 0.$$

It licenses linear replacement inside any compatible context in the resulting skein module. It does not assert $T_+ \approx T_-$, $T_+ \approx T_0$, or ambient isotopy of the three tangles.

> Boundary. Equal closure under one receiver is weaker than contextual equivalence. Contextual equivalence is weaker than causal-soul equivalence unless the receiver family is jointly conservative. A skein quotient can identify distinct knots or retain torsion, and its coefficients may be signed or complex. Skein reduction is therefore not automatically positive, metric-decreasing, unique, or computationally cheaper.

A tangle exposes a marked boundary, so an exterior context $C[-]$ can hold that boundary fixed while replacing the interior. Rational tangles provide a particularly sharp ratio face: their twist words give continued fractions, and their isotopy classes are classified by $\mathbb{Q} \cup \{\infty\}$ [12]. This is a real relation among projective ratios, crossing interiors, and boundary-relative topology, not evidence that every rational is secretly a knot.

For links L_+, L_-, L_0 which differ only in one local disk, a skein equation such as

$$\nabla_{L_+}(z) - \nabla_{L_-}(z) = z\nabla_{L_0}(z)$$

is a linear relation among nonidentical alternatives. A skein module quotients the free module of link presentations by such declared local relations [13]. It does not declare the three links isotopic, and the plus sign in L_+ records crossing orientation rather than positivity of a quadratic form.

The exact compression principle is consequently contextual:

$$T \underset{\mathcal{R}}{\approx} T' \quad \Rightarrow \quad R(C[T]) = R(C[T']) \ \text{ for every declared } \ C \ \text{ and } \ R.$$

One matching closure or invariant value is insufficient. This is the topological form of preserving every admitted future consequence while factoring an inactive interior.

VI.8 A skein-shaped RH route has one additional obligation

Corollary 15 — An exhaustive positive completed-zeta skein carrier would imply RH Let $\mathcal{Z}$ be a boundary-typed involutive linear tangle category whose objects and morphisms present the complete completed-zeta test currents, prime admissions, support changes, archimedean returns, and receiver recharts. Suppose:

1. proved local relations give a presentation-independent and exhaustive quotient of those currents;
2. an involution-preserving evaluation functor sends gluing to operator composition and closure to the complete Weil quadratic response;
3. each generating local relation supplies, independently of the desired sign, the natural contractive filler between the established amplitude functors X and Y; and

4. these local fillers are compatible with every relation, so they descend to one natural transformation Γ on the quotient.

Then

$$Y_j = \Gamma_j X_j, \quad \|\Gamma_j\| \le 1$$

for every completed test object j. Hence

$$Q_{W(f)} = \|X_j f\|^2 - \|Y_j f\|^2 \ge 0$$

on the exhaustive Weil test space, and the Riemann Hypothesis follows by the Weil criterion.

Demonstration. Presentation independence makes the component assigned to a current independent of its braid, diagram, or reduction path. Compatibility with gluing and receiver arrows makes the components natural. The assumed local contractivity therefore gives the natural contraction required by the natural-contraction RH corollary, which yields Weil positivity and RH.

Boundary. No such completed-zeta skein category or positive evaluation functor has been constructed here. Ordinary Reidemeister, Markov, Alexander, Conway, Jones, HOMFLY, or Kauffman relations do not satisfy the stated zeta trace identity by default. A standard skein relation may have signed or complex coefficients and therefore supplies no positivity on its own. The missing object remains the arithmetic–archimedean natural contraction, now with a possible local presentation grammar.

The completed-zeta programme can use this grammar only after supplying an arithmetic interpretation. Its objects would be admitted test currents with prime, support, Mellin, and archimedean boundaries. Its local substitutions would have to be actual Euler, support-growth, and Poisson–Mellin identities. Closing and evaluating the complete construction would have to return exactly Q_W, independently of the selected presentation.

Ordinary skein recursion does not give the missing sign: its coefficients may be signed, Laurent, or complex. A proof-bearing evaluation must additionally be star-compatible and completely positive or contractive, so that its local substitutions independently construct components of the natural filler Γ rather than assuming $\|Yf\| \le \|Xf\|$.

Knot-causal topology therefore sharpens the local-to-global architecture but does not replace the present wall. The next decisive knot-shaped object is one exact Euler/support/Poisson–Mellin substitution square whose evaluation on the already-derived dyadic character cell is both presentation-coherent and contractive.

VI.9 Higher linking supplies relations which orbit length forgets

Arithmetic topology makes the networked-prime intuition rigorous in another direction. Knots correspond to prime ideals; linking numbers correspond to power-residue symbols; higher Milnor invariants correspond to higher residue and Massey data; and knot-complement coverings correspond to ramified number field extensions. Morishita gives examples in which three primes form the arithmetic analogue of a Borromean link: pairwise linking vanishes while the triple relation does not [14].

This supplies genuine candidate interior for a prime-orbit body. Scalar lengths $\log p$ determine the ordinary Euler factors, but they do not encode all pairwise and higher monodromy of the ambient arithmetic topology. Any use of that richer topology in an RH proof nevertheless owes a trace reduction: after every link and boundary contribution is included, the returned distribution must be exactly the classical explicit formula. Additional topology cannot be admitted merely because it is expressive.

VI.10 The sharpened proof-bearing object

The proof target can now be written as one joined construction rather than a choice between analogy and positivity:

$$\Big(E \to B_{\text{arith}} \times B_{\text{arch}} \times B_{\text{recv}}, \big\{T_{(i,j)}\big\}, M_{123}, H, \Theta, \{\gamma_p\}, \mathrm{Tr}\Big).$$

It must satisfy all of the following:

1. γ_p is a primitive closed orbit of length $\log p$, and its repeated traversals give every prime-power term in (“ORBIT”).
2. The archimedean place supplies the required fixed-point or boundary contribution.
3. Θ is constructed before its spectrum is read, and the trace identity identifies its complete nontrivial spectrum with the completed zeros.
4. H is independently positive and begins with the known Sonin/prolate amplitude (“BASE”).
5. The arithmetic, archimedean, and receiver lineages meet through actual incidences; their cross-frame maps have declared domains and compose to M_{123} without a master parameter.
6. Every finite-place, support, completion, and formulation transition satisfies (“COVARIANCE”), including its aperture connection and seam terms, and every closed triangular return preserves H.
7. The nested support induction covers every compact admissible current, and the spectral correspondence loses no zero-bearing mode.
8. Any null radical of the base form is covariantly invariant and contains no omitted zero-bearing mode, or else the completed form is proved nondegenerate.

If this body is constructed, the positive-covariant-monodromy theorem places the spectrum of $\Theta - \frac{1}{2}$ on the imaginary axis and proves RH. The first unsupported construction is no longer “some knot interpretation,” nor a single map moving through an enlarged total field. It is a common three-frame incidence carrying the finite-place continuation $A_{S,R}$ and the cross-frame transports whose returned composition satisfies (“TRACE”), (“COVARIANCE”), and (“SEAM”), with controlled radical, while the same trace body reproduces (“ORBIT”).

VII The explicit formula is the population-level bridge

VII.1 Fix one sign convention

On $G = \mathbb{R}^{\times}_{>0}$, let $h = f * f^{\sharp}$ as above. Define the finite-place face

$$W_{\text{fin}(h)} = \sum_p (\log p) \sum_{m \geq 1} \big(h(p^m) + p^{-m} h(p^{-m})\big).$$

The archimedean face in the convention used by the laboratory record is

$$W_{\infty(h)} = (\log(4\pi) + \gamma_E) h(1) + \int_1^{\infty} \frac{h(x) + x^{-1} h(x^{-1}) - 2x^{-1} h(1)}{x - x^{-1}} \, \mathrm{d}x.$$

Then put

$$Q_{W(f)} = \mathcal{M}h(0) + \mathcal{M}h(1) - W_{\text{fin}(h)} - W_{\infty(h)}.$$

For a standard admissible test class and the prescribed symmetric zero sum, the completed explicit formula gives

$$Q_{W(f)} = \sum_{\rho} \mathcal{M}h(\rho).$$

Equivalent sources place signs and endpoint terms differently; a proof must fix one convention and carry every term. The content is invariant: prime powers plus the infinite place equal the completed zero response [15], [16], [17].

VII.2 Weil positivity is the exact RH closure

On RH, every nontrivial zero has $\rho = \frac{1}{2} + i\gamma$, so the Mellin-return lemma gives

$$\mathcal{M}h(\rho) = \left|A_{f(i\gamma)}\right|^2 \geq 0.$$

Conversely, Weil's criterion states—under the full admissible family—that

$$\text{RH} \quad \leftrightarrow \quad Q_{W(f)} \geq 0 \quad \text{for every admissible } f.$$

This is the proof obligation. It is neither a statistical trend nor an instruction to verify more zeros. It asks for the completed arithmetic distribution to have no negative direction on the complete adjoint-return space.

The displayed inequality is written in one fixed chart. Parametrically, it asks whether the Hermitian section W_λ is positive semidefinite in every anchored formulation fiber representing that test space. A family with $q_{\lambda(f_\lambda)} < 0$ is entirely possible before RH is known. Finding a different g_λ with positive response does not answer it; one must either control the original direction or transport it covariantly to a proved-positive base cell.

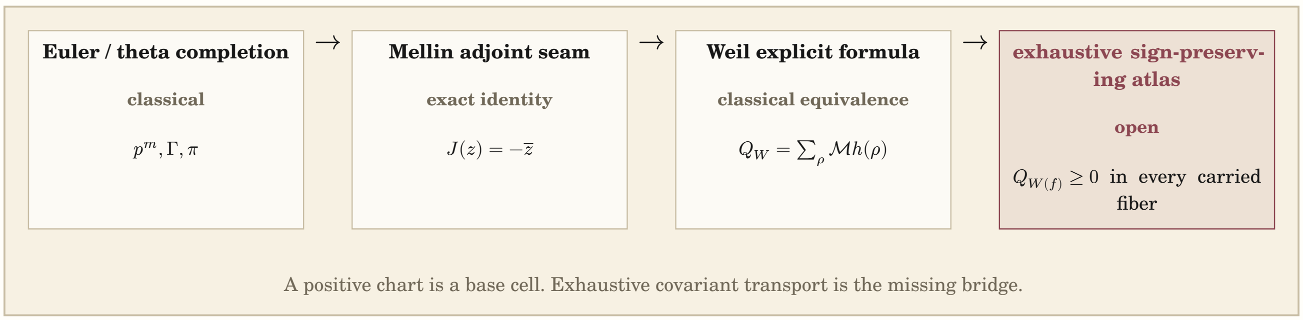

Figure 7: The proof spine and its stopping condition. Everything to the left of the final arrow is either classical or derived exactly in this paper. The final global sign is the coordinate expression of an exhaustive transported-cone statement and remains unproved.

VII.3 One exact triangular probe

Let $N > 1$, $R = \log N$, and take the logarithmic box

$$a_{N(u)} = 1_{[-\frac{R}{2}, \frac{R}{2}]}(u).$$

Put

$$f_{N(x)} = x^{-\frac{1}{2}} a_{N(\log x)}.$$

Then

$$h_{N(x)} = x^{-\frac{1}{2}}(R - |\log x|)_+$$

and

$$A_{N(z)} = \frac{2\sinh\left(z\frac{R}{2}\right)}{z}.$$

The finite prime face is exactly

$$W_{\text{fin}(h_N)} = 2 \sum_{p^m < N} (\log p) p^{-\frac{m}{2}} (R - m \log p).$$

On the seam,

$$A_{N(i\gamma)} = \frac{2\sin\left(\gamma\frac{R}{2}\right)}{\gamma}.$$

This is a useful geometric tent: support in log scale becomes a triangular return, and only finitely many prime powers cross it. It is not smooth at the boundary and the one-parameter family is not the full Weil test family. Passing every such finite probe would not by itself prove RH.

VII.4 What the bounded machine worlds established

The laboratory has several exact finite instruments. Their proper evidential role is summarized here without scalar spectacle.

World	Capability established	Boundary retained
Prime spectral	Exact wheel refinement, prime powers, ordered gaps, and spectral receiver faces can cross one machine boundary while order and orientation survive.	Finite Ramanujan modes are not zeta zeros; no RH implication.
Formula ecology and growth	A finite generator can grow the Euler exponential and preserve localized OPEN residuals when a repeated-prime term is withheld.	Source laws were supplied; no analytic continuation or formula discovery.
Theta/Mellin and Poisson tail	Direct/reciprocal involution, fixed seam, positive finite metric, moving aperture, and generative remainder survive causal replacement.	The Gaussian transform was inherited; no completed Weil carrier.
Interval explicit formula	Exact prime-power sections and certified critical-line zero waves share a receiver with oriented residual and ordered aperture.	A finite critical-line population cannot constrain unobserved zeros.
Windowed explicit formula	Changing support adds and removes exact arithmetic material; the same aperture after intervening passages reuses constituents along new paths.	Pointwise finite residual and finite Weil faces remain distinct; universal completion is open.

The interval worlds also gave an important negative result: adding more critical-line modes did not make the finite pointwise residual decrease monotonically. Waves reinforce or oppose according to phase. Any argument which selects the "best" truncation by a scalar residual discards the ordered construction it is meant to explain.

VIII Support and place form a dynamic receiver

VIII.1 The support filtration supports induction, not finite verification

In logarithmic scale define

$$\mathcal{A}_R = \left\{a \in C_c^{\infty(\mathbb{R})} : \mathrm{supp}(a) \subset \left[-\frac{R}{2}, \frac{R}{2}\right]\right\}, \quad a^{\star(u)} = \overline{a(-u)}.$$

Then

$$\mathcal{A}_R^\star = \mathcal{A}_R, \quad \mathcal{A}_R * \mathcal{A}_S \subset \mathcal{A}_{R+S},$$

and

$$\operatorname{supp}(a * a^{\star}) \subset [-R, R].$$

Thus the apertures form one filtration of an involutive convolution algebra. An individual aperture is not closed under composition, and every aperture is still infinite-dimensional. This rules out finite enumeration, but it does not rule out mathematical induction: a uniform theorem relating one complete aperture to its successor is precisely the useful structure.

A prime power enters the returned arithmetic face exactly at

$$m \log p \leq R.$$

The entry events are discrete; the test aperture and archimedean face vary continuously.

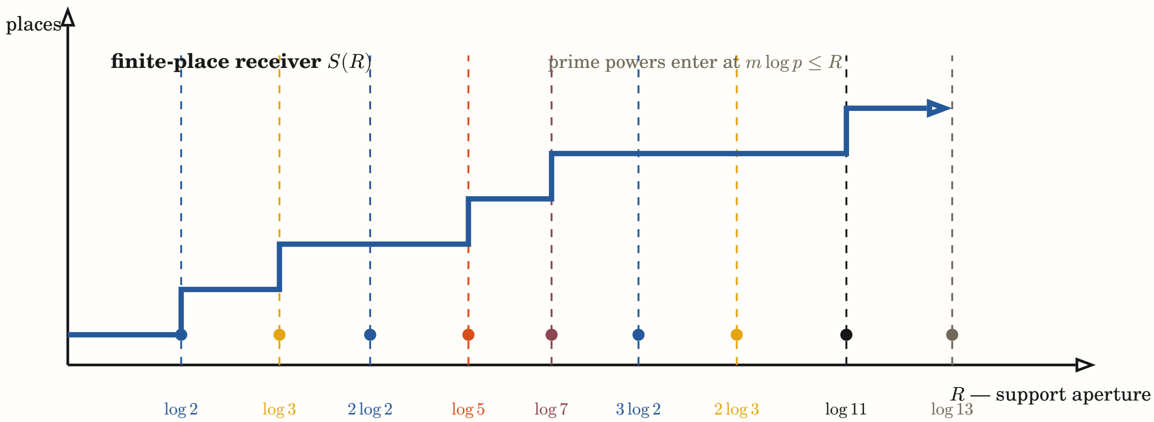

Figure 8: Support and place form two coupled axes. Increasing R admits discrete prime-power incidences but also continuously enlarges an infinite-dimensional test aperture. The staircase records which finite places are required; it is not a finite-dimensional proof.

VIII.2 Why the static enlarged-field picture was rejected

An earlier draft represented a new place as a block appended to an unchanged old Hermitian form. That picture is useful only after a fixed ambient decomposition and metric have already been supplied. Here the semilocal Hilbert structure, orthogonality, and receiving aperture all depend on the finite place set.

The correct event is therefore

$$(H_S, g_S, P_S) \xrightarrow{J_{S,p}} \big(H_{S\cup\{p\}}, g_{S\cup\{p\}}, P_{S\cup\{p\}}\big).$$

The arrow is transport between complete receiver frames. Writing the successor in predecessor coordinates does not mean the predecessor survives as an unchanged crystalline block.

Even this corrected arrow is only the arithmetic/place edge of the full event. The archimedean completion and the receiving current/aperture have their own lineages. Treating them as fixed background while S changes is a valid local cell only when that cut is declared; it is not yet the closed three-frame return.

VIII.3 The published semilocal body

For a finite place set S containing ∞, the semilocal construction uses

$$A_S = \prod_{v \in S} \mathbb{Q}_v, \quad X_S = \frac{A_S}{\Gamma_S},$$

and a receiver Hilbert space

$$H_S = L^2(X_S)^{K_S}.$$

It carries additive Fourier transform, a unitary positive-scale action U_S, and Sonin subspaces

$$\mathrm{Son}_{S,\lambda} = \{\xi \in H_S : \xi = 0 \ \text{ and } \ \mathcal{F}_S\xi = 0 \ \text{ on the phase-space hole } \ |x|_S < \lambda\}.$$

Let $P_{S,\lambda}$ be the corresponding native orthogonal projection.

Connes, Consani, and Moscovici construct bounded Hilbertian intertwiners θ_S from the real receiver to these semilocal receivers and prove stability of the Sonin spaces as finite places are added [18]. Connes and Consani separately identify the finite Euler factors in the quasi-inner/Hardy presentation and show that the useful compact property belongs to the co-present product including the archimedean factor, not to an isolated finite place [19].

For a compact log-current a, the integrated scale action is

$$\Theta_{S(a)} = \int_{\mathbb{R}} a(u) U_{S(u)} \,\mathrm{d}u.$$

Whenever the following operator is trace class,

$$B_{S,\lambda}(a) = \Theta_{S(a)} P_{S,\lambda} \Theta^*_{S(a)} \geq 0$$

and

$$\Sigma_{S,\lambda}(a) = \mathrm{Tr}\big(B_{S,\lambda}(a)\big) \geq 0.$$

This sign is noncircular: it is an ordinary adjoint square compressed by an independently defined Sonin projection. The open question is whether the completed Weil response dominates it at every relevant support and place.

VIII.4 The published maps already close the space-level induction

Theorem 16 — Prime admission is an exact dual Sonin induction Let S be a finite set of places containing ∞, let $p \notin S$, and put $S' = S \cup \{p\}$, $a_p = p^{-\frac{1}{2}}$, and $\ell_p = \log p$. Use the logarithmic translation convention

$$\big(U_p h\big)(u) = h\big(u - \ell_p\big).$$

The published semilocal Sonin maps identify, for every $\lambda > 0$,

$$\theta_S : \mathrm{Son}_{\infty,\lambda} \to \mathrm{Son}_{S,\lambda} \quad \text{isomorphically.}$$

In the common real logarithmic chart, adjoining p acts by

$$\Delta_p = I - a_p U_p, \quad \theta_{S'}\theta_S^{-1} = \Delta_p. \quad \text{(ADMISSION)}$$

With the opposite translation convention, U_p is replaced by U_p^* and none of the following identities changes.

The dual semilocal population return is

$$\mathcal{E}_p = \big(\Delta_p^*\big)^{-1} = \sum_{m \geq 0} a_p^{m(U_p^*)^m}, \quad \Delta_p^* \mathcal{E}_p = I. \quad \text{(DUAL RETURN)}$$

Hence

$$\langle \Delta_p f, \mathcal{E}_p g \rangle = \langle f, g \rangle. \quad \text{(PAIRING)}$$

The difference arm and the prime-power sum are therefore dual faces of one admission event; neither is an independently signed contribution.

The maps in ("ADMISSION") are independent of λ. Consequently, for $\mu \geq \lambda$, the nested-aperture square is the identity

$$\theta_{S,\lambda}\iota_{\infty,\mu,\lambda} = \iota_{S,\mu,\lambda}\theta_{S,\mu},$$

commutes. Distinct prime additions commute as well:

$$\Delta_p\Delta_q = \Delta_q\Delta_p. \quad \text{(INDUCTION COHERENCE)}$$

On the four-cell dyadic section, let R be the core–shell involution and $P_+ = \frac{I+R}{2}$. The admission metric is

$$G_2 = \Delta_2^*\Delta_2 = (1+a^2)I - aR, \quad a = 2^{-\frac{1}{2}}.$$

Therefore the local Poisson filler is the lower-metric spectral face

$$A^*_{\mathrm{Pois},2}A_{\mathrm{Pois},2} = \frac{\log 2}{2}((1+a+a^2)I - G_2) = \frac{\log 2}{\sqrt{2}}P_+. \quad \text{(SPECTRAL FACE)}$$

It is not a restriction of either Δ_2 or $\mathcal{E}_2$: both are invertible, whereas

$$\ker\big(A_{\mathrm{Pois},2}\big) = P_-\mathcal{V}_4$$

has dimension two.

Demonstration. Proposition 4.6 and Theorem 4.6 of the cited semilocal construction give

$$w_{S'}\theta_{S'}f = g(\lambda) - p^{-\frac{1}{2}}g\left(\frac{\lambda}{p}\right)$$

and identify every real Sonin space with its semilocal counterpart. Passing to $u = \log\lambda$ gives ("ADMISSION"). Since $\|a_pU_p\| < 1$, the Neumann series in ("DUAL RETURN") converges in operator norm. The pairing identity follows immediately. It is the relative form of the published equality

$$\langle\theta_S f, \eta_S g\rangle = \langle f, g\rangle.$$

The same θ_S serves every aperture, so restriction to nested Sonin subspaces commutes. Translations by $\log p$ and $\log q$ commute, proving ("INDUCTION COHERENCE").
On the dyadic character cell, R has eigenvalue $+1$ on P_+ and -1 on P_-. Thus

$$(1+a+a^2)I - G_2 = a(I+R) = 2aP_+.$$

Multiplication by $\frac{\log 2}{2}$ proves ("SPECTRAL FACE"). Finally, $a < 1$ makes Δ_2 and its adjoint inverse injective, while the periodization theorem gives the displayed nonzero kernel. An injective map cannot restrict to that quotient.

Boundary. The place and support **spaces** already form a coherent inductive system. The dyadic periodization is a quadratic defect face of its admission metric, not the successor map itself. These identities do not sign the completed Weil form. What remains is a form-level support successor which couples the old positive body, the newly admitted support shell, and—at a prime step—the dual admission/return pair.

The wheel–Euler theorem above supplies the arithmetic source of this transport; the published Sonin maps supply its complete receiver rebase. This changes the target materially. For every λ, the same θ_S identifies the real Sonin space with the semilocal one. In a common logarithmic chart, admitting p is

$$\Delta_p = I - p^{-\frac{1}{2}}U_p,$$

while the dual population return is $\big(\Delta_p^*\big)^{-1}$. Their pairing is exactly conserved. These maps are independent of the aperture, so their squares with nested Sonin inclusions commute; distinct prime additions commute as well.

The prime and support **spaces** therefore do not await a common-domain naturality construction. What remains is the Hermitian form carried across that already-existing square.

IX The one-prime Euler successor

IX.1 The successor is an exact two-arm transport

Let $p \notin S$, $S' = S \cup \{p\}$, and define

$$J_{S,p} = \theta_{S'} \theta_S^{-1}.$$

In the multiplicative spectral chart it is the multiplier

$$D_{p(t)} = 1 - p^{-\frac{1}{2}-it}.$$

In log time, with $T_L a(u) = a(u-L)$,

$$\Delta_p = I - p^{-\frac{1}{2}} T_{\log p}.$$

This is not an isolated prime node. It is a present arm and one translated arm acting on the complete current.

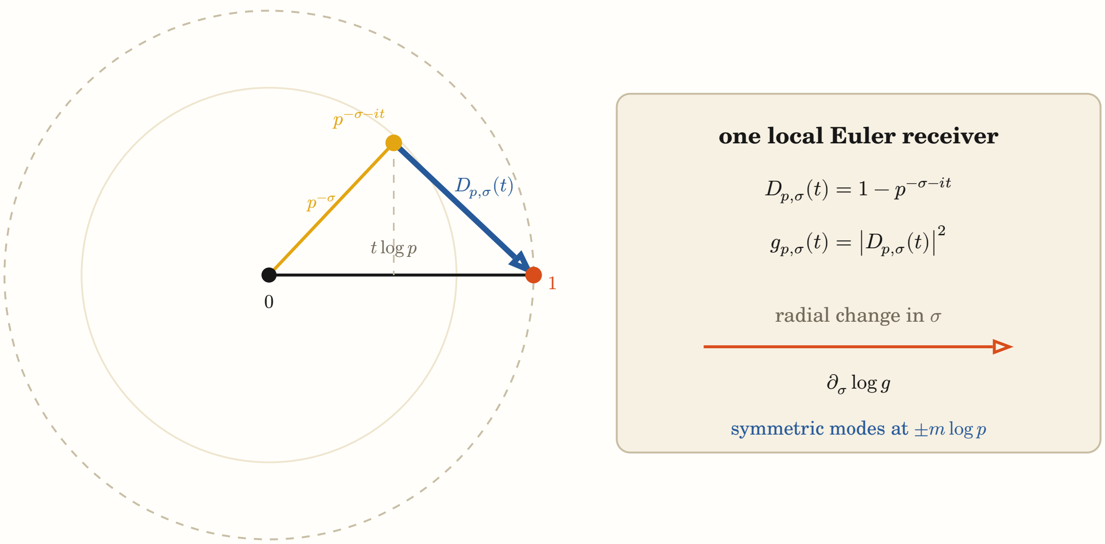

Figure 9: The Euler successor metric is a squared chord. Its endpoint rotates by $t \log p$ while σ changes the radius. The logarithmic normal derivative of this chord metric expands into the complete repeated-prime current. This is the paper's central geometric derivation.

Theorem 17 — One-prime Euler successor geometry Let S be a finite set of places containing ∞, let $S' = S \cup \{p\}$, and suppose the semilocal Sonin intertwiners θ_S and $\theta_{S'}$ are bounded Hilbert-space isomorphisms. Define

$$J_{S,p} = \theta_{S'} \theta_S^{-1}.$$

In the logarithmic spectral chart this successor is multiplication by

$$D_{p(t)} = 1 - p^{-\frac{1}{2}-it},$$

equivalently $\Delta_p = I - a_p U_p$ with $a_p = p^{-\frac{1}{2}}$ and $U_p = T_{\log p}$ unitary. Its pulled-back metric is

$$G_p = J_{S,p}^* J_{S,p} = \left(1 + a_p^2\right) I - a_{p(U_p + U_p^*)},$$

and obeys

$$\left(1 - a_p\right)^2 I \le G_p \le \left(1 + a_p\right)^2 I.$$

If P is the predecessor Sonin projection and the successor Sonin range pulls back to $\mathrm{range}(P)$, then the pulled-back successor orthogonal projection is

$$\Pi_p = \left((PG_pP)|_{\mathrm{range}(P)}\right)^{-1} PG_p.$$

For every successor operator $A' = J_{S,p} A J_{S,p}^{-1}$,

$$J_{S,p}^{-1} A'^* J_{S,p} = G_p^{-1} A^* G_p.$$

> **Demonstration.** The spectral multiplier gives the log-time finite difference directly. Expanding $\Delta_p^*\Delta_p$ gives G_p. The spectral range of $U_p + U_p^*$ lies in $[-2, 2]$, which yields the two operator bounds and bounded invertibility. The displayed Π_p is the unique projection onto $\mathrm{range}(P)$ whose kernel is G_p-orthogonal to that range. The adjoint identity follows by substituting $A' = J_{S,p} A J_{S,p}^{-1}$ and $G_p = J_{S,p}^* J_{S,p}$.

> Boundary. The semilocal spaces, intertwiners, and Euler multiplier are inherited from Connes–Consani–Moscovici. The pulled-back metric, projection, and adjoint package is the laboratory's receiver-covariant synthesis. It proves neither a sign for the completed Weil form nor the Riemann Hypothesis.

The theorem packages three levels which the historical record had kept scattered:

1. the published semilocal transport $J_{S,p}$;
2. the exact receiver metric $G_p = J_{S,p}^* J_{S,p}$; and
3. the receiving projection and adjoint law induced by that metric.

The bounds show that the transition is bounded and invertible. It is not unitary in the native predecessor metric because the Euler chord has nonconstant length.

IX.2 The aperture turns with the receiver

Let $P = P_{S,\lambda}$ and let

$$\Pi_p = J_{S,p}^{-1} P_{S',\lambda} J_{S,p}.$$

The pulled-back successor range is $\mathrm{range}(P)$, but its orthogonal complement is computed with G_p, not with the old metric. Consequently

$$\Pi_p = \left((PG_pP)|_{\mathrm{range}(P)}\right)^{-1} PG_p.$$

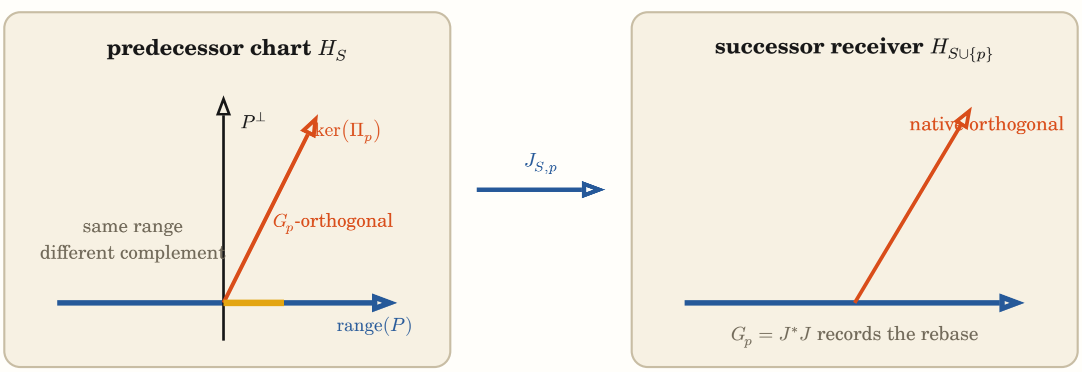

Figure 10: A new prime does not append an untouched block. The whole receiver is transported. In the predecessor chart the Sonin range can be the same vector subspace while its orthogonal complement turns under the pulled-back Euler metric.

Writing

$$H_S = \mathrm{range}(P) \oplus P^\perp H_S$$

and

$$G_p = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix},$$

one obtains

$$\Pi_p = \begin{pmatrix} I & A^{-1}B \\ 0 & 0 \end{pmatrix}$$

and therefore

$$\Pi_p - P = \begin{pmatrix} 0 & A^{-1}B \\ 0 & 0 \end{pmatrix}.$$

For the Euler metric,

$$B = -p^{-\frac{1}{2}} P\big(U_p + U_p^*\big)(I - P).$$

This is the exact cross-cut current by which the successor metric turns the receiving aperture. It is analogous to a second fundamental term: it measures how a direction transverse to the admitted Sonin range is sent into that range. It is not by itself curvature and it has no predecessor-metric sign.

IX.3 Independent place additions are endpoint-flat

For distinct $p, q \notin S$,

$$J_{S\cup\{p\},q} J_{S,p} = \theta_{S\cup\{p,q\}} \theta_S^{-1} = J_{S\cup\{q\},p} J_{S,q}.$$

The arithmetic square commutes because the Euler multipliers commute. Intermediate receivers and discovery chronology can still differ. The statement is zero endpoint holonomy for this particular finite-place transport, not a general claim that every founding event commutes.

IX.4 Prime powers are the metric's normal recurrence

Theorem 18 — Prime-power recurrence is the normal derivative of the Euler metric For $\sigma > 0$, put

$$D_{p,\sigma}(t) = 1 - p^{-\sigma - it}$$

and let $g_{p,\sigma}(t) = \big|D_{p,\sigma}(t)\big|^2$. Then

$$\partial_\sigma \log g_{p,\sigma}(t) = 2 \log p \sum_{m\geq 1} p^{-m\sigma} \cos(mt \log p).$$

At $\sigma = \frac{1}{2}$ this is the Fourier transform, under $\hat{\nu}(t) = \int e^{-itu}\, \mathrm{d}\nu(u)$, of the symmetric prime-power measure

$$\nu_p = \log p \sum_{m\geq 1} p^{-\frac{m}{2}} \big(\delta_{m \log p} + \delta_{-m \log p}\big).$$

> **Demonstration.** Write $z = p^{-\sigma - it}$. Since $|z| < 1$,
> $$\partial_\sigma \log|1 - z|^2 = 2\,\mathrm{Re}\bigg((\log p)\frac{z}{1-z}\bigg).$$
> Expanding $\frac{z}{1-z} = \sum_{m\geq 1} z^m$ gives the cosine series. The second statement follows by Fourier transforming the two atoms at $\pm m \log p$ term by term.

> Boundary. This identifies the complete repeated-prime current inside one exact receiver deformation. It does not compare that local deformation with the archimedean term, endpoint terms, or the completed Weil sign.

This theorem is the most direct answer to the laboratory's geometric intuition about recurring prime strings. The prime-power population is not being inferred from a plotted shape. It is literally the Fourier population of the normal logarithmic deformation of one prime's Euler chord metric.

At $\sigma = \frac{1}{2}$, the local metric is

$$g_{p(t)} = 1 + p^{-1} - 2p^{-\frac{1}{2}} \cos(t \log p).$$

Its radial derivative contains atoms at every two-sided repetition $\pm m \log p$, weighted by $(\log p)p^{-\frac{m}{2}}$. The critical exponent $\frac{1}{2}$ supplies the self-dual half-density weight which also appears in the explicit formula.

This is a real connection between the repeated-prime current and the critical seam. It is not yet the completed relation: W_∞, endpoint terms, the full test algebra, and the sign of the receiver coupling remain.

IX.5 The squeezed remainder reveals the centered sheets

Lemma 19 — A geometric remainder is squeezed while its logarithmic path remains unbounded Let

$$w = \alpha + i\beta, \quad \alpha > 0, \quad z = e^{-w}.$$

For

$$G_N = \sum_{k=0}^{N} z^k \quad \text{and} \quad G = \frac{1}{1-z},$$

the exact remainder is

$$R_N = G - G_N = \frac{z^{N+1}}{1-z}.$$

It obeys the two-sided squeeze

$$\frac{|z|^{N+1}}{1+|z|} \leq |R_N| \leq \frac{|z|^{N+1}}{1-|z|},$$

and hence $R_N \to 0$.

At the same time,

$$-\log|R_N| = (N+1)\alpha + \log|1 - e^{-w}|,$$

while any continuous phase lift satisfies

$$\operatorname{Arg}(R_N) = -(N+1)\beta - \operatorname{Arg}(1 - e^{-w}).$$

Thus Euclidean convergence to one boundary point is uniform linear translation and winding in the logarithmic residual chart.

The dyadic occurrence is the exact specialization

$$\sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N}, \quad R_N = 2^{-N}, \quad -\log R_N = N \log 2.$$

Demonstration. The finite geometric identity gives

$$G_N = \frac{1 - z^{N+1}}{1-z}$$

and therefore the displayed remainder. The triangle and reverse triangle inequalities give

$$1 - |z| \leq |1 - z| \leq 1 + |z|.$$

Since $|z| = e^{-\alpha} < 1$, division yields the two bounds, both of which tend to zero. Taking magnitude and phase of the exact remainder gives the logarithmic formulas. The dyadic row follows by substituting $z = \frac{1}{2}$ and omitting the $k = 0$ term.

Boundary. The squeeze requires $\alpha > 0$. At $\alpha = 0$ the upper denominator can lose its positive lower bound and the series terms no longer decay. Analytic continuation or an Abel boundary value is another transformation law, not the same convergent

> lineage. The scalar MVT may represent a real projection of one finite increment, but it neither causes convergence nor supplies one common witness for the complex path.

The dyadic path makes the geometry elementary:

$$S_N = \sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N}, \quad R_N = 2^{-N}.$$

The partial sums converge in their ordinary coordinate, while

$$-\log R_N = N \log 2$$

travels by equal steps without bound. The endpoint is therefore one point in the completed Euclidean receiver and an infinite ray in the logarithmic residual chart.

For the complex character $z = e^{-\alpha - i\beta}$ with $\alpha > 0$, the exact geometric remainder is

$$R_N = \frac{z^{N+1}}{1-z}.$$

It is squeezed by

$$\frac{|z|^{N+1}}{1+|z|} \le |R_N| \le \frac{|z|^{N+1}}{1-|z|}.$$

At the same time its logarithmic magnitude grows linearly with $(N+1)\alpha$ and its phase turns linearly with $-(N+1)\beta$. Convergence has not erased the path; it has compressed that path into a boundary face.

The Squeeze Theorem owns the convergence statement. The MVT has a narrower role. Along a declared real projection it can represent one finite increment by a path-relative tangent witness. In the log-polar chart the radial and phase velocities are already constant, and a complex or bundle-valued path does not generally admit one common MVT point.

Theorem 20 — A centered prime character has circular phase, hyperbolic normal sheets, and flat local transport Let p be prime, $L = \log p$, and

$$s = \frac{1}{2} + \varepsilon + it, \quad J(s) = 1 - \overline{s} = \frac{1}{2} - \varepsilon + it.$$

Put $x = \varepsilon L$, $\theta = tL$, and define the normalized functional partners

$$A_+ = p^{\frac{1}{2}-s} = e^{-x-i\theta}, \quad A_- = p^{\frac{1}{2}-J(s)} = e^{x-i\theta}.$$

Their common and oriented-normal faces are

$$C = \frac{A_+ + A_-}{2} = e^{-i\theta} \cosh x,$$

$$N = \frac{A_- - A_+}{2} = e^{-i\theta} \sinh x.$$

Consequently

$$C^2 - N^2 = e^{-2i\theta}, \quad N = 0 \ \text{ iff } \ \varepsilon = 0.$$

Pulling the Euclidean metric of $\mathbb{C}$ back along the two character sheets gives

$$g_+ = e^{-2x}(\mathrm{d}x^2 + \mathrm{d}\theta^2), \quad g_- = e^{2x}(\mathrm{d}x^2 + \mathrm{d}\theta^2).$$

Both metrics are locally flat. Their average and oriented half-difference are

$$\frac{g_+ + g_-}{2} = \cosh(2x)(\mathrm{d}x^2 + \mathrm{d}\theta^2),$$

$$\frac{g_- - g_+}{2} = \sinh(2x)(\mathrm{d}x^2 + \mathrm{d}\theta^2).$$

In the (C, N) carrier, the difference of the two ambient Euclidean forms is the exact Lorentzian parameter metric

$$|\mathrm{d}C|^2 - |\mathrm{d}N|^2 = \mathrm{d}\theta^2 - \mathrm{d}x^2.$$

If $a = p^{-\frac{1}{2}}$ and

$$D_+(x, \theta) = 1 - aA_+(x, \theta),$$

then its Euler metric is

$$G_+(x, \theta) = 1 + a^2 e^{-2x} - 2ae^{-x}\cos\theta.$$

At the fixed seam,

$$\partial_\varepsilon \log G_+ \,|_{\varepsilon=0} = 2L \sum_{m\geq 1} a^m \cos(m\theta),$$

with the tail controlled by the geometric-remainder squeeze. The partner sheet has the opposite normal derivative.

Demonstration. The formulas for A_+ and A_- follow from $p^{-s} = e^{-s\log p}$. Adding and subtracting them gives the hyperbolic factors, and $\cosh^2 x - \sinh^2 x = 1$ gives the complex-square identity. Differentiation gives

$$\left|\mathrm{d}A_+\right|^2 = e^{-2x}(\mathrm{d}x^2 + \mathrm{d}\theta^2)$$

and the analogous lower sheet. Each conformal exponent is affine in x, so its two-dimensional Gaussian curvature is zero. Directly differentiating C and N gives the Lorentzian difference.
Expanding $\left|1 - ae^{-x-i\theta}\right|^2$ gives G_+. At $x = 0$, logarithmic differentiation gives

$$2L\,\mathrm{Re}\left(\frac{z}{1-z}\right), \quad z = ae^{-i\theta}.$$

Since $a < 1$, the squeeze lemma licenses $\frac{z}{1-z} = \sum_{m\geq 1} z^m$ and controls every finite truncation remainder. Replacing x by $-x$ reverses the normal derivative.

Boundary. The Lorentzian metric is an exact signature on the parameter carrier, not a physical spacetime metric. A single prime character supplies flat local sheets and no intrinsic triangular holonomy. In the semilocal receiver, the first nontrivial connection is instead the turn of the receiving projection when it fails to commute with the Euler metric. The theorem does not extend the Euler product into the critical strip, identify zeta zeros, or sign the completed Weil response.

For one prime, put

$$x = \varepsilon \log p, \quad \theta = t \log p.$$

After removing the common half-density, the completed functional partners are

$$A_+ = e^{-x-i\theta}, \quad A_- = e^{x-i\theta}.$$

Their common and normal faces are

$$C = e^{-i\theta}\cosh x, \quad N = e^{-i\theta}\sinh x.$$

Thus circular phase and hyperbolic normal rebase are not two imported analogies. They are the two exact factors of the centered exponential character, with

$$C^2 - N^2 = e^{-2i\theta}, \quad N = 0 \ \text{ iff } \ \varepsilon = 0.$$

The two character sheets carry the induced metrics

$$g_+ = e^{-2x}(\mathrm{d}x^2 + \mathrm{d}\theta^2), \quad g_- = e^{2x}(\mathrm{d}x^2 + \mathrm{d}\theta^2).$$

Both are locally flat. Their average and oriented half-difference are

$$g_{\text{common}} = \cosh(2x)(\mathrm{d}x^2 + \mathrm{d}\theta^2),$$

$$g_{\text{coupling}} = \sinh(2x)(\mathrm{d}x^2 + \mathrm{d}\theta^2).$$

In the (C, N) carrier, the same construction has the exact indefinite parameter metric

$$|\mathrm{d}C|^2 - |\mathrm{d}N|^2 = \mathrm{d}\theta^2 - \mathrm{d}x^2.$$

This is a mathematical Lorentzian signature on the declared parameter surface, not a physical spacetime assertion.

The flatness result is consequential. One local prime character cannot source the curved triangular holonomy by itself. Its squeezed series supplies recurrence; its exponential supplies the two sheets; its Euler difference supplies the changed metric. The first nontrivial receiver connection remains the aperture turn $\Pi_p - P$, where that metric mixes the admitted Sonin range with its complement.

X The aperture connection and completed defect

X.1 The positive trace changes by a connection term

Define

$$\Sigma_{S(f)} = \mathrm{Tr}\Big(\Theta_{S(f)} P_S \Theta^*_{S(f)}\Big).$$

In the predecessor chart,

$$\Sigma_{S\cup\{p\}}(f) = \mathrm{Tr}\Big(\Theta_{S(f)} \Pi_p \Theta^*_{S(f)}\Big).$$

The exact trace change is

$$\kappa_{S,p}(f) = \mathrm{Tr}\Big(\Theta_{S(f)}(\Pi_p - P_S)\Theta^*_{S(f)}\Big).$$

It is a connection term because it depends on the contemporary receiver S, its aperture, and the order of previous place additions. It is not an intrinsic scalar assigned to p.

Strict positivity of G_p does not sign κ. The finite exact control

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad \Pi = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}$$

has positive G, yet suitable commuting symmetric operators give either positive or negative trace change. This does not model the arithmetic Sonin operator. It proves the logical point: metric positivity alone cannot order the aperture connection.

X.2 The partial completed response changes by one prime current

Let $Q_{S(f)}$ contain the endpoint and archimedean terms together with the finite-place terms already admitted by S. Then

$$Q_{S\cup\{p\}}(f) - Q_{S(f)} = -W_{p(f*f^\sharp)},$$

where

$$W_{p(h)} = \log p \sum_{m\geq 1} (h(p^m) + p^{-m} h(p^{-m})).$$

Define the contemporary defect

$$D_{S(f)} = Q_{S(f)} - \Sigma_{S(f)}.$$

Theorem 21 — Receiver-covariant completed-defect recurrence Let

$$\Sigma_{S(f)} = \mathrm{Tr}\Big(\Theta_{S(f)} P_S \Theta^*_{S(f)}\Big)$$

be a defined trace-class Sonin return, let $Q_{S(f)}$ be the completed explicit-formula response with the finite places in S, and set

$$D_{S(f)} = Q_{S(f)} - \Sigma_{S(f)}.$$

For $S' = S \cup \{p\}$ define the aperture connection

$$\kappa_{S,p}(f) = \Sigma_{S'}(f) - \Sigma_{S(f)}.$$

If

$$W_{p(h)} = \log p \sum_{m\geq 1} (h(p^m) + p^{-m} h(p^{-m}))$$

for $h = f * f^\sharp$, then

$$D_{S'}(f) = D_{S(f)} - W_{p(h)} - \kappa_{S,p}(f).$$

In the predecessor chart,

$$\kappa_{S,p}(f) = \mathrm{Tr}\Big(\Theta_{S(f)}(\Pi_p - P_S)\Theta^*_{S(f)}\Big).$$

Demonstration. The explicit formula changes by $Q_{S'} - Q_S = -W_p$. Similarity invariance of the trace and the transported successor projection give $\Sigma_{S'} - \Sigma_S = \kappa_{S,p}$. Subtracting these two exact differences yields the recurrence.

Boundary. The recurrence has no sign by itself. In particular, the off-diagonal aperture turn $\Pi_p - P_S$ is not positive in the predecessor metric. Proving nonnegativity of the completed successor defect for the full admissible family is the RH-bearing open step, not a consequence of this identity.

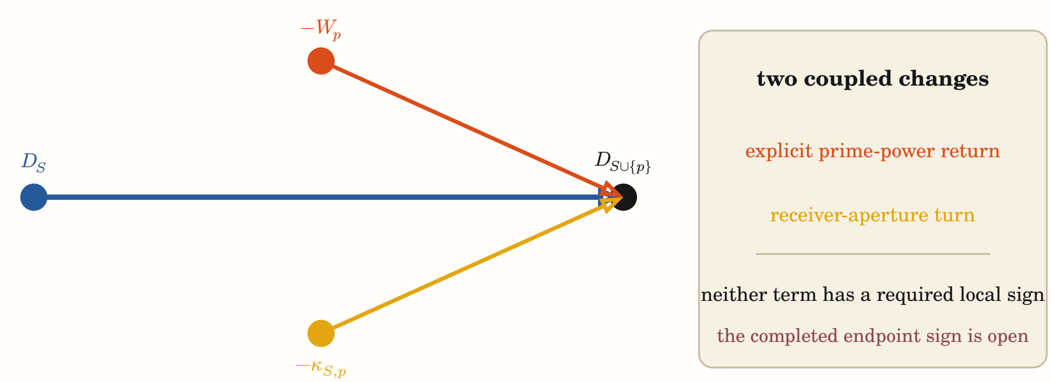

Figure 11: The exact one-prime balance. Adjoining p changes both the explicit arithmetic response and the receiver which measures the positive Sonin return. The missing theorem must control their completed balance; positivity of either local ingredient alone is insufficient.

This is the strongest integrated theorem in the paper. It says what the “lightning” has become mathematically at the RH boundary:

- G_p is the potential/conductance deformation induced by the Euler difference;
- $\Pi_p - P_S$ is the cross-cut aperture current;
- W_p is the explicit repeated-prime return;
- $\kappa_{S,p}$ is the receiver's response to being turned by that return; and
- $D_{S\cup\{p\}}$ is the completed consequence still owed after both changes.

The analogy is now subordinate to exact operators. No electrical physics is being asserted.

X.3 The proof-bearing object is the uniform form-level successor

Conditional corollary 22 — Completed receiver domination would imply RH Suppose every admissible compactly supported returned current f admits a finite contemporary receiver S_f for which

$$Q_{W(f)} \geq \Sigma_{S_f}(f) \geq 0,$$

with Q_W equal to the completed Weil form on that current. Then the Riemann Hypothesis holds.

> **Demonstration.** The assumed inequality gives $Q_{W(f)} \geq 0$ for every admissible f. Weil's positivity criterion is equivalent to the Riemann Hypothesis.

> Boundary. The hypothesis is not established. Neither bounded experiments, critical-line symmetry, positivity of an Euler metric, nor trace-class transport proves the domination inequality.

The previous laboratory record phrased the gap as the scalar domination

$$Q_{W(f)} \geq \Sigma_{S_f}(f) \geq 0$$

for every admissible f in a sufficiently complete contemporary receiver. That remains one sufficient local chart, but it is not the only geometrically possible proof form.

In that chart, a local prime-admission step would require

$$D_{S(f)} \geq W_{p(f*f^\sharp)} + \kappa_{S,p}(f)$$

whenever p enters. No term on the right is individually required to be positive. No theorem audited here proves this inequality.

Conditional corollary 23 — An exhaustive sign-preserving formulation atlas would imply RH Suppose a transport groupoid of formulation charts has the following properties:

- every admissible returned Weil current is represented by an anchored vector in some fiber;
- every orbit meets a base fiber on which the anchored Weil form is proved positive semidefinite without assuming RH;
- every transition is receiver-exact and carries the Hermitian response by a positive congruence; and
- transition composition, adjoint return, boundary terms, and holonomy preserve that covariance.

Then the completed Weil form is nonnegative on every admissible returned current, and the Riemann Hypothesis holds.

> **Demonstration.** Take any admissible returned current. Exhaustiveness places it in one represented fiber and the orbit condition connects it to a positive base fiber. Repeated application of positive-congruence covariance preserves its sign along the complete path. Hence the completed Weil response is nonnegative for every admissible current. Weil's criterion then implies RH.

> Boundary. Merely finding a positive vector, a positive chart, or some path to a positive value is insufficient. The path must carry the same anchored test occurrence, its response, and all completion terms. Exhaustiveness and sign-preserving holonomy are the unproved obligations.

This corollary remains a valid global reformulation. It permits many nonidentical formulations, curved parameter trajectories, and changing receivers. It does not demand that every local contribution or every intermediate evaluation be positive. It demands four relational invariants:

1. **coverage:** every admissible returned current occurs in the atlas;
2. **anchoring:** transition maps carry the same completed explicit-formula occurrence rather than merely reaching an equal endpoint;
3. **positive covariance:** the Hermitian response changes only by a positive congruence, or by an explicitly controlled nonnegative defect; and
4. **coherent return:** composition and holonomy cannot carry a positive cone into a negative one.

This is not a vague enlarged total field, and it is not the method required for the proof. Every compact current belongs to one finite support stage. The induction theorem above is sufficient; an atlas is a way to organize its coherence under alternate prime orders and receiver recharts. Its local comparisons may be spans or partial maps; only invertible exact recharts form a groupoid. The global obligation is that these incidences and their transition arcs cover the complete admissible family without losing the response. A single positive orientation in each neighborhood is insufficient; a complete anchored correspondence atlas whose closed returns preserve the response would be sufficient.

Conditional corollary 24 — A noncircular squeeze by positive returned receivers would imply RH Suppose that for every admissible returned current f there is a sequence of independently positive receiver responses

$$\Sigma_{n(f)} \geq 0$$

and explicitly derived errors $E_{n(f)} \geq 0$ such that

$$\left|Q_{W(f)} - \Sigma_{n(f)}\right| \leq E_{n(f)} \quad \text{and} \quad E_{n(f)} \to 0.$$

Then $Q_{W(f)} \geq 0$ for every admissible f, and the Riemann Hypothesis holds.

Demonstration. The assumed bounds give

$$Q_{W(f)} \geq \Sigma_{n(f)} - E_{n(f)} \geq -E_{n(f)}.$$

Letting $n \to \infty$ and applying the squeeze theorem yields $Q_{W(f)} \geq 0$. Weil's positivity criterion then implies RH.

Boundary. The approximation index is not automatically a prime count. A compact current already meets only finitely many finite places; the useful sequence may instead refine the final receiver, prolate/Sonin resolution, or completed formulation chart. The error must include every endpoint, archimedean, finite-place, aperture-connection, and holonomy term without defining itself from the desired sign. Finite positive samples without such an error squeeze establish nothing universal.

The Squeeze Theorem supplies a third exact proof form. For every admissible returned current, it would suffice to construct independently positive receiver responses $\Sigma_{n(f)}$ and complete errors $E_{n(f)}$ satisfying

$$\left|Q_{W(f)} - \Sigma_{n(f)}\right| \leq E_{n(f)}, \quad E_{n(f)} \to 0.$$

Then $Q_{W(f)} \geq -E_{n(f)}$ for every n, so $Q_{W(f)} \geq 0$ in the limit.

This does not authorize an infinite prime sweep. A compact current already meets only finitely many finite places. The useful approximation index would refine the final receiver—through prolate/Sonin resolution, aperture, certified Galerkin transport, or completed formulation charts—and its error must retain endpoint, archimedean, finite-place, aperture-connection, and holonomy terms. No such complete error squeeze is currently derived.

X.4 The error has a vanishing resolution face and a persistent structural face

Lemma 25 — A negative compact witness survives every larger support aperture Let $\mathcal{T}_R \subseteq \mathcal{T}_{R'}$ for $R \leq R'$ be nested test-current fibers, and suppose their Hermitian responses are compatible:

$$q_{R'}|_{\mathcal{T}_R} = q_R.$$

Define the spectral floor

$$\mu_R = \inf_{0\neq f\in\mathcal{T}_R} \frac{q_{R(f)}}{\|f\|^2}.$$

Then

$$\mu_{R'} \le \mu_R.$$

More strongly, if one $f \in \mathcal{T}_R$ satisfies $q_{R(f)} < 0$, and numbers $e_{R'} \ge 0$ obey

$$q_{R'}(g) \ge -e_{R'}\|g\|^2 \quad \text{for every } g \in \mathcal{T}_{R'},$$

then every later aperture satisfies

$$e_{R'} \ge -\frac{q_{R(f)}}{\|f\|^2} > 0.$$

Consequently no lower-error bound indexed only by enlargement of a compatible support aperture can converge to zero after a negative compact witness has entered.

Demonstration. The infimum defining $\mu_{R'}$ is taken over a set containing the set defining μ_R, so $\mu_{R'} \le \mu_R$. Compatibility carries the same witness and the same response into every later fiber. Substituting that witness into the assumed lower bound gives

$$q_{R(f)} = q_{R'}(f) \ge -e_{R'}\|f\|^2,$$

which rearranges to the displayed positive lower bound for $e_{R'}$.

Boundary. This is a statement about genuine nested aperture growth. A finite-rank resolution sequence inside one fixed receiver is a different axis and may have a vanishing discretization tail. The lemma neither asserts that a negative witness exists nor signs the completed Weil response.

This rules out one tempting but false use of the Squeeze Theorem. If an admissible compact current has negative completed response at aperture R, the same current and response remain present at every larger compatible aperture. Enlarging support cannot dilute that occurrence. In the finite-aperture spectral notation, the lower floors therefore satisfy

$$\mu_{R'} \le \mu_R \quad \text{when } R' \ge R.$$

The associated deficit $(-\mu_R)_+$ can only stay fixed or grow. It is not a vanishing squeeze parameter.

Now hold the final receiver (S, R) fixed and resolve only its positive Sonin return. Put

$$A_{S(f)} = \Theta_{S(f)}P_S.$$

For increasing finite-rank output projections $L_n \to I$, define

$$\Sigma_{S,n}(f) = \left\|L_n A_{S(f)}\right\|^2_{\mathrm{HS}}.$$

Then the exact Hilbert–Schmidt tail is

$$T_{S,n}(f) = \left\|(I - L_n)A_{S(f)}\right\|^2_{\mathrm{HS}} \to 0$$

and

$$Q_{W(f)} - \Sigma_{S,n}(f) = D_{S(f)} + T_{S,n}(f).$$

Thus receiver resolution converges to Σ_S, not automatically to Q_W. The vanishing tail exposes the structural defect; it does not erase it.

Theorem 26 — A positive receiver remainder makes the resolution squeeze exact Fix one final receiver (S, R). Let

$$A_{S(f)} = \Theta_{S(f)}P_S$$

be Hilbert–Schmidt, so that

$$\Sigma_{S(f)} = \left\|A_{S(f)}\right\|_{\mathrm{HS}}^2.$$

Suppose the completed defect has an independently constructed positive carrier: there are a Hilbert space $\mathcal{K}_{S,R}$ and a linear map $C_{S,R}$ such that

$$Q_{W(f)} - \Sigma_{S(f)} = \left\|C_{S,R} f\right\|^2 \quad \text{for every admitted } f.$$

Let L_n and M_n be increasing finite-rank orthogonal projections converging strongly to the identities on H_S and $\mathcal{K}_{S,R}$. Define

$$\hat{\Sigma}_{n(f)} = \left\|L_n A_{S(f)}\right\|_{\mathrm{HS}}^2 + \left\|M_n C_{S,R} f\right\|^2$$

and

$$E_{n(f)} = \left\|(I - L_n) A_{S(f)}\right\|_{\mathrm{HS}}^2 + \left\|(I - M_n) C_{S,R} f\right\|^2.$$

Then

$$\hat{\Sigma}_{n(f)} \geq 0, \quad Q_{W(f)} - \hat{\Sigma}_{n(f)} = E_{n(f)} \geq 0, \quad E_{n(f)} \to 0.$$

If the receiver is enlarged by one prime and

$$c_{S,p}(f) = W_{p(f * f^\sharp)} + \kappa_{S,p}(f),$$

then its remainder form obeys

$$K_{S \cup \{p\}, R} = K_{S,R} - C_{S,p},$$

where these symbols denote the Hermitian forms representing the corresponding quadratic responses. The positive carrier propagates exactly when

$$C_{S,p} \leq K_{S,R}$$

in form order.

Demonstration. Orthogonality of L_n and $I - L_n$ in the output space gives the Hilbert–Schmidt Pythagorean identity

$$\left\|A_{S(f)}\right\|_{\mathrm{HS}}^2 = \left\|L_n A_{S(f)}\right\|_{\mathrm{HS}}^2 + \left\|(I - L_n) A_{S(f)}\right\|_{\mathrm{HS}}^2.$$

The same identity for M_n and $C_{S,R} f$, added to the assumed positive-remainder factorization, gives the exact equality with E_n. Strong convergence of finite-rank projections gives convergence of the vector tail, and gives convergence of the Hilbert–Schmidt tail after summing over any orthonormal basis.

The final formula is the completed-defect recurrence $D_{S \cup \{p\}} = D_S - c_{S,p}$ written in Hermitian-form language. Its successor is nonnegative exactly when $c_{S,p}(f) \leq D_{S(f)}$ for every admitted f, which is the displayed form order. The form $C_{S,p}$ need not itself be nonnegative.

Boundary. The theorem constructs the complete vanishing error once the positive remainder carrier exists; it does not manufacture that carrier. Finite-rank refinement of the Sonin square alone converges to Σ_S, leaving the structural defect $D_S = Q_W - \Sigma_S$ unchanged. Proving $C_{S,p} \leq K_{S,R}$ through every support and place transition is the RH-bearing semilocal domination problem. A covariant coarea carrier may be nonnegative at both contemporary endpoints while this signed connection/seam difference points in either direction; the theorem does not impose monotonicity per prime.

This theorem corrects the previous description of the missing object. The complete error E_n is automatic after one constructs a positive remainder carrier

$$D_{S(f)} = \left\|C_{S,R} f\right\|^2.$$

The genuinely proof-bearing object is $C_{S,R}$, or equivalently a nonnegative remainder form $K_{S,R}$. Under one prime admission its exact law is

$$K_{S\cup\{p\},R} = K_{S,R} - C_{S,p},$$

where

$$C_{S,p}(f) = W_{p(f*f^\sharp)} + \kappa_{S,p}(f).$$

Consequently the archimedean positive base propagates precisely when

$$C_{S,p} \le K_{S,R}$$

in Hermitian-form order. This is the same unproved semilocal domination relation, now separated from all discretization tails. It does not say that $C_{S,p}$ is positive. A covariant carrier may remain a norm square at both receivers while this signed difference points in either direction.

X.5 One finite error squeeze is already available

Corollary 27 — A certified finite archimedean tail supplies an exact bounded squeeze On one fixed finite Galerkin band, suppose Hermitian matrices Q_T and Q_∞ obey

$$0 \le \Delta_T := Q_\infty - Q_T \le B_T I, \quad B_T \to 0.$$

If the finite matrix

$$Q_T + B_T I$$

is independently certified positive semidefinite, then for every coefficient vector v the response

$$\Sigma_{T(v)} = v^*(Q_T + B_T I)v$$

is nonnegative and

$$\left| v^* Q_\infty v - \Sigma_{T(v)} \right| \le B_T \|v\|^2 \to 0.$$

Demonstration. The tail order gives

$$-B_T I \le \Delta_T - B_T I \le 0.$$

Since

$$Q_\infty - (Q_T + B_T I) = \Delta_T - B_T I,$$

evaluation on v gives the stated absolute bound. Positivity of Σ_T is precisely the independent finite-matrix certificate.

Boundary. This closes the archimedean cutoff axis on one already finite frequency band. It does not prove the shifted certificate for every band, control the Galerkin limit, enlarge the support aperture, or sign the semilocal remainder. The index T is an archimedean integration cutoff, not a count of primes and not the full receiver resolution.

The finite Guinand–Weil tail theorem supplies, on each fixed finite frequency band, an exact positive archimedean increment

$$0 \le Q_\infty - Q_T \le B_T I, \quad B_T \to 0.$$

Once the finite shifted matrix $Q_T + B_T I$ is certified positive, it is an independently nonnegative approximant and $B_T\|v\|^2$ is the requested vanishing error. This genuinely closes the archimedean integration-cutoff axis. What it does not close is the frequency-band limit, support growth, or the prime-induced remainder form.

XI The remainder carrier as a covariant coarea sweep

The phrase "positive remainder carrier" should not be read as an instruction to append a positive block to the receiver. A carrier is a contemporary geometric interior: a measured field of response amplitudes whose squared norm is read at one receiver. When that receiver changes, its complete norm remains nonnegative while its derivative and seam jumps may be signed.

This distinction is forced by the defect recurrence. The local change

$$-W_{p(f*f^\sharp)} - \kappa_{S,p}(f)$$

has no required sign. Demanding a positive constituent from every prime would replace the established receiver rebase by a false monotonicity law.

XI.1 Change of variables carries a half-Jacobian

Definition 28 — A covariant coarea carrier is a receiver-local field of squared amplitudes Fix one contemporary receiver ρ. A *covariant coarea carrier* for an admitted current space $\mathcal{H}_\rho$ consists of:

- a measured parameter region (Λ_ρ, ν_ρ);
- a measurable field of Hilbert fibers $\mathcal{K}_{\rho,\xi}$ over $\xi \in \Lambda_\rho$; and
- an independently specified amplitude law $B_{\rho,\xi} : \mathcal{H}_\rho \to \mathcal{K}_{\rho,\xi}$

for which

$$(C_\rho f)(\xi) = B_{\rho,\xi} f$$

is square-integrable. Its carried quadratic response is

$$D_{\rho(f)} = \left\|C_\rho f\right\|^2 = \int_{\Lambda_\rho} \left\|B_{\rho,\xi} f\right\|^2 \,\mathrm{d}\nu_{\rho(\xi)}.$$

When a regular map $I : M \to B$ supplies the parameter fibers, the coarea theorem may disintegrate this response over $I^{-1}(c)$. The normal Jacobian, induced fiber measure, orientation, and any critical strata are part of the carrier. They may not be discarded after the scalar integral is evaluated.

> Boundary. The carrier belongs to the contemporary receiver; it is not a fixed enlarged field containing all earlier and later receivers. Its nonnegativity follows only after the amplitude law and measure have been constructed independently. Defining C_ρ as an abstract square root of a desired nonnegative defect would assume the sign it is meant to establish. Critical points where the coarea Jacobian loses rank are typed seams, not regular fibers with a zero silently divided away.

Let $\Phi : U \to V$ be a regular coordinate change. If $b(x)$ is a carrier amplitude on V, then the amplitude representing the same quadratic response on U is

$$\tilde{b}(u) = |\det D\Phi(u)|^{\frac{1}{2}} R_u b(\Phi(u)),$$

where R_u carries the output fiber unitarily. Consequently

$$\int_V \|b(x)\|^2 \,\mathrm{d}x = \int_U \left\|\tilde{b}(u)\right\|^2 \mathrm{d}u.$$

The Jacobian belongs to amplitude before the square is evaluated. It is not a scalar correction pasted onto a completed answer.

Theorem 29 — The half-Jacobian preserves a coarea carrier and its covariant FTC has signed seams First let $\Phi : U \to V$ be a regular change of carrier coordinates, let

$$J_{\Phi(u)} = |\det D\Phi(u)|,$$

and let R_u be a unitary identification of the corresponding output fibers. If $b(x)$ is a square-integrable carrier section on V, then the pulled section

$$\tilde{b}(u) = J_{\Phi(u)}^{\frac{1}{2}} R_u b(\Phi(u))$$

obeys

$$\int_V \|b(x)\|^2 \, \mathrm{d}x = \int_U \left\|\tilde{b}(u)\right\|^2 \mathrm{d}u.$$

Now let $\rho(t)$ be a piecewise smooth path of contemporary receivers, let f_t be one anchored current transported along that path, and let

$$\Psi(t) = C_{\rho(t)} f_t$$

be the resulting section of the carrier-output Hilbert bundle. For a metric-compatible connection ∇ and seams $t_1, ..., t_m$,

$$\|\Psi(b)\|^2 - \|\Psi(a)\|^2 = 2 \operatorname{Re} \sum_j \int_{I_j} \langle \Psi(t), \nabla_{\partial_t} \Psi(t) \rangle \, \mathrm{d}t + \sum_k \left(\left\|\Psi(t_k^+)\right\|^2 - \|\Psi(t_k^-)\|^2 \right).$$

Thus every contemporary response is nonnegative while its interior connection current and discrete seam changes may have either sign.

Demonstration. The first identity is the ordinary change-of-variables theorem; unitary fiber transport preserves the pointwise norm and the square-root Jacobian converts the volume element into amplitude.
On each smooth stratum, metric compatibility gives

$$\partial_t \|\Psi(t)\|^2 = 2 \operatorname{Re} \langle \Psi(t), \nabla_{\partial_t} \Psi(t) \rangle.$$

Integrate this scalar identity on every stratum and telescope the one-sided endpoint values. The unmatched values at the seams are exactly the displayed jumps.

Boundary. Positivity belongs to each complete carrier norm, not to every derivative or seam increment. A monotone tail integral is one special case in which the connection density and jumps have a chosen sign; it is not a general law of changing receivers. The theorem supplies no carrier amplitude for the completed Weil defect and does not sign the prime-admission connection term.

For a carrier section $\Psi(t) = C_{\rho(t)} f_t$ along a path of contemporary receivers, the exact covariant FTC is

$$\|\Psi(b)\|^2 - \|\Psi(a)\|^2 = 2\Re \sum_j \int_{I_j} \langle \Psi, \nabla_{\partial_t} \Psi \rangle \, \mathrm{d}t + \sum_k \left(\left\|\Psi(t_k^+)\right\|^2 - \|\Psi(t_k^-)\|^2 \right).$$

Every endpoint norm is nonnegative. Neither the connection density nor a discrete seam difference is required to be nonnegative. This is the precise calculus relation between a positive interior and a signed loss/current signal.

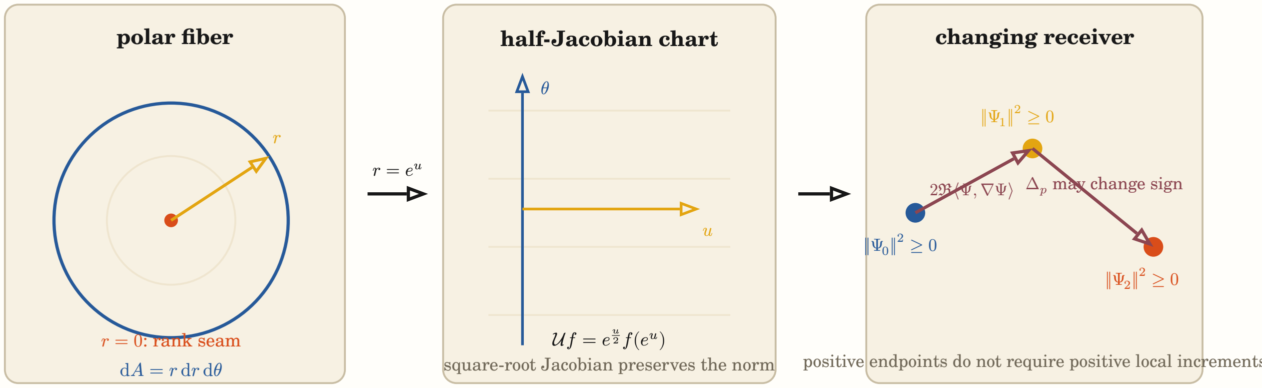

Figure 12: The coarea carrier has three inseparable faces. A coordinate fiber carries its induced measure; a chart change moves the Jacobian into the amplitude as a half-density; and a path of contemporary receivers carries nonnegative endpoint norms through generally signed connection currents and discrete seams.

XI.2 Cartesian and polar coordinates give the finite-dimensional control

For

$$x = r\cos\theta, \quad y = r\sin\theta,$$

one has

$$\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$$

and therefore

$$\int_{\mathbb{R}^2} |f(x,y)|^2\,\mathrm{d}x\,\mathrm{d}y = \int_0^\infty \int_0^\Theta |f(r,\theta)|^2 r\,\mathrm{d}\theta\,\mathrm{d}r.$$

The amplitude chart carries $r^{\frac{1}{2}}$. If

$$f(r,\theta) = \sum_{n\in\mathbb{Z}} f_{n(r)} e^{in\theta},$$

then angular Parseval gives

$$\|f\|^2 = \|f_0\|^2 + \sum_{n\neq 0} \|f_n\|^2.$$

Reading only the rotationally symmetric face leaves the nonzero winding modes as a literal positive remainder carrier.

This example also displays the required seams. Away from $r = 0$ and after choosing an angular branch, the chart is locally reversible. Complete-turn lifts can present the same Cartesian point with different winding. At $r = 0$, the Jacobian loses rank and every angular direction collapses to one point. That critical locus cannot be handled by silently setting the regular fiber density to zero.

The radial FTC

$$V(R) = \int_0^R \int_0^\Theta r\,\mathrm{d}\theta\,\mathrm{d}r, \quad V'(R) = \Theta R$$

shows why a boundary face alone is insufficient. The disk is carried by the ordered family of circular fibers, their normal direction, induced measure, and sweep.

XI.3 Logarithmic transport makes the critical half-density exact

Lemma 30 — The critical half-density is the square-root Jacobian of logarithmic transport Let $r = e^u$ and define

$$(\mathcal{U}F)(u) = e^{\frac{u}{2}} F(e^u).$$

Then $\mathcal{U}$ is unitary from $L^2(\mathbb{R}_{>0}, \mathrm{d}r)$ to $L^2(\mathbb{R}, \mathrm{d}u)$ because

$$\int_0^\infty |F(r)|^2 \,\mathrm{d}r = \int_{-\infty}^\infty \left|e^{\frac{u}{2}} F(e^u)\right|^2 \mathrm{d}u.$$

With the Mellin convention

$$\mathcal{M}F(s) = \int_0^\infty F(r) r^s \frac{\mathrm{d}r}{r},$$

the ordinary Fourier transform of $\mathcal{U}F$ is

$$\widehat{\mathcal{U}F}(t) = \mathcal{M}F\left(\frac{1}{2} - it\right).$$

Moreover, for

$$F^{\sharp(r)} = r^{-1}\overline{F(r^{-1})},$$

one has

$$\mathcal{U}\big(F^\sharp\big)(u) = \overline{(\mathcal{U}F)(-u)}.$$

Demonstration. Substitute $r = e^u$, so $\mathrm{d}r = e^u \,\mathrm{d}u$. The factor $e^{\frac{u}{2}}$ is the square root of that Jacobian and gives the norm identity. Substitution in the Fourier integral gives

$$\int_{\mathbb{R}} e^{\frac{u}{2}} F(e^u) e^{-itu} \,\mathrm{d}u = \int_0^\infty F(r) r^{\frac{1}{2} - it} \frac{\mathrm{d}r}{r}.$$

Applying the same logarithmic change to $F^\sharp$ gives the final reflection formula.

Boundary. On multiplicative Haar measure $\mathrm{d}\frac{r}{r}$, the logarithmic chart already has unit Jacobian. The half-density appears when that multiplicative geometry is related to the additive $L^2(\mathrm{d}r)$ normalization. The identity explains the unitary status of the line $\mathrm{Re}(s) = \frac{1}{2}$; it does not locate the zeros of zeta.

Lemma 31 — The displayed Mellin midpoint is the half-weight of the transported measure For $\beta \in \mathbb{R}$, give $\mathbb{R}_{>0}$ the weighted multiplicative measure

$$\mathrm{d}\mu_{\beta(r)} = r^\beta \frac{\mathrm{d}r}{r}.$$

Under $r = e^u$, define

$$\big(\mathcal{U}_\beta F\big)(u) = e^{\beta\frac{u}{2}} F(e^u).$$

Initially for compactly supported smooth F, and then in the Plancherel sense on the corresponding L^2 completion,

$$\mathcal{U}_\beta : L^2\big(\mathbb{R}_{>0}, \mathrm{d}\mu_\beta\big) \to L^2(\mathbb{R}, \mathrm{d}u)$$

is unitary, and with the Mellin convention

$$\mathcal{M}F(s) = \int_0^\infty F(r) r^s \frac{\mathrm{d}r}{r},$$

one has

$$\widehat{\mathcal{U}_\beta F}(t) = \mathcal{M}F\left(\frac{\beta}{2} - it\right). \quad \text{(WEIGHTED HALF)}$$

The weighted reciprocal-conjugate

$$F^{\sharp_\beta}(r) = r^{-\beta}\overline{F(r^{-1})}$$

becomes ordinary reflected conjugation:

$$\mathcal{U}_{\beta\left(F^{\sharp_\beta}\right)}(u) = \overline{(\mathcal{U}_\beta F)(-u)}.$$

Its Mellin involution is

$$J_{\beta(s)} = \beta - \overline{s}, \quad \mathrm{Fix}\big(J_\beta\big) = \left\{s : \mathrm{Re}(s) = \frac{\beta}{2}\right\}. \quad \text{(WEIGHTED SEAM)}$$

For another weight β', multiplication by

$$\big(R_{\beta\to\beta'}F\big)(r) = r^{\frac{\beta-\beta'}{2}}F(r)$$

is a unitary rechart

$$L^2\big(\mathrm{d}\mu_\beta\big) \to L^2\big(\mathrm{d}\mu_{\beta'}\big)$$

satisfying

$$\mathcal{U}_{\beta'}R_{\beta\to\beta'} = \mathcal{U}_\beta.$$

It transports the seam $\mathrm{Re}(s) = \frac{\beta}{2}$ to $\mathrm{Re}(s') = \frac{\beta'}{2}$ by the affine spectral rebase

$$s = s' + \frac{\beta - \beta'}{2}.$$

Demonstration. On the stated dense domain, substitution $r = e^u$ gives

$$\mathrm{d}\mu_\beta = e^{\beta u}\,\mathrm{d}u,$$

so $e^{\beta\frac{u}{2}}$ is exactly its square-root density and proves unitarity. The same substitution in the Fourier integral gives ("WEIGHTED HALF"). Direct substitution in $F^{\sharp_\beta}$ gives the reflected-conjugate identity, while changing variables $r \to r^{-1}$ in its Mellin transform gives

$$\mathcal{M}\big(F^{\sharp_\beta}\big)(s) = \overline{\mathcal{M}F(\beta - \overline{s})}.$$

The fixed seam follows. The norm of $R_{\beta\to\beta'}F$ against $\mathrm{d}\mu_{\beta'}$ equals the norm of F against $\mathrm{d}\mu_\beta$, and the displayed logarithmic and spectral identities follow by collecting powers of r.

Boundary. The rational glyph $\frac{\beta}{2}$ is typed by a measure and reciprocal law. It is not a numeral-radix effect and it is unrelated to the finite character factor $\frac{1}{q}$ unless another construction supplies a relation. Transporting the function, measure, and involution together is a rechart; changing only the measure or only the function is a different receiver problem. The completed zeta normalization in the present paper uses $\beta = 1$. This lemma explains the resulting $\frac{1}{2}$ chart but does not place any zero on its fixed seam.

With $r = e^u$, define

$$(\mathcal{U}F)(u) = e^{\frac{u}{2}}F(e^u).$$

Then

$$\int_0^\infty |F(r)|^2\,\mathrm{d}r = \int_{-\infty}^\infty |(\mathcal{U}F)(u)|^2\,\mathrm{d}u$$

and ordinary Fourier transformation gives

$$\widehat{\mathcal{U}F}(t) = \mathcal{M}F\left(\frac{1}{2} - it\right).$$

Thus the critical half has an exact change-of-variable provenance: it is the square-root Jacobian carrying the additive $L^2(\mathrm{d}r)$ response into logarithmic scale. On multiplicative Haar measure $\mathrm{d}\frac{r}{r}$, the logarithmic chart already has unit Jacobian; the half-density transports between these two normalizations.

More generally, for

$$\mathrm{d}\mu_{\beta(r)} = r^{\beta}\frac{\mathrm{d}r}{r},$$

the transported amplitude is $e^{\beta\frac{u}{2}}F(e^u)$ and Fourier transformation lands on the Mellin line $\mathrm{Re}(s) = \frac{\beta}{2}$. Reciprocal conjugation becomes $J_{\beta(s)} = \beta - \bar{s}$. Moving from β to β' while carrying the function and measure together conjugates this fixed seam; changing only one of them changes the problem. The completed zeta relation used here is the $\beta = 1$ member. Its displayed $\frac{1}{2}$ therefore belongs to this weighted involution and half-density, not to base ten and not to the $\frac{1}{q}$ of a finite character projector.

Under

$$F^{\sharp(r)} = r^{-1}\overline{F(r^{-1})},$$

the same chart gives

$$\mathcal{U}\big(F^{\sharp}\big)(u) = \overline{(\mathcal{U}F)(-u)}.$$

Scale inversion has become reflection of logarithmic position, and the Mellin fixed seam is exactly where the returned pair is a norm square.

XI.4 Hyperbolic coordinates separate common energy from oriented residual

The Riemannian hyperbolic plane has

$$\mathrm{d}s^2 = \mathrm{d}\rho^2 + \sinh^2(\rho)\,\mathrm{d}\theta^2, \quad \mathrm{d}A = \sinh(\rho)\,\mathrm{d}\rho\,\mathrm{d}\theta.$$

Its radial sweep obeys

$$A(R) = \Theta\sinh R, \quad V(R) = \Theta(\cosh R - 1), \quad V'(R) = A(R).$$

This is a positive coarea geometry: $\sinh\rho$ records the expansion of successive boundary fibers.

Lorentzian hyperbolic coordinates carry a different relation:

$$X = \rho\cosh\eta, \quad T = \rho\sinh\eta,$$

$$\mathrm{d}X^2 - \mathrm{d}T^2 = \mathrm{d}\rho^2 - \rho^2\,\mathrm{d}\eta^2, \quad X \pm T = \rho e^{\pm\eta}.$$

The reciprocal factors $e^{\pm\eta}$ are the same hyperbolic species as the centered prime sheets $e^{\pm x - i\theta}$. Their common metric face is proportional to $\cosh(2x)$ and is positive; their oriented half-difference is proportional to $\sinh(2x)$ and changes sign across the fixed seam. The Lorentzian difference records normal hand. It is not itself a positive carrier.

This supplies the correct division:

$$\text{positive common/Gram interior} \quad + \quad \text{signed hyperbolic normal and connection current.}$$

XI.5 The Euler metric supplies an operator half-Jacobian

Theorem 32 — The square root of a pulled metric gives the successor aperture an exact positive complement Let $J : H \to H'$ be boundedly invertible and put

$$G = J^*J.$$

Its polar decomposition is

$$J = UG^{\frac{1}{2}}, \quad U = JG^{-\frac{1}{2}},$$

with U unitary. If Π is a G-orthogonal projection, then

$$\tilde{P} = G^{\frac{1}{2}}\Pi G^{-\frac{1}{2}}$$

is an ordinary orthogonal projection. Consequently, for every $v \in H$,

$$\|Jv\|^2 = \left\|G^{\frac{1}{2}}\Pi v\right\|^2 + \left\|G^{\frac{1}{2}}(I-\Pi)v\right\|^2.$$

The same identity holds after summing over an orthonormal source basis whenever the corresponding operators are Hilbert–Schmidt.

Demonstration. Since G is strictly positive, its bounded positive square root and inverse exist. Direct calculation gives $U^*U = \mathbb{U}^* = I$. The G-self-adjoint identity $\Pi^*G = G\Pi$ implies

$$\tilde{P}^* = \tilde{P},$$

while $\Pi^2 = \Pi$ implies $\tilde{P}^2 = \tilde{P}$. Hence $\tilde{P}$ is orthogonal. Apply ordinary Pythagoras to $G^{\frac{1}{2}}v$ and use

$$\tilde{P}G^{\frac{1}{2}} = G^{\frac{1}{2}}\Pi$$

together with $\|Jv\| = \left\|G^{\frac{1}{2}}v\right\|$.

Boundary. $G^{\frac{1}{2}}$ is an operator-valued half-metric analogous to a scalar square-root Jacobian. The positive complement belongs to the complete successor metric at that event; it is not an untouched predecessor block and is not automatically the completed Weil defect. Identifying that complement, or another independently constructed coarea carrier, with the exact defect remains the RH-bearing open identity.

For one receiver transition J, put $G = J^*J$. Its polar decomposition is

$$J = UG^{\frac{1}{2}},$$

with U unitary. Hence

$$\|Jv\|^2 = \left\|G^{\frac{1}{2}}v\right\|^2.$$

The positive stretch $G^{\frac{1}{2}}$ is the operator analogue of the scalar half-Jacobian.

If Π is the pulled-back successor projection, which is orthogonal in the G-metric, then

$$\tilde{P} = G^{\frac{1}{2}}\Pi G^{-\frac{1}{2}}$$

is an ordinary orthogonal projection. Pythagoras now gives the exact successor decomposition

$$\|Jv\|^2 = \left\|G^{\frac{1}{2}}\Pi v\right\|^2 + \left\|G^{\frac{1}{2}}(I-\Pi)v\right\|^2.$$

The complement is genuinely positive, but it belongs to the complete successor metric. It is neither an untouched predecessor block nor, without another identity, the completed Weil remainder.

XI.6 The archimedean remainder already has a real amplitude

Theorem 33 — The conditioned archimedean remainder has an explicit spectral amplitude Let

$$I_2 = \left[-\frac{\log(2)}{2}, \frac{\log(2)}{2}\right], \quad \mathcal{H}_2 = L^2(I_2, \mathrm{d}x),$$

and let $\xi_0 = (\log 2)^{-\frac{1}{2}}$ be the normalized constant vector. Write P_0 for the orthogonal projection onto $\mathcal{H}_0 = \xi_0^\perp$.

Connes and Consani independently construct a self-adjoint operator

$$N_2 = -2\varepsilon'(1^+)(I - K_2)$$

on $\mathcal{H}_2$, where K_2 is the Hilbert–Schmidt integral operator determined by the prolate/Sonin kernel $Q\varepsilon$. Their spectral estimate gives

$$\langle \xi, N_2 \xi \rangle \leq \gamma \|(I - P_0)\xi\|^2.$$

Hence

$$B_2 = -P_0 N_2 P_0|_{\mathcal{H}_0} \geq 0$$

and its positive square root

$$A_2 = B_2^{\frac{1}{2}}$$

is defined without using the desired Weil defect.

Let $g \in C_c^{\infty(\mathbb{R}_{>0})}$ have support in $\left[2^{-\frac{1}{2}}, 2^{\frac{1}{2}}\right]$ and satisfy, in the final-theorem convention of the source,

$$\hat{g}\left(-\frac{i}{2}\right) = 0, \quad \hat{g}(0) = 0.$$

Define

$$k_{g(u)} = u^{\frac{1}{2}} \int_0^u v^{-\frac{1}{2}} g(v) \mathrm{d}^* v, \quad \xi_{g(x)} = k_{g(\mathrm{e}^x)}.$$

Then $\xi_g \in \mathcal{H}_0$ and the difference between the archimedean Weil response and the positive Sonin trace is exactly

$$W_{\infty(g*g^*)} - \mathrm{Tr}\left(\Theta_{\infty(g)} \boldsymbol{S} \Theta^*_{\infty(g)}\right) = \left\|A_2 \xi_g\right\|^2.$$

If E_2 is the spectral resolution of B_2, the same carrier has the direct-integral face

$$\left\|A_2 \xi_g\right\|^2 = \int_{[0,\infty)} \lambda \, \mathrm{d}\langle \xi_g, E_2(\lambda) \xi_g \rangle.$$

Demonstration. The source proves

$$\mathrm{Tr}\left(\Theta_{\infty(g)} \boldsymbol{S} \Theta^*_{\infty(g)}\right) = W_{\infty(g*g^*)} + E(g * g^*)$$

and, after the displayed logarithmic lift,

$$E(g * g^*) = \langle \xi_g, N_2 \xi_g \rangle.$$

Its identities

$$\hat{k_g}(0) = -2\hat{g}(0)$$

and $\hat{g}\left(-\frac{i}{2}\right) = 0$ put ξ_g respectively in $\xi_0^\perp$ and inside I_2. The quoted spectral estimate therefore makes the compression $-P_0 N_2 P_0$ positive. The positive square-root theorem gives the norm identity, and the spectral theorem gives its direct-integral form.

Boundary. This is a genuine independently sourced remainder carrier, not an abstract square root of a defect declared positive in advance. It is proved only at the one-place support aperture and on the conditioned current subspace. It does not construct the corresponding remainder operator after support growth or admission of a finite place.

This improves the status of the carrier problem. At the one-place aperture, the remainder is not merely known to be nonnegative. Connes and Consani construct the operator producing it:

$$N_2 = -2\varepsilon'(1^+)(I - K_2),$$

where K_2 comes from the prolate/Sonin kernel $Q\varepsilon$. After the constant mode is removed, $-N_2$ is positive. Its spectral square root therefore gives the exact amplitude

$$A_2 = (-P_0 N_2 P_0)^{\frac{1}{2}}$$

and, on the source's conditioned current family,

$$W_{\infty(g*g^*)} - \Sigma_{\infty(g)} = \left\| A_2 \xi_g \right\|^2.$$

This square root is not circular: N_2 was constructed from the archimedean trace remainder before the inequality was read. Its spectral resolution is an actual direct-integral carrier field. The open problem has therefore moved from "find any arithmetic amplitude" to "continue this amplitude through support growth and finite-place rebase."

XI.7 Prime powers enter as translated aperture intersections

Theorem 34 — Finite-prime currents are translated aperture intersections Let $R > 1$, put $L = \log R$ and

$$I_R = \left[-\frac{L}{2}, \frac{L}{2}\right].$$

Let $a \in L^2(\mathbb{R})$ be supported in I_R and define

$$g(x) = x^{-\frac{1}{2}} a(\log x), \quad h(x) = \int_0^\infty g(xy)\overline{g(y)}\, \mathrm{d}y.$$

For logarithmic translation

$$(U_t a)(v) = a(v+t),$$

one has

$$h(e^t) = e^{-\frac{t}{2}} \langle a, U_t a \rangle.$$

Consequently the finite-place response is

$$W_{p(h)} = 2 \log p \sum_{m \geq 1} p^{-\frac{m}{2}} \mathfrak{R} \langle a, U_{m \log p} a \rangle.$$

If P_R denotes multiplication by 1_{I_R}, then

$$\langle a, U_t a \rangle = \langle a, P_R U_t P_R a \rangle$$

and

$$P_R U_t P_R = 0 \quad \text{whenever} \quad |t| \geq L.$$

Thus the m-th prime-power current meets the aperture exactly when $p^m < R$; equality is a measure-zero boundary contact. For the box current $a = 1_{I_R}$,

$$W_{p(h)} = 2 \sum_{p^m < R} (\log p) p^{-\frac{m}{2}} (\log R - m \log p).$$

> **Demonstration.** Substitute $y = e^v$ and the half-density expression for g into the definition of h. The factors from the two copies of g and the measure $\mathrm{d}y = e^v\, \mathrm{d}v$ leave the single factor $e^{-\frac{t}{2}}$ and the translated inner product. Applying the same identity at $-t$ turns the two explicit-formula arms into twice the real part.
> The kernel of $P_R U_t P_R$ is supported on the intersection $I_R \cap (I_R - t)$. Its length is $\max(0, L - |t|)$, which proves the incidence criterion. For the box current, the translated inner product is exactly that overlap length.

Boundary. The theorem identifies when and where finite-prime recurrence contacts a bounded current. It does not sign the resulting correlation and does not combine it with the archimedean or Sonin-aperture connection terms. In particular it supplies the finite-place current in the defect recurrence, not the positive semilocal remainder carrier.

For a logarithmic current a supported on an interval of length $\log R$, the finite-place response is exactly

$$W_{p(h)} = 2\log p \sum_{m\geq 1} p^{-\frac{m}{2}} \Re\langle a, U_{m\log p} a\rangle.$$

Each term lives on the overlap between the aperture and its translate by $m\log p$. It is present precisely when $p^m < R$. For the box current, the overlap length is $\log R - m\log p$, recovering the earlier exact prime-population formula without treating the support cutoff as an external enumeration rule.

Theorem 35 — Prime-power hinge cells recur; conditioning creates their cross-axis Gram Define the finite-prime part of the zeta screw kernel by

$$g_{\mathrm{fin}(t)} = \sum_{m\geq 2} c_m(|t| - \ell_m)_+, \quad c_m = \frac{\Lambda(m)}{\sqrt{m}}, \quad \ell_m = \log m. \quad \text{(HINGE POPULATION)}$$

The sum is locally finite. For $R > 1$, put $L_R = \log R$, $I_R = \left[-\frac{L_R}{2}, \frac{L_R}{2}\right]$, and rescale

$$x = L_R\xi, \quad y = L_R\eta, \quad (\xi,\eta) \in C = \left[-\frac{1}{2}, \frac{1}{2}\right]^2.$$

A prime-power hinge $m = p^k$ meets this receiver exactly when $m < R$. With

$$\alpha_{m(R)} = \frac{\log m}{\log R},$$

its nonzero support in C is the paired triangular region

$$T^+_{m,R} = \left\{(\xi,\eta) : \xi - \eta > \alpha_{m(R)}\right\},$$

$$T^-_{m,R} = \left\{(\xi,\eta) : \eta - \xi > \alpha_{m(R)}\right\}. \quad \text{(PAIRED CELL)}$$

Each triangle has normalized leg $1 - \alpha_{m(R)}$ and physical leg $\log\left(\frac{R}{m}\right)$. On their union the hinge is the affine face

$$c_m L_R\left(|\xi - \eta| - \alpha_{m(R)}\right).$$

Let

$$\mathcal{P}_R = \{m \geq 2 : \Lambda(m) > 0, m < R\}.$$

The lines $\xi - \eta = \pm\alpha_{m(R)}$ for $m \in \mathcal{P}_R$ are parallel and divide C into exactly

$$2\,\mathrm{card}(\mathcal{P}_R) + 1$$

connected open strata. At the integer support successor $I_n \subset I_{n+1}$, the incidence poset changes if and only if $n = p^k$ is a prime power. In that case one new pair of corner cells is born; otherwise every existing cell only deforms. At the endpoint I_{n+1}, the number of traversals on the prime axis p is

$$N_{p(n+1)} = \left\lfloor \frac{\log n}{\log p} \right\rfloor,$$

and hence

$$\mathrm{card}(\mathcal{P}_{n+1}) = \sum_{p\leq n} \left\lfloor \frac{\log n}{\log p} \right\rfloor. \quad \text{(AXIS CENSUS)}$$

Along one prime axis, the cell positions and weights are

$$\ell_{p^k} = k \log p, \quad c_{p^k} = (\log p) p^{-\frac{k}{2}}. \quad \text{(LOG HARMONICS)}$$

Thus recurrence is equally spaced in logarithmic displacement and geometrically weighted in half-density.

Distributionally,

$$g_{\mathrm{fin}''}(t) = \sum_{m \geq 2} c_m \Big(\delta_{\ell_m} + \delta_{-\ell_m} \Big). \quad \text{(CROSSING MEASURE)}$$

Let $v \in H_0^1(I_R)$ and extend it by zero to the real line. With

$$(U_\ell v)(x) = v(x - \ell),$$

the $m = p^k$ hinge contributes to the localized Weil form exactly

$$q_{m,R}(v) = -2(\log p) p^{-\frac{k}{2}} \operatorname{Re}\langle U_{k \log p} v, v \rangle. \quad \text{(CROSSING ENERGY)}$$

Equivalently, if

$$\Delta_m = I - m^{-\frac{1}{2}} U_{\log m},$$

then

$$q_{m,R}(v) = (\log p) \Big(\|\Delta_m v\|^2 - (1 + m^{-1}) \|v\|^2 \Big). \quad \text{(SIGNED EDGE)}$$

The first term is a squared difference edge and the second is its exact diagonal potential. Neither may be dropped. Every active raw hinge has both signs: for a sufficiently small bump f with disjoint translates,

$$v_+ = f + U_{\log m} f, \quad v_- = f - U_{\log m} f$$

give respectively negative and positive values of $q_{m,R}$.

The raw additivity does not survive support conditioning as independent cellwise energy. Fix a successor $n \to n+1$, decompose its complete form on a common domain as

$$q_{n+1} = \sum_{\alpha \in \mathcal{A}_n} q_{\alpha, n+1}, \quad \mathcal{A}_n = \{\infty\} \cup \mathcal{P}_{n+1},$$

where ∞ denotes the complete non-finite-prime face. Let $\tilde{J}_n y$ be a conditioned quotient lift. Assume the component cross functionals

$$\ell_{\alpha,y}(x) = q_{\alpha,n+1}(x, \tilde{J}_n y)$$

descend continuously to the old energy completion $\mathcal{E}_n$, and let $c_{\alpha,y}$ be their Riesz carriers. Then the carrier of the complete cross is

$$c_y = \sum_{\alpha \in \mathcal{A}_n} c_{\alpha,y},$$

and the exact conditioned short is

$$s_{n(y)} = \sum_{\alpha \in \mathcal{A}_n} q_{\alpha,n+1}(\tilde{J}_n y) - \left\| \sum_{\alpha \in \mathcal{A}_n} c_{\alpha,y} \right\|_{\mathcal{E}_n}^2. \quad \text{(COHERENT SHORT)}$$

Expanding the last term gives

$$s_{n(y)} = \sum_\alpha q_{\alpha,n+1}(\tilde{J}_n y) - \sum_{\alpha,\beta} \langle c_{\alpha,y}, c_{\beta,y}\rangle_{\mathcal{E}_n}. \quad \text{(CARRIER GRAM)}$$

Hence moment rebase and old-interior elimination create exact cross-axis incidences between distinct prime-power and archimedean carriers even though their raw threshold lines are parallel.

Demonstration. Since $|x-y| \le L_R$ on I_R^2, the m-th hinge is nonzero exactly when $\ell_m < L_R$. After the displayed rescaling, its two strict inequalities cut congruent right triangles from the two opposed corners of C, proving ("PAIRED CELL"). Distinct positive threshold values produce parallel diagonal segments, so each pair adds two connected strata. At $R = n$ the active integers satisfy $m < n$, whereas at $R = n+1$ they satisfy $m \le n$. This proves the successor statement and ("AXIS CENSUS"). The identities in ("LOG HARMONICS") follow from $m = p^k$ and $\Lambda(p^k) = \log p$.

The second distributional derivative of $(|t| - \ell)_+$ is $\delta_\ell + \delta_{-\ell}$, proving ("CROSSING MEASURE"). Integrating twice by parts against $v'(y)\overline{v'(x)}$ gives ("CROSSING ENERGY"). Since translation is unitary on the zero-extended whole-line carrier,

$$\|(I - aU)v\|^2 = (1+a^2)\|v\|^2 - 2a \operatorname{Re}\langle Uv, v\rangle.$$

Substitution of $a = m^{-\frac{1}{2}}$ proves ("SIGNED EDGE"). Choose f so that $f, U_{\log m} f, U_{2\log m} f$ have pairwise disjoint supports and the first two lie in I_R. Then

$$\operatorname{Re}\langle U_{\log m} v_+, v_+\rangle = \|f\|^2,$$

while the same expression for v_- is $-\|f\|^2$. This proves the asserted raw indefiniteness.

Finally, linearity of the form makes the complete cross functional the sum of its component cross functionals. Uniqueness in the Riesz representation theorem therefore makes its carrier the sum of the component carriers. Substitute that sum into the invariant form-short identity from the conditioned-support theorem and expand its squared norm. This proves ("COHERENT SHORT") and ("CARRIER GRAM").

Boundary. The prime-power incidence combinatorics now scales exactly; it is not the missing RH theorem. The paired cells supply a signed graph-Schrödinger law—difference energy together with diagonal potential—not independent positive prime constituents. Conditioning then couples every active constituent through the common old energy carrier. Therefore a factorization of the form $\sum_\alpha \partial_\alpha^* W_\alpha \partial_\alpha$ with independent $W_\alpha \ge 0$ cannot be inferred from the hinge cells alone.

The precise remaining inequality is nonnegativity of the effective tension after the old body relaxes:

$$\sum_\alpha q_{\alpha,n+1}(\tilde{J}_n y) \ge \Big\| \sum_\alpha c_{\alpha,y} \Big\|_{\mathcal{E}_n}^2 \quad \text{for every admitted } y,$$

together with cross descent when the old form has a radical. At $n = 2$ this is exactly $S_{2,3} \ge 0$. A source-derived proof must couple the archimedean kernel, all active prime-power translations, the moment graph, and the old energy inverse. Further cell enumeration cannot decide that sign.

This closes the combinatorial scaling question. In the normalized aperture square, every active p^k cuts a pair of affine triangular hinge cells. Passing from I_n to I_{n+1} changes the cell incidence exactly when n is a prime power; otherwise the same cells deform. The number of traversals on one prime axis at the successor endpoint is

$$\left\lfloor \frac{\log n}{\log p} \right\rfloor.$$

The corresponding form is not a sum of positive prime tiles. Each hinge is exactly a weighted squared translation difference minus its diagonal potential. More importantly, the conditioned short subtracts the squared norm of the **sum** of all old-energy carriers. Its expansion therefore contains cross-terms between distinct prime powers and the archimedean face. The raw threshold lines are parallel; their consequential intersections are created by their common old carrier.

XI.8 The Euler resolvent folds the complete prime tower into one return

Lemma 36 — The normalized Euler resolvent folds every prime-power return into one amplitude Let U be unitary on a complex Hilbert space, let $0 < a < 1$, and put

$$J_a = I - aU, \quad G_a = J_a^* J_a, \quad d_a = \sqrt{1-a^2}.$$

The normalized Euler resolvent

$$\mathcal{R}_a = d_a J_a^{-1} = d_a \sum_{n\geq 0} a^n U^n$$

is independently defined by an operator-norm convergent Neumann series. Its returned metric is

$$\mathcal{R}_a^* \mathcal{R}_a = (1-a^2) G_a^{-1} = I + \sum_{m\geq 1} a^{m(U^m + (U^*)^m)}. \quad \text{(POISSON)}$$

For a prime p, take

$$a = p^{-\frac{1}{2}}, \quad \ell_p = \log p, \quad U = U_{\ell_p}.$$

Whenever the logarithmic current x represents a returned test current $h = f * f^\sharp$, the finite-place response is exactly

$$W_{p(h)} = \ell_{p\left(\|\mathcal{R}_p x\|^2 - \|x\|^2\right)}. \quad \text{(PRIME AMPLITUDE)}$$

The same resolvent is the fixed-state response of the lossless two-port operator

$$\mathcal{J}_a = \begin{pmatrix} aU & d_a I \\ d_a I & -aU^* \end{pmatrix}, \quad \mathcal{J}_a^* \mathcal{J}_a = I.$$

Indeed, the state equation

$$s = aUs + d_a x$$

has the unique solution $s = \mathcal{R}_a x$.

This resolvent is also the exact inverse of prime admission in the semilocal Poisson population. For a finite receiver S let

$$\mathcal{M}_S = \{m \geq 1 : q \nmid m \ \text{ for every finite } \ q \in S\}$$

and, wherever the sum converges absolutely, define

$$(\mathcal{E}_S f)(x) = |x|^{\frac{1}{2}} \sum_{m\in\mathcal{M}_S} f(mx).$$

If $p \notin S$ and $\left(V_p \varphi\right)(x) = \varphi(px)$ on $L^2(\mathbb{R}_+^*, \mathrm{d}^* x)$, then

$$\mathcal{E}_{S\cup\{p\}} = \left(I - p^{-\frac{1}{2}} V_p\right) \mathcal{E}_S. \quad \text{(ADMISSION)}$$

Therefore, with $J_p = I - p^{-\frac{1}{2}} V_p$,

$$\mathcal{R}_p \mathcal{E}_{S\cup\{p\}} = \sqrt{1-p^{-1}} \mathcal{E}_S. \quad \text{(RETURN)}$$

Demonstration. Since $\|aU\| = a < 1$, the Neumann series for J_a^{-1} converges in operator norm. The operator J_a is normal because it is a polynomial in the unitary U. Hence

$$\mathcal{R}_a^* \mathcal{R}_a = (1-a^2)(J_a^* J_a)^{-1}.$$

Expanding both resolvents gives

$$(1-a^2) \sum_{r,s\geq 0} a^{r+s} U^{(s-r)}.$$

The coefficient of U^m and of U^{-m} is a^m for every $m \geq 1$, while the coefficient of I is one. This proves ("POISSON") in operator norm.

Taking the quadratic response of ("POISSON") gives

$$\|\mathcal{R}_a x\|^2 - \|x\|^2 = 2\sum_{m\geq 1} a^m \operatorname{Re}\langle x, U^m x\rangle.$$

The prime-power aperture-incidence theorem identifies the right side, after multiplication by ℓ_p, with $W_{p(h)}$. Direct block multiplication proves that $\mathcal{J}_a$ is unitary because $a^2 + d_a^2 = 1$ and d_a commutes with U. Solving its first-row fixed state equation gives the displayed resolvent.
Finally,

$$\mathcal{M}_S = \mathcal{M}_{S\cup\{p\}} \cup p\mathcal{M}_S, \quad \text{(disjoint)}.$$

Splitting the defining sum for $\mathcal{E}_S$ along this disjoint union gives

$$\mathcal{E}_S f = \mathcal{E}_{S\cup\{p\}} f + p^{-\frac{1}{2}} V_p \mathcal{E}_S f,$$

which is ("ADMISSION"). Applying $d_p J_p^{-1}$ gives ("RETURN").

Boundary. This is a positive endpoint factorization of the entire signed prime-power response; it does not say that W_p is positive. For the translation representation, whose spectrum contains one,

$$\|\mathcal{R}_a\|^2 = \frac{1+a}{1-a} > 1.$$

Thus the prime channel is not itself contractive. The local Poisson metric in ("POISSON") and the monoid partition in ("ADMISSION") do not by themselves supply the global additive Poisson summation formula. That further relation conjugates additive Fourier transform with multiplicative inversion and fixes the compatible arithmetic principal-value normalization. It is the missing source of cross-channel coupling, not an interchangeable name for the resolvent identity.

Put $a = p^{-\frac{1}{2}}$ and $J_p = I - aU_{\log p}$. The normalized inverse

$$\mathcal{R}_p = \sqrt{1-p^{-1}}\, J_p^{-1}$$

has the exact Poisson-kernel metric

$$\mathcal{R}_p^* \mathcal{R}_p = I + \sum_{m\geq 1} p^{-\frac{m}{2}} \big(U_{m\log p} + U_{m\log p}^*\big).$$

Consequently the **whole** prime-power response, including every translated overlap which the current can meet, is one endpoint norm change:

$$W_{p(f*f^\sharp)} = \log p\Big(\big\|\mathcal{R}_p L_R f\big\|^2 - \|L_R f\|^2\Big).$$

The signed prime current is therefore not a detached negative or positive block. It is the difference between two positive amplitudes before and after one returned Euler passage.

The same operator is already present in the global arithmetic population. If

$$(\mathcal{E}_S f)(x) = |x|^{\frac{1}{2}} \sum_{m\in\mathcal{M}_S} f(mx),$$

where $\mathcal{M}_S$ contains the positive integers prime to the finite places of S, then partitioning that population into terms divisible and not divisible by a new prime gives

$$\mathcal{E}_{S\cup\{p\}} = J_p \mathcal{E}_S, \quad \mathcal{R}_p \mathcal{E}_{S\cup\{p\}} = \sqrt{1-p^{-1}}\,\mathcal{E}_S.$$

Prime admission removes the p-divisible sheet; the resolvent returns the predecessor population from the admitted one. The powers p^m are the successive feedback turns of this single relation. The associated two-port Julia operator is unitary, so the return can be embedded in a lossless colligation even though its observed prime channel is not contractive.

Corollary 37 — A finite place cannot enter through the fixed archimedean carrier aperture
At the base aperture I_2, for every prime $p \geq 2$ and every $m \geq 1$,

$$P_2 U_{m\log p} P_2 = 0.$$

Hence for the one-prime Euler difference

$$J_p = I - p^{-\frac{1}{2}} U_{\log p}$$

one has

$$P_2 J_p P_2 = P_2.$$

Extending the base amplitude form $B_2 = -P_0 N_2 P_0$ by zero outside I_2 therefore gives

$$P_2 J_p^* B_2 J_p P_2 = B_2.$$

A fixed-aperture conjugation of the archimedean carrier cannot produce either the prime-power response or the receiver-aperture connection.

After aperture growth, a semilocal carrier would instead have to be independently defined so that, in one anchored comparison chart,

$$B_{S\cup\{p\},R} - B_{S,R} = -\mathcal{W}_{p,R} - \mathcal{K}_{S,p,R}$$

as Hermitian forms, where the two forms on the right represent W_p and $\kappa_{S,p}$. This is precisely the operator form of the completed-defect recurrence.

> **Demonstration.** The interval I_2 has length $\log 2$, while $m \log p \geq \log 2$. The translated-aperture incidence theorem therefore makes every compressed shift zero. Expanding the two displayed compressions gives the fixed-carrier identities. The required successor equation is the completed-defect recurrence written after polarizing its quadratic responses.

> Boundary. This corollary does not assert that semilocal transport is impossible. It proves that the published base remainder cannot be propagated by keeping its support domain fixed. The semilocal intertwiners construct the receiver, Euler metric, Sonin aperture, and connection term; the audited sources do not construct the expanded remainder operator $B_{S,R}$ or prove the displayed form identity.

The published base interval has length $\log 2$. Every prime translation is at least $\log 2$, so its interior overlap is empty. This is why the archimedean theorem excludes finite primes geometrically, and why applying the Euler difference while compressing back into the same base interval cannot propagate the remainder: the translated arm is erased before it can meet the current.

The needed successor is therefore not a conjugate of one fixed N_2. It is an expanded, receiver-relative operator $B_{S,R}$ whose polarized form obeys

$$B_{S\cup\{p\},R} - B_{S,R} = -\mathcal{W}_{p,R} - \mathcal{K}_{S,p,R}.$$

The first term is now the explicit translated-overlap form; the second is the already-derived Sonin-aperture connection. The semilocal sources construct the receiving space and its Sonin transport, but do not construct this remainder operator or prove this equality.

XI.9 The first dyadic contact has two polarizations

Theorem 38 — The first dyadic Euler contact is the polarization of two positive boundary channels Put

$$\ell = \log 2, \quad a = 2^{-\frac{1}{2}}, \quad I_4 = [-\ell, \ell],$$

and divide I_4 into the four half-prime cells

$$A_1 = \left[-\ell, -\frac{\ell}{2}\right], \quad A_2 = \left[-\frac{\ell}{2}, 0\right], \quad A_3 = \left[0, \frac{\ell}{2}\right], \quad A_4 = \left[\frac{\ell}{2}, \ell\right].$$

Let $U = U_\ell$ be logarithmic translation, $(Uf)(v) = f(v + \ell)$. Choose normalized smooth vectors e_3, e_4 supported in the interiors of A_3, A_4, respectively, and set

$$e_1 = Ue_3, \quad e_2 = Ue_4, \quad \mathcal{V}_4 = \operatorname{span}\{e_1, e_2, e_3, e_4\}.$$

These vectors are orthonormal. If P_4 is multiplication by 1_{I_4}, then the compressed translation

$$T = (P_4 U P_4)|_{\mathcal{V}_4}$$

satisfies

$$Te_3 = e_1, \quad Te_4 = e_2, \quad Te_1 = Te_2 = 0, \quad T^2 = 0,$$

and therefore has the matrix

$$T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

in the displayed basis.

Let

$$\mathcal{R}_2 = a(I - aU)^{-1} = a \sum_{n \geq 0} a^n U^n$$

be the normalized $p = 2$ Euler resolvent, and put

$$r_4 = (P_4 \mathcal{R}_2 P_4)|_{\mathcal{V}_4}, \quad z_4 = ((I - P_4)\mathcal{R}_2 P_4)|_{\mathcal{V}_4}.$$

Since $a = \sqrt{1 - a^2}$ and $T^2 = 0$,

$$r_4 = a(I + aT). \quad \text{(INTERIOR RETURN)}$$

For $x = \sum_{j=1}^{4} x_j e_j$, define the two exact boundary amplitudes

$$B_- x = (ax_1 - a^2 x_3, ax_2 - a^2 x_4),$$

$$B_+ x = (ax_1 + a^2 x_3, ax_2 + a^2 x_4).$$

Then

$$I - r_4^* r_4 = B_-^* B_-, \quad z_4^* z_4 = B_+^* B_+, \quad \text{(TWO CHANNELS)}$$

and orthogonal aperture decomposition gives

$$P_4 \mathcal{R}_2^* \mathcal{R}_2 P_4 - r_4^* r_4 = z_4^* z_4. \quad \text{(DEPARTURE)}$$

Consequently

$$I - P_4 \mathcal{R}_2^* \mathcal{R}_2 P_4 = B_-^* B_- - B_+^* B_+ = -a(T + T^*). \quad \text{(POLARIZATION)}$$

Whenever x is the logarithmic current of an admitted returned test current h, the complete $p = 2$ response on this cell is therefore

$$W_2(h) = \ell\Big(\big\|B_+ x\big\|^2 - \|B_- x\|^2\Big). \quad \text{(PRIME FACE)}$$

Demonstration. Translation by ℓ carries A_3 to A_1 and A_4 to A_2. A second translation leaves I_4 up to measure-zero boundary, proving the matrix for T and $T^2 = 0$. Compressing the Neumann series therefore gives ("INTERIOR RETURN").
Direct multiplication gives

$$r_4^* r_4 = a^2 I + a^3(T + T^*) + a^4 T^* T.$$

Since $a^2 = \frac{1}{2}$, the restriction of $I - r_4^* r_4$ to each translated pair (e_1, e_3) and (e_2, e_4) is

$$\begin{pmatrix} a^2 & -a^3 \\ -a^3 & a^4 \end{pmatrix} = \begin{pmatrix} a \\ -a^2 \end{pmatrix} (a \ \ -a^2).$$

This is $B_-^* B_-$.

Put $Q_4 = I - P_4$. Since P_4 and Q_4 are orthogonal,

$$P_4 \mathcal{R}_2^* \mathcal{R}_2 P_4 = r_4^* r_4 + z_4^* z_4.$$

The two departing tails satisfy

$$z_4 e_3 = a z_4 e_1, \quad z_4 e_4 = a z_4 e_2, \quad \|z_4 e_1\|^2 = \|z_4 e_2\|^2 = a^2.$$

Hence the Gram matrix of z_4 on either translated pair is

$$\begin{pmatrix} a^2 & a^3 \\ a^3 & a^4 \end{pmatrix} = \begin{pmatrix} a \\ a^2 \end{pmatrix} (a \ \ a^2),$$

proving (“TWO CHANNELS”) and (“DEPARTURE”).

The Poisson-kernel metric of $\mathcal{R}_2$, compressed to $\mathcal{V}_4$, is

$$P_4 \mathcal{R}_2^* \mathcal{R}_2 P_4 = I + a(T + T^*),$$

because no higher prime-power translate overlaps this cell. Subtracting the two positive Gram matrices proves (“POLARIZATION”). Taking its quadratic form and applying the normalized Euler-resolvent response identity proves (“PRIME FACE”).

Boundary. This theorem does not make the prime response positive. It identifies its sign-indefinite part exactly: the same core/shell incidence has a difference polarization B_- retained by the compressed return and a sum polarization B_+ carried by the departing tail. In the completed defect, B_- contributes on the positive side and B_+ on the negative side.

The equality $a^2 = 1 - a^2 = \frac{1}{2}$ is special to the first prime. For the analogous $R = p^2$ two-cell chain, the determinant of $I - r^* r$ is $a^2(2a^2 - 1)$; it vanishes at $p = 2$ and is negative for $p > 2$. This dyadic half is an exact normalization balance, not by itself the critical-line theorem.

The calculation also fixes one ambient logarithmic Hilbert fiber, one aperture P_4, one translation law, and one receiver metric. It is therefore an exact local edge face, not a complete three-frame counting return. Arithmetic place admission, archimedean Fourier–Poisson–Mellin transport, and receiver/aperture change still owe an actual common incidence and the other cross-frame transports. Recasting this cell as a triangle without constructing those maps would only rename the missing relation.

Global Poisson summation intertwines additive Fourier return with multiplicative inversion on a constrained Schwartz domain, and the semilocal Sonin isomorphism intertwines Fourier at fixed support. Neither established map carries the outgoing amplitude B_+ into the independently constructed base remainder amplitude A_2, nor extends A_2 from I_2 to I_4. The remaining proof-bearing datum is therefore a mixed support/place intertwiner coupling this explicit departing channel to the enlarged archimedean/Sonin carrier.

The smallest joint support/place cell is now explicit. Set $\ell = \log 2$ and enlarge the logarithmic aperture from

$$I_2 = \left[-\frac{\ell}{2}, \frac{\ell}{2}\right]$$

to

$$I_4 = [-\ell, \ell].$$

“Joint” here means that one fixed logarithmic receiver admits both the first support growth and the first prime translation. It does not mean that the arithmetic, archimedean, and receiving lineages have already been constructed as one asynchronous triangle. The ambient L^2 fiber, translation, aperture, and inner product remain fixed for this local calculation.

Dividing I_4 into four consecutive half-ℓ cells turns translation by ℓ into two incidence chains:

$$e_3 \to e_1 \to \text{outside}, \quad e_4 \to e_2 \to \text{outside}.$$

Thus the compressed translation has square zero. Because the Euler weight and the lossless-port normalization agree at the first prime,

$$a^2 = 2^{-1} = 1 - a^2,$$

the normalized compressed return has a positive rank-two defect $B_-^*B_-$. The discarded continuation is not nothing: its exact Gram operator is the second positive rank-two channel $B_+^*B_+$. Their difference is the full prime face:

$$-W_2(h) = \ell\Big(\|B_-x\|^2 - \|B_+x\|^2\Big).$$

This is the first concrete cross-face which the earlier scalar recurrence hid. The minus and plus channels are not separate facts about the prime. They are the two polarizations of the same translated core/shell incidence:

$$|au + a^2v|^2 - |au - a^2v|^2 = 4a^3\,\mathrm{Re}(\overline{u}v).$$

The finite-prime term is therefore an interference or cross term between two occurrences of one current, not a scalar property attached to either occurrence.

Theorem 39 — The dyadic prime defect is a (C_2)-character signature and forces a symmetric filler In the dyadic cell $\mathcal{V}_4 = \mathrm{span}\{e_1, e_2, e_3, e_4\}$, define the core–shell exchange

$$Re_1 = e_3, \quad Re_3 = e_1, \quad Re_2 = e_4, \quad Re_4 = e_2.$$

Then $R = R^* = R^{-1}$ and, for the compressed translation T of the dyadic polarization theorem,

$$R = T + T^*.$$

Its two exact character projectors are

$$P_+ = \frac{I+R}{2}, \quad P_- = \frac{I-R}{2}. \quad \text{(CHARACTERS)}$$

Their ranges are respectively spanned by

$$s_1 = \frac{e_1+e_3}{\sqrt{2}}, \quad s_2 = \frac{e_2+e_4}{\sqrt{2}},$$

and

$$a_1 = \frac{e_1-e_3}{\sqrt{2}}, \quad a_2 = \frac{e_2-e_4}{\sqrt{2}}.$$

Put $a = 2^{-\frac{1}{2}}$, $\ell = \log 2$, and let $\mathcal{D}_2$ be the Hermitian operator contributed by $-W_2$ to the completed-defect recurrence on this cell. Then

$$\mathcal{D}_2 = \ell(B_-^*B_- - B_+^*B_+) = -\ell aR = c(P_- - P_+), \quad c = \frac{\log 2}{\sqrt{2}}. \quad \text{(SIGNATURE)}$$

Thus the symmetric character space is exactly the two-dimensional negative subspace and the antisymmetric character space is exactly the two-dimensional positive subspace. This inertia is invariant under every invertible rechart.

Let an anchored linear section of admitted test currents carry $x \in \mathcal{V}_4$ to a current f_x whose logarithmic prime channel is x. Suppose the remaining predecessor and aperture terms define a bounded Hermitian operator F_4 by

$$\langle x, F_4x\rangle = D_{S(f_x)} - \kappa_{S,2}(f_x).$$

Positivity of the compressed successor defect requires

$$F_4 + \mathcal{D}_2 \geq 0. \quad \text{(SUCCESSOR)}$$

In particular,

$$P_+F_4P_+ \geq cP_+. \quad \text{(SYMMETRIC DEMAND)}$$

Relative to $\mathcal{V}_4 = P_+\mathcal{V}_4 \oplus P_-\mathcal{V}_4$, write

$$F_4 = \begin{pmatrix} F_{++} & F_{+-} \\ F_{-+} & F_{--} \end{pmatrix}.$$

If $F_{--} + cI$ is strictly positive, ("SUCCESSOR") is equivalent to the Schur-complement condition

$$F_{--} + cI > 0, \quad F_{++} - cI \geq F_{+-}(F_{--} + cI)^{-1}F_{-+}. \quad \text{(MIXED DEMAND)}$$

Finally, among positive operators $H \geq 0$ used to repair the isolated dyadic contribution,

$$\mathcal{D}_2 + H \geq 0,$$

every such filler obeys

$$P_+HP_+ \geq cP_+, \quad \mathrm{Tr}(H) \geq 2c.$$

The unique minimum-trace filler is

$$H_{\min} = cP_+, \quad \mathcal{D}_2 + H_{\min} = cP_-. \quad \text{(MINIMUM FILLER)}$$

Demonstration. The displayed action of T gives $Te_3 = e_1, Te_4 = e_2$, and zero on e_1, e_2; its adjoint reverses those arrows. Hence $T + T^* = R$. The exchange is a self-adjoint involution, so ("CHARACTERS") are its orthogonal spectral projectors and the displayed symmetric and antisymmetric vectors are their bases.
The dyadic polarization theorem gives

$$B_-^*B_- - B_+^*B_+ = -a(T + T^*).$$

Multiplying by ℓ and substituting $R = P_+ - P_-$ proves ("SIGNATURE"). The signature statement and its invariance follow from the spectral decomposition and Hermitian inertia.
Compressing ("SUCCESSOR") to $P_+\mathcal{V}_4$ gives ("SYMMETRIC DEMAND"). In the character decomposition, the successor operator is

$$\begin{pmatrix} F_{++} - cI & F_{+-} \\ F_{-+} & F_{--} + cI \end{pmatrix},$$

whose positivity is equivalent to ("MIXED DEMAND") by the ordinary Schur-complement theorem under the stated strict-positivity hypothesis.
If $\mathcal{D}_2 + H \geq 0$, compression to the negative character space gives $P_+HP_+ \geq cP_+$ and therefore $\mathrm{Tr}(H) \geq 2c$. The operator $H_{\min} = cP_+$ reaches the bound and leaves cP_-. Equality of the trace forces the P_- compression of a positive H to vanish; positivity then forces its cross blocks to vanish as well, proving uniqueness.

Boundary. This theorem does not construct F_4, the Poisson–Sonin passage, or a global positive carrier. It gives a necessary local signature which every such construction must satisfy after anchoring to this dyadic section. The minimum filler is minimal only among additive positive repairs of the isolated prime contribution; the actual semilocal relation may be larger and may mix the two character spaces, in which case the Schur term is owed as well.

The $\frac{1}{2}$ in the projectors $\frac{I \pm R}{2}$ is finite C_2 character normalization. The equality $a^2 = \frac{1}{2}$ is the separate dyadic Euler balance. Neither is the Mellin half-density merely because their rational glyphs agree.

The radix/character distinction makes this boundary sharper. The core–shell exchange

$$R = T + T^*$$

is an exact C_2 action. Its symmetric and antisymmetric projectors are

$$P_+ = \frac{I + R}{2}, \quad P_- = \frac{I - R}{2}.$$

The contribution of $-W_2$ to the completed defect is

$$\mathcal{D}_2 = \frac{\log 2}{\sqrt{2}}(P_- - P_+).$$

Thus the missing passage does not merely owe "some positivity." After anchoring to this cell, its predecessor-remainder and aperture contribution F_4 must satisfy

$$P_+ F_4 P_+ \geq \frac{\log 2}{\sqrt{2}} P_+.$$

If the passage mixes character modes, it also owes the exact Schur complement recorded in the theorem. A positive filler which repairs only the isolated prime contribution has unique minimum trace at

$$H_{\min} = \frac{\log 2}{\sqrt{2}} P_+.$$

This does not construct the semilocal filler, but it rules out every candidate which misses the symmetric core–shell modes or treats the four cells independently.

XI.10 The positive half is an oriented boundary factor

Theorem 40 — The positive half-density factor is an oriented boundary carrier Work first in the logarithmic coordinate $u = \log \rho$ and put

$$D = \partial_u, \quad \mathcal{L}_+ = D + \frac{1}{2}, \quad \mathcal{L}_- = -D + \frac{1}{2}. \quad \text{(ORIENTED FACTORS)}$$

The laboratory symbol $\mathcal{L}_+$ names this positively oriented half-density factor. It is not the L_+ member of a knot skein triple. On compactly supported smooth currents,

$$Q = -D^2 + \frac{1}{4} = \mathcal{L}_- \mathcal{L}_+ = \mathcal{L}_+^* \mathcal{L}_+ \geq 0. \quad \text{(POSITIVE INTERIOR)}$$

On a finite interval $[a, b]$, before the endpoint values are suppressed, one instead has the exact boundary balance

$$\left\| \mathcal{L}_+ h \right\|^2 = \|Dh\|^2 + \frac{1}{4}\|h\|^2 + \frac{1}{2}(|h(b)|^2 - |h(a)|^2). \quad \text{(POSITIVE CROSSING)}$$

Reversing the orientation replaces $\mathcal{L}_+$ by $\mathcal{L}_-$ and reverses only the signed endpoint term.

Let additive convolution have involution $g^*(u) = \overline{g(-u)}$. Then

$$Q(g * g^*) = (\mathcal{L}_+ g) * (\mathcal{L}_+ g)^*. \quad \text{(COMMUTING POSITIVE SQUARE)}$$

Thus the same first-order crossing which creates the signed boundary flux creates a positive Gram interior after the crossing is closed with its adjoint.

If g is supported in $\left[-\frac{A}{2}, \frac{A}{2}\right]$, then both sides of ("COMMUTING POSITIVE SQUARE") are supported in $[-A, A]$. Under $\rho = e^u$, this is the exact multiplicative support passage

$$\left[e^{-\frac{A}{2}}, e^{\frac{A}{2}}\right] * \left[e^{-\frac{A}{2}}, e^{\frac{A}{2}}\right] \to \left[e^{-A}, e^{A}\right]. \quad \text{(HALF SUPPORT)}$$

At $A = \log 2$, its root and closed supports are respectively

$$\left[2^{-\frac{1}{2}}, 2^{\frac{1}{2}}\right] \quad \text{and} \quad \left[\frac{1}{2}, 2\right].$$

> **Demonstration.** With compact support, $D^* = -D$, so
>
> $$\mathcal{L}_+^* = -D + \frac{1}{2} = \mathcal{L}_-.$$
>
> Multiplying the two factors cancels their mixed first-order terms and proves ("POSITIVE INTERIOR"). Expanding the norm on $[a, b]$ gives

$$\int_a^b \left| Dh + \frac{h}{2} \right|^2 \mathrm{d}u = \int_a^b \left(|Dh|^2 + \frac{|h|^2}{4} \right) \mathrm{d}u + \int_a^b \mathrm{Re}\Big(Dh\overline{h} \Big)\, \mathrm{d}u.$$

The last integral is $\frac{|h(b)|^2-|h(a)|^2}{2}$, proving ("POSITIVE CROSSING"). The reverse factor changes the sign of this cross term.

Fourier transformation sends D to multiplication by it. Hence

$$Q\widehat{(g * g^*)}(t) = \left(t^2 + \frac{1}{4}\right)|\hat{g}(t)|^2 = \left| \left(it + \frac{1}{2} \right) \hat{g}(t) \right|^2,$$

which is the transform of $(\mathcal{L}_+ g) * (\mathcal{L}_+ g)^*$. Convolution adds supports, while a differential operator does not enlarge them, proving ("HALF SUPPORT").

Boundary. The $\frac{1}{2}$ here is not a dimensionless assertion that every geometric surface has raw area $\frac{1}{2}$. It is the coefficient which splits a full Jacobian or endpoint flux between an amplitude and its adjoint, and it is the exponent which halves logarithmic support before a positive convolution closure. A geometric half-area statement requires a normalized measure and an actual crossing surface.

The factorization constructs the support-preserving positive operator used by the archimedean vanishing ideal. It does not by itself identify that positive square with the complete Weil response after finite-prime admission.

The symbol L_+ had been used too loosely in the preceding discussion. Hereafter the laboratory crossing operator is written $\mathcal{L}_+$ to keep it separate from the positive crossing diagram in a knot skein triple. In the logarithmic coordinate it is

$$\mathcal{L}_+ = \partial_u + \frac{1}{2}, \quad \mathcal{L}_- = -\partial_u + \frac{1}{2},$$

and the differential operator which imposes the two Mellin vanishing characters factors exactly as

$$-\partial_u^2 + \frac{1}{4} = \mathcal{L}_- \mathcal{L}_+ = \mathcal{L}_+^* \mathcal{L}_+.$$

This is the same support-preserving operator used by Connes and Consani to describe the ideal of test functions vanishing at the two half-characters [20].

The factorization gives the positive $\frac{1}{2}$ an ontological role which a sign label does not. On a finite oriented interval,

$$\left\| \mathcal{L}_+ h \right\|^2 = \|\partial_u h\|^2 + \frac{1}{4}\|h\|^2 + \frac{1}{2}(|h(b)|^2 - |h(a)|^2).$$

The interior closes positively; the direction of crossing remains visible as one signed boundary flux. Reversing the receiver reverses this boundary term without changing the closed second-order interior.

The same factor commutes with positive convolution closure:

$$\left(-\partial_u^2 + \frac{1}{4} \right)(g * g^*) = (\mathcal{L}_+ g) * (\mathcal{L}_+ g)^*.$$

Thus $\frac{1}{2}$ is not inserted after the fact to repair a sign. It is the first-order connection coefficient whose pairing with the opposite orientation creates the positive interior. It simultaneously halves logarithmic support:

$$\mathrm{supp}(g) \subset \left[-\frac{A}{2}, \frac{A}{2} \right] \quad \Rightarrow \quad \mathrm{supp}(g * g^*) \subset [-A, A].$$

For $A = \log 2$, exponentiation gives the exact passage

$$\left[2^{-\frac{1}{2}}, 2^{\frac{1}{2}}\right] \to \left[\frac{1}{2}, 2\right]$$

used in the published archimedean positive cell. The coefficient $\frac{1}{2}$, the square-root support endpoints, and the critical Mellin chart are therefore three faces of one amplitude/adjoint closure, not three coincident glyphs.

XI.11 Convexity turns the half into a crossing surface

Theorem 41 — A convex overlap field has an oriented half-crossing surface Let $K \subset \mathbb{R}^n$ be a convex body with volume $V_K > 0$, and define its overlap field

$$c_{K(x)} = \mathrm{Vol}(K \cap (K + x)) = (1_K * 1_K^*)(x). \quad \text{(OVERLAP FIELD)}$$

Then c_K is even and positive definite,

$$\mathrm{supp}(c_K) = K - K, \quad \widehat{c_K} = \left|\widehat{1_K}\right|^2,$$

and $c_K^{\frac{1}{n}}$ is concave on $K - K$.

For $0 < t < 1$, put

$$\Omega_t = \left\{x : c_{K(x)} \geq tV_K\right\}. \quad \text{(OVERLAP STRATUM)}$$

Every Ω_t is centrally symmetric and convex. The laboratory positive-half crossing is the outward-oriented regular face

$$L_+(K) = \partial\Omega_{\frac{1}{2}} = \left\{x : c_{K(x)} = \frac{V_K}{2}\right\}. \quad \text{(HALF CROSSING)}$$

A receiver direction selects its positive branch; the opposite branch is the orientation-reversed $L_-(K)$.

Whenever c_K is continuously differentiable on the swept regular strata, neighboring overlap boundaries obey the exact coarea transport

$$\mathrm{Vol}(\Omega_\alpha) - \mathrm{Vol}(\Omega_\beta) = \int_\alpha^\beta \left(\int_{c_K^{-1}(tV_K)} \frac{V_K}{\left\|\nabla c_{K(x)}\right\|} \, \mathrm{d}\mathcal{H}^{n-1}(x)\right) \mathrm{d}t \quad \text{(BOUNDARY SWEEP)}$$

for $0 < \alpha < \beta < 1$. Thus the changing interior between two neighboring shape boundaries is carried by their receiver-relative surface field; it is not reconstructed from a final contour.

In one logarithmic dimension, for $K = \left[-\frac{\ell}{2}, \frac{\ell}{2}\right]$,

$$c_{K(x)} = (\ell - |x|)_+, \quad L_+(K) = \left\{\frac{\ell}{2}\right\}, \quad L_-(K) = \left\{-\frac{\ell}{2}\right\}. \quad \text{(DYADIC FACE)}$$

Hence the positive crossing has normalized overlap $\frac{c_{K(\frac{\ell}{2})}}{c_{K(0)}} = \frac{1}{2}$. When $\ell = \log 2$, exponentiation sends the two branches to $2^{\frac{1}{2}}$ and $2^{-\frac{1}{2}}$, exactly the two boundary points of the multiplicative convolution-root aperture.

> **Demonstration.** Formula (“OVERLAP FIELD”) is an autocorrelation. Fourier transformation therefore gives the displayed modulus square, proving positive definiteness and evenness. Its support is the difference body $K - K$.
> For $x, y \in K - K$ and $0 \leq \lambda \leq 1$,
> $$(1 - \lambda)(K \cap (K + x)) + \lambda(K \cap (K + y)) \subset K \cap (K + ((1 - \lambda)x + \lambda y)).$$
> Brunn–Minkowski applied to this inclusion proves concavity of $c_K^{\frac{1}{n}}$. Superlevel sets of a concave function are convex; evenness makes them centrally symmetric. This proves (“OVERLAP STRATUM”) and the existence of the half-crossing boundary.
> Apply the coarea theorem to the scalar map $\frac{c_K}{V_K}$ on the region between $\Omega_\beta \subset \Omega_\alpha$. Its normal Jacobian is $\frac{\|\nabla c_K\|}{V_K}$, which gives (“BOUNDARY SWEEP”).

For an interval of length ℓ, translating by x leaves an overlap of length $(\ell - |x|)_+$. Substitution gives ("DYADIC FACE") and its exponential image.

Boundary. What equals $\frac{1}{2}$ is the receiver-normalized overlap on the crossing face, not the unnormalized Hausdorff area of that face. The weighted surface measure in ("BOUNDARY SWEEP") is the derivative of swept interior volume and depends on the local curvature through $\|\nabla c_K\|$. At a singular half level, the convex boundary remains defined, while the classical smooth integrand must be replaced by the corresponding measure-theoretic coarea statement.

Convexity supplies a real positive crossing surface and a lawful change-of-boundary calculus. It does not identify every completed-zeta current with one convex-body covariogram.

The geometric face is supplied by an autocorrelation rather than by drawing a preferred contour. For a convex body K, define

$$c_{K(x)} = \mathrm{Vol}(K \cap (K + x)) = (1_K * 1_K^*)(x).$$

This is a genuine field of co-present interior: its value measures what remains common while one boundary is transported relative to the other. Its Fourier transform is $\left|\widehat{1_K}\right|^2$, and its support is the difference body $K - K$; these are the standard covariogram relations [21]. Brunn–Minkowski makes $c_K^{\frac{1}{n}}$ concave [22]. Consequently every normalized overlap stratum

$$\Omega_t = \left\{x : c_{K(x)} \geq t c_{K(0)}\right\}$$

is convex and centrally symmetric.

The positively oriented half crossing is now an actual receiver-relative surface:

$$L_+(K) = \partial\Omega_{\frac{1}{2}} = \left\{x : c_{K(x)} = \frac{c_{K(0)}}{2}\right\}.$$

The opposite receiver orientation sees $L_-(K)$. Neighboring level surfaces carry the change of interior by coarea:

$$\mathrm{Vol}(\Omega_\alpha) - \mathrm{Vol}(\Omega_\beta) = \int_\alpha^\beta \int_{c_K^{-1}\left(t c_{K(0)}\right)} \frac{c_{K(0)}}{\|\nabla c_K\|} \,\mathrm{d}\mathcal{H}^{n-1}\,\mathrm{d}t.$$

This is the missing change-of-boundary calculus. Curvature is present in the rate at which the level surface sweeps volume, through both its area measure and the normal derivative of c_K.

The literal half must be stated at the correct type. The raw, dimension-carrying Hausdorff area of $L_+(K)$ is not universally $\frac{1}{2}$. What is exactly $\frac{1}{2}$ is the normalized co-present interior on that boundary. For the logarithmic interval $K = \left[-\frac{\ell}{2}, \frac{\ell}{2}\right]$,

$$c_{K(x)} = (\ell - |x|)_+,$$

so the two oriented crossings occur at $x = \pm\frac{\ell}{2}$ and each retains exactly half of the original interval. At $\ell = \log 2$, those points exponentiate to $2^{\pm\frac{1}{2}}$. This realizes the user's proposed "surface equal to one half" as a half-overlap boundary, while coarea records the actual surface measure which transports one neighboring shape into the next.

XI.12 Dyadic Poisson return fills the local positive half

Theorem 42 — Dyadic periodization constructs the minimum positive half-filler Retain the dyadic logarithmic cell

$$\mathcal{V}_4 = \mathrm{span}\{e_1, e_2, e_3, e_4\}, \quad \ell = \log 2,$$

of the boundary-polarization theorem, with $e_1 = U_\ell e_3$ and $e_2 = U_\ell e_4$. Let

$$F=\left[-\frac{\ell}{2},\frac{\ell}{2}\right]$$

be a fundamental domain for the lattice $\ell\mathbb{Z}$, and let

$$(\Pi_\ell h)(u)=\sum_{k\in\mathbb{Z}}h(u+k\ell),\quad u\in F,\quad \text{(PERIODIZATION)}$$

be logarithmic periodization. On $\mathcal{V}_4$, there are orthonormal, disjointly supported $q_1,q_2\in L^2(F)$ such that

$$\Pi_\ell e_1=\Pi_\ell e_3=q_1,\quad \Pi_\ell e_2=\Pi_\ell e_4=q_2.$$

Therefore

$$\Pi_\ell^*\Pi_\ell=2P_+,\quad \ker(\Pi_\ell)=P_-\mathcal{V}_4,\quad \overline{\Pi}_\ell=\frac{\Pi_\ell}{\sqrt{2}},\quad \overline{\Pi}_\ell^*\overline{\Pi}_\ell=P_+.\quad \text{(POSITIVE HALF QUOTIENT)}$$

Here $P_+=\frac{I+R}{2}$ is the symmetric core–shell character projector. Its rank is two of four, so its normalized operator area is exactly $\operatorname{Tr}\frac{P_+}{\operatorname{Tr}}(I)=\frac{1}{2}$.

Put

$$a=2^{-\frac{1}{2}},\quad c=\ell a=\frac{\log 2}{\sqrt{2}}.$$

The source-normalized dyadic Poisson amplitude

$$A_{\mathrm{Pois},2}=\sqrt{c}\Pi_\ell\quad \text{(POISSON AMPLITUDE)}$$

has the exact Gram operator

$$A_{\mathrm{Pois},2}^*A_{\mathrm{Pois},2}=cP_+=H_{\min}.\quad \text{(LOCAL FILLER)}$$

Consequently the isolated dyadic completed-defect cell closes as

$$\mathcal{D}_2+A_{\mathrm{Pois},2}^*A_{\mathrm{Pois},2}=cP_-\geq 0.\quad \text{(CLOSED LOCAL CELL)}$$

Formula (“PERIODIZATION”) is an actual Poisson relation: with $\hat{h}(\xi)=\int_{\mathbb{R}}h(u)e^{-i\xi u}\,\mathrm{d}u$,

$$\Pi_\ell h(u)=\frac{1}{\ell}\sum_{n\in\mathbb{Z}}\hat{h}\left(2\pi\frac{n}{\ell}\right)e^{2\pi in\frac{u}{\ell}}.\quad \text{(POISSON SERIES)}$$

The coefficient a in (“POISSON AMPLITUDE”) is independently fixed by the $n=2$ arm of the global scaling map,

$$e^{\frac{u}{2}}f(2e^u)=2^{-\frac{1}{2}}h(u+\ell),$$

while ℓ is its logarithmic Euler weight. The remaining factor $\frac{1}{\sqrt{2}}$ is the exact normalization of the two-to-one quotient in (“POSITIVE HALF QUOTIENT”).

Demonstration. Only one translate of each e_j meets the fundamental domain. Since $e_1=U_\ell e_3$ and $e_2=U_\ell e_4$, periodization identifies each translated pair. Thus, for $x=\sum_j x_je_j$,

$$\Pi_\ell x=(x_1+x_3)q_1+(x_2+x_4)q_2.$$

Its Gram matrix is the direct sum of two matrices

$$\begin{pmatrix}1&1\\1&1\end{pmatrix}=2\begin{pmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&\frac{1}{2}\end{pmatrix},$$

proving (“POSITIVE HALF QUOTIENT”). The kernel is the antisymmetric character space.
Multiplying the normalized quotient by $\sqrt{c}$ gives (“LOCAL FILLER”). The dyadic character theorem gives

$$\mathcal{D}_2=c(P_--P_+),$$

so addition cancels exactly the symmetric negative component and proves ("CLOSED LOCAL CELL"). Formula ("POISSON SERIES") is the ordinary Poisson summation formula on the lattice $\ell\mathbb{Z}$. The half-density calculation of the scaling map gives the displayed coefficient a; differentiating its scale character supplies the factor ℓ.

Boundary. This theorem closes the unique minimum-trace filler for the isolated four-cell dyadic defect by an actual lattice quotient. It cannot be the restriction of the semilocal Sonin successor: the quotient has the nonzero kernel $P_-\mathcal{V}_4$, whereas prime admission and its dual return are invertible. The semilocal Sonin theorem already makes prime admission commute with every nested aperture.

The remaining RH-bearing statement is therefore form-level. The support successor must couple the old positive body to the newly admitted shell, realize this local quotient as a defect face of that coupling, and pay the resulting cross-character shorted-shell term.

The four-cell defect no longer stops at an abstract minimum projector. Let Π_ℓ periodize the logarithmic current over the lattice $\ell\mathbb{Z}$. On the established dyadic basis it identifies precisely the two translated pairs:

$$\Pi_\ell e_1 = \Pi_\ell e_3, \quad \Pi_\ell e_2 = \Pi_\ell e_4.$$

Therefore

$$\Pi_\ell^* \Pi_\ell = 2P_+, \quad \overline{\Pi}_\ell = \frac{\Pi_\ell}{\sqrt{2}}, \quad \overline{\Pi}_\ell^* \overline{\Pi}_\ell = P_+.$$

This gives the symmetric character projector a geometric meaning. It is the part of the four-cell current which descends to the logarithmic quotient; the antisymmetric part is the orientation-sensitive fiber killed by periodization. Its rank is two in a four-dimensional cell, hence its normalized operator area is exactly $\frac{1}{2}$.

The scale is not fitted from a numerical deficit. The global scaling map has dyadic arm

$$e^{\frac{u}{2}} f(2e^u) = 2^{-\frac{1}{2}} h(u + \log 2),$$

and the Euler normal derivative contributes $\log 2$. Thus

$$c = (\log 2) 2^{-\frac{1}{2}} = \frac{\log 2}{\sqrt{2}}$$

is fixed before the quotient is evaluated. The normalized Poisson amplitude

$$A_{\mathrm{Pois},2} = \sqrt{c}\,\Pi_\ell$$

then satisfies

$$A_{\mathrm{Pois},2}^* A_{\mathrm{Pois},2} = cP_+ = H_{\min}.$$

It cancels exactly the negative symmetric face:

$$\mathcal{D}_2 + A_{\mathrm{Pois},2}^* A_{\mathrm{Pois},2} = cP_- \geq 0.$$

This closes the isolated dyadic minimum-filler problem by an actual periodization map. It is a construction, not a restatement of the required inequality.

The earlier draft drew the wrong conclusion from this cell. Poisson summation does give

$$\mathcal{E}(\mathcal{F}f) = \mathcal{I}\mathcal{E}(f)$$

on the two-vanishing-condition Schwartz domain [23], and the semilocal Sonin maps are Hilbertian isomorphisms on every aperture [18]. But $A_{\mathrm{Pois},2}$ cannot be a restriction of that successor. The successor and its dual return are invertible; the periodization amplitude kills the entire antisymmetric space $P_-\mathcal{V}_4$.

The quotient is instead an exact spectral face of the admission metric. With $a = 2^{-\frac{1}{2}}$,

$$G_2 = (1+a^2)I - aR$$

and therefore

$$A^*_{\mathrm{Pois},2} A_{\mathrm{Pois},2} = \frac{\log 2}{2}((1+a+a^2)I - G_2).$$

It records the lower-metric symmetric defect of the dyadic contact, not the whole transition.

The remaining object is consequently form-level: when support grows, couple the already-positive old body to the newly admitted shell, identify the actual cross term, and construct the positive part of the shell left after that cross term is shorted through the old carrier. The dyadic quotient is one required face of this successor, not the successor map itself.

XI.13 The proof architecture is ordinary mathematical induction

Theorem 43 — Conditioning returns every new shell direction through the old face Put

$$a_R = \frac{\log R}{2}, \quad I_R = (-a_R, a_R),$$

and fix $\sigma^2 = 1$ according to the Fourier convention. In the final convention of the archimedean source, $\sigma = -1$; reflection of the interval exchanges the two signs. Define

$$M_R g = \left(\int_{I_R} g(u)\,\mathrm{d}u, \int_{I_R} e^{\sigma \frac{u}{2}} g(u)\,\mathrm{d}u \right), \quad \mathcal{T}_R = \ker(M_R) \subset L^2(I_R). \quad \text{(MOMENT FIBER)}$$

These, rather than $\mathcal{Q}C_c^{\infty(I_R)}$, are the Hilbert amplitude fibers carrying the source's conditions $\hat{g}(0) = \hat{g}(\sigma \frac{i}{2}) = 0$. The differential operator

$$\mathcal{Q} = -\left(\rho \partial_\rho\right)^2 + \frac{1}{4}$$

instead imposes the two half-character conditions on a returned convolution.

For the first successor let

$$I_o = I_2, \quad I = I_3, \quad S = I \setminus I_o, \quad \mathcal{H}_o = L^2(I_o), \quad \mathcal{H}_s = L^2(S).$$

Write M_o, M_s for the restrictions of M_3 to these two regions and

$$G_o = M_o M_o^*.$$

The two moment vectors $1, e^{\sigma \frac{u}{2}}$ are independent on I_o, so G_o is positive definite. With $\alpha = \frac{\log(2)}{2}$, its exact matrix is

$$G_o = \begin{pmatrix} \log 2 & 4\sinh\left(\frac{\log(2)}{4}\right) \\ 4\sinh\left(\frac{\log(2)}{4}\right) & \frac{1}{\sqrt{2}} \end{pmatrix}. \quad \text{(OLD MOMENT GRAM)}$$

Define

$$L = -M_o^* G_o^{-1} M_s, \quad Jy = (Ly) \oplus y. \quad \text{(CONDITIONED LIFT)}$$

Then the amplitude successor has the exact orthogonal decomposition

$$\mathcal{T}_3 = \mathcal{T}_2 \oplus J\mathcal{H}_s \quad \text{inside } L^2(I_3), \quad J^*J = I + M_s^* G_o^{-1} M_s. \quad \text{(CONDITIONED QUOTIENT)}$$

Thus shell restriction identifies the quotient $\frac{\mathcal{T}_3}{\mathcal{T}_2}$ with $\mathcal{H}_s$, but a conditioned representative is never the detached shell $0 \oplus y$: it carries the old compensating face Ly.

Let a Hermitian operator or form on the raw split $\mathcal{H}_o \oplus \mathcal{H}_s$ have, on a common core, the block expression

$$K = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}.$$

Let P_o be the orthogonal projection of $\mathcal{H}_o$ onto $\mathcal{T}_2$. In the conditioned coordinates $\mathcal{T}_2 \oplus J\mathcal{H}_s$, the same object has blocks

$$H_2 = P_o A|_{\mathcal{T}_2}, \quad C = P_{o(AL+B)},$$

$$D_{\text{cond}} = L^* A L + L^* B + B^* L + D. \quad \text{(CONDITIONED BLOCK)}$$

In particular, if a newly admitted prime is raw off-diagonal,

$$K_p = \begin{pmatrix} 0 & B_p \\ B_p^* & 0 \end{pmatrix},$$

then

$$C_p = P_o B_p, \quad D_{p,\text{cond}} = L^* B_p + B_p^* L. \quad \text{(RETURNED PRIME FACE)}$$

The prime has zero diagonal only before the moment fiber is imposed.

Now let q_3 be the completed Weil form on its conditioned form domain and let $q_2 = q_3|_{\mathcal{T}_2} \geq 0$. Shell restriction identifies the form-domain quotient with a dense subspace $\mathcal{B}_2^{\bigcirc}$ of the Hilbert quotient above. Choose any form-domain section

$$\tilde{J} : \mathcal{B}_2^{\bigcirc} \to \operatorname{dom}(q_3) \cap \mathcal{T}_3.$$

It need not equal the sharp L^2 section J, because a sharp old–shell cut need not preserve the form domain. Put

$$\mathcal{N}_2 = \{x : q_2(x) = 0\}$$

and let $\mathcal{E}_2$ be the completion of $\frac{\mathcal{T}_2}{\mathcal{N}_2}$ in the norm $\sqrt{q_2}$. For a quotient direction y, define

$$\ell_{y(x)} = q_3\big(x, \tilde{J}y\big).$$

The successor is nonnegative exactly when:

- ℓ_y annihilates $\mathcal{N}_2$ and is continuous in the q_2-norm;
- its Riesz carrier $c_y \in \mathcal{E}_2$, defined by

  $$\ell_{y(x)} = \langle [x], c_y \rangle_{\mathcal{E}_2},$$

  exists for every admitted y; and
- the invariant shorted shell

  $$s_2(y) = q_3\big(\tilde{J}y\big) - \left\| c_y \right\|_{\mathcal{E}_2}^2 \quad \text{(FORM SHORT)}$$

  is nonnegative.

This criterion does not depend on the chosen form-domain lift. Replacing $\tilde{J}y$ by $\tilde{J}y + k_y$ with $k_y \in \mathcal{T}_2$ translates c_y by $\left[k_y\right]$ and leaves $s_2(y)$ exact.

In a bounded operator chart with old carrier $H_2 \geq 0$ and cross operator C_2, continuity of every cross face is the range condition

$$\operatorname{Ran}(C_2) \subset \operatorname{Ran}\left(H_2^{\frac{1}{2}}\right). \quad \text{(RANGE)}$$

A bounded carrier Y_2 with $C_2 = H_2^{\frac{1}{2}} Y_2$ exists exactly when, for some finite c,

$$C_2 C_2^* \leq c H_2. \quad \text{(DOUGLAS)}$$

The short is then $D_{\text{cond}} - Y_2^* Y_2$. The weaker radical identity $C_2^* \ker(H_2) = 0$ is not sufficient when $\operatorname{Ran}\left(H_2^{\frac{1}{2}}\right)$ is not closed.

There is a source-native continuous-kernel chart. Let $D = i\frac{\mathrm{d}}{\mathrm{d}}u$ with Dirichlet boundary conditions and let G_R be the localized screw-kernel operator. For $v \in H_0^1(I_R)$ and $w = Dv$,

$$q_{R(v)} = \langle G_R w, w\rangle.$$

The two amplitude moments become three exact current moments:

$$\mathcal{K}_R = \left\{ w : \int w = 0, \int u w(u)\, \mathrm{d}u = 0, \int e^{\sigma \frac{u}{2}} w(u)\, \mathrm{d}u = 0 \right\}. \quad \text{(CURRENT FIBER)}$$

Repeating (“CONDITIONED LIFT”) with the three current-moment vectors gives the exact first current lift J_D, cross operator $C_{2,3}$, and conditioned shell operator $D_{2,3}$.

The published base is positive definite, and the localized form has compact resolvent. Hence its attained ground value satisfies

$$q_2(v) \geq \lambda_2 \|v\|^2, \quad \lambda_2 > 0. \quad \text{(BASE GAP)}$$

The localized successor form is closed and lower-semibounded. These two facts force every first-step cross functional to be q_2-continuous. In the bounded current chart this proves, rather than assumes,

$$\operatorname{Ran}(C_{2,3}) \subset \operatorname{Ran}\left(G_2^{\frac{1}{2}}\right), \quad C_{2,3} C_{2,3}^* \leq c G_2 \quad \text{for some finite } c. \quad \text{(FIRST CROSS CARRIED)}$$

Let $Y_{2,3}$ be the Douglas reduced solution $C_{2,3} = G_2^{\frac{1}{2}} Y_{2,3}$. The only remaining sign in the first induction cell is

$$S_{2,3} = D_{2,3} - Y_{2,3}^* Y_{2,3} \geq 0. \quad \text{(FIRST RESIDUAL)}$$

Equivalently, for every form-domain quotient direction,

$$\langle S_{2,3} y, y\rangle = \inf_x q_3\left(x + \tilde{J} y\right), \quad x \in \operatorname{dom}(q_2). \quad \text{(CONDITIONAL ENERGY)}$$

Demonstration. The matrix in (“OLD MOMENT GRAM”) consists of the three elementary integrals of $1, e^{\sigma\frac{u}{2}}, e^{\sigma u}$. It is positive definite because the first two functions are linearly independent. Moreover,

$$M_o L = -M_s,$$

so Jy is conditioned. Every conditioned (x, y) decomposes uniquely as

$$(x, y) = (x - Ly, 0) + Jy.$$

The first term lies in $\mathcal{T}_2$, while Ly lies in $\operatorname{Ran}(M_o^*) = \mathcal{T}_2^{\perp}$. This proves the orthogonal decomposition and the metric formula.

Substitute $(x + Ly, y)$ into the raw quadratic form. Collecting the old, cross, and shell terms gives (“CONDITIONED BLOCK”). Setting $A = D = 0$ proves (“RETURNED PRIME FACE”).

For the form statement, Riesz representation gives c_y precisely when the cross functional descends through the old radical and is continuous in the old form norm. Completing the square in $\mathcal{E}_2$ gives

$$q_3\left(x + \tilde{J} y\right) = \left\|[x] + c_y\right\|^2 + s_2(y).$$

This proves necessity and sufficiency. The same identity proves invariance under changing the quotient lift. Douglas’s factorization theorem gives the equivalence of (“RANGE”) and (“DOUGLAS”) with the bounded factorization.

Finally, integration by parts with v zero at both endpoints gives
$$\int Dv = 0, \quad \int uDv = -i\int v, \quad \int e^{\sigma\frac{u}{2}}Dv = -\frac{i\sigma}{2}\int e^{\sigma\frac{u}{2}}v.$$
This proves ("CURRENT FIBER"). The published screw representation then gives the bounded current chart. Positive definiteness of the attained base ground state proves ("BASE GAP"). If $q_3(z) \geq -\beta\|z\|^2$, its shifted form
$$a_3 = q_3 + (\beta+1)\langle\cdot,\cdot\rangle$$
is positive. For x in the old domain and fixed successor direction z, Cauchy–Schwarz in a_3 gives
$$|q_3(x,z)| \leq K_z\sqrt{q_2(x)},$$
because ("BASE GAP") controls the old L^2 norm. For $z = D^{-1}J_Dy$, boundedness of J_D, D^{-1}, and the continuous screw kernel makes $K_z \leq K\|y\|$. Riesz representation followed by Douglas factorization proves ("FIRST CROSS CARRIED"). Completing the square gives ("CONDITIONAL ENERGY"), so the first successor is positive exactly when ("FIRST RESIDUAL") is positive.

Boundary. The quotient, cross carriage, and shorting law are closed here; their missing datum is no longer an unspecified "positive successor." What remains is the single analytic theorem ("FIRST RESIDUAL") for the explicit zeta screw kernel. Published Sonin transport identifies the nested spaces, and the explicit formula identifies the raw prime cross, but neither source proves that this conditional shell energy is nonnegative. Proving it would close the first nontrivial $P(2) \to P(3)$ cell. A Moore–Penrose inverse or a raw-shell calculation does not determine its sign.

The source imposes its two conditions on the amplitude:

$$\hat{g}(0) = 0, \quad \hat{g}\left(\sigma\frac{i}{2}\right) = 0.$$

The operator

$$\mathcal{Q} = -\left(\rho\partial_\rho\right)^2 + \frac{1}{4}$$

belongs to a different face: it imposes the two half-character zeros on the returned convolution. Treating $\mathcal{Q}C_c^{\infty(I_n)}$ as the canonical amplitude space had silently mixed those faces [20].

In the logarithmic chart, the corrected Hilbert amplitude fiber is the codimension-two moment kernel

$$\mathcal{T}_n = \ker\left[g \mapsto \left(\int_{I_n} g(u)\,\mathrm{d}u, \int_{I_n} e^{\sigma\frac{u}{2}}g(u)\,\mathrm{d}u\right)\right], \quad I_n = \left(-\frac{\log(n)}{2}, \frac{\log(n)}{2}\right).$$

Its returned current is supported in $[n^{-1}, n]$. Define $P(n)$ to mean that the completed Weil form is nonnegative on the whole conditioned fiber. The published archimedean theorem supplies $P(2)$.

Moment conditioning changes the geometry of support growth. With $I_3 = I_2 \cup S$, let M_o, M_s be the old and shell moment maps and put

$$L = -M_o^*(M_oM_o^*)^{-1}M_s, \quad Jy = (Ly) \oplus y.$$

Then

$$\mathcal{T}_3 = \mathcal{T}_2 \oplus JL^2(S), \quad J^*J = I + M_s^*(M_oM_o^*)^{-1}M_s. \quad \text{(CONDITIONED QUOTIENT)}$$

The sum is orthogonal in the ambient L^2 receiver, but Jy is not a detached shell: its old component is exactly what preserves both moments.

Consequently a raw block

$$K = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$$

becomes, in conditioned coordinates,

$$H_2 = P_oA|_{\mathcal{T}_2}, \quad C = P_{o(AL+B)},$$

$$D_{\text{cond}} = L^*AL + L^*B + B^*L + D. \quad \text{(RECHART)}$$

This is why the quotient metric and the crossings must be derived together, rather than declaring a shell and later imposing constraints.

Theorem 44 — RH reduces to one uniform conditioned support-successor law For every integer $n \geq 2$, let $\mathcal{T}_n$ be the Mellin-conditioned amplitude fiber

$$\mathcal{T}_n = \left\{ g \in L^2(I_n) : \int g = 0, \int e^{\sigma \frac{u}{2}} g(u) \, \mathrm{d}u = 0 \right\}, \quad I_n = \left(-\frac{\log(n)}{2}, \frac{\log(n)}{2} \right), \quad \sigma^2 = 1. \quad \text{(AMPLITUDE FIBER)}$$

Let q_n be the completed Weil Hermitian form restricted to its natural dense domain in $\mathcal{T}_n$, and put

$$P(n) : q_{n(g)} \geq 0 \ \text{ for every admitted } g \in \mathcal{T}_n.$$

The archimedean theorem supplies $P(2)$ under these two amplitude conditions. If $h = g * g^*$, multiplicative convolution gives

$$\operatorname{supp}(h) \subset \left[n^{-1}, n \right].$$

The inclusion $\mathcal{T}_n \subset \mathcal{T}_{n+1}$ is not a detached old–shell direct sum. Moment conditioning identifies its quotient by a graph section

$$J_n : \mathcal{B}_n \to \mathcal{T}_{n+1},$$

whose new-support face is accompanied by an old-region compensator. The exact section at $n = 2$ is `theorem:conditioned-support-quotient-shorting`; the same finite-moment construction applies for arbitrary n.

Assume $P(n)$, and choose a form-domain lift $\tilde{J}_n$ of the Hilbert quotient section. Let

$$\mathcal{N}_n = \left\{ x : q_{n(x)} = 0 \right\}$$

and let $\mathcal{E}_n$ be the completion of $\frac{\mathcal{T}_n}{\mathcal{N}_n}$ in the q_n-norm. For every quotient direction $y \in \mathcal{B}_n$, define the cross functional

$$\ell_{n,y}(x) = q_{n+1}\left(x, \tilde{J}_n y\right).$$

Then

$$P(n) \to P(n+1) \quad \text{(SUCCESSOR)}$$

holds exactly when:

- every $\ell_{n,y}$ annihilates $\mathcal{N}_n$ and is continuous in the q_n-norm, hence has a Riesz carrier $c_{n,y} \in \mathcal{E}_n$; and
- the conditioned shorted shell

$$s_{n(y)} = q_{n+1}\left(\tilde{J}_n y\right) - \left\| c_{n,y} \right\|_{\mathcal{E}_n}^2 \quad \text{(SHORTED SHELL)}$$

  is nonnegative for every y.

Equivalently, whenever the carriers assemble into linear maps, there are Hilbert spaces and source-derived maps Z_n, Y_n, R_n such that

$$q_{n(x)} = \|Z_n x\|^2, \quad c_{n,y} = Y_n y, \quad s_{n(y)} = \|R_n y\|^2,$$

and therefore

$$q_{n+1}\left(x + \tilde{J}_n y\right) = \|Z_n x + Y_n y\|^2 + \|R_n y\|^2. \quad \text{(POSITIVE SUCCESSOR)}$$

In an auxiliary bounded operator chart, with conditioned block

$$H_{n+1} = \begin{pmatrix} H_n & C_n \\ C_n^* & D_n \end{pmatrix},$$

cross continuity is

$$\operatorname{Ran}(C_n) \subset \operatorname{Ran}\left(H_n^{\frac{1}{2}}\right). \quad \text{(RANGE)}$$

A bounded factor $C_n = H_n^{\frac{1}{2}} Y_n$ exists exactly when

$$C_n C_n^* \le c_n H_n$$

for some finite c_n, and the remaining condition is

$$D_n - Y_n^* Y_n \ge 0. \quad \text{(OPERATOR SHORT)}$$

Only when the relevant range is closed may this be rewritten using a bounded Moore–Penrose inverse. Radical annihilation by itself is not the successor theorem.

In the logarithmic interval receiver, a prime translation by $\ell_p = \log p$ has normalized raw overlap

$$\omega_{p(R)} = \frac{(\log R - \log p)_+}{\log R},$$

so

$$\omega_{p(p^2)} = \frac{1}{2}. \quad \text{(HALF CROSSING)}$$

This is the midpoint of that prime's support continuation, not its first admission.

The first induction cell is $P(2) \to P(3)$. The raw $p = 2$ incidence has overlap

$$\log 3 - \log 2 = \log\left(\frac{3}{2}\right), \quad \omega_2(3) = \frac{\log\left(\frac{3}{2}\right)}{\log 3}. \quad \text{(FIRST STEP)}$$

It is off-diagonal in the unconditioned old–shell split, but moment conditioning returns it through the old face and generally contributes to both the conditioned cross and conditioned shell blocks. At $R = 4$, the dyadic arm reaches half-overlap while the prime 3 is already present; the isolated dyadic character cell is a calibration, not the complete $P(4)$ receiver.

The finite-prime support combinatorics itself is uniform. In the normalized square I_{n+1}^2, the thresholds

$$|x - y| = \log\left(p^k\right)$$

form parallel paired hinge cells. The successor $n \to n + 1$ changes their incidence poset exactly when n is a prime power. Along one prime axis their positions are $k \log p$ and their half-density weights are $(\log p) p^{-\frac{k}{2}}$.

This closure of the cell recurrence does not make the form cellwise positive. If the complete conditioned cross is decomposed into its archimedean and active prime-power carriers

$$c_{n,y} = \sum_\alpha c_{\alpha,n,y},$$

then the short contains their complete Gram:

$$s_{n(y)} = \sum_\alpha q_{\alpha,n+1}\left(\tilde{J}_n y\right) - \sum_{\alpha,\beta} \langle c_{\alpha,n,y}, c_{\beta,n,y} \rangle_{\mathcal{E}_n}. \quad \text{(COHERENT GRAM)}$$

Distinct raw hinge lines are parallel, but eliminating the common old interior creates their cross-axis terms.

One uniform source-derived proof of ("RANGE") and ("OPERATOR SHORT"), or their form-level equivalents, for every $n \geq 2$ proves RH. Ordinary induction gives every $P(n)$, every compactly supported admissible amplitude lies in some $\mathcal{T}_n$, and Weil's criterion then gives RH.

Demonstration. The support statement follows because the quotient of two members of $\left[n^{-\frac{1}{2}}, n^{\frac{1}{2}}\right]$ lies in $[n^{-1}, n]$. The base case is the published archimedean positivity theorem with its independently constructed Sonin/prolate remainder amplitude.
The conditioned quotient theorem constructs J_n by solving the two moment equations in the old region. If the cross functional descends to the old energy completion, Riesz representation and completion of the square give

$$q_{n+1}(x + \tilde{J}_n y) = \left\|[x] + c_{n,y}\right\|^2 + s_{n(y)}.$$

This proves necessity and sufficiency of the two form-level conditions, as well as ("POSITIVE SUCCESSOR"). Douglas's factorization theorem gives the bounded operator version.
The overlap formula is the length of the intersection of an interval of length $\log R$ with its translate by $\log p$, divided by $\log R$. Substitution of $R = p^2$ and $(p, R) = (2, 3)$ gives the two displayed values. The raw-to-conditioned block formula proves that raw off-diagonality does not survive the moment rechart.
Finally, ordinary induction establishes the form on every integer aperture. Compact support places each individual Weil test amplitude at one finite aperture, completing the criterion.

Boundary. This theorem fixes the induction architecture but does not assert the missing positivity. The first unresolved cell is now explicit in the continuous screw-kernel chart. Positive definiteness and compact resolvent of the published $P(2)$ base already carry the conditioned $I_2 \to I_3$ cross through the old energy space. What remains is positivity of that cell's single shorted-shell operator. Published Sonin transport supplies coherent space transport, and the explicit formula supplies the raw prime incidence; neither signs this conditional energy.

A uniform proof must then show that the same construction survives every support successor. Prime stages admit a new Euler difference/return pair; higher prime powers expose another traversal of an existing place; other stages move only the support boundary. None of those distinctions changes the one form-level obligation above. The prime-power cell recurrence and its exact census are now closed. What remains is the effective-tension sign represented in coordinates by ("COHERENT GRAM"), not a further combinatorial enumeration or a sum of independently positive prime cells.

The proof still has the form of ordinary induction, but its source-native statement is a shorting theorem for closed forms, not a bounded Moore–Penrose calculation. Assume $P(n)$, quotient the radical of q_n, and complete in the q_n-norm to obtain $\mathcal{E}_n$. Choose a form-domain lift $\tilde{J}_n$ of the Hilbert quotient. For a quotient direction y, the cross functional

$$\ell_{n,y}(x) = q_{n+1}(x, \tilde{J}_n y)$$

must descend to a continuous functional on $\mathcal{E}_n$. If its Riesz carrier is $c_{n,y}$, the exact remaining face is

$$s_{n(y)} = q_{n+1}(\tilde{J}_n y) - \left\|c_{n,y}\right\|^2. \quad \text{(FORM SHORT)}$$

Then $P(n) \to P(n+1)$ holds exactly when every cross functional descends and $s_n \geq 0$. In a bounded chart this becomes

$$\operatorname{Ran}(C_n) \subset \operatorname{Ran}\left(H_n^{\frac{1}{2}}\right), \quad D_n - Y_n^* Y_n \geq 0, \quad C_n = H_n^{\frac{1}{2}} Y_n. \quad \text{(INDUCTION)}$$

Douglas factorization rewrites the first clause as $C_n C_n^* \leq c_n H_n$ for some finite c_n. Merely annihilating $\ker(H_n)$ is weaker when the square-root range is not closed. This matters here: the localized Weil form is naturally a closed lower-semibounded form with an unbounded Friedrichs operator, while its derivative-current representation is a compact continuous-kernel operator [24].

Every step adds support. If n is prime, the exact dual $\frac{\Delta_n}{(\Delta_n^*)^{-1}}$ pair is also admitted. If n is a higher prime power, the shell exposes another traversal of a place already present. Thus prime powers do not require a separate induction principle.

The interval receiver also locates the positive half precisely. A prime shift by $\log p$ occupies the normalized overlap

$$\omega_{p(R)} = \frac{(\log R - \log p)_+}{\log R},$$

so $\omega_{p(p^2)} = \frac{1}{2}$. The L_+ half-overlap is therefore a canonical midpoint of the support continuation of the p-arm, not the instant at which the place first enters. For $p = 2$, this is exactly the passage from $R = 2$ to the dyadic $R = 4$ contact.

That dyadic contact is not the first induction step. The actual base successor is $P(2) \to P(3)$, where $p = 2$ enters with overlap

$$\log\left(\frac{3}{2}\right), \quad \omega_2(3) = \frac{\log\left(\frac{3}{2}\right)}{\log 3}.$$

At $R = 4$, the prime 3 has also entered the integer filtration. The isolated $p = 2$, $R = 4$ character block is therefore a valuable calibration face, but it is not the complete $P(4)$ receiver.

Theorem 45 — A newly admitted prime is raw old–shell cross incidence Let $p \geq 2$, put

$$\ell = \log p, \quad r = \log(p+1), \quad \delta = \frac{r - \ell}{2} = \frac{1}{2}\log\frac{p+1}{p},$$

and define the old aperture, successor aperture, and new shell by

$$I_p = \left[-\frac{\ell}{2}, \frac{\ell}{2}\right], \quad I_{p+1} = \left[-\frac{r}{2}, \frac{r}{2}\right], \quad B_p = \left\{u \in I_{p+1} : u \notin I_p\right\}.$$

Let P and B denote multiplication by 1_{I_p} and 1_{B_p} in $L^2\left(I_{p+1}\right)$, and let

$$\left(U_p f\right)(u) = f(u - \ell).$$

Since $p + 1 \leq p^2$, the compressed prime shift has no old–old or shell–shell incidence:

$$PU_pP = 0, \quad BU_pB = 0, \quad (P+B)U_{p(P+B)} = PU_pB + BU_pP. \quad \text{(PURE CROSS)}$$

More precisely, translation carries

$$\left[-\frac{r}{2}, -\frac{\ell}{2}\right] \overset{+\ell}{\to} \left[\ell - \frac{r}{2}, \frac{\ell}{2}\right]$$

from the left shell to the right old boundary strip, and

$$\left[-\frac{\ell}{2}, \frac{r}{2} - \ell\right] \overset{+\ell}{\to} \left[\frac{\ell}{2}, \frac{r}{2}\right]$$

from the left old boundary strip to the right shell. All four intervals have length δ.

At the support step $R = p + 1$, no higher power of the newly admitted prime is present because $p^2 > p + 1$. Thus its complete first finite-place response on the **raw** split is the Hermitian off-diagonal form

$$\mathcal{W}_{p,\text{first}} = (\log p)p^{-\frac{1}{2}}\left(U_p + U_p^*\right) \quad \text{compressed to } I_{p+1}, \quad \text{(PRIME CROSS)}$$

with zero diagonal blocks relative to the unconditioned decomposition $L^2\left(I_p\right) \oplus L^2\left(B_p\right)$.

Demonstration. The old interval has length ℓ, so its translate by ℓ meets it only at one endpoint, a null set. This proves $PU_pP = 0$.
A source point contributes to the compressed shift exactly when it lies in

$$I_{p+1} \cap \left(I_{p+1} - \ell\right) = \left[-\frac{r}{2}, \frac{r}{2} - \ell\right].$$

The point $-\frac{\ell}{2}$ divides this interval into the two displayed pieces. Translating them by ℓ gives the two target pieces. Their common length is $\frac{r-\ell}{2}$. Since $r \leq 2\ell$, neither source shell can land in the opposite shell, proving $BU_pB = 0$ and ("PURE CROSS").
The prime-power aperture formula has coefficient $(\log p)p^{-\frac{m}{2}}$ on the Hermitian shift $U_p^m + \left(U_p^*\right)^m$. At $R = p+1 < p^2$, only $m = 1$ meets the aperture. This proves ("PRIME CROSS").

Boundary. This theorem computes the finite-prime incidence before the source's Mellin moments are imposed; it does not provide the conditioned successor block. A conditioned shell direction carries an old-region compensator, so recharting ("PRIME CROSS") generally creates a shell-diagonal term. The conditioned quotient and that returned prime face are calculated in `theorem:conditioned-support-quotient-shorting`. A raw pure-cross form still takes either sign as the relative phase changes, and no argument assigning an independent positive constituent to the new prime can prove the induction step.

The first prime arm can now be located in the **raw** support split exactly. At $P(p) \to P(p+1)$, translation by $\log p$ is the full width of the old aperture. Consequently

$$P_{\text{old}} U_p P_{\text{old}} = 0, \quad P_{\text{shell}} U_p P_{\text{shell}} = 0.$$

The left shell translates into a right old boundary strip, while the opposite old boundary strip translates into the right shell. Both have width

$$\frac{1}{2} \log \frac{p+1}{p}.$$

Because $p^2 > p+1$, no repeated traversal is present yet. The newly admitted prime has zero diagonal in the unconditioned $L^2\left(I_p\right) \oplus L^2(S)$ split. After conditioning, ("RECHART") gives

$$C_p = P_o B_p, \quad D_{p,\text{cond}} = L^* B_p + B_p^* L.$$

The previous claim that the prime remained "only cross" in the actual successor was therefore false: the old compensator returns that incidence into the conditioned shell diagonal. This does not make the prime an independent positive constituent. Its returned diagonal and the archimedean/Sonin response must close one joint shorted form.

Once ("INDUCTION") is constructed uniformly, ordinary induction proves every $P(n)$. Every compact admissible current lies in some $\mathcal{T}_n$, so Weil positivity and RH follow. "Every support" is the consequence of compactness and induction, not a demand for exhaustive search.

XI.14 The successor proof is one cross-channel contraction

Theorem 46 — A semilocal successor carrier is exactly a cross-channel defect contraction Fix one receiver S, support aperture R, and prime $p \notin S$. Let $\mathcal{T}_{S,R}$ be a Hilbert space of admitted currents. Suppose the predecessor completed defect has an independently constructed amplitude

$$C_{S,R} : \mathcal{T}_{S,R} \to \mathcal{K}_{S,R}, \quad D_{S(f)} = \left\|C_{S,R} f\right\|^2.$$

Let $L_R f$ be the logarithmic current entering the exact prime-amplitude identity, and write $\Theta_{S(f)}$ for the Hilbert–Schmidt scale-action current. Assume the displayed current-to-amplitude maps are bounded and linear on $\mathcal{T}_{S,R}$.

In the receiver chart, let P_S be the predecessor orthogonal Sonin projection, G_p the Euler successor metric, Π_p the G_p-orthogonal successor projection, and

$$\tilde{P}_p = G_p^{\frac{1}{2}} \Pi_p G_p^{-\frac{1}{2}}.$$

Then $\tilde{P}_p$ is orthogonal. If $\Theta_{S(f)}$ commutes with G_p, the aperture connection has the exact endpoint-amplitude form

$$\kappa_{S,p}(f) = \left\|\Theta_{S(f)} \tilde{P}_p\right\|_{\text{HS}}^2 - \left\|\Theta_{S(f)} P_S\right\|_{\text{HS}}^2. \quad \text{(APERTURE)}$$

Put $\ell_p = \log p$, let $\mathcal{R}_p$ be the normalized Euler resolvent on the logarithmic-current channel, and define the two direct-sum amplitude maps

$$X_{S,p,R}f = \left(C_{S,R}f, \sqrt{\ell_p}L_R f, \Theta_{S(f)}P_S\right),$$

$$Y_{S,p,R}f = \left(\sqrt{\ell_p}\mathcal{R}_p L_R f, \Theta_{S(f)}\tilde{P}_p\right).$$

Their exact balance is

$$\left\|X_{S,p,R}f\right\|^2 - \left\|Y_{S,p,R}f\right\|^2 = D_{S(f)} - W_{p(f*f^\sharp)} - \kappa_{S,p}(f) = D_{S\cup\{p\}}(f). \quad \text{(BALANCE)}$$

The following are equivalent:

- $D_{S\cup\{p\}}(f) \geq 0$ for every admitted current f;
- $\left\|Y_{S,p,R}f\right\| \leq \left\|X_{S,p,R}f\right\|$ for every f;
- the rule

$$\Gamma_{S,p,R}\left(X_{S,p,R}f\right) = Y_{S,p,R}f$$

  is well-defined on $\operatorname{range}\left(X_{S,p,R}\right)$ and extends to a contraction on its closure; and
- there is a successor amplitude $C_{S\cup\{p\},R}$ satisfying

$$\left\|C_{S\cup\{p\},R}f\right\|^2 = D_{S\cup\{p\}}(f).$$

When the contraction is constructed independently, the canonical successor amplitude is

$$C_{S\cup\{p\},R} = \left(I - \Gamma^*_{S,p,R}\Gamma_{S,p,R}\right)^{\frac{1}{2}} X_{S,p,R}. \quad \text{(DEFECT)}$$

Demonstration. Because $\tilde{P}_p$ is orthogonal,

$$\left\|\Theta_{S(f)}\tilde{P}_p\right\|^2_{\mathrm{HS}} = \mathrm{Tr}\left(\Theta_{S(f)}\tilde{P}_p\Theta^*_{S(f)}\right).$$

Trace cyclicity and commutation of $\Theta^*_{S(f)}\Theta_{S(f)}$ with G_p turn this into

$$\mathrm{Tr}\left(\Theta_{S(f)}\Pi_p\Theta^*_{S(f)}\right).$$

Subtracting the predecessor Hilbert–Schmidt square proves ("APERTURE").
The normalized Euler-resolvent lemma gives

$$\ell_{p\left(\|\mathcal{R}_p L_R f\|^2 - \|L_R f\|^2\right)} = W_{p(f*f^\sharp)}.$$

Adding this identity to ("APERTURE") and the predecessor carrier norm proves ("BALANCE") by the completed-defect recurrence.
The first two equivalent statements are ("BALANCE"). The norm inequality implies that $Xf = 0$ forces $Yf = 0$, so the displayed rule for Γ is well-defined and contractive on the range of X; continuity extends it to the closure. Conversely, a contractive Γ gives the norm inequality.
Finally, the positive defect operator

$$D_\Gamma = (I - \Gamma^*\Gamma)^{\frac{1}{2}}$$

obeys

$$\|D_\Gamma X f\|^2 = \|Xf\|^2 - \|\Gamma X f\|^2 = \|Xf\|^2 - \|Yf\|^2.$$

This proves ("DEFECT") and the successor-amplitude equivalence.

Boundary. The theorem does not construct $\Gamma_{S,p,R}$. Defining it only after assuming the norm inequality would restate the desired positivity. Since $\mathcal{R}_p$ is expansive on part of the translation spectrum and the aperture endpoint may change in either direction, a successful Γ cannot be a block-diagonal passage which treats the prime, Sonin, and predecessor-remainder channels independently. It must mix them in one arithmetic colligation.

The global Poisson map is the known source relation joining local Euler factors, additive Fourier return, multiplicative inversion, and principal-value normalization; the isolated Euler and Sonin operators do not supply the required mixing blocks. Moreover, this theorem assumes a predecessor carrier at the same support R. The published archimedean carrier supplies it only at its bounded base aperture, so a compatible support-growth contraction remains a second axis of the proof.

The turned Sonin aperture also has an endpoint-amplitude expression. With

$$\tilde{P}_p = G_p^{\frac{1}{2}} \Pi_p G_p^{-\frac{1}{2}}$$

orthogonal in the fixed chart,

$$\kappa_{S,p}(f) = \left\| \Theta_{S(f)} \tilde{P}_p \right\|_{\mathrm{HS}}^2 - \left\| \Theta_{S(f)} P_S \right\|_{\mathrm{HS}}^2.$$

Assume the predecessor remainder has already been constructed as $D_{S(f)} = \left\| C_{S,R} f \right\|^2$. Collect its three outgoing amplitudes and the two successor amplitudes as

$$Xf = \left(C_{S,R} f, \sqrt{\log p} L_R f, \Theta_{S(f)} P_S \right),$$

$$Yf = \left(\sqrt{\log p} \mathcal{R}_p L_R f, \Theta_{S(f)} \tilde{P}_p \right).$$

The completed recurrence is now exactly

$$D_{S \cup \{p\}}(f) = \|Xf\|^2 - \|Yf\|^2.$$

Thus the successor is positive for every admitted current exactly when the rule

$$\Gamma_{S,p,R}(Xf) = Yf$$

extends to a contraction. If that contraction is independently constructed, the successor amplitude is not guessed:

$$C_{S \cup \{p\},R} = \left(I - \Gamma_{S,p,R}^* \Gamma_{S,p,R} \right)^{\frac{1}{2}} X.$$

This is the fixed-support, prime-admission face of the induction step. Constructing Γ after assuming the inequality would be circular. It also cannot be a block-diagonal collection of the known channels: $\mathcal{R}_p$ expands some translation modes, while the aperture connection can have either sign. The required contraction must mix the predecessor remainder, prime return, and Sonin aperture through one arithmetic relation.

The global Poisson formula is the natural source of that mixing because it joins the semilocal Euler population to additive Fourier return, multiplicative inversion, and the common principal-value normalization [23]. The elementary admission identity above supplies its Euler arm; it does not yet supply the Fourier/Sonin off-diagonal blocks.

XI.15 Form-level coherence remains a naturality condition

The local factorization is an instance of Douglas' operator majorization/factorization principle [25]. Category theory sharpens the coherence of the induction without changing its sign obligation. Let $\mathcal{J}$ have the actual support inclusions, prime admissions, Poisson–Sonin returns, and anchored receiver recharts as arrows. The published Sonin spaces and prime maps already supply the underlying space-level functor. The completed forms and amplitude sides must respect it:

$$\mathcal{T}, \mathcal{K}_X, \mathcal{K}_Y : \mathcal{J} \to \mathrm{Hilb},$$

$$X : \mathcal{T} \Rightarrow \mathcal{K}_X, \quad Y : \mathcal{T} \Rightarrow \mathcal{K}_Y.$$

Theorem 47 — Coherent contractive fillers form a convex feasibility set Let $\mathcal{J}$ be a small category of support, prime, and receiver transitions. Let $\mathcal{T}, \mathcal{K}_X, \mathcal{K}_Y : \mathcal{J} \to \mathrm{Hilb}$ be Hilbert-space functors and let

$$X : \mathcal{T} \to \mathcal{K}_X, \quad Y : \mathcal{T} \to \mathcal{K}_Y$$

be natural amplitude transformations.

Define $\mathcal{C}(X,Y)$ to be the families of bounded maps $\Gamma_j : \mathcal{K}_{X,j} \to \mathcal{K}_{Y,j}$ satisfying

$$\Gamma_j X_j = Y_j, \quad \|\Gamma_j\| \le 1,$$

and, for every $a : j \to k$,

$$\mathcal{K}_{Y(a)}\Gamma_j = \Gamma_k \mathcal{K}_{X(a)}.$$

Then $\mathcal{C}(X,Y)$ is convex and weak-operator closed. In finite dimensions its contraction condition is the linear matrix inequality

$$\begin{pmatrix} 1 & \Gamma_j^* \\ \Gamma_j & 1 \end{pmatrix} \ge 0.$$

If every finite collection of equations (“F”), (“N”), and (“LMI”) has a simultaneous solution, then $\mathcal{C}(X,Y)$ is nonempty. For every $\Gamma \in \mathcal{C}(X,Y)$ and every j,

$$\|X_j f\|^2 - \|Y_j f\|^2 = \left\|(1-\Gamma_j^*\Gamma_j)^{\frac{1}{2}} X_j f\right\|^2 \ge 0.$$

Demonstration. Factorization and naturality are affine linear equations in the components Γ_j. Operator unit balls are convex, so their intersection with those equations is convex. The unit ball is weak-operator compact and the displayed equations are weak-operator closed. Their product over j is compact. Feasibility of every finite collection gives the finite-intersection property, hence a simultaneous family exists.
In finite dimensions, the Schur complement identifies the displayed block inequality with $1 - \Gamma_j^*\Gamma_j \ge 0$ in the operator order. Substituting $Y_j = \Gamma_j X_j$ yields the displayed defect identity. Componentwise, the equivalence between norm domination and contractive factorization is the Douglas factorization principle.

Boundary. Pointwise contractions need not satisfy naturality. Finite numerical feasibility does not establish feasibility for every finite constraint family, and approximate feasibility does not establish the exact factorization. The compactness clause reduces a coherent infinite construction to exact finite compatibility; it does not supply the arithmetic reason those finite feasible sets are nonempty. Because $\Gamma_j X_j = Y_j$ is inhomogeneous when $Y_j \ne 0$, this set is generally not a cone: scaling a filler need not leave it feasible.

The desired successor passage is then one natural transformation

$$\Gamma : \mathcal{K}_X \Rightarrow \mathcal{K}_Y,$$

not an unrelated contraction selected at every aperture. For each arrow $a : j \to k$, it owes the square

$$\mathcal{K}_{Y(a)}\Gamma_j = \Gamma_k \mathcal{K}_{X(a)}.$$

At every object it owes

$$\Gamma_j X_j = Y_j, \quad \begin{pmatrix} I & \Gamma_j^* \\ \Gamma_j & I \end{pmatrix} \ge 0.$$

For a finite subdiagram these are affine equalities intersected with positive-semidefinite cones, hence a convex feasibility problem. If every finite collection of exact constraints is simultaneously feasible, weak-operator compactness produces a coherent global family. This is not a license to infer the infinite result from sampled computations: the arithmetic construction must prove all finite compatibilities or supply a monotone extension law.

Theorem 48 — Anchored positive forms descend across lawful perspectives Let $\mathcal{R}$ be a receiver rechart groupoid, let $V : \mathcal{R} \to \mathrm{Hilb}_{\mathrm{iso}}$ assign a test space to every chart, and let $T_u = V(u)$ for $u : \rho \to \sigma$. Suppose a Hermitian form Q_ρ is supplied in every chart and satisfies the anchored covariance law

$$Q_{\sigma(T_u f, T_u g)} = Q_{\rho(f,g)} \quad \text{for every } u : \rho \to \sigma.$$

Then positive semidefiniteness and Hermitian inertia are constant on each receiver orbit. If the charts cover one glued test bundle and their transitions obey the cocycle law, the compatible family (Q_ρ) descends to one global Hermitian form Q. The descended form is positive semidefinite if and only if every local representative is positive semidefinite.

Demonstration. For an invertible transition T_u, the covariance law is the congruence $Q_\rho = T_u^* Q_\sigma T_u$. Therefore $Q_{\rho(f,f)} = Q_{\sigma(T_u f, T_u f)}$, proving sign equivalence, while Sylvester inertia is preserved by invertible congruence. On chart overlaps, the covariance law makes the value assigned to a represented vector independent of the chosen chart. The transition cocycle makes this pairwise identification consistent on triple overlaps, so the local forms define one form on the glued bundle. Every global vector has a local representative, which proves the final equivalence.

Boundary. The theorem does not turn a negative direction positive by changing orientation. "Some positive perspective exists" is insufficient unless it carries the same anchored form through the same cocycle and the charts cover every admitted direction. The set of positive semidefinite Hermitian forms is a convex cone; for an indefinite form, the set of vectors on which its quadratic value is nonnegative need not itself be convex.

This also states exactly what a proof "by perspectives" can mean. Compatible positive receiver forms descend when they are congruent presentations of the same anchored response and their transition cocycle closes. Invertible recharting preserves Hermitian inertia, so no orientation can make a genuine negative direction positive. The useful convex object is the cone of compatible positive forms—or, here, the natural contraction feasible set—not the set of favorable drawings of one indefinite form.

Conditional corollary 49 — A complete natural contraction of the zeta amplitudes implies RH Suppose the completed zeta test currents, prime/support apertures, and lawful receiver recharts form an exhaustive index category $\mathcal{J}$. Suppose the existing predecessor, Euler, and aperture amplitudes assemble into the natural transformations X and Y of the natural-contraction theorem, with

$$Q_{W(f)} = \|X_j f\|^2 - \|Y_j f\|^2$$

for every represented completed Weil current. If $\mathcal{C}(X, Y)$ is nonempty, then $Q_{W(f)} \geq 0$ for every admissible test current and the Riemann Hypothesis holds.

Demonstration. A natural contraction gives nonnegativity of every represented response by the defect identity of the preceding theorem. Exhaustiveness covers the complete Weil test space. Weil's positivity criterion is equivalent to RH, so RH follows.

Boundary. This is not a new positivity carrier under another name. It identifies the exact coherence which the already-open contraction must satisfy across perspectives, prime additions, and support growth. Proving that the candidate feasibility set is nonempty from the completed arithmetic relation remains the proof-bearing obligation.

This is the same RH wall in a stricter form, not a renamed carrier. The new information is that pointwise positivity is insufficient even if it is proved at every named chart: the component fillers must be one natural family across the prime, support, and receiver arrows which assemble the completed response.

XI.16 The contraction must extend over the full carrier atlas

Conditional corollary 50 — An exact covariant coarea carrier for every completed defect would imply RH Suppose that for every final receiver (S, R) there is an independently constructed covariant coarea carrier $C_{S,R}$ satisfying

$$Q_{W(f)} - \Sigma_{S(f)} = \|C_{S,R} f\|^2$$

for every admitted current. Require the carrier charts to preserve this response by the half-Jacobian law and require every prime seam, after anchoring the same current in the two contemporary receivers, to obey

$$\|C_{S\cup\{p\},R} f\|^2 - \|C_{S,R} f\|^2 = -W_{p(f * f^\sharp)} - \kappa_{S,p}(f).$$

Then $Q_{W(f)} \geq 0$ for the complete admissible family and the Riemann Hypothesis holds.

Demonstration. The Sonin response is a Hilbert–Schmidt norm square and the assumed carrier identity gives

$$Q_{W(f)} = \Sigma_{S(f)} + \|C_{S,R}f\|^2 \geq 0.$$

The chart and seam laws ensure that this is the same anchored completed response under every lawful reparameterization and prime transition, rather than a positive value obtained by changing the occurrence. Weil's criterion then implies RH.

Boundary. The prime seam on the right may have either sign; the corollary does not impose a positive contribution per prime or a monotone receiver history. The archimedean base carrier is independently constructed by `theorem:archimedean-remainder-amplitude`. The unproved statement is its expanded semilocal identity, including endpoint, finite-place, support, aperture-connection, and critical-stratum terms. Constructing $C_{S,R}$ from the desired defect by an abstract square root would be circular.

The RH-bearing target can now be stated without treating $K_{S,R}$ as an opaque positive operator. For every final receiver, construct an amplitude

$$B_{S,R,\xi} : \mathcal{T}_{S,R} \to \mathcal{K}_{S,R,\xi}$$

and measure $\nu_{S,R}$ such that

$$Q_{W(f)} - \Sigma_{S(f)} = \int_{\Lambda_{S,R}} \|B_{S,R,\xi}f\|^2 \, \mathrm{d}\nu_{S,R}(\xi).$$

The field must come independently from the Gaussian–Poisson–Mellin completion, the archimedean prolate remainder, finite Euler transport, and the moving support/aperture geometry.

Under one prime transition, the endpoint norms—not positive prime pieces—obey

$$\|C_{S\cup\{p\},R}f\|^2 - \|C_{S,R}f\|^2 = -W_{p(f*f^\sharp)} - \kappa_{S,p}(f).$$

Critical Jacobian strata owe explicit seam terms. Chart changes owe the half-Jacobian. If this exact defect identity is established over the full admissible family, then

$$Q_{W(f)} = \Sigma_{S(f)} + \|C_{S,R}f\|^2 \geq 0$$

and Weil's criterion gives RH.

What remains open is not the base arithmetic amplitude field or the prime-power endpoint factorization. It is an independently constructed cross-channel contraction $\Gamma_{S,p,R}$, compatible with global Poisson return and support growth, and then its assembly over the complete admitted place/support atlas. The polar, hyperbolic, Euler, Sonin, and archimedean constituents are explicit; the missing off-diagonal arithmetic coupling may not simply be declared contractive.

XII Relation to current external strategies

XII.1 Archimedean positivity is a genuine base cell

Connes and Consani prove, under their stated bounded support and vanishing conditions at the archimedean place, that the Weil functional dominates a positive Sonin trace [20]. This is not a heuristic. It is a closed positive cell built from prolate/Toeplitz geometry.

The theorem does not automatically propagate through every finite place or support aperture. The successor metric and aperture connection derived here state exactly what such a propagation theorem would have to control.

XII.2 Semilocal and adelic work supplies the body, not the final sign

Connes's adelic trace formula and later semilocal constructions supply the arithmetic space, scale action, local terms, and finite-place structure [23], [26]. Burnol's conductor operator uniformly realizes local explicit-formula terms and proves positivity under support restrictions [27].

These results mean that the desired carrier is not missing wholesale. The unclosed part is the positive completed comparison after the finite-place metric has turned the receiving aperture.

XII.3 Spectral triples and finite Galerkin certificates are adjacent routes

The 2025 zeta spectral-triple programme constructs finite self-adjoint approximants whose spectra closely track low zeta zeros; rigorous convergence would establish RH, and that convergence remains open [28].

The 2026 finite Guinand–Weil dictionary gives exact finite test functions and an archimedean tail certification budget for truncated quadratic forms, while explicitly making no RH claim [29]. It can serve as a bounded falsification and diagnostic instrument for projections of the defect recurrence. It cannot replace a theorem over the continuum test family.

XII.4 Selberg and dynamical analogies specify obligations

Selberg's trace formula is a rigorous world in which a self-adjoint spectrum and closed geodesics meet through one trace relation [30]. Deninger's dictionary identifies primes with primitive periodic orbits and $\log p$ with orbit length, while leaving the natural global Riemann system open [5].

These are valuable because they specify what an RH carrier owes: a space, flow or operator, domain, primitive paths, repetitions, archimedean term, trace identity, and positive pairing. Renaming Eros arcs as orbits does not supply those objects.

Nyman–Beurling gives another exact equivalent density problem [31]. A Hilbert completion built from the Weil distribution under RH describes post-positivity geometry rather than an independent source of the sign [32]. Equivalent formulations are maps of the boundary; they do not remove it.

XIII The present proof boundary

The former target was malformed. The dyadic periodization cannot be a restriction of the common Sonin successor because it has a kernel and the successor is invertible. The published maps already settle the space-level prime/support square. The absent object is the Hermitian form carried through that square.

XIII.1 Construct the source-derived conditioned short

One bounded Euler calibration has been resolved. On

$$(\{\infty\}, R = 2) \to (\{\infty, 2\}, R = 4),$$

the $p = 2$ return is the difference of the positive interior polarization B_- and the positive departing polarization B_+. This rules out treating the first prime arm as one unsigned scalar or as one independently contractive block. It does not close the first induction step, which is

$$\mathcal{T}_2 \to \mathcal{T}_3$$

with $p = 2$ admitted at support $R = 3$.

For arbitrary n, the quotient is represented by the conditioned graph

$$\mathcal{T}_{n+1} = \mathcal{T}_n \oplus J_n \mathcal{B}_n.$$

Choose a form-domain lift $\tilde{J}_n$ of this Hilbert quotient. The calculation must derive, from the semilocal trace and explicit formula, the cross functional against $\tilde{J}_n y$ and the shell value on $\tilde{J}_n y$. It must then prove that the cross descends continuously through the old energy completion and that the resulting form short is nonnegative. This puts the new support directions, their forced old-region compensation, additive Fourier return, multiplicative inversion, the explicit B_+ departure, and the two Sonin projections on one declared dense domain.

At a prime step $n = p$, it must also use the already-proved dual transport

$$\Delta_p = I - p^{-\frac{1}{2}} U_p, \quad \mathcal{E}_p = \left(\Delta_p^*\right)^{-1}, \quad \Delta_p^* \mathcal{E}_p = I,$$

together with the globally normalized Poisson intertwiner. At a composite step no new place map is needed; the moving shell and any newly exposed prime-power traversal remain.

The dyadic quotient supplies a necessary calibration of this short: its symmetric metric-defect face must be $\frac{\log 2}{\sqrt{2}} P_+$. It does not determine the conditioned cross. Defining R_n from the square root of a form already assumed positive would be circular, and no finite zero check can manufacture the identity.

The result sought is one formula valid for symbolic n. Once it is proved, ordinary induction and compact support finish the Weil criterion. There is no separate requirement to construct or inspect every aperture.

The symbolic prime-power part of that request is now closed. Let $\mathcal{A}_n$ contain the archimedean face and every active prime-power hinge, and let $c_{\alpha,n,y}$ be the old-energy carrier of component α against a conditioned quotient direction. The exact successor residual is

$$s_{n(y)} = \sum_\alpha q_{\alpha,n+1}\big(\tilde{J}_n y\big) - \left\| \sum_\alpha c_{\alpha,n,y} \right\|_{\mathcal{E}_n}^2. \quad \text{(GLOBAL GRAM DEFICIT)}$$

The prime-power cells determine the index set, their log-displacements, and their weights. They do not sign ("GLOBAL GRAM DEFICIT"). The remaining uniform theorem is precisely that this coupled expression is nonnegative after the source's moment rebase—not a further count of cells and not an independent positive factor for each prime.

The first irreducible instance is now completely scoped in a source-native continuous-kernel chart. Suzuki represents the localized Weil form by

$$q_{R(v)} = \langle G_R Dv, Dv \rangle, \quad D = i\frac{\mathrm{d}}{\mathrm{d}} u,$$

where G_R is the compression of the explicit zeta screw kernel $g_{\zeta(x-y)}$ [24]. The published $P(2)$ base agrees with Yoshida's verified positive interval [33]. For a conditioned amplitude v, integration by parts places $w = Dv$ in

$$\mathcal{K}_R = \left\{ w : \int w = 0, \int u w(u)\, \mathrm{d}u = 0, \int e^{\sigma \frac{u}{2}} w(u)\, \mathrm{d}u = 0 \right\}. \quad \text{(CURRENT FIBER)}$$

Let J_D be the three-moment graph lift from the new current shell into $\mathcal{K}_3$, and write the raw restriction of G_3 to old and shell current regions as

$$\begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \quad A|_{\mathcal{K}_2} = G_2.$$

Then the actual first cross and shell operators are

$$C_{2,3} = P_{\mathcal{K}_2}(A L_D + B),$$

$$D_{2,3} = L_D^* A L_D + L_D^* B + B^* L_D + D. \quad \text{(FIRST CONDITIONED BLOCK)}$$

The $p = 2$ translation contributes zero raw diagonal, but contributes $L_D^* B_2 + B_2^* L_D$ to this conditioned shell. That is the concrete correction to the previous first-successor formula.

Because G_2 is compact in this current chart, a closed square-root range cannot be assumed merely from its kernel. The source nevertheless settles the first cross by a different route: the $P(2)$ form is positive definite, its compact-resolvent ground value is attained, and hence

$$q_2(v) \geq \lambda_2 \|v\|^2, \quad \lambda_2 > 0.$$

Lower-semibounded closedness of q_3 then makes every cross functional q_2-continuous. In the current chart, Riesz representation and Douglas factorization give

$$C_{2,3} = G_2^{\frac{1}{2}} Y_{2,3}, \quad C_{2,3} C_{2,3}^* \leq c G_2$$

for a finite c. Thus the first crossing already has finite energy in the old positive carrier.

XIII.2 The conditioned residual is effective tension

The exact first obligation has narrowed to one sign:

$$S_{2,3} = D_{2,3} - Y_{2,3}^* Y_{2,3} \geq 0, \quad \text{(FIRST SUCCESSOR)}$$

or equivalently

$$\langle S_{2,3} y, y \rangle = \inf_x q_3\big(x + \tilde{J} y\big) \geq 0.$$

It is the genuinely new conditional interior left after the carried old component is removed. The sources construct the kernel, base gap, Sonin spaces, raw prime incidence, and cross carrier. They do not sign this conditioned residual on $I_2 \to I_3$. This is the first genuine wall, now one specific operator inequality rather than an unspecified atlas or a request for more examples.

Theorem 51 — Shorting returns the shell's effective tension after the old body relaxes Let a densely defined closed Hermitian form q be written, after one conditioned quotient lift, on an old fiber $\mathcal{V}_o$ and a quotient fiber $\mathcal{B}$ as

$$q\big(x + \tilde{J} y\big) = \langle Ax, x \rangle + 2 \operatorname{Re} \langle Cy, x \rangle + \langle Dy, y \rangle. \quad \text{(BLOCK ENERGY)}$$

Suppose the old operator has a strict gap

$$A \geq \lambda_o I, \quad \lambda_o > 0. \quad \text{(OLD GAP)}$$

Then the old response to a supplied quotient direction is the unique relaxed configuration

$$x_y = -A^{-1} C y. \quad \text{(RELAXATION)}$$

The energy remaining after that response is represented by

$$S = D - C^* A^{-1} C, \quad s(y) = \langle Sy, y \rangle = \inf_{x \in \mathcal{V}_o} q\big(x + \tilde{J} y\big). \quad \text{(EFFECTIVE TENSION)}$$

More precisely,

$$q\big(x + \tilde{J} y\big) = \left\| A^{\frac{1}{2}} \big(x + A^{-1} C y\big) \right\|^2 + \langle Sy, y \rangle. \quad \text{(RELAXED INTERIOR)}$$

If S is read as an operator from quotient displacement to its residual generalized force, then

$$\tau_{\mathrm{eff}(y)} = Sy$$

is the effective tension seen at that quotient after every old-interior direction has equilibrated. It is the nonlocal Dirichlet-to-Neumann or Steklov face of the declared decomposition: supplied boundary data go in; the unexplained returned force comes out. This statement uses only the variational relation and does not require a local differential equation.

The block form is congruent to the relaxed diagonal form:

$$\begin{pmatrix} A & C \\ C^* & D \end{pmatrix} = \begin{pmatrix} I & 0 \\ C^*A^{-1} & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} I & A^{-1}C \\ 0 & I \end{pmatrix}. \quad \text{(INERTIA SPLIT)}$$

Consequently all negative and null directions not already present in the strictly positive old body belong exactly to S. In particular,

$$q \geq 0 \quad \leftrightarrow \quad S \geq 0. \quad \text{(STABILITY)}$$

The reduced boundary energy

$$E_{\mathrm{eff}(y)} = \inf_x q\big(x + \tilde{J}y\big) = \langle Sy, y \rangle$$

has Hessian $2S$. Thus positive, null, and negative tension are exactly convex, flat, and concave directions of the relaxed response. Along a parameterized family, a zero mode of S is the discriminant at which one reduced direction can pass between convexity and concavity.

Shorting is associative. For nested old interiors $\mathcal{V}_0 \subset \mathcal{V}_1 \subset \mathcal{V}_2$, whenever the displayed relaxations exist,

$$\frac{\frac{q}{\mathcal{V}_0}}{\frac{\mathcal{V}_1}{\mathcal{V}_0}} = \frac{q}{\mathcal{V}_1}, \quad \text{(ASSOCIATIVE SHORT)}$$

because both sides minimize q over the same complete $\mathcal{V}_1$ interior. Thus a completed lower-grain response may depart while its effective boundary law becomes the next standing form.

There is a necessary distinction between effective tension and an arbitrary equilibrium Hessian. If $Av = \lambda v$ with $\|v\| = 1$ is a ground state, the constrained second variation on $v^\perp$ is

$$2\langle (A - \lambda I)\eta, \eta \rangle. \quad \text{(MODE HESSIAN)}$$

Replacing A by $A - cI$ replaces λ by $\lambda - c$ and leaves ("MODE HESSIAN") unchanged. Therefore standing-wave modes, nodes, antinodes, and local equilibrium stability cannot by themselves determine whether the absolute ground value is positive. The zero of the Weil response must remain calibrated by the explicit formula.

Apply this theorem to the first conditioned Weil successor. Let A_2 represent the strictly positive $P(2)$ form, let $C_{2,3}$ be the already-carried old/shell cross, and let $D_{2,3}$ be the conditioned shell block of the exact zeta screw kernel. Then

$$S_{2,3} = D_{2,3} - C_{2,3}^* A_2^{-1} C_{2,3} = D_{2,3} - Y_{2,3}^* Y_{2,3}. \quad \text{(FIRST TENSION)}$$

Here $C_{2,3} = A_2^{\frac{1}{2}} Y_{2,3}$ is the reduced cross carrier. Hence $S_{2,3}$ is not an additional global Gram object. The component Gram expansion is one coordinate expansion of the energy absorbed by the single old response $A_2^{-1}C_{2,3}y$. The first RH-support obligation is exactly that the returned effective tension ("FIRST TENSION") have no negative mode.

Demonstration. Differentiate ("BLOCK ENERGY") in an arbitrary old direction. The stationarity equation is

$$Ax + Cy = 0,$$

and ("OLD GAP") makes its solution unique, proving ("RELAXATION"). Substitute that solution, or complete the square, to obtain ("EFFECTIVE TENSION") and ("RELAXED INTERIOR"). Differentiating the effective quadratic form in a quotient direction gives Sy, which is precisely the residual force after the old stationarity equation has been solved.

Direct multiplication proves (“INERTIA SPLIT”). Congruence by an invertible triangular operator preserves inertia, proving (“STABILITY”). For (“ASSOCIATIVE SHORT”), write a representative as $x_0 + x_1 + y$ and observe

$$\inf_{x_1} \inf_{x_0} q(x_0 + x_1 + y) = \inf_{x \in \mathcal{V}_1} q(x + y).$$

The Lagrange multiplier equation for the unit sphere is $Av = \lambda v$. Its second variation is (“MODE HESSIAN”). The displayed scalar shift cancels between the operator and its ground value, proving the calibration warning. In the first Weil cell, the published strict base gap makes A_2^{-1} bounded on the old carrier. The conditioned-support theorem supplies $C_{2,3} = A_2^{\frac{1}{2}} Y_{2,3}$. Substitution into (“EFFECTIVE TENSION”) gives (“FIRST TENSION”), while the inertia split proves its exact equivalence to the first successor sign.

Boundary. This theorem replaces the phrase “global Gram dominance” with the invariant object it was trying to name: effective tension after internal relaxation. It does not prove that $S_{2,3} \geq 0$. It proves that no separate knot energy, favorable perspective, equilibrium picture, or componentwise prime estimate can decide that sign. A proof must bound the source-derived zeta screw kernel’s conditioned Dirichlet-to-Neumann operator at its explicit-formula zero level.

This identifies the invariant geometry behind the earlier component Gram expansion. A supplied shell direction first induces the unique old-body response $-A_2^{-1} C_{2,3} y$. Only the residual force after that relaxation is visible at the new boundary. Congruence of the full block form with $A_2 \oplus S_{2,3}$ proves that every new negative or null direction belongs exactly to this effective tension. Sequential shorting is associative, so a completed interior may depart while this returned boundary law becomes the next standing form.

This also makes the convexity intuition exact. The relaxed boundary energy $E_{\mathrm{eff}(y)} = \inf_x q_3\big(x + \tilde{J}y\big)$ has Hessian $2S_{2,3}$. Positive, null, and negative modes are respectively convex, flat, and concave directions of the returned response. If a lowest mode reaches zero while the aperture changes, that event is the discriminant between the two local curvature classes—not a favorable change of coordinates.

The same theorem also limits the knot analogy precisely. Nodes, antinodes, and constrained equilibrium modes describe the Hessian $A - \lambda_{\min} I$. A scalar shift changes the absolute ground value while leaving that Hessian unchanged. Tension geometry therefore becomes proof-bearing only after the explicit formula fixes the zero of energy; a generic knot energy cannot supply the Weil sign.

Existing Galerkin evidence lands on precisely this wall. Connes–Consani observe numerically that the archimedean block loses positivity after the prime-free boundary, while the exact $p = 2$ contribution restores the combined sign through the first prime-active window ending at $R = 3$ [34]. Their computation supports $S_{2,3} \geq 0$ but does not prove the continuum conditional-energy inequality. The present reduction identifies what that numerical restoration is measuring: not an independently positive prime term, but closure of the conditioned old–new short.

XIII.3 Positivity already crosses the first prime seam

Corollary 52 — Strict base positivity continues through a genuine first-prime interval Let q_a be the localized Weil form on $(-a, a)$ and let

$$\lambda(a) = \inf_{v \neq 0} \frac{q_{a(v)}}{\|v\|^2}$$

be its attained lowest spectral value. Put

$$a_2 = \frac{\log 2}{2}, \quad a_3 = \frac{\log 3}{2}.$$

The published base theorem gives

$$\lambda(a_2) > 0,$$

and the published screw-function theorem gives continuity of $a \mapsto \lambda(a)$. Therefore there is a

$$\delta \in (0, a_3 - a_2)$$

such that

$$\lambda(a) > 0 \quad \text{for every} \quad a \in [a_2, a_2 + \delta]. \quad \text{(PRIME-ACTIVE CONTINUATION)}$$

Equivalently, with

$$R_* = \exp(2(a_2 + \delta)) = 2e^{2\delta} > 2,$$

Weil positivity is proved on every aperture

$$2 \leq R \leq R_*.$$

Every interior aperture in this interval contains the newly active $p = 2$ crossing and no $p = 3$ crossing. Hence the result crosses the prime-free boundary; it is not merely another statement inside the archimedean cell.

If RH is false, continuity and small-aperture positivity imply the existence of a first degeneracy parameter

$$a_* = \inf\{a \geq a_2 : \lambda(a) \leq 0\}$$

with

$$\lambda(a_*) = 0.$$

Relative to any earlier strictly positive aperture used as the old body, the conditioned effective tension at a_* has a nonzero null mode. Thus the first possible failure of RH appears geometrically as a zero-tension standing mode of the source-derived support continuation, not as a change of receiver coordinates.

Demonstration. Since $\lambda(a_2) > 0$ and λ is continuous, choose $\delta > 0$ small enough that

$$|\lambda(a) - \lambda(a_2)| < \frac{\lambda(a_2)}{2}$$

on $[a_2, a_2 + \delta]$, and also $\delta < a_3 - a_2$. This proves ("PRIME-ACTIVE CONTINUATION"). The prime-power threshold law says that $p = 2$ becomes active as soon as $2a > \log 2$, whereas $p = 3$ does not become active before $2a = \log 3$.

If RH fails, Weil's criterion supplies some aperture with a negative lowest value. Continuity, together with positivity at the base, gives the first zero a_*. The inertia split in the effective-tension theorem carries that first new null direction entirely into the shorted continuation after the earlier positive interior has relaxed.

Boundary. The corollary proves a real prime-active interval but does not give a numerical value of δ and therefore does not reach a_3. Closing the full first cell now has a precise quantitative form: control the source-derived effective tension, or an equivalent modulus for $\lambda(a)$, throughout

$$\left[\frac{\log(2)}{2}, \frac{\log(3)}{2}\right].$$

Continuity alone cannot provide that bound.

This is a genuine advance over the former all-or-nothing phrasing. If $a = \frac{\log R}{2}$, the published strict base gap gives $\lambda(a_2) > 0$ at $a_2 = \frac{\log 2}{2}$, and Suzuki proves continuity of the attained lowest localized eigenvalue [24], [33]. Hence some $\delta > 0$ satisfies

$$\lambda(a) > 0 \quad \text{on} \quad [a_2, a_2 + \delta],$$

with $a_2 + \delta < a_3 = \frac{\log 3}{2}$. The corresponding apertures have $2 < R \leq 2e^{2\delta}$: the $p = 2$ crossing is actually present, while the $p = 3$ crossing is not.

Continuity does not provide the size of δ. The first-cell obligation is therefore quantitative rather than existential: bound the source-derived effective tension, or equivalently control the modulus of $\lambda(a)$, all the way across

$$\left[\frac{\log 2}{2}, \frac{\log 3}{2}\right].$$

If positivity first fails at some later aperture, the effective tension has a nonzero null mode there. This is the exact zero-tension standing mode which the tension intuition was approaching.

XIII.4 The fixed-domain wall contains one finite-place hinge

Lemma 53 — Before the third prime enters, the finite-place operator is one exact boundary-strip swap Put

$$a_2 = \frac{\log 2}{2}, \quad a_3 = \frac{\log 3}{2}, \quad h(a) = \frac{\log 2}{a}, \quad \alpha_2 = \frac{\log 2}{\sqrt{2}}.$$

For $a \in (a_2, a_3)$ one has $1 < h(a) < 2$. In $\mathcal{H} = L^2(-1,1)$ define the disjoint boundary-strip fibers

$$\mathcal{H}_{L(a)} = L^2(-1, 1-h(a)), \quad \mathcal{H}_{R(a)} = L^2(-1+h(a), 1),$$

and let $V_a : \mathcal{H}_{L(a)} \to \mathcal{H}_{R(a)}$ be translation by $h(a)$, extended by zero on the orthogonal complement. Then

$$T_a = V_a + V_a^*, \quad T_a^2 = P_{L(a)} + P_{R(a)}, \quad \operatorname{spec}(T_a) = \{-1, 0, 1\}. \quad \text{(HINGE SWAP)}$$

Its positive and negative faces are the paired boundary configurations

$$f \oplus V_a f \quad \text{and} \quad f \oplus (-V_a f),$$

while the central fiber is its zero face.

Let r and the closed form $\mathcal{L}$ be the archimedean objects in the published fixed-domain expansion of the zeta screw function, and define

$$\langle \mathcal{R}_a w, w\rangle = \int_{-1}^{1}\int_{-1}^{1} r''(a(x-y))w(y)\overline{w(x)}\,\mathrm{d}x\,\mathrm{d}y.$$

If

$$A_\zeta = \frac{1}{2}(\log(2\pi) + \gamma - 1),$$

the localized Weil form transported to $(-1,1)$ is exactly

$$\bar{q}_{a(w)} = \mathcal{L}(w) - \big(\log a + 2A_\zeta + 1\big)\|w\|^2 - \alpha_2\langle T_a w, w\rangle - a\langle \mathcal{R}_a w, w\rangle. \quad \text{(ONE-PRIME FORM)}$$

At $a = a_2$ the two strips have zero measure and the hinge is born; at $a = a_3$ the prospective $p = 3$ strip still has zero measure. Hence no other finite-place operator occurs in the complete first open cell.

Define the remaining archimedean operator form

$$\mathcal{H}_a = \mathcal{L} - \big(\log a + 2A_\zeta + 1\big)I - a\mathcal{R}_a.$$

Positivity through the first prime-active interval is therefore exactly the source inequality

$$\mathcal{H}_a - \alpha_2 T_a \geq 0 \quad \text{for} \quad a \in [a_2, a_3]. \quad \text{(FIXED-DOMAIN WALL)}$$

In particular,

$$|\langle T_a w, w\rangle| \leq \|w\|^2,$$

but this sharp hinge bound alone does not prove (“FIXED-DOMAIN WALL”).

Demonstration. For $a \in (a_2, a_3)$, the prime-power threshold $\log n \leq 2a$ admits only $n = 2$. Specializing the published scaled screw-form identity to that one term makes its two overlap integrals exactly

$$\langle (V_a + V_a^*)w, w\rangle.$$

Because $h(a) > 1$, the source and target strips are disjoint and V_a is a partial isometry with

$$V_a^* V_a = P_{L(a)}, \quad V_a V_a^* = P_{R(a)}.$$

This proves ("HINGE SWAP") and its three spectral faces. The coefficient of the $n = 2$ von Mangoldt term is $\frac{\Lambda(2)}{\sqrt{2}} = \alpha_2$. The remaining terms of the published fixed-domain formula give $\mathcal{H}_a$, proving ("ONE-PRIME FORM") and ("FIXED-DOMAIN WALL"). The quadratic bound follows from $\|T_a\| = 1$.

Boundary. This lemma removes all finite-place combinatorial ambiguity from the first cell. It does not show that the archimedean form $\mathcal{H}_a$ dominates the positive hinge face. The smooth-kernel dependence and the barely logarithmic form domain yield continuity of the lowest value, but the published argument supplies no quantitative modulus reaching a_3. A valid next theorem must estimate this declared operator family, not replace it with a finite prime census or an independently chosen knot energy.

Suzuki's fixed-domain formula makes the quantitative obligation still more explicit [24]. Throughout the open interval before (p=3) enters, the entire finite-place operator is

$$\alpha_2 T_a, \quad \alpha_2 = \frac{\log 2}{\sqrt{2}},$$

where T_a swaps two disjoint boundary strips and has exact spectral faces $-1, 0, 1$. The changing strip width carries the new combinatorics completely. Everything not in this one hinge is the declared archimedean logarithmic form and smooth screw-kernel remainder.

Thus the first-cell problem is not missing another prime relation. In the fixed receiver it is the continuum domination

$$\mathcal{H}_a - \alpha_2 T_a \geq 0 \quad \text{for} \quad a \in \left[\frac{\log 2}{2}, \frac{\log 3}{2}\right].$$

The sharp bound $\|T_a\| = 1$ is insufficient because the archimedean body does not commute with the moving strip swap. Proving the displayed inequality requires a source-specific lower estimate, comparison principle, or quantitative ground-value bound.

XIII.5 The first-prime cross completes to one open capacitor

Theorem 54 — Passive incidence survives internal balance, gluing, and shorting Let an oriented incidence complex have vertex fiber

$$\mathcal{V} = \mathcal{V}_I \oplus \mathcal{V}_B,$$

edge fiber $\mathcal{E}$, boundary operator $\partial : \mathcal{E} \to \mathcal{V}$, and coboundary $d = \partial^* : \mathcal{V} \to \mathcal{E}$. Let $W \geq 0$ be an edge constitutive operator and let $G \geq 0$ be a vertex-storage operator. On their natural common form domain define

$$\mathcal{E}(v) = \left\|W^{\frac{1}{2}} dv\right\|^2 + \left\|G^{\frac{1}{2}} v\right\|^2 = \langle Lv, v\rangle, \quad L = d^* W d + G. \quad \text{(PASSIVE INCIDENCE)}$$

Here $\partial_{k-1}\partial_k = 0$ types legal incidence when adjacent chain degrees are present; nonnegativity comes from W and G, not from that topological identity alone.

Write the response operator in interior/boundary blocks as

$$L = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix}$$

and suppose $A \geq \lambda_I I$ for some $\lambda_I > 0$. Supplying boundary potential y and imposing zero interior injection gives

$$Ax + Cy = 0, \quad x_y = -A^{-1}Cy. \quad \text{(INTERIOR BALANCE)}$$

The returned boundary current is

$$i_B = C^* x_y + Dy = Sy, \quad S = D - C^* A^{-1} C. \quad \text{(PORT RESPONSE)}$$

Moreover,

$$\langle Sy, y \rangle = \inf_x \mathcal{E}(x \oplus y) \geq 0. \quad \text{(PASSIVE SHORT)}$$

Thus the Dirichlet-to-Neumann response and the Kron reduction are the same returned boundary law, and internal balance cannot turn a passive incidence network into a negative terminal response.

Parallel composition adds edge and storage forms. Gluing two ports identifies their potentials and sums their oriented currents. Both operations preserve (“PASSIVE INCIDENCE”). Eliminating any completed set of internal vertices then preserves (“PASSIVE SHORT”), and nested eliminations are associative. Consequently a support induction would be complete if every source successor supplied, before the sign was assumed, an enlarged incidence/storage form with nonnegative constitutive operators whose Kron return is the next Weil form.

In a capacitive reading, $\rho = Gv$ is stored charge and $\dot{\rho}$ is displacement current. The complete balance law is

$$\partial j + \dot{\rho} = s. \quad \text{(STORED BALANCE)}$$

A nonzero port imbalance at one cut can therefore be the rate of change of internal storage. Conservation applies to conduction plus storage over the complete event; it does not require the visible terminal currents to cancel instantaneously.

> **Demonstration.** Positivity in (“PASSIVE INCIDENCE”) is immediate from the two squared norms. The interior Euler equation is (“INTERIOR BALANCE”). Substitution gives (“PORT RESPONSE”), while completing the square gives
>
> $$\mathcal{E}(x \oplus y) = \left\| A^{\frac{1}{2}}(x + A^{-1} C y) \right\|^2 + \langle Sy, y \rangle.$$
>
> This proves (“PASSIVE SHORT”). Direct sums preserve squared norms, and port gluing is restriction to a closed linear relation, so neither can create negative energy. Associativity follows because successive shorts minimize over the same total interior. Finally (“STORED BALANCE”) is the vertex balance obtained by adjoining the time derivative of the stored vertex quantity to the incident edge current.

> Boundary. This theorem says exactly what a circuit proof of a Weil-support successor would owe. A signed off-diagonal coupling, a conservation identity, or a Schur complement is not by itself such a proof. The source must also supply the nonnegative edge self-terms or storage whose complete form is being shorted. Factoring an operator only after assuming that operator nonnegative would be circular.

The circuit reading is exact at the level at which it is useful. An oriented incidence matrix takes vertex potentials to edge differences; a nonnegative constitutive law turns those differences into current or stored tension; and internal current balance returns the Schur complement at the exposed ports. In electrical-network language this is Kron reduction [35]. Incidence, storage, and exposed ports also give the compositional core of graph port-Hamiltonian systems [36].

The theorem separates three facts which had been running together:

- $\partial^2 = 0$ makes adjacent incidences compose lawfully;
- conservation includes the change of stored charge, so visible terminal current need not cancel at one instantaneous cut; and
- passivity additionally owes a nonnegative constitutive/storage form.

Only the third fact signs the returned Dirichlet-to-Neumann response. Therefore conservation over an order of time can carry a temporary port deficit as stored charge, but cannot make an indefinite mutual coupling positive without the corresponding self-storage.

Lemma 55 — The first-prime cross is an open capacitor with an exact endpoint-storage debit Retain the first-cell notation

$$a_2 = \frac{\log 2}{2}, \quad a_3 = \frac{\log 3}{2}, \quad \alpha_2 = \frac{\log 2}{\sqrt{2}},$$

together with the left and right strip projections $P_{L(a)}, P_{R(a)}$ and their translation partial isometry V_a. Put

$$P_a = P_{L(a)} + P_{R(a)}, \quad T_a = V_a + V_a^*,$$

and define the oriented prime difference

$$\delta_a w = P_{R(a)} w - V_a P_{L(a)} w. \quad \text{(PRIME INCIDENCE)}$$

Then

$$\delta_a^* \delta_a = P_a - T_a. \quad \text{(CAPACITOR COMPLETION)}$$

Hence the exact first-cell Weil form admits the identity

$$\overline{q}_a = \mathcal{H}_a - \alpha_2 T_a = \alpha_2 \delta_a^* \delta_a + \mathcal{K}_a, \quad \mathcal{K}_a = \mathcal{H}_a - \alpha_2 P_a. \quad \text{(OPEN CAPACITOR)}$$

The prime contribution supplies the mutual cross term of a capacitor. Completing it to the nonnegative branch energy $\alpha_2 \|\delta_a w\|^2$ exposes the equal endpoint-storage debit $\alpha_2 \|P_a w\|^2$. The original prime term is therefore not a passive branch by itself.

The archimedean form occurring here already has the exact Beurling–Deny incidence/storage representation

$$\mathcal{L}(w) = \frac{1}{4} \int_{-1}^{1} \int_{-1}^{1} \frac{|w(x) - w(y)|^2}{|x-y|} \,\mathrm{d}x \,\mathrm{d}y + \frac{1}{2} \int_{-1}^{1} (-\log(1-x^2)) |w(x)|^2 \,\mathrm{d}x. \quad \text{(ARCHIMEDEAN NETWORK)}$$

If

$$c_a = \log a + 2A_\zeta + 1$$

and $\mathcal{R}_a$ is the smooth-kernel operator in the fixed-domain screw form, define

$$\mathcal{E}_{a(w)} = \mathcal{L}(w) + \alpha_2 \|\delta_a w\|^2,$$

$$\mathcal{D}_{a(w)} = c_a \|w\|^2 + a \langle \mathcal{R}_a w, w \rangle + \alpha_2 \|P_a w\|^2. \quad \text{(STORAGE AND LOAD)}$$

Then, exactly,

$$\overline{q}_{a(w)} = \mathcal{E}_{a(w)} - \mathcal{D}_{a(w)}. \quad \text{(FIRST-CELL BALANCE)}$$

The first-cell sign is consequently equivalent to the one source-derived relative passivity inequality

$$\mathcal{D}_{a(w)} \leq \mathcal{E}_{a(w)} \quad \text{for every } w \quad \text{and every } a \in [a_2, a_3]. \quad \text{(UNIT-GAIN WALL)}$$

The closed positive form $\mathcal{E}_a$ has a strict lower gap and compact resolvent. The load form $\mathcal{D}_a$ is a bounded Hermitian perturbation. Therefore

$$\Gamma_a = \mathcal{E}_a^{-\frac{1}{2}} \mathcal{D}_a \mathcal{E}_a^{-\frac{1}{2}} \quad \text{(RELATIVE LOAD)}$$

is compact and self-adjoint, and

$$\overline{q}_a \geq 0 \quad \leftrightarrow \quad \lambda_{\max(\Gamma_a)} \leq 1. \quad \text{(RELATIVE PASSIVITY)}$$

If equality first occurs, its mode w satisfies the complete weak balance

$$\mathcal{D}_{a(w,z)} = \mathcal{E}_{a(w,z)} \quad \text{for every } z. \quad \text{(UNIT-GAIN STANDING)}$$

Thus the possible zero is a unit-gain standing mode of the whole archimedean–prime circuit, not a scalar failure of one isolated prime edge.

On the paired strips, the aligned face $f + V_a f$ lies in $\ker(\delta_a)$, whereas the opposed face $f - V_a f$ has

$$\left\|\delta_{a(f-V_a f)}\right\|^2 = 2\|f - V_a f\|^2.$$

The prime branch therefore spends no difference energy on a transported recurrence and stiffens the opposed configuration by $2\alpha_2$. Any negative part of $\mathcal{K}_a$ can be repaired only through such an opposed component. In particular,

$$\mathcal{K}_a|_{\ker \delta_a} \geq 0 \quad \text{(COHERENT NECESSITY)}$$

is necessary for first-cell positivity, while the stronger global assertion $\mathcal{K}_a \geq 0$ is sufficient but not necessary.

Demonstration. Since $V_a^* V_a = P_{L(a)}$ and $V_a V_a^* = P_{R(a)}$, direct multiplication of (“PRIME INCIDENCE”) gives

$$\delta_a^* \delta_a = P_{R(a)} + P_{L(a)} - V_a - V_a^* = P_a - T_a.$$

Rearranging proves (“OPEN CAPACITOR”). Substituting the published formula for $\mathcal{H}_a$ and the exact difference-energy representation of $\mathcal{L}$ proves (“FIRST-CELL BALANCE”).

The two terms in (“ARCHIMEDEAN NETWORK”) are nonnegative. If both vanish, the jump term makes w constant almost everywhere, while the boundary-storage term then forces that constant to be zero. Closedness and compact embedding therefore give a strict lower gap for $\mathcal{L}$, hence also for $\mathcal{E}_a$. The scalar, projection, and continuous-kernel terms in $\mathcal{D}_a$ are bounded. The compact resolvent of $\mathcal{E}_a$ makes (“RELATIVE LOAD”) compact and self-adjoint. Conjugating (“FIRST-CELL BALANCE”) by $\mathcal{E}_a^{-\frac{1}{2}}$ proves (“RELATIVE PASSIVITY”) and its equality equation.

The aligned and opposed formulas follow directly from the partial isometry relations. Restricting (“OPEN CAPACITOR”) to $\ker(\delta_a)$ proves (“COHERENT NECESSITY”).

Boundary. This identity locates the missing diagonal exactly. The source supplies positive nonlocal archimedean incidence and boundary storage, but after its scalar and smooth-kernel loads are included it has not yet been proved to pay the endpoint debit on the complete first-cell domain. Treating $\mathcal{K}_a$ as positive would add a stronger assumption and is not the proof. The honest remaining obligation is (“UNIT-GAIN WALL”), equivalently the largest relative-load eigenvalue staying at most one through the complete first prime interval.

For the first prime, define the transported strip difference

$$\delta_a w = P_{R(a)} w - V_a P_{L(a)} w.$$

The exact identity

$$\delta_a^* \delta_a = P_{L(a)} + P_{R(a)} - T_a$$

turns the signed prime cross into

$$-\alpha_2 \langle T_a w, w \rangle = \alpha_2 \|\delta_a w\|^2 - \alpha_2 \left\| \left(P_{L(a)} + P_{R(a)}\right) w \right\|^2. \quad \text{(OPEN CAPACITOR)}$$

The first term is the complete positive energy of a branch joining corresponding strip values. The second is the endpoint-storage debit which must be paid by the co-present archimedean body. The prime is therefore neither an independently negative constituent nor an independently passive capacitor. It is the mutual face of a coupled archimedean–prime circuit.

This completion is especially concrete because Suzuki’s $\mathcal{L}$ is already a genuine nonlocal Dirichlet form [24]:

$$\mathcal{L}(w) = \frac{1}{4} \int \int \frac{|w(x) - w(y)|^2}{|x - y|} \,\mathrm{d}x \,\mathrm{d}y + \frac{1}{2} \int (-\log(1 - x^2)) |w(x)|^2 \,\mathrm{d}x.$$

The first integral is a continuum of positive pair incidences and the second is positive boundary-to-ground storage. After adjoining the completed prime branch, write their total as $\mathcal{E}_a$ and place the scalar shift, smooth screw-kernel term, and exposed endpoint debit in the Hermitian load $\mathcal{D}_a$. Then

$$\bar{q}_a = \mathcal{E}_a - \mathcal{D}_a$$

exactly. Since $\mathcal{E}_a$ has a strict gap and compact resolvent, the relative load

$$\Gamma_a = \mathcal{E}_a^{-\frac{1}{2}} \mathcal{D}_a \mathcal{E}_a^{-\frac{1}{2}}$$

is compact self-adjoint, and the first wall becomes

$$\lambda_{\max(\Gamma_a)} \le 1 \quad \text{through} \quad a \in \left[\frac{\log 2}{2}, \frac{\log 3}{2}\right]. \quad \text{(RELATIVE PASSIVITY)}$$

This is narrower than asking for a positive remainder. Requiring $\mathcal{H}_a - \alpha_2(P_L + P_R) \ge 0$ everywhere would be sufficient, but it is stronger than the Weil sign: the positive difference branch can compensate negative load in opposed strip modes. It cannot compensate an aligned mode because $\delta_{a(f+V_a f)} = 0$. Hence the irreducible necessary condition is positivity of the remainder on the coherent kernel, while the full statement is the unit-gain bound above. At equality the candidate zero is a standing generalized eigenmode

$$\mathcal{D}_{a(w,z)} = \mathcal{E}_{a(w,z)} \quad \text{for every } z.$$

That is the exact capacitive form of the remaining continuum estimate.

XIII.6 The Euler, Gamma, and endpoint faces are one normal return field

Theorem 56 — The completed local metrics form one normal return field Normalize the completed zeta function by

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + z\right).$$

For $s = \sigma + it$ with $\sigma > 1$, define the positive local return metrics

$$\mathcal{K}_0(s) = |s(s-1)|^2, \quad \mathcal{K}_{\infty(s)} = \pi^{-\sigma} \left|\Gamma\left(\frac{s}{2}\right)\right|^2,$$

$$\mathcal{K}_{p(s)} = |1 - p^{-s}|^{-2}. \quad \text{(LOCAL METRICS)}$$

The Euler product gives the exact factorization

$$|\xi(s)|^2 = \frac{1}{4} \mathcal{K}_0(s) \mathcal{K}_{\infty(s)} \prod_p \mathcal{K}_{p(s)}. \quad \text{(COMPLETED RETURN METRIC)}$$

Its outward normal logarithmic current is

$$\mathcal{V}(\sigma, t) = \partial_\sigma \log |\xi(s)|^2 = 2 \operatorname{Re} \frac{\xi'(s)}{\xi(s)}$$

$$= 2 \operatorname{Re}\left(\frac{1}{s} + \frac{1}{s-1}\right) + \operatorname{Re} \psi\left(\frac{s}{2}\right) - \log \pi$$

$$-2 \sum_p \sum_{m \ge 1} (\log p) p^{-m\sigma} \cos(mt \log p). \quad \text{(LOCAL-TO-COMPLETED FLUX)}$$

Thus the endpoint, Gamma/archimedean, and complete prime-power responses are not separate loads: in their common half-plane they are the three local faces of one covector $d \log |\xi|^2$.

Center the seam by $z = x + it$ and put

$$F(z) = \frac{\Xi'(z)}{\Xi(z)}.$$

Then

$$\mathcal{V}\left(\frac{1}{2} + x, t\right) = 2\operatorname{Re} F(x + it). \quad \text{(CENTERED NORMAL FIELD)}$$

If the normal coordinate is reparameterized by $x = x(\tau)$, its reported component becomes

$$\mathcal{V}_\tau = \dot{x}(\tau)\mathcal{V}_x. \quad \text{(NORMAL COVARIANCE)}$$

Positivity below is therefore oriented toward the declared half-plane $x > 0$; reversing the receiver normal reverses the reported current.

The following statements are equivalent:

- every nontrivial zero of ζ lies on $\operatorname{Re}(s) = \frac{1}{2}$;
- F is holomorphic on $x > 0$ and

$$\operatorname{Re} F(z) \geq 0 \quad \text{for every } \operatorname{Re}(z) > 0; \quad \text{(POSITIVE-REAL FIELD)}$$

- for any $c > 0$, the Cayley scattering face

$$\mathcal{S}_{c(z)} = \frac{F(z) - c}{F(z) + c} = \frac{\Xi'(z) - c\Xi(z)}{\Xi'(z) + c\Xi(z)} \quad \text{(SCATTERING FACE)}$$

  is a Schur function on the right half-plane and is inner at almost every regular point of its boundary;
- the genus-zero entire function

$$H(w) = \frac{\Xi(\sqrt{w})}{\Xi(0)} \quad \text{(SQUARED-AXIS RETURN)}$$

  has only negative real zeros; and
- its impedance

$$Z(w) = \frac{H'(w)}{H(w)}, \quad F(z) = 2zZ(z^2), \quad \text{(STIELTJES IMPEDANCE)}$$

  is a Stieltjes transform of a positive measure on $[0, \infty)$.

Under these equivalent statements, if the critical ordinates are γ with multiplicities m_γ, then

$$Z(w) = \sum_{\gamma > 0} \frac{m_\gamma}{w + \gamma^2}, \quad \operatorname{Re} F(x + it) = \sum_{\gamma} m_\gamma \frac{x}{x^2 + (t - \gamma)^2}. \quad \text{(POISSON RETURN)}$$

Hence

$$\mathrm{d}\mu_{x(t)} = \frac{1}{\pi} \operatorname{Re} F(x + it)\, \mathrm{d}t$$

is a positive receiver measure, and it converges weakly as $x \to 0^+$ to the complete zero-counting measure

$$\mathrm{d}\mu_0 = \sum_\gamma m_\gamma \delta_\gamma. \quad \text{(BOUNDARY RETURN)}$$

For an admissible current f, let

$$A_{f(z)} = \mathcal{M}f\left(\frac{1}{2} + z\right).$$

The completed Weil response is then the boundary normal energy of the same return metric:

$$Q_{W(f)} = \lim_{x\to 0^+} \frac{1}{2\pi} \int_{\mathbb{R}} \left|A_{f(it)}\right|^2 \partial_x \log|\Xi(x+it)|^2 \,\mathrm{d}t$$

$$= \sum_{\gamma} m_\gamma \left|A_{f(i\gamma)}\right|^2. \quad \text{(WEIL AS BOUNDARY FLUX)}$$

Moment conditioning, support restriction, and old-body relaxation act on this one quadratic form. In particular, whenever the normal field is positive, every conditioned support short is nonnegative because it is the infimum of a nonnegative boundary-flux energy. Conversely, nonnegativity of the complete nested family of conditioned shorts gives Weil positivity and therefore the first equivalent statement.

There is an exact coefficient receiver which does not enumerate zeros. Write

$$H(w) = \sum_{n\ge 0} h_n w^n, \quad h_0 = 1, \quad Z(w) = \sum_{n\ge 0} c_n w^n, \quad \mu_n = (-1)^n c_n.$$

The relation $H' = HZ$ gives the source recursion

$$c_n = (n+1)h_{n+1} - \sum_{k=1}^{n} h_k c_{n-k}. \quad \text{(CONNECTED RETURN)}$$

The RH statements above are further equivalent to the simultaneous positive-semidefiniteness, for every N, of the two Hankel populations

$$\left[\mu_{i+j}\right]_{0\le i,j\le N} \ge 0, \quad \left[\mu_{i+j+1}\right]_{0\le i,j\le N} \ge 0. \quad \text{(STIELTJES GRAM LAW)}$$

The first connected members are

$$\mu_0 = h_1, \quad \mu_1 = h_1^2 - 2h_2,$$

$$\mu_2 = h_1^3 - 3h_1h_2 + 3h_3,$$

$$\mu_3 = h_1^4 - 4h_1^2h_2 + 2h_2^2 + 4h_1h_3 - 4h_4. \quad \text{(FIRST CONNECTED MEMBERS)}$$

Thus the remaining sign can be attacked as one uniform Gram factorization of the connected return sequence derived from the archimedean theta moments, rather than by checking zeros or support cells one at a time.

Demonstration. For $\sigma > 1$, take absolute squares in the completed Euler product. Logarithmic differentiation is justified by absolute convergence. The endpoint and Gamma factors give

$$\partial_\sigma \log \mathcal{K}_0 = 2\,\mathrm{Re}\left(\frac{1}{s} + \frac{1}{s-1}\right),$$

$$\partial_\sigma \log \mathcal{K}_\infty = \mathrm{Re}\,\psi\left(\frac{s}{2}\right) - \log\pi.$$

For a finite place, the norm-convergent geometric series gives

$$\partial_\sigma \log \mathcal{K}_p = -2(\log p) \sum_{m\ge 1} p^{-m\sigma} \cos(mt\log p).$$

Their sum proves (“LOCAL-TO-COMPLETED FLUX”). The chain rule proves (“NORMAL COVARIANCE”). The functional equation and reality give $\Xi(-z) = \Xi(z)$ and $\Xi(\overline{z}) = \overline{\Xi(z)}$. Pairing opposite zeros in the Hadamard product removes the genus-one exponentials:

$$\Xi(z) = \Xi(0) \prod_{a \bmod \pm} \left(1 - \frac{z^2}{a^2}\right).$$

If RH holds, every representative is $a = i\gamma$, so

$$F(z) = \sum_{\gamma>0} m_\gamma \left(\frac{1}{z - i\gamma} + \frac{1}{z + i\gamma}\right).$$

Every summand has positive real part for $\mathrm{Re}(z) > 0$, proving ("POSITIVE-REAL FIELD") and ("POISSON RETURN"). Conversely, a zero of Ξ in the right half-plane is a pole of F whose real part changes sign in every punctured neighborhood. Positive-realness excludes such a zero. Evenness then excludes zeros in the left half-plane, so all zeros lie on the fixed seam. The Cayley transform carries the right half-plane to the unit disk; boundary reality of $\Xi(it)$ makes its regular boundary values unimodular. This proves the scattering equivalence.
Evenness makes H an entire function of $w = z^2$ of genus zero. The centered zeros $z = \pm i\gamma$ become the negative roots $w = -\gamma^2$. Logarithmic differentiation gives ("STIELTJES IMPEDANCE"). Conversely, a Stieltjes logarithmic derivative can have singularities only on the negative real axis, while every zero of H is a pole of $\frac{H'}{H}$ with positive integral residue. Hence the Stieltjes property forces all zeros of H to be negative real.
The Poisson-kernel approximation to the identity proves ("BOUNDARY RETURN"). Since a compactly supported smooth Mellin current has rapidly decreasing seam amplitude, it may be used as the test function in that weak limit. The Mellin adjoint-return identity and the completed explicit formula then give ("WEIL AS BOUNDARY FLUX"). Nonnegative forms remain nonnegative under restriction and form shorting. The converse through nested support is the established Weil support-induction reduction.
Finally, $H' = HZ$ proves ("CONNECTED RETURN"). If RH holds, then

$$\mu_n = \sum_{\gamma>0} m_\gamma \gamma^{-2n-2},$$

so both matrices in ("STIELTJES GRAM LAW") are Gram matrices of the positive measure $\sum_{\gamma>0} m_\gamma \gamma^{-2}\delta_{\gamma^{-2}}$. Conversely, the Stieltjes moment theorem applied to the two Hankel populations produces a positive measure on $[0,\infty)$. Cauchy bounds for the Taylor series of Z make this measure compactly supported, and

$$Z(w) = \int \frac{1}{1+tw}\,\mathrm{d}\nu(t)$$

near the origin. Analytic continuation then makes Z a Stieltjes function, proving RH by the preceding equivalence. Expanding ("CONNECTED RETURN") gives ("FIRST CONNECTED MEMBERS").

Boundary. The theorem identifies the single proof-bearing field; it does not assume that positivity of each local metric signs its normal derivative. The source-side task is now precise: factor every connected Hankel population in ("STIELTJES GRAM LAW"), or equivalently construct the completed scattering face as an inner contraction, from the Euler–Gamma–theta data without importing the zero locations.

If RH is false, a non-fixed zero orbit z and $J(z) = -\bar{z}$ contributes

$$2\,\mathrm{Re}\Big(A_{f(z)}\overline{A_{f(J(z))}}\Big)$$

to the returned Weil response. Its two-arm Gram has one positive and one negative direction. This is the exact unabsorbed interior residue: not a missing scalar load, but a pole of the completed normal field between the Euler half-plane and the fixed seam.

The preceding induction is a support-local chart of a more global object. Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\Big(\frac{s}{2}\Big)\zeta(s), \quad \Xi(z) = \xi\Big(\frac{1}{2}+z\Big).$$

For $s = \sigma + it$ with $\sigma > 1$, the positive scalar metrics

$$|s(s-1)|^2, \quad \pi^{-\sigma}\Big|\Gamma\Big(\frac{s}{2}\Big)\Big|^2, \quad |1-p^{-s}|^{-2}$$

multiply to $4|\xi(s)|^2$. Their logarithmic normal derivatives add:

$$\partial_\sigma \log|\xi(s)|^2 = 2\,\mathrm{Re}\Big(\frac{1}{s}+\frac{1}{s-1}\Big) + \mathrm{Re}\,\psi\Big(\frac{s}{2}\Big) - \log\pi$$

$$-2\sum_p\sum_{m\ge1}(\log p)p^{-m\sigma}\cos(mt\log p). \quad \text{(COMPLETED NORMAL CURRENT)}$$

The term previously identified as the Euler return derivative is therefore not merely analogous to the archimedean term. In the common convergence region they are exact local factors of the same metric variation. The endpoint factor completes the third face.

Centering by $z = x + it$ gives

$$F(z) = \frac{\Xi'(z)}{\Xi(z)}, \quad \partial_x \log|\Xi(z)|^2 = 2\,\mathrm{Re}\,F(z). \quad \text{(RETURN IMPEDANCE)}$$

This makes the orientation issue precise. The value is a normal covector, not an absolute scalar: under $x = x(\tau)$ it acquires the factor $\dot{x}(\tau)$, and reversing the receiver normal reverses its sign. The RH orientation is the outward direction into $x > 0$.

The exact proof statement is now:

$$\text{RH} \quad \leftrightarrow \quad F \;\; \text{is positive real on } x > 0. \quad \text{(POSITIVE-REAL CLOSURE)}$$

Under RH, Hadamard pairing at the fixed seam gives

$$\mathrm{Re}\,F(x+it) = \sum_{\gamma} m_\gamma \frac{x}{x^2 + (t-\gamma)^2}.$$

This is a field of Poisson kernels. It is positive at every interior receiver and converges at the seam to the complete zero-counting measure. Consequently, for every admissible Mellin amplitude A_f,

$$Q_{W(f)} = \lim_{x \to 0^+} \frac{1}{2\pi} \int_{\mathbb{R}} \left|A_{f(it)}\right|^2 \partial_x \log|\Xi(x+it)|^2 \,\mathrm{d}t. \quad \text{(WEIL BOUNDARY ENERGY)}$$

The moment-conditioned support short is the Dirichlet-to-Neumann short of this same boundary energy. It does not owe a second positivity mechanism. Once ("POSITIVE-REAL CLOSURE") is established, restriction, moment compensation, and old-body relaxation all preserve its sign. Conversely, the nested support shorts recover the complete boundary energy and hence the same closure.

An off-line zero identifies the exact failure. The completed involution in the centered chart is $J(z) = -\bar{z}$. A non-fixed orbit contributes

$$2\,\mathrm{Re}\Big(A_{f(z)}\overline{A_{f(J(z))}}\Big)$$

rather than a square. Its two-arm Gram has eigenvalues of opposite sign. Equivalently, F has an interior pole whose real part changes sign around the crossing. The missing deed is therefore not another archimedean scalar or prime correction. It is the exclusion of non-fixed poles from the normal transport between the Euler half-plane and the seam.

XIII.7 The squared axis turns the proof into a connected Gram law

Evenness of Ξ lets the normal and phase axes fold into

$$H(w) = \frac{\Xi(\sqrt{w})}{\Xi(0)}.$$

The RH is exactly the statement that every zero of H lies on the negative real axis. Its logarithmic impedance

$$Z(w) = \frac{H'(w)}{H(w)}, \quad F(z) = 2zZ(z^2)$$

must therefore be a Stieltjes transform:

$$Z(w) = \sum_{\gamma > 0} \frac{m_\gamma}{w + \gamma^2}. \quad \text{(STIELTJES RETURN)}$$

This is the squared-axis version of the revolving receiver image: the opposed poles $\pm i\gamma$ become one negative radial address $-\gamma^2$, while the original phase crossings remain recoverable through $F(z) = 2zZ(z^2)$.

This formulation has a source-native coefficient face. The classical Riemann theta kernel is positive and gives

$$H(w) = \sum_{n\geq 0} h_n w^n, \quad h_n = \frac{\int_0^\infty \Phi(u)u^{2n}\,\mathrm{d}u}{(2n)!\int_0^\infty \Phi(u)\,\mathrm{d}u} > 0. \quad \text{(THETA MOMENTS)}$$

No zero location occurs in these coefficients. Write

$$\frac{H'(w)}{H(w)} = \sum_{n\geq 0}(-1)^n \mu_n w^n.$$

Then

$$\mu_0 = h_1, \quad \mu_1 = h_1^2 - 2h_2, \quad \mu_2 = h_1^3 - 3h_1h_2 + 3h_3,$$

and all later members follow from the exact connected recursion in the theorem. Stieltjes's moment criterion yields the proof-equivalent family

$$\left[\mu_{i+j}\right]_{0\leq i,j\leq N} \geq 0, \quad \left[\mu_{i+j+1}\right]_{0\leq i,j\leq N} \geq 0 \quad \text{for every } N. \quad \text{(CONNECTED GRAM LAW)}$$

This is not a proposal to check determinants one by one. The construction sought is a single factorization or positive incidence law which produces both Hankel populations for symbolic N. It is the coefficient-level version of constructing the positive-real return field.

There is a direct neighboring result in the literature. Pólya's coefficient/Jensen formulation makes RH equivalent to hyperbolicity of the complete Jensen family; Griffin, Ono, Rolen, and Zagier prove broad asymptotic hyperbolicity and all degrees through eight [37]. A very recent coefficient result proves a uniform tail wedge of the related Pólya-frequency Toeplitz minors [38]. Those are direct-coefficient faces. The present μ_n are the **connected** coefficients of $\frac{H'}{H}$: logarithmic differentiation removes composite products and retains primitive returned incidence. A structural passage from the theta-moment geometry of h_n to the Stieltjes Gram geometry of μ_n is therefore a precise proof construction, not an enumeration.

Nakamura and Suzuki independently locate the same connected object in the zeta screw function g_ζ [39]. Their exact Laplace identity is

$$\int_0^\infty g_{\zeta(u)}e^{izu}\,\mathrm{d}u = \frac{1}{z^2}\frac{\xi'\left(\frac{1}{2}-iz\right)}{\xi\left(\frac{1}{2}-iz\right)} \quad \text{for } \operatorname{Im}(z) > \frac{1}{2}.$$

They prove that RH is equivalent to $\exp(g_\zeta)$ being the characteristic function of an infinitely divisible distribution. Under RH its Lévy–Khintchine measure is

$$\sum_\gamma \frac{m_\gamma}{\gamma^2}\delta_{-\gamma}.$$

Thus the prime tents, archimedean completion, Stieltjes impedance, connected Hankel moments, and conditioned support energy are five transported faces of one return law. The proof-bearing construction is to obtain the positive Lévy/Stieltjes measure—or equivalently the inner scattering contraction— directly from the Euler–Gamma–theta source expression.

XIII.8 The finite scattering face exposes an exact source successor

Theorem 57 — The finite normal holonomy has one exact bipolar Hankel successor Retain

$$\Xi(z) = \xi\left(\frac{1}{2} + z\right), \quad F(z) = \frac{\Xi'(z)}{\Xi(z)},$$

and, for $\omega > 0$, define the two-face scattering ratio

$$\Theta_{\omega(z)} = \frac{\xi\left(\frac{1}{2} - \omega - iz\right)}{\xi\left(\frac{1}{2} + \omega - iz\right)} = \frac{\Xi(\omega + iz)}{\Xi(\omega - iz)}. \quad \text{(FINITE NORMAL HOLONOMY)}$$

The second equality is the completed reflection $\Xi(-q) = \Xi(q)$. If $z = u + iv$, $0 < v < \omega$, and the displayed normal segment avoids a zero, then

$$-\log\left|\Theta_{\omega(u+iv)}\right| = \int_{\omega-v}^{\omega+v} \operatorname{Re} F(x - iu)\,\mathrm{d}x. \quad \text{(NORMAL SWEEP)}$$

Consequently

$$\operatorname{Re} F(\omega - iu) = -\lim_{v \to 0^+} \frac{\log\left|\Theta_{\omega(u+iv)}\right|}{2v}. \quad \text{(INFINITESIMAL RETURN)}$$

For $v \geq \omega$, ("NORMAL SWEEP") remains the corresponding principal-value or interval-split identity. Oddness of $\operatorname{Re} F(x - iu)$ in the normal coordinate cancels the opposed portion of the sweep.

The Riemann Hypothesis is equivalent to

$$\Theta_\omega \quad \text{being inner on } \mathbb{C}_+ \quad \text{for every } \omega > 0. \quad \text{(INNER FAMILY)}$$

Thus F is the infinitesimal normal connection and Θ_ω is its finite receiver holonomy. They are not two candidate proof objects.

This finite holonomy has an explicit arithmetic–archimedean source kernel. Put

$$c_{\omega(n)} = n^\omega \prod_{p|n} \left(1 - p^{-2\omega}\right), \quad c_{\omega(1)} = 1, \quad \text{(INTEGER AMPLITUDE)}$$

and, for $0 < r < 1$,

$$g_{\omega(r)} = \frac{2\pi^\omega}{\Gamma(\omega)} [r^{2-\omega} \left(1 - r^2\right)^{\omega-1}$$
$$-\omega r^{\omega-1} \int_{r^2}^{1} t^{\frac{1}{2}-\omega} (1-t)^{\omega-1}\,\mathrm{d}t],$$

with $g_{\omega(r)} = 0$ for $r > 1$. Define

$$h_{\omega(x)} = \begin{cases} \frac{1}{x} \sum_{1 \leq n \leq x} c_{\omega(n)} g_{\omega\left(\frac{n}{x}\right)} & x > 1 \\ 0 & 0 < x < 1. \end{cases} \quad \text{(ONE-SIDED SOURCE)}$$

Its Mellin transform in the initial convergence half-plane is exactly Θ_ω. On $L^2(0, a)$ let

$$\left(\boldsymbol{H}_{\omega,a} f\right)(x) = \int_0^a h_{\omega(xy)} f(y)\,\mathrm{d}y. \quad \text{(SOURCE HANKEL CUT)}$$

The local weak singularities of h_ω give a compact self-adjoint realization for every finite a and $\omega > 0$.

The following statements are equivalent:

- RH;

- for every $\omega > 0$ and $a > 0$,

$$\left\|\boldsymbol{H}_{\omega,a}\right\| \leq 1; \quad \text{(ALL CUTS CONTRACT)}$$

- for every $\omega > 0$, $a > 0$, and $\varepsilon \in \{\pm 1\}$,

$$I + \varepsilon \boldsymbol{H}_{\omega,a} \geq 0. \quad \text{(BIPOLAR GRAM LAW)}$$

There is a necessary distinction between boundary scattering and causal source conduct. On the real boundary, completion gives

$$\left|\Theta_{\omega(u)}\right| = 1, \quad \Theta_{\omega(u)}\Theta_{\omega(-u)} = 1.$$

Consequently

$$\mathcal{W}_\omega = \mathcal{F}_{\frac{1}{2}}^{-1} M_{\Theta_\omega} \mathcal{R} \mathcal{F}_{\frac{1}{2}} \quad \text{(BOUNDARY INVOLUTION)}$$

is a unitary involution on $L^2(0,\infty)$ for every $\omega > 0$, where $\mathcal{R}\Phi(u) = \Phi(-u)$. This fact is unconditional. Innerness is the stronger assertion that this all-pass boundary response preserves the future Hardy half-space and is realized there by the one-sided arithmetic–archimedean source above.

That causal assertion has an exact scalar residual. Define

$$g_\omega^{\langle 1\rangle}(r) = \int_r^1 \sqrt{\frac{y}{r}}\, g_{\omega(y)} \frac{\mathrm{d}y}{y}, \quad 0 < r < 1,$$

and

$$h_\omega^{\langle 1\rangle}(x) = \frac{1}{x} \sum_{1 \leq n \leq x} c_{\omega(n)} g_\omega^{\langle 1\rangle}\left(\frac{n}{x}\right)$$

for $x > 1$, with zero support below 1. Equivalently,

$$h_\omega^{\langle 1\rangle}(x) = \int_1^x \sqrt{\frac{y}{x}}\, h_{\omega(y)} \frac{\mathrm{d}y}{y}. \quad \text{(INTEGRATED SOURCE)}$$

Put

$$J_{\omega(x)} = \sqrt{x}\, h_\omega^{\langle 1\rangle}(x) = \int_1^x h_{\omega(y)} \frac{\mathrm{d}y}{\sqrt{y}},$$

and

$$R_{\omega(x)} = x^{-\frac{1}{2}} - h_\omega^{\langle 1\rangle}(x) = x^{-\frac{1}{2}}\left(1 - J_{\omega(x)}\right). \quad \text{(CAUSAL RETURN RESIDUAL)}$$

Then the same RH equivalence is

$$\mathrm{RH} \leftrightarrow \int_1^\infty \left|1 - J_{\omega(x)}\right|^2 \frac{\mathrm{d}x}{x} < \infty \quad \text{for every } \omega > 0. \quad \text{(LOG-SCALE CLOSURE)}$$

It is enough to require the displayed law for $0 < \omega < \frac{1}{2}$. In that open range, another equivalent source face is eventual nonnegativity of $h_\omega^{\langle 1\rangle}$ for every ω. This is not positivity of R_ω: the residual may change orientation.

The half-density in (“CAUSAL RETURN RESIDUAL”) is forced by the multiplicative receiver measure:

$$\|R_\omega\|_{L^2(\mathrm{d}x)}^2 = \int_1^\infty \left|1 - J_{\omega(x)}\right|^2 \frac{\mathrm{d}x}{x}. \quad \text{(HALF-DENSITY REBASE)}$$

Thus the critical $\frac{1}{2}$ is the exact Jacobian which carries ordinary source amplitude into scale-invariant logarithmic amplitude; it is not an absolute sign convention.

The whole source is a deterministic log-time convolution. Put

$$\tau = \log x, \quad j_{\omega(\tau)} = J_{\omega(e^\tau)},$$

$$w_{\omega(n)} = \frac{c_{\omega(n)}}{\sqrt{n}}, \quad \varphi_{\omega(u)} = e^{-\frac{u}{2}} g_{\omega(e^{-u})} \mathbf{1}_{u>0},$$

and

$$\Phi_{\omega(u)} = \int_0^u \varphi_{\omega(v)}\, \mathrm{d}v.$$

Then

$$j_{\omega'}(\tau) = \sum_{\log n \le \tau} w_{\omega(n)} \varphi_{\omega(\tau - \log n)}, \quad \text{(LOG-TIME CURRENT)}$$

and

$$j_{\omega(\tau)} = \sum_{\log n \le \tau} w_{\omega(n)} \Phi_{\omega(\tau - \log n)}. \quad \text{(CAUSAL INTEGER SUPERPOSITION)}$$

Equivalently, if

$$\nu_\omega = \sum_{n \ge 1} w_{\omega(n)} \delta_{\log n},$$

then $j_{\omega'} = \nu_\omega * \varphi_\omega$ and $j_\omega = \nu_\omega * \Phi_\omega$ on the additive logarithmic ray. Every integer is admitted at its exact time $\log n$; the half-density cell weight is its amplitude and the same Gamma kernel is its local response. Moreover

$$\mathrm{RH} \leftrightarrow 1 - j_\omega \in L^2(0, \infty; \mathrm{d}\tau) \quad \text{for every } 0 < \omega < \frac{1}{2}. \quad \text{(ADDITIVE CAUSAL CLOSURE)}$$

An off-seam zero makes the causal failure explicit. Suppose

$$\Xi(q_0) = 0, \quad q_0 = \alpha + i\gamma, \quad \alpha > 0.$$

For $0 < \omega < \alpha$, except at isolated numerator-cancellation values, Θ_ω has the upper-half-plane pole

$$z_0 = -\gamma + i(\alpha - \omega).$$

Moving the source inversion from its convergence line toward the real boundary crosses that pole and contributes

$$C_{\omega,q_0} x^{\alpha - \omega - \frac{1}{2} + i\gamma} P_{q_0}(\log x) \quad \text{(OFF-SEAM CAUSAL MODE)}$$

to $h_\omega^{\langle 1 \rangle}$, where $C_{\omega,q_0} \neq 0$ and P_{q_0} has degree one less than the uncancelled pole order. Its squared radial size contains

$$x^{2(\alpha-\omega)-1} \left| P_{q_0}(\log x) \right|^2,$$

which is not integrable on $(1, \infty)$ because $\alpha > \omega$. The obstruction is therefore a concrete supercritical causal tail, not loss of boundary unitarity.

The source is locally finite in the aperture variable. If

$$\kappa_{\omega(r)} = r^{-1} g_{\omega(r^{-1})} \mathbf{1}_{r>1}$$

and $\boldsymbol{G}_\omega$ is the Hankel kernel $\kappa_{\omega(xy)}$, then

$$h_{\omega(r)} = \sum_{n\geq 1} \frac{c_{\omega(n)}}{n} \kappa_{\omega(\frac{r}{n})}. \quad \text{(INTEGER DILATION CELLS)}$$

For the unitary dilation

$$(\mathcal{D}_\lambda f)(x) = \lambda^{\frac{1}{2}} f(\lambda x),$$

this becomes the locally finite operator-kernel identity

$$\boldsymbol{H}_{\omega,a} = P_a \left[\sum_{n<a^2} \frac{c_{\omega(n)}}{\sqrt{n}} \mathcal{D}_{n^{-\frac{1}{2}}} \boldsymbol{G}_\omega \mathcal{D}_{n^{\frac{1}{2}}} \right] P_a. \quad \text{(ARITHMETIC--ARCHIMEDEAN FACTORIZATION)}$$

Hence a new integer cell becomes available exactly when a^2 crosses that integer. Its coefficient is

$$\frac{c_{\omega(n)}}{\sqrt{n}} = n^{\omega-\frac{1}{2}} \prod_{p|n} (1 - p^{-2\omega}). \quad \text{(HALF-DENSITY CELL WEIGHT)}$$

The coefficient population is multiplicative. In particular,

$$c_{\omega(p^k)} = p^{k\omega}(1 - p^{-2\omega}),$$

while $c_{\omega(pn)} = p^\omega c_{\omega(n)}$ when $p \mid n$. Prime axes, their repeated traversals, and composites therefore enter the same integer ecology without being identified with one another.

The first admitted cell gives a genuine, but only local, induction base. If

$$1 < a^2 < 2, \quad L = \log a,$$

only $n = 1$ is present. The inactive interval $\left(0, \frac{1}{a}\right)$ is a zero summand. On the active interval $\left(\frac{1}{a}, a\right)$, the unitary logarithmic lift $f(e^t) \mapsto e^{\frac{t}{2}} f(e^t)$ gives the Hankel kernel

$$\varphi_{\omega(t+s)}, \quad -L < t, s < L. \quad \text{(FIRST SOURCE CELL)}$$

Reflection of one variable and Young's inequality give the exact bound

$$\|\boldsymbol{H}_{\omega,a}\| \leq M_{\omega(L)} := \int_0^{2L} \left| e^{-\frac{u}{2}} g_{\omega(e^{-u})} \right| \mathrm{d}u. \quad \text{(FIRST-CELL BOUND)}$$

Since

$$M_{\omega(L)} \sim \frac{(2\pi)^\omega}{\omega\Gamma(\omega)} (2L)^\omega \quad \text{as } L \to 0^+,$$

every fixed $\omega > 0$ has a nonempty exact aperture interval above $a = 1$ on which $\|\boldsymbol{H}_{\omega,a}\| < 1$. This crosses the rest boundary without assuming RH. It does not establish contraction on the whole $n = 1$ interval, still less across later integer cells.

The exact induction cell is bipolar shorting. For $0 < a < b$, split

$$L^2(0, b) = L^2(0, a) \oplus L^2(a, b)$$

and write

$$\boldsymbol{H}_{\omega,b} = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix}, \quad A = \boldsymbol{H}_{\omega,a}. \quad \text{(NESTED SOURCE CUT)}$$

If $\|A\| < 1$, then ("BIPOLAR GRAM LAW") at b is equivalent to the two simultaneous effective-tension inequalities

$$S_\varepsilon = I + \varepsilon D - C^*(I + \varepsilon A)^{-1}C \geq 0, \quad \varepsilon \in \{\pm 1\}. \quad \text{(BIPOLAR SUCCESSOR)}$$

If an old polarity is only semidefinite, the same statement holds with the Moore–Penrose inverse and the compatibility condition

$$\operatorname{ran} C \subseteq \operatorname{ran} (I + \varepsilon A)^{\frac{1}{2}}. \quad \text{(NULL COMPATIBILITY)}$$

The cross-incidence C therefore consumes capacity in both polarities; D is read with the corresponding orientation. A one-sided positive remainder cannot substitute for this pair.

There is also an exact fixed-domain crossing form. The unitary

$$(\mathcal{U}_a f)(x) = \sqrt{a} f(ax)$$

carries $\boldsymbol{H}_{\omega,a}$ to the operator on $L^2(0,1)$ with kernel

$$\hat{h}_{\omega,a}(x,y) = a h_{\omega(a^2xy)}. \quad \text{(FIXED RECEIVER)}$$

Away from an integer admission seam its derivative kernel is

$$\mathcal{X}_{\omega,a}(x,y) = h_{\omega(r)} + 2r h_{\omega'}(r), \quad r = a^2xy. \quad \text{(CROSSING CURRENT)}$$

On a null mode

$$\left(I + \varepsilon \widehat{\boldsymbol{H}}_{\omega,a}\right)v = 0,$$

the first variation of the corresponding polarity is

$$\varepsilon \int_0^1 \int_0^1 \mathcal{X}_{\omega,a}(x,y) v(y) \overline{v(x)} \, \mathrm{d}y \, \mathrm{d}x. \quad \text{(NULL-MODE CROSSING)}$$

Integer seams use the one-sided distributional or finite-difference form of the same source kernel. This crossing current diagnoses a transverse loss; the full Schur pair ("BIPOLAR SUCCESSOR") retains tangencies and higher-order contact.

Finally, if Ψ denotes the completed source screw function normalized by

$$\frac{F(q)}{q^2} = \int_0^\infty \Psi(t) e^{-qt} \, \mathrm{d}t$$

in its source convergence half-plane, then

$$-\lim_{v \to 0^+} \frac{\log\left|\Theta_{\omega(u+iv)}\right|}{2v} = \operatorname{Re}\left[(\omega - iu)^2 \int_0^\infty \Psi(t) e^{-(\omega - iu)t} \, \mathrm{d}t\right]. \quad \text{(SCREW--SCATTERING JOIN)}$$

The screw Gram, positive-real field, Stieltjes impedance, connected moment Grams, and bipolar Hankel successor are therefore exact transforms of the same completed return.

Demonstration. Evenness of Ξ proves ("FINITE NORMAL HOLONOMY"). Reality of the completed function gives

$$|\Xi(\omega - v + iu)| = |\Xi(\omega - v - iu)|.$$

Integrating $\partial_x \log|\Xi(x - iu)| = \operatorname{Re} F(x - iu)$ proves ("NORMAL SWEEP"), and division by $2v$ followed by $v \to 0^+$ proves ("INFINITESIMAL RETURN"). Evenness of Ξ makes the real normal field odd, giving the interval-split extension when the sweep crosses the fixed seam.

Suzuki's inner-function criterion states that RH is equivalent to ("INNER FAMILY"). Under innerness, the whole source Hankel transform is an isometry on $L^2(0,\infty)$, so every compression $\boldsymbol{H}_{\omega,a}$ is a contraction. Conversely, suppose ("ALL CUTS CONTRACT") holds. For compactly supported f, choose a beyond its support. The functions $P_a \boldsymbol{H}_\omega f$ are compatible as a grows and have norm at most $\|f\|$. Monotone convergence therefore supplies a bounded whole-source transform. After multiplicative inversion this is Suzuki's L^2 convolution criterion for Θ_ω to be inner. Applying this for every $\omega > 0$ proves RH. Self-adjointness gives

$$\|\boldsymbol{H}_{\omega,a}\| \le 1 \leftrightarrow -I \le \boldsymbol{H}_{\omega,a} \le I,$$

which is exactly ("BIPOLAR GRAM LAW").

The completed functional equation and reality on conjugate points give the two boundary identities for Θ_ω. Multiplication by its boundary value and reflection are therefore unitary, and their product squares to the identity, proving ("BOUNDARY INVOLUTION"). Suzuki's integrated-source criterion gives

$$\Theta_\omega \quad \text{inner} \leftrightarrow R_\omega \in L^2(1,\infty;\mathrm{d}x).$$

Multiplying ("INTEGRATED SOURCE") by $\sqrt{x}$ proves the formula for J_ω, and substitution proves ("HALF-DENSITY REBASE"). Combining this criterion with ("INNER FAMILY") proves ("LOG-SCALE CLOSURE"). Suzuki's weighted-summatory theorem gives the stated eventual-sign equivalent for $0 < \omega < \frac{1}{2}$.

Differentiate $j_{\omega(\tau)} = J_{\omega(e^\tau)}$ and substitute the finite source sum:

$$j_{\omega'}(\tau) = e^{\frac{\tau}{2}} h_{\omega(e^\tau)} = \sum_{n \le e^\tau} \frac{c_{\omega(n)}}{\sqrt{n}} e^{-\frac{\tau - \log n}{2}} g_{\omega(e^{-(\tau-\log n)})}.$$

This is ("LOG-TIME CURRENT"). Integration from the exact rest value $j_{\omega(0)} = 0$ proves ("CAUSAL INTEGER SUPERPOSITION"), and $\mathrm{d}\frac{x}{x} = \mathrm{d}\tau$ turns ("LOG-SCALE CLOSURE") into ("ADDITIVE CAUSAL CLOSURE").

If $q_0 = \alpha + i\gamma$ is an off-seam zero, solving $\omega - iz_0 = q_0$ gives the displayed pole. Its numerator is $\Xi(2\omega - q_0)$, which can vanish only at isolated ω unless it vanishes identically. Mellin inversion contains $x^{-\frac{1}{2}-iz}$; the residue at z_0 is therefore a nonzero multiple of

$$x^{-\frac{1}{2}-iz_0} = x^{\alpha-\omega-\frac{1}{2}+i\gamma}.$$

Higher pole order supplies the polynomial in $\log x$. Squaring the radial size proves its failure of $L^2(\mathrm{d}x)$ and hence ("OFF-SEAM CAUSAL MODE").

Substituting the definitions gives

$$\frac{c_{\omega(n)}}{n} \kappa_{\omega(\frac{r}{n})} = \frac{c_{\omega(n)}}{r} g_{\omega(\frac{n}{r})}$$

whenever $n < r$, and zero otherwise. Summation proves ("INTEGER DILATION CELLS"). Conjugating a Hankel kernel by $\mathcal{D}_\lambda$ changes it to

$$\lambda \kappa_{\omega(\lambda^2 xy)}.$$

Taking $\lambda = n^{-\frac{1}{2}}$ proves ("ARITHMETIC–ARCHIMEDEAN FACTORIZATION"). The formula for c_ω proves multiplicativity and the prime-power recurrences.

If $1 < a^2 < 2$, local finiteness leaves only the $n = 1$ summand. The source vanishes whenever either variable is at most $\frac{1}{a}$. Direct logarithmic conjugation on the remaining interval gives ("FIRST SOURCE CELL"). Reflecting one argument turns it into a truncated convolution, so Young's inequality proves ("FIRST-CELL BOUND"). The endpoint asymptotic for g_ω gives the displayed asymptotic for $M_{\omega(L)}$, which tends to zero for every $\omega > 0$.

Apply the ordinary block positivity theorem to

$$I + \varepsilon \boldsymbol{H}_{\omega,b} = \begin{pmatrix} I + \varepsilon A & \varepsilon C \\ \varepsilon C^* & I + \varepsilon D \end{pmatrix}.$$

Its Schur complement is exactly S_ε, because $\varepsilon^2 = 1$. This proves ("BIPOLAR SUCCESSOR"); the standard semidefinite block criterion gives ("NULL COMPATIBILITY").

Direct conjugation by $\mathcal{U}_a$ proves ("FIXED RECEIVER"). Differentiating $a h_{\omega(a^2xy)}$ at a regular source point gives ("CROSSING CURRENT"). Restriction to the null spectral subspace is the Hellmann–Feynman crossing form, proving ("NULL-MODE CROSSING"). The final identity follows by substituting the source Laplace representation of F into ("INFINITESIMAL RETURN").

Boundary. This theorem changes the proof construction in one material way. The former request for a "uniform source-side Gram factorization" now has an explicit nested cell:

$$C^*(I + \varepsilon A)^{-1} C \le I + \varepsilon D \quad \text{for both } \varepsilon = \pm 1.$$

Every entry is determined by the locally finite source $c_{\omega(n)} g_{\omega(\frac{n}{x})}$, and every fixed ω begins from the exact rest cut $\boldsymbol{H}_{\omega,a} = 0$ for $a \le 1$. The first-cell estimate proves a nonempty strict-contraction interval above that cut. A symbolic factorization of the two Schur complements across the remaining aperture and every integer-admission cell would prove all finite contractions and hence RH. No zero enumeration or floating approximation is involved.

The sign itself has not been proved for $0 < \omega < \frac{1}{2}$. Positivity of the canonical Hamiltonian

$$\mathrm{diag}\Big(m_{\omega(a)}^{-2}, m_{\omega(a)}^{2}\Big)$$

in the range where its Fredholm construction is already justified is an output of this contraction structure, not a source proof of its continuation. Likewise, nonnegativity of the first crossing form alone does not exclude a higher-order exit. The unresolved statement is exactly the bipolar source inequality above. Equivalently, it is the scale-tail statement

$$1 - J_\omega \in L^2\bigg((1,\infty), \mathrm{d}\frac{x}{x}\bigg)$$

for every $0 < \omega < \frac{1}{2}$. These are two exact charts of the same RH obstruction: the former resolves every finite shell and both orientations, while the latter exposes the long causal residue and the nonintegrable mode an off-seam zero would create. The local $n = 1$ cell alone cannot decide that tail.

The positive-real field has a finite receiver holonomy which makes the source induction explicit. Following Suzuki [40], put

$$\Theta_{\omega(z)} = \frac{\xi\big(\frac{1}{2} - \omega - iz\big)}{\xi\big(\frac{1}{2} + \omega - iz\big)} = \frac{\Xi(\omega + iz)}{\Xi(\omega - iz)}.$$

For $z = u + iv$ and $0 < v < \omega$,

$$-\log\Big|\Theta_{\omega(u+iv)}\Big| = \int_{\omega - v}^{\omega + v} \mathrm{Re}\, F(x - iu)\,\mathrm{d}x. \quad \text{(FINITE NORMAL SWEEP)}$$

Thus ω is the receiver's normal offset from the fixed seam, v is the thickness of the comparison, and

$$\mathrm{Re}\, F(\omega - iu) = -\lim_{v \to 0^+} \frac{\log\Big|\Theta_{\omega(u+iv)}\Big|}{2v}.$$

The ratio is not a second speculative object. It is the finite transport whose infinitesimal connection is the completed field already derived above. Suzuki proves that RH is equivalent to Θ_ω being inner for every $\omega > 0$.

More importantly for construction, the inverse Mellin face of Θ_ω is not a black box. It is a one-sided kernel

$$h_{\omega(x)} = \frac{1}{x} \sum_{1 \le n \le x} c_{\omega(n)} g_{\omega(\frac{n}{x})}, \quad c_{\omega(n)} = n^\omega \prod_{p \mid n} (1 - p^{-2\omega}),$$

where g_ω is the explicit Gamma/endpoint kernel recorded in the theorem. Its finite Hankel cut is

$$\big(\boldsymbol{H}_{\omega,a} f\big)(x) = \int_0^a h_{\omega(xy)} f(y)\,\mathrm{d}y.$$

The exact proof equivalence is

$$\begin{gathered} \mathrm{RH} \quad \leftrightarrow \quad I + \varepsilon \boldsymbol{H}_{\omega,a} \ge 0 \quad \text{for every} \quad \omega > 0, \\ a > 0, \\ \varepsilon = \pm 1. \quad \text{(BIPOLAR SOURCE LAW)} \end{gathered}$$

This does more than rename the connected Gram requirement. For fixed ω, the source begins at exact rest:

$$\boldsymbol{H}_{\omega,a} = 0 \quad \text{when } a \le 1.$$

It is locally finite in a, and the integer n enters precisely when a^2 crosses n, with half-density weight

$$\frac{c_{\omega(n)}}{\sqrt{n}} = n^{\omega - \frac{1}{2}} \prod_{p \mid n} (1 - p^{-2\omega}). \quad \text{(INTEGER CELL)}$$

This is the requested causal counting topology. A prime axis, a repeated prime traversal, and a composite are distinct cells, while multiplicativity records their exact common construction.

If the cut at b is split into its old interval $(0,a)$ and new shell (a,b),

$$\boldsymbol{H}_{\omega,b} = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix}, \quad A = \boldsymbol{H}_{\omega,a},$$

then the whole successor is contractive exactly when the two residual tensions

$$I + \varepsilon D - C^*(I+\varepsilon A)^{-1} C \geq 0, \quad \varepsilon = \pm 1, \quad \text{(BIPOLAR SUCCESSOR CELL)}$$

both hold. The same cross C loads both orientations. This is why a single positive "remainder" was never enough: one must keep both faces of the scattering circuit passive at once.

The fixed-domain derivative further identifies the critical and inflection data. After the exact rebase $x \mapsto ax$, the kernel is $ah_{\omega(a^2xy)}$, and its regular crossing current is

$$h_{\omega(r)} + 2rh_{\omega'}(r), \quad r = a^2xy.$$

At an eigenvalue ± 1, its restriction to the null eigenspace is the spectral crossing form. It reveals a transverse exit, while the Schur pair above also retains higher-order tangency. Integer thresholds contribute their one-sided distributional crossings.

The remaining statement is now concrete: derive ("BIPOLAR SUCCESSOR CELL") from the displayed source kernel, uniformly in the symbolic integer cell and for $0 < \omega < \frac{1}{2}$. This would induct from $a \leq 1$ through every finite cut and prove RH. It is still the complete RH sign, but unlike the earlier phrase "continuum operator estimate," its old body, cross-incidence, shell, two orientations, and arithmetic admission law are all explicit.

XIII.8.1 The source event enters through the receiver boundary

Theorem 58 — The completed source enters a shifted receiver through one bipolar boundary port Fix $\omega > 0$ and retain the source cut $\widehat{\boldsymbol{H}}_{\omega,a}$ on $L^2(0,1)$ with kernel $ah_{\omega(a^2xy)}$. Put

$$\tau = 2\log a$$

and let

$$(\mathcal{V}f)(s) = e^{-\frac{s}{2}} f(e^{-s}), \quad \mathcal{V} : L^2(0,1) \to L^2(0,\infty).$$

Then $\mathcal{V}$ is unitary, and

$$\mathcal{H}_{\omega,\tau} = \mathcal{V}\widehat{\boldsymbol{H}}_{\omega,a}\mathcal{V}^{-1}$$

is the causal triangular Hankel operator

$$\big(\mathcal{H}_{\omega,\tau}F\big)(s) = \int_0^\infty \psi_{\omega(\tau-s-t)} F(t)\,\mathrm{d}t, \quad \text{(LOGARITHMIC RECEIVER)}$$

where

$$\psi_{\omega(u)} = e^{\frac{u}{2}} h_{\omega(e^u)} = j_{\omega'}(u) = \sum_{\log n \leq u} w_{\omega(n)} \varphi_{\omega(u-\log n)} \quad \text{(ACTION CURRENT)}$$

on the positive logarithmic ray and is zero for $u < 0$. Consequently

$$\mathcal{H}_{\omega,\tau} = \sum_{\log n < \tau} w_{\omega(n)} \mathcal{K}_{\omega,\tau-\log n}, \quad \text{(INTEGER EVENT POPULATION)}$$

in its locally finite form sense, where $\mathcal{K}_{\omega,\rho}$ has kernel $\varphi_{\omega(\rho-s-t)}$ and support

$$s+t<\rho. \quad \text{(EVENT TRIANGLE)}$$

An integer is therefore not an isolated value attached to an old operator. At $\tau=\log n$ its response triangle is born at the receiver boundary $(s,t)=(0,0)$ and subsequently grows through the same source current. For $\tau\leq 0$,

$$\mathcal{H}_{\omega,\tau}=0, \quad \mathcal{M}^{\varepsilon}_{\omega,\tau}:=I+\varepsilon\mathcal{H}_{\omega,\tau}=I, \quad \varepsilon\in\{\pm 1\}. \quad \text{(INHERITED REST)}$$

Thus the zero source face is not an empty mathematical object: the receiver space and both identity standing forms are already present.

The old body is carried exactly through every later event. For $\tau_1<\tau_2$, put

$$\ell=\frac{\tau_2-\tau_1}{2}$$

and let $\mathcal{S}_\ell$ be the right-shift isometry

$$(\mathcal{S}_\ell F)(s)=\begin{cases}0 & 0\leq s<\ell\\ F(s-\ell) & s\geq \ell.\end{cases}$$

Directly from (“LOGARITHMIC RECEIVER”),

$$\mathcal{S}_\ell^*\mathcal{H}_{\omega,\tau_2}\mathcal{S}_\ell=\mathcal{H}_{\omega,\tau_1},$$

$$\mathcal{S}_\ell^*\mathcal{M}^{\varepsilon}_{\omega,\tau_2}\mathcal{S}_\ell=\mathcal{M}^{\varepsilon}_{\omega,\tau_1}. \quad \text{(INHERITED SHIFT)}$$

Let

$$\mathcal{B}_\ell=L^2(0,\ell)$$

be the exposed boundary strip and use the unitary

$$\mathcal{W}_{\ell(F\oplus b)}=\mathcal{S}_\ell F+b$$

from $L^2(0,\infty)\oplus\mathcal{B}_\ell$ onto $L^2(0,\infty)$. There are source-determined cross and boundary operators C_e, D_e such that

$$\mathcal{W}_\ell^*\mathcal{H}_{\omega,\tau_2}\mathcal{W}_\ell=\begin{pmatrix}\mathcal{H}_{\omega,\tau_1} & C_e\\ C_e^* & D_e\end{pmatrix}. \quad \text{(FINITE EVENT)}$$

Whenever the inherited polarity is invertible, the exact event factorization is

$$\mathcal{W}_\ell^*\mathcal{M}^{\varepsilon}_{\omega,\tau_2}\mathcal{W}_\ell=L^*_{\varepsilon,e}\begin{pmatrix}\mathcal{M}^{\varepsilon}_{\omega,\tau_1} & 0\\ 0 & S_{\varepsilon,e}\end{pmatrix}L_{\varepsilon,e}, \quad \text{(EVENT CONGRUENCE)}$$

with

$$L_{\varepsilon,e}=\begin{pmatrix}I & \varepsilon\left(\mathcal{M}^{\varepsilon}_{\omega,\tau_1}\right)^{-1}C_e\\ 0 & I\end{pmatrix},$$

$$S_{\varepsilon,e}=I_{\mathcal{B}_\ell}+\varepsilon D_e-C_e^*\left(\mathcal{M}^{\varepsilon}_{\omega,\tau_1}\right)^{-1}C_e. \quad \text{(BOUNDARY RETURN)}$$

This is the bipolar successor as one situated event: the old standing body is shifted intact, while the new strip, its old–new cross, and its returned short are co-present.

The corresponding infinitesimal law is a source-derived boundary-port identity and uses no inverse or assumed factorization. At a regular τ, for F in a common smooth form core,

$$\partial_\tau\langle \mathcal{H}_{\omega,\tau}F, F\rangle = \mathrm{Re}\langle F', \mathcal{H}_{\omega,\tau}F\rangle + \mathrm{Re}\Big[\overline{F(0)}\big(\mathcal{H}_{\omega,\tau}F\big)(0)\Big]. \quad \text{(SOURCE GREEN IDENTITY)}$$

Put

$$u = F(0), \quad y_\varepsilon = \big(\mathcal{M}^\varepsilon_{\omega,\tau}F\big)(0), \quad v_\varepsilon = y_\varepsilon - u = \varepsilon\big(\mathcal{H}_{\omega,\tau}F\big)(0).$$

After subtracting the inherited right-shift transport $\partial_\tau F = -\frac{F'}{2}$, the standing energy

$$E_{\varepsilon(\tau,F)} = \langle \mathcal{M}^\varepsilon_{\omega,\tau}F, F\rangle$$

obeys

$$\nabla_\tau E_\varepsilon = \frac{1}{2}\big(|y_\varepsilon|^2 - |v_\varepsilon|^2\big). \quad \text{(BIPOLAR PORT BALANCE)}$$

The two orientations share the same boundary occurrence. If $z = \big(\mathcal{H}_{\omega,\tau}F\big)(0)$, then

$$y_+ = u + z, \quad y_- = u - z, \quad \nabla_\tau E_+ + \nabla_\tau E_- = |u|^2. \quad \text{(ONE CURRENT TWO FACES)}$$

In particular, an inherited shifted state has $u = 0$ and equal incoming and outgoing port sizes, so its standing energy is unchanged. If a nonzero mode satisfies $\mathcal{M}^\varepsilon_{\omega,\tau}F = 0$, then its receiver-visible crossing is

$$\nabla_\tau E_\varepsilon = -\frac{1}{2}|F(0)|^2. \quad \text{(NULL PORT)}$$

A first singularity is therefore either a transverse outward event visible at the boundary or a port-dark null mode with $F(0) = 0$.

The published Fredholm chart supplies the same event as an explicit bipolar differential connection. On an aperture interval where both $I + \varepsilon \boldsymbol{H}_{\omega,a}$ are invertible, let $\phi^\varepsilon_{\omega,a}$ solve

$$\phi^{\varepsilon(x)}_{\omega,a} + \varepsilon\int_0^a h_{\omega(xy)}\phi^{\varepsilon(y)}_{\omega,a}\,\mathrm{d}y = h_{\omega(ax)}, \quad \varepsilon\in\{\pm 1\}. \quad \text{(BOUNDARY RESPONSE)}$$

Put

$$\delta_x = x\partial_x + \frac{1}{2}, \quad \mu_{\omega(a)} = a\phi^+_{\omega,a}(a) + a\phi^-_{\omega,a}(a). \quad \text{(SHARED BOUNDARY FIELD)}$$

In the continuous-kernel range $\omega > 1$, Suzuki's source calculation gives

$$\Big(a\partial_a + \frac{1}{2} + \varepsilon\mu_{\omega(a)}\Big)\phi^{\varepsilon(x)}_{\omega,a} = \delta_x\phi^{-\varepsilon}_{\omega,a}(x). \quad \text{(BIPOLAR RESOLVENT CONNECTION)}$$

Thus the two polarities are not independently propagated estimates. Dilation of either response turns into the oppositely oriented response, while their boundary values form one common connection coefficient. With

$$m_{\omega(a)} = \exp\left(\int_1^a \mu_{\omega(b)}\frac{\mathrm{d}b}{b}\right), \quad m_{\omega(1)} = 1, \quad \text{(ACCUMULATED CONNECTION)}$$

the gauged return satisfies the canonical system with Hamiltonian

$$\operatorname{diag}\Big(m_{\omega(a)}^{-2}, m_{\omega(a)}^{2}\Big). \quad \text{(CANONICAL INTERIOR)}$$

For $\omega > 1$ this agrees with

$$m_{\omega(a)} = \frac{\det\big(I + \boldsymbol{H}_{\omega,a}\big)}{\det\big(I - \boldsymbol{H}_{\omega,a}\big)}. \quad \text{(DETERMINANT RETURN)}$$

The identity is source-derived on its invertibility interval; it does not assume either standing form positive.

Demonstration. Substitute $x = e^{-s}$ and $y = e^{-t}$ in the fixed-receiver kernel. The two half-density factors give

$$ae^{-\frac{s+t}{2}} h_{\omega(a^2 e^{-(s+t)})} = e^{\frac{\tau-s-t}{2}} h_{\omega(e^{\tau-s-t})} = \psi_{\omega(\tau-s-t)}.$$

The log-time current formula in `theorem:completed-scattering-hankel-successor` proves ("ACTION CURRENT") and ("INTEGER EVENT POPULATION"). Its one-sided support gives ("EVENT TRIANGLE") and ("INHERITED REST").
For $\ell = \frac{\tau_2-\tau_1}{2}$,

$$\psi_{\omega(\tau_2-(s+\ell)-(t+\ell))} = \psi_{\omega(\tau_1-s-t)}.$$

This proves ("INHERITED SHIFT"). Splitting the target into the shifted range and its orthogonal boundary strip gives ("FINITE EVENT"). Ordinary block elimination gives ("EVENT CONGRUENCE") and ("BOUNDARY RETURN"); no sign is used in that algebra.
Away from the locally finite admission seams,

$$\partial_\tau \psi_{\omega(\tau-s-t)} = -\frac{1}{2}(\partial_s + \partial_t)\psi_{\omega(\tau-s-t)}.$$

Integrate the two derivatives by parts on the positive quadrant. The two exposed faces at $s = 0$ and $t = 0$ give the boundary term in ("SOURCE GREEN IDENTITY"), while the interior derivatives give $\operatorname{Re}\langle F', \mathcal{H}F\rangle$. The weak source seams use the corresponding one-sided form identity.
Since

$$\operatorname{Re}\langle F', F\rangle = -\frac{1}{2}|F(0)|^2,$$

replacing $\varepsilon\mathcal{H}$ by $\mathcal{M}^\varepsilon - I$ gives

$$\partial_\tau E_\varepsilon = \operatorname{Re}\langle F', \mathcal{M}^\varepsilon F\rangle + \operatorname{Re}(\overline{u} y_\varepsilon) - \frac{1}{2}|u|^2.$$

The state derivative $\partial_\tau F = -\frac{F'}{2}$ cancels the first term. Finally,

$$\operatorname{Re}(\overline{u} y_\varepsilon) - \frac{1}{2}|u|^2 = \frac{1}{2}\Big(|y_\varepsilon|^2 - |y_\varepsilon - u|^2\Big),$$

proving ("BIPOLAR PORT BALANCE"). The parallelogram identity for $u + z$ and $u - z$ proves ("ONE CURRENT TWO FACES"), and setting $y_\varepsilon = 0$ proves ("NULL PORT").
Finally, differentiate ("BOUNDARY RESPONSE") with respect to a and apply δ_x. The product-kernel identity

$$x\partial_x h_{\omega(xy)} = y\partial_y h_{\omega(xy)}$$

moves the Euler derivative to the response variable. Integration by parts exposes the two endpoint values in ("SHARED BOUNDARY FIELD"). Subtracting the equations with opposed signs and using uniqueness of the Fredholm solutions gives ("BIPOLAR RESOLVENT CONNECTION"). Gauging its common scalar connection by ("ACCUMULATED CONNECTION") gives ("CANONICAL INTERIOR"). Differentiation of the two Fredholm determinants gives their boundary resolvent values, hence ("DETERMINANT RETURN").

Boundary. This theorem removes the initialization circle. The receiver does not arise from the first integer cell: it begins as the nonempty identity standing pair, carries every old direction by the exact shift ("INHERITED SHIFT"), and admits new source material only through the boundary strip. The integer population, continuous Gamma response, old–new cross, and receiver motion are different faces of that one event.

The theorem does not prove the remaining causal inequality. RH is now equivalently the assertion that both port relations are passive on every finite logarithmic interval:

$$\int |v_\varepsilon|^2 \, d\tau \le \int |y_\varepsilon|^2 \, d\tau + 2E_{\varepsilon(\text{initial})} \quad \text{for } \varepsilon = \pm 1, \quad \text{(BIPOLAR CAUSAL PASSIVITY)}$$

or, event by event, that every explicit $S_{\varepsilon,e}$ in ("BOUNDARY RETURN") is nonnegative. Boundary unitarity gives an all-time equality but does not imply this causal inequality. A proof must derive ("BIPOLAR CAUSAL PASSIVITY") from the convolution current $\psi_\omega = \nu_\omega * \varphi_\omega$ without first inverting or taking a square root of the desired positive form. The remaining analytic alternatives are correspondingly precise: rule out every boundary-visible unit mode and every port-dark null mode, or construct the two causal port contractions directly.

The canonical system does not close this gap by notation. Its published unconditional construction above is in the regular range $\omega > 1$, where innerness is already known. Extending the boundary response, determinant return, and coupled connection through every finite aperture for $0 < \omega < \frac{1}{2}$ without assuming innerness is exactly the target-range continuation problem. If a determinant vanishes, the corresponding response and μ_ω become singular; declaring the positive Hamiltonian past that point would simply assume the desired result.

The logarithmic rebase corrects the apparent initialization problem. With $\tau = 2\log a$, the fixed source cut has kernel

$$\psi_{\omega(\tau-s-t)}, \quad \psi_\omega = j_{\omega'} = \nu_\omega * \varphi_\omega.$$

The receiver at $\tau \le 0$ is not absent: its two standing forms are both the identity. When τ grows, the previous body is carried by an exact right-shift isometry. Every new integer response is born as a triangle at the exposed boundary and grows from there. Thus neither the receiver nor a prime cell is created from a blank state.

The same identity supplies a lossless boundary-port balance for each polarity. Interior right-shift is pure transport; only the incoming and outgoing boundary amplitudes can alter the standing energy. This turns the remaining RH obligation into simultaneous causal passivity of the two port relations. It does not assume the old positive body in order to manufacture a factor: the inherited body, the action current, and the boundary through which it enters are all explicit before the sign is asked.

Suzuki's Fredholm response makes the connection literal in its established regular range. The two solutions of $(I + \varepsilon \boldsymbol{H}_{\omega,a})\phi_a^\varepsilon = h_{\omega(a\cdot)}$ obey

$$\left(a\partial_a + \frac{1}{2} + \varepsilon\mu_{\omega(a)}\right)\phi_a^\varepsilon = \delta_x \phi_a^{-\varepsilon}.$$

The response therefore swings between the two polarities through one shared boundary field; accumulating that field gives the determinant ratio and the diagonal canonical Hamiltonian. For the RH range $0 < \omega < \frac{1}{2}$, the honest remaining task is to continue this same source connection without allowing either Fredholm determinant to vanish—not to posit a new receiver after a failure.

XIII.8.2 The two polarities close through one causal frame

Theorem 59 — The bipolar source defect is the Gram deficit of one causal action-current frame Fix $\omega > 0$ and a logarithmic cut $\tau > 0$. On

$$\mathcal{E}_\tau = L^2(0,\tau)$$

retain the completed source current

$$\psi_{\omega(u)} = j_{\omega'}(u) = \sum_{\log n \le u} w_{\omega(n)}\varphi_{\omega(u-\log n)}, \quad \psi_{\omega(u)} = 0 \ \text{ for } u < 0.$$

Define causal transport and receiver reflection by

$$(\mathcal{V}_{\omega,\tau}F)(r) = \int_0^r \psi_{\omega(r-s)}F(s)\,\mathrm{d}s, \quad \text{(CAUSAL TRANSPORT)}$$

and

$$(\mathcal{R}_\tau F)(s) = F(\tau - s). \quad \text{(RECEIVER REFLECTION)}$$

Then $\mathcal{R}_\tau$ is a self-adjoint unitary and the logarithmic Hankel cut from `theorem:completed-source-event-port-transport` factors as

$$\mathcal{H}_{\omega,\tau} = \mathcal{R}_\tau \mathcal{V}_{\omega,\tau}. \quad \text{(REFLECTED CURRENT)}$$

Self-adjointness of $\mathcal{H}_{\omega,\tau}$ consequently gives

$$\mathcal{R}_\tau \mathcal{V}_{\omega,\tau} \mathcal{R}_\tau = \mathcal{V}_{\omega,\tau}^*,$$

$$\mathcal{H}_{\omega,\tau}^2 = \mathcal{V}_{\omega,\tau}^* \mathcal{V}_{\omega,\tau}. \quad \text{(ROUND TRIP)}$$

Hence the two standing orientations have one exact shared defect:

$$\mathcal{D}_{\omega,\tau} := \big(I - \mathcal{H}_{\omega,\tau}\big)\big(I + \mathcal{H}_{\omega,\tau}\big) = I - \mathcal{V}_{\omega,\tau}^* \mathcal{V}_{\omega,\tau}. \quad \text{(BIPOLAR DEFECT)}$$

Since $\mathcal{H}_{\omega,\tau}$ is compact and self-adjoint, the following are equivalent:

$$I + \varepsilon \mathcal{H}_{\omega,\tau} \geq 0 \quad \text{for both } \varepsilon = \pm 1; \quad \text{(TWO ORIENTATIONS)}$$

$$\mathcal{D}_{\omega,\tau} \geq 0; \quad \text{(ROUND-TRIP PASSIVITY)}$$

$$\big\|\mathcal{V}_{\omega,\tau}\big\| \leq 1. \quad \text{(CAUSAL CONTRACTION)}$$

This defect has a source-native continuous Gram factorization. For $0 < r < \tau$, let

$$k_{\omega,r}(s) = \overline{\psi_{\omega(r-s)}} \mathbf{1}_{0<s<r}. \quad \text{(ACTION-CURRENT FACE)}$$

Then

$$\big(\mathcal{V}_{\omega,\tau} F\big)(r) = \langle F, k_{\omega,r} \rangle,$$

and define the rank-one current operator by

$$\mathcal{P}_{\omega,r} F = \langle F, k_{\omega,r} \rangle k_{\omega,r}.$$

Then

$$\mathcal{V}_{\omega,\tau}^* \mathcal{V}_{\omega,\tau} = \int_0^\tau \mathcal{P}_{\omega,r} \, \mathrm{d}r \quad \text{(CURRENT GRAM)}$$

as a weak operator integral. Therefore the complete bipolar law at the cut is exactly the continuous Bessel inequality

$$\int_0^\tau \big|\big\langle F, k_{\omega,r} \big\rangle\big|^2 \, \mathrm{d}r \leq \|F\|^2 \quad \text{for every } F \in \mathcal{E}_\tau. \quad \text{(ACTION-CURRENT CAPACITY)}$$

No detached edge, independently selected mode, or later positivity carrier has entered: each $k_{\omega,r}$ is the whole arithmetic–Gamma current visible at causal time r.

The causal factorization also gives a noncircular strict departure from rest. Young's inequality yields

$$\big\|\mathcal{V}_{\omega,\tau}\big\| \leq \int_0^\tau \big|\psi_{\omega(u)}\big| \, \mathrm{d}u. \quad \text{(LOCAL SOURCE BOUND)}$$

The locally integrable endpoint behavior of the completed source makes the right side tend to zero as $\tau \to 0^+$. Hence every fixed $\omega > 0$ has a nonempty interval $0 < \tau < \tau_\omega$ on which the bipolar defect is strictly positive. This uses the actual source current and no hypothesis about its whole-ray continuation.

The former split between boundary-visible and port-dark null modes is not a split in the remaining proof object. If

$$\big(I + \varepsilon \mathcal{H}_{\omega,\tau}\big) F = 0 \quad (F \neq 0),$$

then

$$\mathcal{V}_{\omega,\tau}F = -\varepsilon\mathcal{R}_\tau F,$$

$$\mathcal{V}_{\omega,\tau}^*\mathcal{V}_{\omega,\tau}F = F, \quad \left\|\mathcal{V}_{\omega,\tau}F\right\| = \|F\|. \quad \text{(UNIT-GAIN SATURATION)}$$

Conversely, every nonzero vector satisfying

$$\mathcal{V}_{\omega,\tau}^*\mathcal{V}_{\omega,\tau}F = F$$

lies in the orthogonal sum of the $+1$ and -1 eigenspaces of $\mathcal{H}_{\omega,\tau}$. When ("ROUND-TRIP PASSIVITY") holds, this is exactly equality in ("ACTION-CURRENT CAPACITY"). Its value at the exposed port determines whether the crossing is visible in the infinitesimal chart; it does not define another kind of obstruction.

Finally let $\mathcal{V}_{\omega,\infty}$ denote causal convolution by ψ_ω on the whole positive logarithmic ray. All finite inequalities ("ACTION-CURRENT CAPACITY") hold if and only if this whole causal transport is a contraction. In the initial convergence half-plane its Laplace transfer is

$$\hat{\psi}_{\omega(p)} = \int_0^\infty \psi_{\omega(u)}e^{-pu}\,\mathrm{d}u = \Theta_{\omega(ip)} = \frac{\Xi(\omega-p)}{\Xi(\omega+p)}. \quad \text{(CAUSAL TRANSFER)}$$

Thus whole-ray contraction is equivalent to this transfer extending as a Schur function on $\operatorname{Re} p > 0$. Under $z = ip$, this is exactly innerness of Θ_ω on the upper half-plane. Requiring it for every $\omega > 0$ is therefore equivalent to RH. More locally, if

$$p = x + iy, \quad 0 < x < \omega,$$

and the displayed normal segment avoids a zero, the finite normal-sweep identity gives

$$-\log\left|\hat{\psi}_{\omega(x+iy)}\right| = \int_{\omega-x}^{\omega+x} \operatorname{Re}\left[\frac{\Xi'(r+iy)}{\Xi(r+iy)}\right]\mathrm{d}r. \quad \text{(NORMAL CAUSAL FLUX)}$$

At regular points, pointwise interior contractivity is therefore the nonnegative orientation of this normal flux. Global Schur conduct additionally requires the pole-free analytic continuation; neither property follows from the unit-modulus boundary face.

Demonstration. The support of ψ_ω reduces the logarithmic Hankel operator to $\mathcal{E}_\tau$. Directly,

$$\left(\mathcal{R}_\tau\mathcal{V}_{\omega,\tau}F\right)(s) = \int_0^{\tau-s} \psi_{\omega(\tau-s-t)}F(t)\,\mathrm{d}t = \left(\mathcal{H}_{\omega,\tau}F\right)(s),$$

which proves ("REFLECTED CURRENT"). Since the source kernel is real, $\mathcal{H}_{\omega,\tau}$ is self-adjoint. Taking the adjoint of $\mathcal{H} = \mathcal{R}\mathcal{V}$ and using $\mathcal{R}^2 = I$ proves the reflected adjoint identity and ("ROUND TRIP"). The two factors $I - \mathcal{H}$ and $I + \mathcal{H}$ commute, so their product is ("BIPOLAR DEFECT"). The spectral theorem for the compact self-adjoint $\mathcal{H}$ proves the equivalence of ("TWO ORIENTATIONS"), ("ROUND-TRIP PASSIVITY"), and ("CAUSAL CONTRACTION").

The definition of $k_{\omega,r}$ gives

$$\mathcal{V}F(r) = \langle F, k_{\omega,r}\rangle.$$

Integrating the squared amplitudes proves ("CURRENT GRAM") weakly and turns nonnegativity of $I - \mathcal{V}^*\mathcal{V}$ into ("ACTION-CURRENT CAPACITY"). Young's convolution inequality proves ("LOCAL SOURCE BOUND"), and local integrability of ψ_ω proves its strict small-cut consequence.

If $(I + \varepsilon\mathcal{H})F = 0$, then $\mathcal{H}F = -\varepsilon F$. Multiplication by $\mathcal{R}$ and $\mathcal{H} = \mathcal{R}\mathcal{V}$ gives $\mathcal{V}F = -\varepsilon\mathcal{R}F$, hence ("UNIT-GAIN SATURATION"). Conversely, equality in the positive operator $I - \mathcal{H}^2$ places a vector in $\ker(I - \mathcal{H}^2) = \ker(I - \mathcal{H}) \oplus \ker(I + \mathcal{H})$.

Zero-extension and monotone convergence show that contraction of the whole causal convolution implies every finite compression and that uniform contraction of all finite compressions defines the whole contraction. The completed source Mellin identity is

$$\Theta_{\omega(z)} = \int_0^\infty \psi_{\omega(u)} e^{izu} \, \mathrm{d}u$$

in its convergence half-plane. Substitution $z = ip$ proves ("CAUSAL TRANSFER"). The Laplace–Hardy correspondence identifies contractive causal convolution with a Schur transfer on the right half-plane. The rotation $z = ip$ carries that half-plane to the upper half-plane, where Suzuki's inner-family criterion is equivalent to RH. Finally, substitute $z = -y + ix$ into the completed normal-sweep identity from `theorem:completed-scattering-hankel-successor`; because $i(x + iy) = -y + ix$, this gives ("NORMAL CAUSAL FLUX").

Boundary. This theorem removes a false continuation. Boundary-visible unit modes and port-dark null modes must not be pursued as separate populations. They are the same possible equality case of one source-derived continuous frame. The exact proof-bearing statement is now

$$I - \int_0^\tau \mathcal{P}_{\omega,r} \, \mathrm{d}r \geq 0 \quad \text{for every } \tau > 0,$$

$$\omega > 0. \quad \text{(SOURCE FRAME DEFICIT)}$$

Its initial body is the identity, not an empty receiver, and its entire load is the causal current $\psi_\omega = \nu_\omega * \varphi_\omega$.

The factorization does not prove ("SOURCE FRAME DEFICIT"). Through ("CAUSAL TRANSFER"), that inequality at every cut is exactly the RH Schur/innerness condition already sought; invoking generic contractive-system realization would merely return the desired sign as an assumption. A genuinely new proof must derive the continuous Bessel bound from the arithmetic admissions $w_{\omega(n)}$ and the common archimedean response φ_ω, or equivalently prove positivity of the source kernel

$$\frac{1 - \hat{\psi}_{\omega(p)} \overline{\hat{\psi}_{\omega(q)}}}{p + \overline{q}} \quad (\operatorname{Re} p, \operatorname{Re} q > 0) \quad \text{(SOURCE SCHUR KERNEL)}$$

without assuming its analytic contractive continuation. This is the present analytic obligation. The local source bound proves that this obligation starts from a genuine strict body; ("NORMAL CAUSAL FLUX") shows exactly what can fail as the receiver grows. Another Schur block, crossing census, or null-mode taxonomy would not advance it.

The apparent choice between excluding a boundary-visible unit mode and excluding a port-dark null mode was false. On the effective interval $L^2(0, \tau)$, let $\mathcal{R}_\tau$ reflect s across the cut and let $\mathcal{V}_{\omega,\tau}$ be causal convolution by the completed source current:

$$\big(\mathcal{V}_{\omega,\tau} F\big)(r) = \int_0^r \psi_{\omega(r-s)} F(s) \, \mathrm{d}s.$$

Then the source Hankel face is exactly

$$\mathcal{H}_{\omega,\tau} = \mathcal{R}_\tau \mathcal{V}_{\omega,\tau}, \quad \mathcal{H}_{\omega,\tau}^2 = \mathcal{V}_{\omega,\tau}^* \mathcal{V}_{\omega,\tau}.$$

The bipolar successor law therefore reduces without discarding either orientation:

$$\big(I - \mathcal{H}_{\omega,\tau}\big)\big(I + \mathcal{H}_{\omega,\tau}\big) = I - \mathcal{V}_{\omega,\tau}^* \mathcal{V}_{\omega,\tau}. \quad \text{(SHARED CAUSAL DEFECT)}$$

At causal time r, the translated source response

$$k_{\omega,r}(s) = \overline{\psi_{\omega(r-s)}} \mathbf{1}_{0<s<r}$$

is one complete arithmetic–Gamma action-current face. Consequently

$$\mathcal{P}_{\omega,r} F = \langle F, k_{\omega,r} \rangle k_{\omega,r}, \quad \mathcal{V}^* \mathcal{V} = \int_0^\tau \mathcal{P}_{\omega,r} \, \mathrm{d}r.$$

The initial identity standing is consumed only by this actual current population. A null mode of either polarity is precisely a vector on which the continuous frame has unit gain. Whether its boundary value vanishes changes the local crossing testimony, not the obstruction itself.

The remaining sign is therefore one exact source inequality,

$$\int_0^\tau \left|\langle F, k_{\omega,r}\rangle\right|^2 \mathrm{d}r \le \|F\|^2. \quad \text{(CAUSAL FRAME CAPACITY)}$$

This is unconditionally strict immediately after rest:

$$\|\mathcal{V}_{\omega,\tau}\| \le \int_0^\tau \left|\psi_{\omega(u)}\right| \mathrm{d}u < 1$$

for every sufficiently small positive τ. The completed source's local integrability makes the integral tend to zero. Thus the construction has a real base; the open issue is continuation, not initialization.

On the whole logarithmic ray its transfer is

$$\hat{\psi}_{\omega(p)} = \Theta_{\omega(ip)} = \frac{\Xi(\omega - p)}{\Xi(\omega + p)}.$$

Thus all finite frame capacities are equivalent to the transfer being Schur in the right half-plane, hence to the innerness condition already equivalent to RH. The factorization materially narrows the construction: another crossing census or separate dark-mode argument cannot close it. What remains is a direct Bessel estimate for the source convolution $\nu_\omega * \varphi_\omega$, or equivalently source-side positivity of its Schur kernel. For $p = x + iy$ with $0 < x < \omega$, away from a zero on the displayed normal segment, this same condition has the normal-flux face

$$-\log\left|\hat{\psi}_{\omega(x+iy)}\right| = \int_{\omega-x}^{\omega+x} \mathrm{Re}\left[\frac{\Xi'(r+iy)}{\Xi(r+iy)}\right] \mathrm{d}r.$$

The unit-modulus boundary supplies equality at $x = 0$; it does not determine the orientation of these interior sweeps.

XIII.8.3 Every pair has a common founding and a reunion

Theorem 60 — Common founding and reunion factor every source crossing, but only the completed current is causal Fix $\omega > 0$ and define the generalized Jordan atom

$$J_{2\omega}(n) = n^{2\omega} \prod_{p|n} (1 - p^{-2\omega}), \quad J_{2\omega}(1) = 1. \quad \text{(DIVISOR ATOM)}$$

The completed source coefficients are exactly

$$c_{\omega(n)} = \frac{J_{2\omega}(n)}{n^\omega}, \quad w_{\omega(n)} = \frac{J_{2\omega}(n)}{n^{\omega+\frac{1}{2}}}. \quad \text{(SOURCE ATOM)}$$

Their divisor cumulative is

$$\sum_{d|n} J_{2\omega}(d) = n^{2\omega}. \quad \text{(CUMULATIVE AXIS)}$$

Consequently, if

$$u_{n(d)} = \sqrt{J_{2\omega}(d)}\mathbf{1}_{d|n},$$

then

$$\langle u_m, u_n\rangle = \gcd(m,n)^{2\omega}, \quad \text{(GCD GRAM)}$$

and the normalized incidence vectors $v_n = n^{-\omega} u_n$ satisfy

$$\langle v_m, v_n\rangle = \left(\frac{\gcd(m,n)}{\operatorname{lcm}(m,n)}\right)^{\omega}. \quad \text{(VALUATION DISTANCE)}$$

Thus the generalized Jordan population is a positive divisor-axis incidence before it becomes a source amplitude.

The source amplitude itself is modular on the divisor lattice:

$$w_{\omega(m)}w_{\omega(n)} = w_{\omega(\gcd(m,n))}w_{\omega(\operatorname{lcm}(m,n))}. \quad \text{(FOUNDING--REUNION LAW)}$$

Equivalently, write

$$d = \gcd(m,n), \quad m = da, \quad n = db, \quad \gcd(a,b) = 1.$$

With

$$\eta_{\omega,d}(a) = a^{\omega-\frac{1}{2}} \prod_{p|a;\, p\nmid d} (1 - p^{-2\omega}), \quad \text{(CONTEXTUAL ARM)}$$

one has

$$w_{\omega(da)} = w_{\omega(d)}\eta_{\omega,d}(a),$$

$$w_{\omega(m)}w_{\omega(n)} = w_{\omega(d)}^2\eta_{\omega,d}(a)\eta_{\omega,d}(b) = w_{\omega(d)}w_{\omega(dab)}. \quad \text{(CROSSING WEIGHT)}$$

The arm therefore depends on which prime axes have already been founded in d; the same integer ratio is not an absolute detached constituent.

This meet–join law gives the exact crossing expansion of the causal energy. Let

$$G_F = \varphi_\omega * F$$

on the positive logarithmic ray, extended by zero to negative time, and for a finite receiver τ put

$$b_n^\tau(r) = w_{\omega(n)}G_{F(r-\log n)}\mathbf{1}_{0<r<\tau}. \quad \text{(ADMITTED EVENT FACE)}$$

Local finiteness gives

$$\mathcal{V}_{\omega,\tau}F = \sum_{n<e^\tau} b_n^\tau.$$

Partitioning every ordered pair by its unique common founding gives

$$\left\|\mathcal{V}_{\omega,\tau}F\right\|^2 = \sum_{d<e^\tau} \mathcal{C}_{d,\tau}(F), \quad \text{(FOUNDING PARTITION)}$$

where

$$\mathcal{C}_{d,\tau}(F) = \sum_{a,b\geq 1;\, \gcd(a,b)=1} \langle b_{da}^\tau, b_{db}^\tau\rangle. \quad \text{(COPRIME ARMS)}$$

Each summand uses the common founding d, the two coprime arms, and the reunion label dab through (“CROSSING WEIGHT”).

A founding layer is not, however, an independently positive energy. Its coprime incidence matrix

$$C_{a,b} = \mathbf{1}_{\gcd(a,b)=1}$$

already has the principal minor on $\{1, p\}$

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det = -1. \quad \text{(INDEFINITE ARM CELL)}$$

The exact resolution is Möbius-signed:

$$\mathcal{C}_{d,\tau}(F) = \sum_{k\geq 1} \mu(k) \left\| \sum_{j\geq 1} b^{\tau}_{dkj} \right\|^2. \quad \text{(MOBIUS FOIL)}$$

The first opposed square occurs at every prime p because $\mu(p) = -1$. Summing all founding layers reconstitutes the one coherent current rather than a population of positive independent cells:

$$\sum_{d} \mathcal{C}_{d,\tau}(F) = \sum_{q\geq 1} \left(\sum_{k|q} \mu(k) \right) \left\| \sum_{j\geq 1} b^{\tau}_{qj} \right\|^2 = \left\| \sum_{j\geq 1} b^{\tau}_{j} \right\|^2. \quad \text{(GLOBAL RECONSTITUTION)}$$

The same inseparability is visible between the arithmetic and archimedean faces. On a finite cut define

$$\big(\mathcal{A}_{\omega,\tau} G\big)(r) = \sum_{\log n \leq r} w_{\omega(n)} G(r - \log n), \quad \text{(ARITHMETIC ADMISSION)}$$

and

$$\big(\mathcal{G}_{\omega,\tau} F\big)(r) = \int_0^r \varphi_{\omega(r-s)} F(s)\, \mathrm{d}s. \quad \text{(ARCHIMEDEAN RESPONSE)}$$

Causality gives the exact cascade

$$\mathcal{V}_{\omega,\tau} = \mathcal{A}_{\omega,\tau} \mathcal{G}_{\omega,\tau}. \quad \text{(COMPLETED CURRENT)}$$

Therefore its shared bipolar defect is the coupled capacitance

$$I - \mathcal{V}^{*}\mathcal{V} = (I - \mathcal{G}^{*}\mathcal{G}) - \mathcal{G}^{*}(\mathcal{A}^{*}\mathcal{A} - I)\mathcal{G}. \quad \text{(COUPLED CAPACITANCE)}$$

Neither parenthesis has been assigned a sign; only their exact difference is the physical source defect.

The arithmetic load is genuinely expansive after the first nontrivial admission. For every $\tau > \log 2$, choose a nonzero nonnegative G supported in $(0, \delta)$ with

$$0 < \delta < \min(\log 2, \tau - \log 2).$$

The $n = 1$ and $n = 2$ response faces are disjoint, while every other admitted face is pointwise nonnegative. Hence

$$\big\| \mathcal{A}_{\omega,\tau} G \big\|^2 \geq \big(1 + w^2_{\omega(2)}\big) \|G\|^2 > \|G\|^2. \quad \text{(ARITHMETIC PRESSURE)}$$

Thus the second term in (“COUPLED CAPACITANCE”) is an actual load, not a sign convention that can be removed by declaring the arithmetic subsystem passive.

Indeed the separation is singular inside the target causal half-plane. In the initial Dirichlet convergence region,

$$\hat{A}_{\omega(p)} := \sum_{n\geq 1} w_{\omega(n)} n^{-p} = \frac{\zeta\big(p + \frac{1}{2} - \omega\big)}{\zeta\big(p + \frac{1}{2} + \omega\big)}, \quad \operatorname{Re} p > \frac{1}{2} + \omega. \quad \text{(ARITHMETIC TRANSFER)}$$

The completed and complementary transfers are

$$\hat{\psi}_{\omega(p)} = \frac{\Xi(p - \omega)}{\Xi(p + \omega)},$$

$$\hat{G}_{\omega(p)} = \frac{\hat{\psi}_{\omega(p)}}{\hat{A}_{\omega(p)}}. \quad \text{(COMPLEMENTARY TRANSFER)}$$

For $0 < \omega < \frac{1}{2}$, put

$$p_0 = \frac{1}{2} - \omega > 0.$$

Meromorphic continuation of ("ARITHMETIC TRANSFER") has a simple zero at p_0 because its denominator is $\zeta(1)$, while $\hat{\psi}_{\omega(p_0)}$ is finite and nonzero. Consequently $\hat{G}_\omega$ has the compensating simple pole:

$$\hat{A}_{\omega(p)} = \zeta(1-2\omega)(p-p_0) + O\big((p-p_0)^2\big),$$

$$\hat{G}_{\omega(p)} = \frac{\hat{\psi}_{\omega(p_0)}}{\zeta(1-2\omega)} \frac{1}{p-p_0} + O(1). \quad \text{(COMPLETION SEAM)}$$

The zero and pole cancel in $\hat{\psi}_\omega = \hat{A}_\omega \hat{G}_\omega$. At $\omega = \frac{1}{2}$ this seam reaches the boundary $p = 0$; for $\omega > \frac{1}{2}$ it lies outside the causal right half-plane.

Demonstration. The first identity in ("SOURCE ATOM") is the definition of c_ω. Dividing once more by $\sqrt{n}$ gives the second. Both $J_{2\omega}$ and the divisor sum in ("CUMULATIVE AXIS") are multiplicative. At $n = p^a$ the sum telescopes:

$$1 + \sum_{j=1}^{a} \big(p^{2\omega j} - p^{2\omega(j-1)}\big) = p^{2\omega a}.$$

This proves ("CUMULATIVE AXIS") and hence the two Gram identities.
At each prime axis, the minimum and maximum of the two valuations are the valuations of the gcd and lcm. If both valuations are positive, each side of

$$J(m)J(n) = J(\gcd(m,n))J(\operatorname{lcm}(m,n))$$

carries the same two factors $1 - p^{-2\omega}$ and the same total exponent. If one valuation is zero, each side carries one such factor. Multiplication over primes and $mn = \gcd(m,n)\operatorname{lcm}(m,n)$ prove ("FOUNDING–REUNION LAW"). Comparing the prime supports of da and d proves ("CONTEXTUAL ARM") and ("CROSSING WEIGHT").
Substitute

$$\psi_\omega = \sum_n w_{\omega(n)} \varphi_{\omega(\cdot - \log n)}$$

into causal convolution. This gives the admitted event sum. Expanding its squared norm and assigning (m, n) to $d = \gcd(m,n)$ proves ("FOUNDING PARTITION") and ("COPRIME ARMS").
The displayed two-arm determinant proves ("INDEFINITE ARM CELL"). Möbius inversion gives

$$\mathbf{1}_{\gcd(a,b)=1} = \sum_{\substack{k|a,\\ k|b}} \mu(k).$$

Substitute this into ("COPRIME ARMS") and put $a = ki$, $b = kj$ to obtain ("MOBIUS FOIL"). Then put $q = dk$ and use

$$\sum_{k|q} \mu(k) = \begin{cases} 1 \\ \quad q = 1 \\ 0 \\ \quad q > 1 \end{cases}$$

to prove ("GLOBAL RECONSTITUTION").
Associativity of causal convolution gives ("COMPLETED CURRENT"). Add and subtract $\mathcal{G}^*\mathcal{G}$ in $I - \mathcal{G}^*\mathcal{A}^*\mathcal{A}\mathcal{G}$ to prove ("COUPLED CAPACITANCE"). For the pulse in ("ARITHMETIC PRESSURE"), its translate by $\log 2$ is disjoint from the original. The remaining translated faces have nonnegative coefficients and nonnegative values, so they cannot reduce the pointwise sum. Pythagoras for the first two faces proves the displayed strict lower bound.
Finally,

$$\sum_{n\geq 1} \frac{J_{2\omega}(n)}{n^s} = \frac{\zeta(s-2\omega)}{\zeta(s)}$$

for $\operatorname{Re} s > 1+2\omega$. Substitution $s = p+\omega+\frac{1}{2}$ proves ("ARITHMETIC TRANSFER"). Evenness of Ξ changes the transfer from $\frac{\Xi(\omega-p)}{\Xi(\omega+p)}$ to the displayed completed ratio. At $p_0 = \frac{1}{2}-\omega$, the denominator argument in ("ARITHMETIC TRANSFER") is one and the numerator argument is $1-2\omega$. The zeta pole has residue one, while zeta has no real zero in $(0,1)$. The completed ratio is finite and nonzero there. The two local Laurent expansions therefore give ("COMPLETION SEAM").

Boundary. The gcd was the correct common-founding coordinate, but positivity cannot be assigned one founding at a time. A cross between the 1-arm and the p-arm belongs to founding d, whereas the self-interaction of the p-arm belongs to founding dp. Möbius parity is the exact foil which prevents those perspectives from being counted twice. The negative square at $k=p$ is therefore not evidence against the whole source; it proves that independent positive common-founding blocks are the wrong proof object.

The completion seam gives the same warning at the continuum boundary. In $0<\omega<\frac{1}{2}$, neither the arithmetic admissions nor the archimedean response defines a bounded causal subsystem on the whole target half-plane: their internal zero and pole cancel only in the completed current. A proof may use the factorization on finite cuts, but it may not prove passivity of the two factors separately and then compose them.

What remains is now narrower than an unspecified Bessel estimate. It is the coupled inequality

$$I - \mathcal{G}^*\mathcal{G} \geq \mathcal{G}^*(\mathcal{A}^*\mathcal{A} - I)\mathcal{G} \quad \text{(COMPLETED CAPACITANCE BOUND)}$$

on every finite cut, with the Möbius-signed common-founding crossings retained inside the right side and the completion seam retained inside the left. A valid square decomposition must pair those two before either is collapsed. The existing identities do not determine that orientation. This is the first genuine signed term reached by the requested source expansion, not a reformulation introduced in place of the calculation.

The integer-event expansion now resolves the crossing geometry exactly. Define

$$J_{2\omega}(n) = n^{2\omega}\prod_{p\mid n}(1-p^{-2\omega}).$$

Then

$$w_{\omega(n)} = \frac{J_{2\omega}(n)}{n^{\omega+\frac{1}{2}}},$$

and every pair satisfies the meet–join identity

$$w_{\omega(m)}w_{\omega(n)} = w_{\omega(\gcd(m,n))}w_{\omega(\operatorname{lcm}(m,n))}. \quad \text{(FOUNDING AND REUNION)}$$

The gcd is the actual common divisor-axis standing; the lcm is the composed reunion label. Moreover,

$$\gcd(m,n)^{2\omega} = \sum_{\substack{d\mid m,\\ d\mid n}} J_{2\omega}(d),$$

so these common axes have an ordinary positive Gram realization.

That does not make each common-founding stratum positive. Writing $m = da$, $n = db$ with $\gcd(a,b) = 1$ assigns the crossing to d, but the coprime-arm incidence on $\{1,p\}$ is

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

whose determinant is -1. Möbius inversion gives the exact orientation:

$$\mathcal{C}_{d,\tau}(F) = \sum_{k\geq 1}\mu(k)\left\|\sum_{j\geq 1} b^{\tau}_{dkj}\right\|^2. \quad \text{(COMMON-FOUNDING FOIL)}$$

The negative prime layer $mu(p) = -1$ is not a negative source energy. It prevents the p-arm's self-interaction—whose founding is dp—from being counted again in the d perspective. Summation over all d

cancels the Möbius layers and reconstitutes the one complete causal current. Therefore the gcd is a valid crossing coordinate but not an independently shortable positive constituent.

The same inseparability occurs between arithmetic admission and the Gamma response. On every finite cut,

$$\mathcal{V} = \mathcal{A}\mathcal{G},$$

and hence

$$I - \mathcal{V}^*\mathcal{V} = (I - \mathcal{G}^*\mathcal{G}) - \mathcal{G}^*(\mathcal{A}^*\mathcal{A} - I)\mathcal{G}. \quad \text{(COMPLETED CAPACITANCE)}$$

This is the precise storage-versus-load relation. Neither side may be assigned a sign before their coupling.

The load is not merely formal. Once $\tau > \log 2$, a sufficiently narrow nonnegative pulse has disjoint $n = 1$ and $n = 2$ images, so

$$\left\|\mathcal{A}_{\omega,\tau}G\right\|^2 \geq \left(1 + w^2_{\omega(2)}\right)\|G\|^2 > \|G\|^2.$$

Arithmetic admission is already expansive at the first nontrivial event; the completed response must carry the compensating storage.

There is a concrete reason. The arithmetic transfer is

$$\hat{A}_{\omega(p)} = \frac{\zeta\left(p + \frac{1}{2} - \omega\right)}{\zeta\left(p + \frac{1}{2} + \omega\right)},$$

while the completed transfer is

$$\hat{\psi}_{\omega(p)} = \frac{\Xi(p - \omega)}{\Xi(p + \omega)}.$$

For $0 < \omega < \frac{1}{2}$, the point $p_0 = \frac{1}{2} - \omega$ lies inside the causal half-plane. At p_0, $\hat{A}_\omega$ has a zero caused by the pole $\zeta(1)$, and the separated Gamma/endpoint response has the compensating pole. Their product is regular. The critical value $\frac{1}{2}$ therefore marks the exact place where a seemingly natural subsystem split becomes singular: at $\omega = \frac{1}{2}$ the cancellation reaches the boundary, and above it the seam leaves the causal half-plane.

The required construction is consequently not a positive prime-by-prime or gcd-by-gcd factorization. It must factor (“COMPLETED CAPACITANCE”) with the Möbius foil and completion seam still co-present.

XIII.8.4 The half-seam retains strict completed capacity

Theorem 61 — The Jordan incidence diverges at the half-seam while the completed seam retains strict capacity Fix

$$0 < \omega < \frac{1}{2}, \quad p_0 = \frac{1}{2} - \omega, \quad s_0 = 2p_0 = 1 - 2\omega. \quad \text{(HALF-SEAM)}$$

Retain the arithmetic, complementary, and completed transfers

$$A_{\omega(p)} = \frac{\zeta\left(p + \frac{1}{2} - \omega\right)}{\zeta\left(p + \frac{1}{2} + \omega\right)},$$

$$G_{\omega(p)} = \frac{\Xi(p - \omega)}{\Xi(p + \omega)} \frac{1}{A_{\omega(p)}},$$

$$S_{\omega(p)} = A_{\omega(p)}G_{\omega(p)} = \frac{\Xi(p - \omega)}{\Xi(p + \omega)}. \quad \text{(SEPARATED AND COMPLETED)}$$

Put $\rho(p) = p - p_0$ and regularize the internal cancellation by

$$\tilde{A}_{\omega(p)} = \frac{A_{\omega(p)}}{\rho(p)}, \quad \tilde{G}_{\omega(p)} = \rho(p) G_{\omega(p)}. \quad \text{(SEAM REBASE)}$$

Both rebased faces are holomorphic and nonzero near p_0, and

$$S_\omega = \tilde{A}_\omega \tilde{G}_\omega,$$

$$\tilde{A}_{\omega(p_0)} = \zeta(s_0), \quad \tilde{G}_{\omega(p_0)} = \frac{2\xi(s_0)}{\zeta(s_0)}. \quad \text{(REBASED CROSSING)}$$

The opposed signs of the two values multiply to the positive completed face: $\zeta(s_0) < 0$ while $\xi(s_0) > 0$.

The positive Jordan incidence does not itself converge to this rebased crossing. At the seam, its raw cutoff is

$$A^{\mathrm{raw}(p_0)}_{\omega,X} = \sum_{n \le X} w_{\omega(n)} n^{-p_0} = \sum_{n \le X} \frac{J_{2\omega}(n)}{n}. \quad \text{(RAW JORDAN SEAM)}$$

Every summand is positive and

$$A^{\mathrm{raw}(p_0)}_{\omega,X} \sim \frac{X^{2\omega}}{2\omega\zeta(1+2\omega)} \to \infty. \quad \text{(JORDAN PRESSURE)}$$

By contrast, meromorphic continuation gives

$$A_{\omega(p_0)} = 0. \quad \text{(CONTINUED ZERO)}$$

Therefore no termwise positive limit of the Jordan gcd Gram can produce the seam. Completion must supply a counterterm of at least the exact leading size in ("JORDAN PRESSURE") before the infinite receiver limit is taken.

This counterterm does not erase the completed capacity. For any scalar transfer T, define its right-half-plane defect kernel

$$\mathcal{K}_{T(p,q)} = \frac{1 - T(p)\overline{T(q)}}{p + \overline{q}}. \quad \text{(TRANSFER DEFECT)}$$

The product law and ("SEAM REBASE") give the regular identity

$$\mathcal{K}_{S_\omega}(p,q) = \mathcal{K}_{\tilde{A}_\omega}(p,q) + \tilde{A}_{\omega(p)} \overline{\tilde{A}_{\omega(q)}} \mathcal{K}_{\tilde{G}_\omega}(p,q). \quad \text{(REGULARIZED CAPACITANCE)}$$

At the seam this becomes

$$\mathcal{K}_{S_\omega}(p_0,p_0) = \frac{1 - \zeta(s_0)^2}{s_0} + \frac{\zeta(s_0)^2 - 4\xi(s_0)^2}{s_0} = \frac{1 - 4\xi(s_0)^2}{s_0} > 0. \quad \text{(STRICT COMPLETED CELL)}$$

The two finite summands do not have fixed signs. As $s_0 \to 1^-$, $\zeta(s_0) \to -\infty$, so the first is negative and the second positive. As $s_0 \to 0^+$,

$$\zeta(s_0) \to -\frac{1}{2}, \quad 2\xi(s_0) \to 1,$$

so the first is positive and the second negative. Their changing orientations are a genuine swing between the two regularized faces; only their completed sum has the fixed positive seam orientation.

Demonstration. The Laurent expansion

$$\zeta(1+\rho) = \frac{1}{\rho} + O(1)$$

gives

$$A_{\omega(p)} = \zeta(s_0)\rho(p) + O(\rho(p)^2).$$

The completed ratio is regular and nonzero at p_0, with

$$S_{\omega(p_0)} = \frac{\Xi\left(\frac{1}{2} - 2\omega\right)}{\Xi\left(\frac{1}{2}\right)} = \frac{\xi(s_0)}{\xi(1)} = 2\xi(s_0).$$

Dividing and multiplying by ρ prove ("REBASED CROSSING"). On $0 < s_0 < 1$, the alternating eta representation gives $\zeta(s_0) < 0$, while the defining completed product gives $\xi(s_0) > 0$.

The generalized Jordan identity

$$J_{2\omega}(n) = \sum_{dm=n} \mu(d) m^{2\omega}$$

gives

$$\sum_{n \le X} \frac{J_{2\omega}(n)}{n} = \sum_{d \le X} \frac{\mu(d)}{d} \sum_{m \le \frac{X}{d}} m^{2\omega - 1}.$$

Since

$$\sum_{m \le Y} m^{2\omega-1} \sim \frac{Y^{2\omega}}{2\omega}$$

and

$$\sum_{d \ge 1} \frac{\mu(d)}{d^{1+2\omega}} = \frac{1}{\zeta(1+2\omega)},$$

partial summation proves ("JORDAN PRESSURE"). Positivity of every summand proves that this is not an ordinary convergent realization of ("CONTINUED ZERO").

For arbitrary scalar functions A, G, direct expansion gives

$$1 - A(p)G(p)\overline{A(q)G(q)} = \left(1 - A(p)\overline{A(q)}\right) + A(p)\overline{A(q)}\left(1 - G(p)\overline{G(q)}\right).$$

Apply this to the two rebased faces and divide by $p + \overline{q}$ to prove ("REGULARIZED CAPACITANCE").

Finally put

$$\theta_+(x) = \sum_{n \ge 1} e^{-\pi n^2 x}.$$

The completed theta integral is

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2} \int_1^\infty \theta_+(x)\left(x^{\frac{s}{2}} + x^{\frac{1-s}{2}}\right)\frac{\mathrm{d}x}{x}.$$

For $0 < s < 1$, its integral is strictly positive and $s(s-1) < 0$, so $\xi(s) < \frac{1}{2}$. Positivity of $\xi(s)$ was established above. Therefore $0 < 2\xi(s_0) < 1$, which proves ("STRICT COMPLETED CELL"). The endpoint limits of zeta and xi give the two asserted orientation reversals.

Boundary. The requested Jordan factorization fails for a precise reason. Its finite gcd Grams are positive, but their seam mass grows like $X^{2\omega}$ and cannot converge to the analytically continued zero. Subtracting that mass only from the arithmetic face would destroy the causal identity of the current. The counterterm belongs to the co-present completion, exactly as the pole in G_ω cancels the zero in A_ω.

The completed cancellation nevertheless leaves the strict scalar capacity ("STRICT COMPLETED CELL") at every $0 < \omega < \frac{1}{2}$. This is an unconditional positive cell inside the target half-plane, not an RH assumption and not a numerical sample. It proves that the half-seam itself is not where completed passivity fails.

A scalar diagonal cell is not yet positivity of the two-variable kernel. The remaining local calculation is the first nontrivial crossing of two nearby receiver coordinates: equivalently, the 2×2 determinant of $\mathcal{K}_{S_\omega}$ as both coordinates depart from p_0. Put

$$L_{\xi(s)} = \frac{\xi'(s)}{\xi(s)}.$$

Since

$$S_{\omega'}(p_0) = 2\xi(s_0)\left(L_{\xi(s_0)} - L_{\xi(1)}\right),$$

its infinitesimal Schwarz–Pick numerator is exactly

$$\mathcal{Q}(s_0) = \left(1 - 4\xi(s_0)^2\right)^2 - 4s_0^2\xi(s_0)^2\left(L_{\xi(s_0)} - L_{\xi(1)}\right)^2. \quad \text{(SEAM CROSSING CURVATURE)}$$

> Nonnegativity of this quantity is the first nearby two-receiver obligation. Its sign is not established here. The Jordan incidence may enter only after the divergent common mode and its completion counterterm are paired; it cannot be inserted as an independently positive infinite Gram.

The internal zero–pole pair can be removed without separating the causal event. Put

$$p_0 = \frac{1}{2} - \omega, \quad \rho(p) = p - p_0,$$

and rebase

$$\tilde{A}_{\omega(p)} = \frac{A_{\omega(p)}}{\rho(p)}, \quad \tilde{G}_{\omega(p)} = \rho(p) G_{\omega(p)}.$$

Both faces are regular at p_0, their product remains the completed transfer, and

$$\tilde{A}_{\omega(p_0)} = \zeta(1-2\omega), \quad \tilde{G}_{\omega(p_0)} = \frac{2\xi(1-2\omega)}{\zeta(1-2\omega)}.$$

This rebase shows why the positive Jordan gcd Gram cannot be inserted by itself. Its raw seam population is

$$\sum_{n \le X} \frac{J_{2\omega}(n)}{n} \sim \frac{X^{2\omega}}{2\omega\zeta(1+2\omega)},$$

which diverges positively. The analytically continued arithmetic face, however, is zero at the same coordinate. Completion must therefore remove this common divergent mode before the infinite receiver limit; no termwise limit of positive divisor atoms can perform that deed.

For the defect kernel

$$\mathcal{K}_{T(p,q)} = \frac{1 - T(p)\overline{T(q)}}{p + \overline{q}},$$

the regularized product law is

$$\mathcal{K}_{S_\omega} = \mathcal{K}_{\tilde{A}_\omega} + \tilde{A}_\omega \overline{\tilde{A}_\omega} \mathcal{K}_{\tilde{G}_\omega}.$$

At the seam, with $s_0 = 1 - 2\omega$,

$$\mathcal{K}_{S_\omega}(p_0, p_0) = \frac{1 - \zeta(s_0)^2}{s_0} + \frac{\zeta(s_0)^2 - 4\xi(s_0)^2}{s_0} = \frac{1 - 4\xi(s_0)^2}{s_0} > 0. \quad \text{(STRICT SEAM CAPACITY)}$$

The theta-integral identity gives $0 < \xi(s_0) < \frac{1}{2}$ without RH. The two summands exchange signs between the limits $s_0 \to 1^-$ and $s_0 \to 0^+$, while their completed sum remains positive. This is a literal orientation swing between the regularized faces.

The half-seam is therefore not the site of completed failure. What remains locally is the first crossing of two nearby seam receivers—the 2×2 defect-kernel determinant—not another scalar evaluation or a termwise positive Jordan limit. With $s_0 = 1 - 2\omega$ and $L_\xi = \frac{\xi'}{\xi}$, its exact infinitesimal numerator is

$$\left(1 - 4\xi(s_0)^2\right)^2 - 4s_0^2\xi(s_0)^2\left(L_{\xi(s_0)} - L_{\xi(1)}\right)^2. \quad \text{(SEAM CROSSING CURVATURE)}$$

This is the first genuinely two-receiver sign left by the regularization.

XIII.8.5 The theta source exposes one phase-current sign

Theorem 62 — The shifted completed kernel is the Laplace pullback of one theta phase current Let

$$\Xi(z) = \xi\left(\frac{1}{2} + z\right)$$

and normalize the even positive Riemann theta kernel Φ by

$$\Xi(z) = c_\Phi \int_{\mathbb{R}} \Phi(t) e^{zt}\,\mathrm{d}t, \quad c_\Phi > 0. \quad \text{(BILATERAL THETA SOURCE)}$$

The super-exponential decay of Φ makes this identity and every differentiation below absolutely convergent for all complex z.

For $\omega > 0$ and $\operatorname{Re} p, \operatorname{Re} q > 0$, define the completed shifted kernel

$$\mathcal{L}_{\omega(p,q)} = \frac{\Xi(p+\omega)\overline{\Xi(q+\omega)} - \Xi(p-\omega)\overline{\Xi(q-\omega)}}{p+\overline{q}}. \quad \text{(SHIFTED COMPLETED KERNEL)}$$

For real a, b, put

$$m = \frac{a+b}{2}, \quad d = \frac{a-b}{2},$$

and define the coordinate kernel

$$\mathcal{D}_{\omega(a,b)} = 2c_\Phi^2 \int_{|m|}^{\infty} \Phi(y+d)\Phi(y-d)\sinh(2\omega y)\,\mathrm{d}y. \quad \text{(SWEPT COORDINATE KERNEL)}$$

Then the complete shifted kernel is its exact bilateral-Laplace pullback:

$$\mathcal{L}_{\omega(p,q)} = \int_{\mathbb{R}}\int_{\mathbb{R}} e^{pa}\mathcal{D}_{\omega(a,b)} e^{\overline{q}b}\,\mathrm{d}a\,\mathrm{d}b. \quad \text{(THETA-TO-COMPLETED PULLBACK)}$$

The coordinate kernel is generated by one source-native phase current. For $\eta \geq 0$, define

$$\mathcal{P}_{\eta(a,b)} = \frac{c_\Phi^2}{2}\int_{|m|}^{\infty} y\cosh(2\eta y)\Phi(y+d)\Phi(y-d)\,\mathrm{d}y. \quad \text{(THETA PHASE CURRENT)}$$

Then

$$\partial_\eta \mathcal{D}_\eta = 8\mathcal{P}_\eta, \quad \mathcal{D}_0 = 0, \quad \mathcal{D}_\omega = 8\int_0^\omega \mathcal{P}_\eta\,\mathrm{d}\eta. \quad \text{(NORMAL SWEEP)}$$

Consequently, positivity of $\mathcal{P}_\eta$ as a kernel for $0 \leq \eta \leq \omega$ implies positivity of $\mathcal{L}_\omega$. This implication uses no zero locations and no analytically continued Euler split.

More sharply, the seam current alone carries the complete RH sign. Put

$$\mathcal{M}(p,q) = \lim_{\omega\to 0^+} \frac{\mathcal{L}_{\omega(p,q)}}{2\omega}.$$

Then

$$\mathcal{M}(p,q) = \frac{\Xi'(p)\overline{\Xi(q)} + \Xi(p)\overline{\Xi'(q)}}{p+\overline{q}} \quad \text{(INFINITESIMAL COMPLETED KERNEL)}$$

and

$$\mathcal{M}(p,q) = 4\int_{\mathbb{R}}\int_{\mathbb{R}} e^{pa}\mathcal{P}_0(a,b)e^{\overline{q}b}\,\mathrm{d}a\,\mathrm{d}b. \quad \text{(INFINITESIMAL SOURCE PULLBACK)}$$

Thus the following are equivalent:

- RH;
- $\mathcal{M}$ is a positive-semidefinite kernel on the right half-plane;
- $\mathcal{P}_0$ is nonnegative on the receiver span generated by the bilateral exponentials $a \mapsto e^{pa}$, $\operatorname{Re} p > 0$.

Positivity of $\mathcal{P}_0$ on a larger closed function space is a sufficient strengthening, not an additional RH requirement.

Demonstration. The half-plane Cauchy kernel has the causal representation

$$\frac{1}{p+\overline{q}} = \int_0^\infty e^{-(p+\overline{q})r}\,\mathrm{d}r.$$

Substitute ("BILATERAL THETA SOURCE") into the numerator of ("SHIFTED COMPLETED KERNEL"). Its two orientations differ by

$$e^{\omega(t+u)} - e^{-\omega(t+u)} = 2\sinh(\omega(t+u)).$$

After inserting the causal integral, set

$$a = t - r, \quad b = u - r.$$

The coefficient of $e^{pa+\overline{q}b}$ is

$$2c_\Phi^2 \int_0^\infty \Phi(a+r)\Phi(b+r)\sinh(\omega(a+b+2r))\,\mathrm{d}r.$$

With $y = m + r$, this is the integral from m to infinity of

$$2c_\Phi^2 \Phi(y+d)\Phi(y-d)\sinh(2\omega y).$$

The product of the two theta faces is even in y, while $\sinh(2\omega y)$ is odd. If $m < 0$, the integral over $[m, -m]$ therefore vanishes. The lower boundary is consequently $|m|$ for either orientation. This proves ("SWEPT COORDINATE KERNEL") and ("THETA-TO-COMPLETED PULLBACK").

Differentiation under the absolutely convergent integral gives

$$\partial_\eta \mathcal{D}_{\eta(a,b)} = 4c_\Phi^2 \int_{|m|}^\infty y\cosh(2\eta y)\Phi(y+d)\Phi(y-d)\,\mathrm{d}y = 8\mathcal{P}_{\eta(a,b)}.$$

Since the hyperbolic sine vanishes at $\eta = 0$, integration in η proves ("NORMAL SWEEP"). Positive kernels remain positive under positive integration and under pullback, proving the first consequence.

Differentiating the analytic numerator at $\omega = 0$ proves ("INFINITESIMAL COMPLETED KERNEL"). Dividing ("NORMAL SWEEP") by 2ω and taking the limit proves ("INFINITESIMAL SOURCE PULLBACK").

Away from zeros, put $F = \frac{\Xi'}{\Xi}$. Then

$$\mathcal{M}(p,q) = \Xi(p)\overline{\Xi(q)}\frac{F(p) + \overline{F(q)}}{p+\overline{q}}.$$

Under RH, `theorem:completed-return-flux` makes F positive real. The positive-real kernel in the last display is positive semidefinite, and multiplication by $\Xi(p)\overline{\Xi(q)}$ is a congruence. Hence $\mathcal{M} \geq 0$.

Conversely, on the diagonal $p = x + iy$,

$$\mathcal{M}(p,p) = \frac{|\Xi(p)|^2}{x}\operatorname{Re} F(p). \quad \text{(NORMAL PHASE ENERGY)}$$

A zero ρ of order k in the open right half-plane would give, along $p = \rho + \varepsilon$ with real ε,

$$\mathcal{M}(p,p) \sim \frac{k|c|^2}{\operatorname{Re}\rho}|\varepsilon|^{2k-2}\varepsilon,$$

where $\Xi(p) = c\varepsilon^k + O(\varepsilon^{k+1})$. This changes sign across the zero while both nearby points remain in the half-plane, contradicting positivity. Thus Ξ is zero-free there, and ("NORMAL PHASE ENERGY") gives $\operatorname{Re} F \geq 0$. The positive-real equivalence in `theorem:completed-return-flux` now gives RH. Finally, ("INFINITESIMAL SOURCE PULLBACK") identifies positivity of $\mathcal{M}$ with positivity of $\mathcal{P}_0$ on the declared exponential receiver span.

Boundary. This theorem replaces the unnecessarily large demand for an independently guessed Gram of every finite shift. One exact source kernel remains:

$$\mathcal{P}_0(a,b) = \frac{c_\Phi^2}{2}\int_{\left|\frac{a+b}{2}\right|}^\infty y\Phi\left(y + \frac{a-b}{2}\right)\Phi\left(y - \frac{a-b}{2}\right)\mathrm{d}y. \quad \text{(OPEN PHASE-CURRENT SIGN)}$$

Proving its nonnegativity on the exponential receiver span proves RH. Proving it on the whole coordinate space is a stronger sufficient theorem.

> Pointwise positivity of Φ does not prove this sign. The two theta factors are coupled through the difference coordinate, while the moving lower boundary is set by the sum coordinate. A fixed-y layer is not a rank-one square, and separating those coordinates discards the crossing which the kernel measures. The required sign belongs to the integrated phase current.
>
> The same coordinate kernel, with the same $y\cosh(2\eta y)$ weight, is studied as a Weyl kernel in [41]. That preprint records exact Volterra reductions and computational evidence but explicitly leaves its original all-parameter positivity and the RH-facing bridge open. ("THETA-TO-COMPLETED PULLBACK") supplies the latter bridge at the level needed here; it does not supply the still-open sign of ("OPEN PHASE-CURRENT SIGN").

The global two-receiver kernel does not have to be guessed from the seam. The bilateral theta representation pulls it back from one exact coordinate kernel. With

$$m = \frac{a+b}{2}, \quad d = \frac{a-b}{2},$$

that kernel is swept by

$$\mathcal{P}_{\eta(a,b)} = \frac{c_\Phi^2}{2}\int_{|m|}^{\infty} y\cosh(2\eta y)\Phi(y+d)\Phi(y-d)\,\mathrm{d}y. \quad \text{(THETA PHASE CURRENT)}$$

The sum coordinate sets the moving boundary and the difference coordinate sets the opposed theta faces. They are not independent axes.

The finite shifted kernel is the positive-parameter integral of this current, followed by bilateral-Laplace pullback. More sharply, its derivative at the unshifted seam is

$$\mathcal{M}(p,q) = \frac{\Xi'(p)\overline{\Xi(q)} + \Xi(p)\overline{\Xi'(q)}}{p+\overline{q}}$$

and is four times the pullback of $\mathcal{P}_0$. Positivity of $\mathcal{M}$ is equivalent to RH: its diagonal is

$$\mathcal{M}(p,p) = \frac{|\Xi(p)|^2}{\operatorname{Re} p}\operatorname{Re}\frac{\Xi'(p)}{\Xi(p)},$$

while any right-half-plane zero makes this diagonal change sign across the zero.

The remaining source theorem is therefore narrower than the earlier global request: prove $\mathcal{P}_0 \geq 0$ on the bilateral-exponential receiver span. Full coordinate-space Weyl positivity would be sufficient but is stronger. The pointwise sign of Φ does not settle it, because a fixed crossing layer is not a square; the sign belongs to the integrated sum/difference current.

XIII.8.6 The theta event has an exact ordered-execution meaning

Theorem 63 — RH is the positive ordered-execution lift of the theta phase current Let

$$\Xi(z) = \xi\left(\frac{1}{2}+z\right), \quad H_{t(x)} = |\Xi(x+it)|^2,$$

and let $\mathcal{V}_{\exp}$ be the finite span of the bilateral-exponential receivers

$$a \mapsto e^{pa}, \quad \operatorname{Re} p > 0.$$

Define the polarized normal-current kernel

$$\mathcal{M}(p,q) = \frac{\Xi'(p)\overline{\Xi(q)} + \Xi(p)\overline{\Xi'(q)}}{p+\overline{q}}. \quad \text{(NORMAL EXECUTION RESIDUAL)}$$

On the diagonal $p = x + it$,

$$\mathcal{M}(p,p) = \frac{\partial_x H_{t(x)}}{2x} = \partial_r H_{t(\sqrt{r})} \quad (r = x^2). \quad \text{(SQUARED-NORMAL CURRENT)}$$

The completed shifted kernels give actual signed finite events between positive receiver boundaries. Their symmetric derivative at the unshifted seam is the identity-carried theta execution current Γ_Ξ whose residual Hermitian form has kernel $\mathcal{M}$. Then the following statements are equivalent:

- the Riemann Hypothesis;
- $\mathcal{M}$ is positive semidefinite on every finite co-present family of right-half-plane receivers;
- Γ_Ξ is a positive ordered-execution current, so its finite outward integrals are arrows of Exec_+;
- the theta phase current

$$\mathcal{P}_0(a,b) = \frac{c_\Phi^2}{2}\int_{\left|\frac{a+b}{2}\right|}^{\infty} y\Phi\left(y+\frac{a-b}{2}\right)\Phi\left(y-\frac{a-b}{2}\right)\mathrm{d}y$$

  is nonnegative on $\mathcal{V}_{\mathrm{exp}}$; and
- there are a Hilbert space $\mathcal{K}$ and a linear residual executor

$$\mathcal{R} : \mathcal{V}_{\mathrm{exp}} \to \mathcal{K}$$

  such that every finite exponential superposition g satisfies

$$4\int_{\mathbb{R}}\int_{\mathbb{R}} g(a)\mathcal{P}_0(a,b)\overline{g(b)}\,\mathrm{d}a\,\mathrm{d}b = \|\mathcal{R}g\|^2. \quad \text{(SOURCE RESIDUAL FACTORIZATION)}$$

Consequently, an explicit source-derived construction of $\mathcal{R}$, together with the displayed identity on the complete declared receiver span, is a constructive proof of RH. Finite outward events then compose by residual addition, or equivalently by the transported fundamental theorem in the squared-normal coordinate $r = x^2$.

Demonstration. `theorem:completed-theta-phase-current-bridge` proves that $\mathcal{M}$ is four times the bilateral-Laplace pullback of $\mathcal{P}_0$, and proves

$$\mathrm{RH} \leftrightarrow \mathcal{M} \geq 0 \leftrightarrow \mathcal{P}_0 \geq 0 \ \text{ on } \mathcal{V}_{\mathrm{exp}}.$$

By `definition:ordered-causal-execution`, an identity-carried tangent is a positive execution current exactly when its residual form is positive. Matrix positivity of $\mathcal{M}$ is precisely positivity under every finite co-present receiver family.

A positive Hermitian form admits a Kolmogorov–GNS factorization, giving $\mathcal{R}$ and ("SOURCE RESIDUAL FACTORIZATION"). Conversely, that factorization makes the quadratic form a squared norm and hence nonnegative. This proves all equivalences.

Finally, for $r_1 \leq r_2$, integration of the normal current gives

$$H_{t(\sqrt{r_2})} - H_{t(\sqrt{r_1})} = \int_{r_1}^{r_2} \partial_r H_{t(\sqrt{r})}\,\mathrm{d}r.$$

Positive residuals are closed under integration and under the residual composition law, so an exact positive infinitesimal executor composes into every finite outward successor.

Boundary. Abstractly invoking a square root, GNS construction, or reproducing kernel does not prove RH: each exists here exactly when the unresolved sign is already known. The proof-bearing construction must derive $\mathcal{R}$ directly from the theta source Φ, the opposed coordinates $\frac{a-b}{2}$, and the moving boundary $\left|\frac{a+b}{2}\right|$.

The theorem defines the exact algorithmic proof object; it does not claim that the present software engine has constructed it. A numerical execution on finitely many receivers is testimony, while an exact rule valid on arbitrary finite exponential superpositions is the proof.

The causal-machine statement is now precise. The supplied theta source and the bilateral-exponential receivers define an infinitesimal event in the signed ordered-execution category. Its carried map is the identity after the receivers have been lawfully anchored, and its residual is $\mathcal{M}$. The event exists without RH. What RH asserts is exactly that this residual is positive on every finite co-present receiver family, so that the same event lifts to the positive-residual subcategory.

This is algorithmic without being computational exhaustion. A finite source-derived rule

$$\mathcal{R} : \mathcal{V}_{\exp} \to \mathcal{K}$$

which proves

$$4\langle g, \mathcal{P}_0 g\rangle = \|\mathcal{R}g\|^2$$

for an arbitrary finite exponential superposition g proves the complete kernel sign. Functorial composition and the transported fundamental theorem then carry that infinitesimal residual through every finite outward receiver arc.

XIII.8.7 An isolated crossing layer cannot be the executor

Lemma 64 — Every isolated theta crossing layer is indefinite For $y > 0$, remove the positive constant from one fixed layer of the theta phase current and put

$$K_{y(a,b)} = \mathbf{1}_{|\frac{a+b}{2}|\le y}\Phi\left(y + \frac{a-b}{2}\right)\Phi\left(y - \frac{a-b}{2}\right). \quad \text{(CROSSING LAYER)}$$

Then K_y is indefinite on the ambient coordinate receiver space. Indeed, for every $s > y$, its restriction to the opposed receivers $\{s, -s\}$ is

$$\begin{pmatrix} 0 & k_{y,s} \\ k_{y,s} & 0 \end{pmatrix}, \quad k_{y,s} = \Phi(y+s)\Phi(y-s) > 0,$$

with eigenvalues $\pm k_{y,s}$.

Hence the phase current cannot be proved positive by treating each y-layer as an independent Gram constituent and then integrating those constituents. The ordered accumulation in y must carry an open off-diagonal crossing until later diagonal storage can participate.

The first accumulated opposed-receiver face makes this explicit. Up to the common positive factor $\frac{c_\Phi^2}{2}$, put

$$S(s) = \int_s^\infty y\Phi(y)^2\,\mathrm{d}y, \quad C(s) = \int_0^\infty y\Phi(y+s)\Phi(y-s)\,\mathrm{d}y.$$

Then

$$\begin{pmatrix} \mathcal{P}_0(s,s) & \mathcal{P}_0(s,-s) \\ \mathcal{P}_0(-s,s) & \mathcal{P}_0(-s,-s) \end{pmatrix} = \frac{c_\Phi^2}{2}\begin{pmatrix} S(s) & C(s) \\ C(s) & S(s) \end{pmatrix},$$

whose symmetric and antisymmetric responses are proportional to $S(s) + C(s)$ and $S(s) - C(s)$. Thus the first ambient bipolar closure law is the exact domination $S(s) \ge C(s)$.

Demonstration. If $s > y$, then the diagonal activation conditions are

$$\left|\frac{s+s}{2}\right| = s > y, \quad \left|\frac{-s-s}{2}\right| = s > y,$$

so both diagonal entries vanish. For the opposed entry,

$$\frac{s+(-s)}{2} = 0, \quad \frac{s-(-s)}{2} = s,$$

and positivity of the even Riemann theta kernel gives

$$K_{y(s,-s)} = \Phi(y+s)\Phi(y-s) > 0.$$

The displayed matrix therefore has one positive and one negative eigenvalue.
Substitution of $(a,b) = (s,s)$ and $(s,-s)$ into the definition of $\mathcal{P}_0$ gives $S(s)$ and $C(s)$ respectively. Symmetry supplies the other two entries, and diagonalization in the vectors $(1,1)$ and $(1,-1)$ gives the final responses.

Boundary. The opposed-point matrix concerns positivity on the full coordinate receiver space, which is stronger than the RH-equivalent requirement on the bilateral-exponential span. It is therefore a diagnostic of the stronger Weyl-kernel route, not a new equivalent criterion for RH.

The lemma does not show that the integrated phase current is indefinite. It shows why a local-in-y, memoryless, one-layer-at-a-time executor cannot supply its positive factorization. Any successful source executor must mix crossing layers, or work directly after the bilateral-exponential pullback.

The obstruction is causal rather than cosmetic. At a fixed y, opposed receivers s and $-s$ with $s > y$ already see each other, while neither diagonal receiver has yet entered the layer. The local matrix is therefore

$$\begin{pmatrix} 0 & k \\ k & 0 \end{pmatrix},$$

with one positive and one negative orientation. Later y-layers introduce the missing diagonal storage. Consequently the source executor cannot be memoryless in the sweep parameter: it must retain the open crossing and let later incidence close or foil it.

On the stronger ambient coordinate face, the first accumulated bipolar law is

$$S(s) - C(s) \geq 0,$$

where S is the diagonal theta tail and C is the opposed crossing integral defined in the lemma. This is not substituted for the RH-equivalent exponential-span condition. It identifies exactly what the full-coordinate Weyl route would have to transport rather than assuming each layer was already a Gram square.

Theorem 65 — The full theta coordinate executor is one bipolar contraction Let $K = \mathcal{P}_0$ act by its quadratic form on a declared dense test space in $L^2(\mathbb{R})$. Since

$$K(-a, -b) = K(a, b),$$

it commutes with parity. On the positive half-line define the same-hand and opposed-hand kernels

$$A(x, y) = K(x, y), \quad B(x, y) = K(x, -y), \quad x, y \geq 0.$$

Under the unitary parity receiver

$$(Uf)_{\pm(x)} = \frac{f(x) \pm f(-x)}{\sqrt{2}},$$

the complete coordinate operator becomes

$$UKU^* = (A + B) \oplus (A - B). \quad \text{(PARITY EXECUTION)}$$

Consequently the following are equivalent:

- $K \geq 0$ on the full coordinate test space;
- $A + B \geq 0$ and $A - B \geq 0$;
- $A \geq 0$ and $-A \leq B \leq A$; and
- on the support of A, there is a self-adjoint contraction T such that

$$B = A^{\frac{1}{2}} T A^{\frac{1}{2}}, \quad \|T\| \leq 1. \quad \text{(BIPOLAR TRANSPORT)}$$

When these conditions hold, the explicit parity residual executor is

$$\mathcal{R}_{\pm} = (1 \pm T)^{\frac{1}{2}} A^{\frac{1}{2}},$$

and

$$\langle f, Kf \rangle = \left\| \mathcal{R}_+ f_+ \right\|^2 + \|\mathcal{R}_- f_-\|^2. \quad \text{(BIPOLAR GRAM CLOSURE)}$$

Demonstration. Simultaneous sign reversal fixes $\left|\frac{a+b}{2}\right|$ and reverses $\frac{a-b}{2}$. Evenness of Φ interchanges the two theta factors, proving $K(-a, -b) = K(a, b)$. Direct substitution of

$$f(x) = \frac{f_+(x) + f_-(x)}{\sqrt{2}}, \quad f(-x) = \frac{f_+(x) - f_-(x)}{\sqrt{2}}$$

into the full-line quadratic form gives ("PARITY EXECUTION").

The first three conditions are then algebraically equivalent. The operator interval $-A \le B \le A$ implies that B vanishes on the null space of A and, by the Douglas factorization principle, has the form

$$A^{\frac{1}{2}} T A^{\frac{1}{2}}$$

for a self-adjoint contraction T on the support of A. Conversely such a contraction gives

$$A \pm B = A^{\frac{1}{2}} (1 \pm T) A^{\frac{1}{2}} \ge 0.$$

Taking their positive square roots proves ("BIPOLAR GRAM CLOSURE").

Boundary. Full coordinate positivity is stronger than the RH-equivalent positivity required only after bilateral-exponential pullback. This theorem therefore identifies one sufficient source executor, not a new equivalent formulation of RH.

The theorem does not construct T. It proves that the correct full-coordinate task is relative: the opposed crossing B need not be positive, but it must be contractively carried by the available same-hand storage A. `lemma:theta-crossing-layer-indefiniteness` shows why no such T exists independently at every fixed sweep layer; it can emerge only after ordered accumulation.

The complete full-line source makes the relative law exact. Simultaneous sign reversal is an actual symmetry of $\mathcal{P}_0$, so the receiver separates into even and odd currents. The same-hand kernel A supplies storage; the opposed-hand kernel B supplies the crossing. Neither $B \ge 0$ nor layer-by-layer positivity is required. The exact requirement is

$$B = A^{\frac{1}{2}} T A^{\frac{1}{2}}, \quad \|T\| \le 1,$$

which says that both polarities of the crossing fit within the same stored current. The two emitted residuals are then

$$\mathcal{R}_\pm = (1 \pm T)^{\frac{1}{2}} A^{\frac{1}{2}}.$$

The exact Volterra reduction in [41] has the causal form forced by this obstruction. Each positive-side theta mode first factors through the common second-order law

$$\varphi_{n(t)} = \left(\partial_t^2 - \frac{1}{4}\right)\left(e^{\frac{t}{2}} e^{-\pi n^2 e^{2t}}\right),$$

and integration along the common ray produces a boundary term plus a Volterra tail. On the paper's Green-minimizer trace quotient, the resulting feature form is a signed Krein square

$$D_{\text{trace}} = G_+^* G_+ - G_-^* G_-.$$

Its positive ordered-execution lift is therefore exactly a source-compatible contraction

$$G_- = T G_+, \quad \|T\| \le 1.$$

In holonic terms, G_+ is available storage, G_- is the opposed open crossing carried by the sweep, and T is the lawful transport which closes that crossing without erasing its hand.

This substantially narrows the executor but does not import a proof from the Volterra paper. Its normalized trace quotient still owes a quotient-to-original-Weyl identification. The completed theta bridge above supplies the direct RH-facing map from the **original** phase current to $\mathcal{M}$, but it does not identify that original current with the normalized Volterra quotient. A proof may therefore either construct the quotient lift and the contraction together, or derive $\mathcal{R}$ directly on $\mathcal{V}_{\text{exp}}$ and bypass the stronger full-coordinate route.

XIII.8.8 Boundary unitarity is not causal closure

The functional equation makes the boundary response deceptively complete: for real u,

$$\left|\Theta_{\omega(u)}\right| = 1, \quad \Theta_{\omega(u)}\Theta_{\omega(-u)} = 1.$$

Thus multiplication by Θ_ω, followed by Mellin reflection, is always a unitary involution. No hypothesis about the zeros is needed. What RH adds is not norm preservation on a frozen boundary. It says that this all-pass response is analytic in the upper half-plane, preserves the future Hardy half-space, and is therefore the boundary face of the explicit one-sided Euler–Gamma source. In the laboratory's language: a reversible boundary face need not yet have a lawful causal interior.

Suzuki's Appendix A gives that distinction a scalar source coordinate [40]. Integrate the source kernel once:

$$h_\omega^{\langle 1\rangle}(x) = \int_1^x \sqrt{\frac{y}{x}}\, h_{\omega(y)} \frac{\mathrm{d}y}{y},$$

and define its accumulated logarithmic current

$$J_{\omega(x)} = \sqrt{x}\, h_\omega^{\langle 1\rangle}(x) = \int_1^x h_{\omega(y)} \frac{\mathrm{d}y}{\sqrt{y}}.$$

Then

$$R_{\omega(x)} = x^{-\frac{1}{2}} - h_\omega^{\langle 1\rangle}(x) = x^{-\frac{1}{2}}\left(1 - J_{\omega(x)}\right)$$

is the exact causal return residual, and

$$\mathrm{RH} \leftrightarrow \int_1^\infty \left|1 - J_{\omega(x)}\right|^2 \frac{\mathrm{d}x}{x} < \infty \quad \text{for every } 0 < \omega < \frac{1}{2}. \quad \text{(CAUSAL TAIL CRITERION)}$$

The factor $x^{-\frac{1}{2}}$ is not a preferred positive orientation. It is the half-density that changes ordinary source amplitude $L^2(\mathrm{d}x)$ into scale-invariant logarithmic amplitude $L^2\left(\mathrm{d}\frac{x}{x}\right)$. This is a precise appearance of the critical half in the multiplicative receiver chart.

In logarithmic time this is a literal causal integer current. Set

$$\tau = \log x, \quad j_{\omega(\tau)} = J_{\omega(e^\tau)}, \quad w_{\omega(n)} = \frac{c_{\omega(n)}}{\sqrt{n}},$$

and

$$\varphi_{\omega(u)} = e^{-\frac{u}{2}} g_{\omega(e^{-u})} \mathbf{1}_{u>0}, \quad \Phi_{\omega(u)} = \int_0^u \varphi_{\omega(v)}\, \mathrm{d}v.$$

Direct substitution into the finite source sum gives

$$j_{\omega'}(\tau) = \sum_{\log n \le \tau} w_{\omega(n)} \varphi_{\omega(\tau - \log n)} \quad \text{(LOG-TIME CURRENT)}$$

and

$$j_{\omega(\tau)} = \sum_{\log n \le \tau} w_{\omega(n)} \Phi_{\omega(\tau - \log n)}. \quad \text{(LOG-TIME STANDING)}$$

Thus the arithmetic source is the atomic measure $\sum_n w_{\omega(n)}\delta_{\log n}$, every integer enters at its exact logarithmic time, and the archimedean face is the common response kernel transported from that event. Prime axes, prime powers, and composites are not separately classified; their multiplicative construction is already carried by $w_{\omega(n)}$. The causal criterion becomes simply

$$1 - j_\omega \in L^2(0, \infty; \mathrm{d}\tau).$$

This is the complete discrete-to-continuous bridge that the local aperture picture did not itself show.

Suzuki's weighted-summatory theorem gives a second exact source statement: RH is equivalent to eventual nonnegativity of $h^{\langle 1\rangle}_{\omega}$ for every $0 < \omega < \frac{1}{2}$ [42]. This does **not** require the causal residual R_ω to be nonnegative. The two signs have different identities: one orients the accumulated source current; the other measures its difference from unit closure.

XIII.8.9 Causal time parity makes the source obligation compositional

Definition 66 — Causal time parity is opposed incidence on one shared time face Let $(C_*(X;A),\partial)$ be an oriented occurrence complex with coefficients in an abelian group A. A causal event grain from a receiving cut Σ_k to a later cut Σ_{k+1} is an oriented chain E_k whose boundary has the form

$$\partial E_k = \Sigma_{k+1} - \Sigma_k + \Gamma_k, \quad \text{(ONE CAUSAL GRAIN)}$$

where Γ_k is its exposed lateral world boundary.

A composable family has **causal time parity** when the output cut of E_k and the input cut of E_{k+1} are the same occurrence-chain with their induced boundary orientations opposed. Consequently,

$$\partial\left(\sum_{k=m}^{n-1} E_k\right) = \Sigma_n - \Sigma_m + \sum_{k=m}^{n-1} \Gamma_k. \quad \text{(CAUSAL SWEEP)}$$

Every interior time face therefore cancels from the boundary of the composed sweep while remaining part of its causal interior.

The same law applies between adjacent chain degrees: an interior $(r-1)$-face shared by oriented r-cells occurs once with each induced hand. It is the incidence content of $\partial_{r-1}\partial_r = 0$ and the discrete boundary face of telescoping and the fundamental theorem.

> Boundary. Causal time parity is not an inverse event, a reconstruction of an earlier occurrence, or a claim that a physical process is reversible. It concerns the opposed incidence of one shared face under forward composition. A conserved quantity additionally requires an additive coefficient-valued current or cocycle; parity supplies the cancellation by which that declared quantity composes.

Theorem 67 — Causal parity carries balance, Kirchhoff duality, and phase return Let $B : C_1(X;A) \to C_0(X;A)$ be the oriented incidence of a finite occurrence complex and let $d = B^*$ be its coboundary in a supplied inner-product chart. Let an additive A-valued current on each causal grain have stored cut-value q_k, integrated edge current j_k, and source or exterior return r_k. Discrete Stokes on ("ONE CAUSAL GRAIN") gives

$$q_{k+1} - q_k + Bj_k = r_k. \quad \text{(EVENT BALANCE)}$$

Causal time parity then gives, for every completed order of time,

$$q_n - q_m + B\sum_{k=m}^{n-1} j_k = \sum_{k=m}^{n-1} r_k. \quad \text{(INTERVAL BALANCE)}$$

Thus zero instantaneous incidence is not required: a nonzero Bj_k can be carried by storage change. If $q_n = q_m$ and the interval is source-free, the integrated current is a cycle.

For a relative node potential φ, put $v = d\varphi$. Every current cycle $c \in \ker B$ satisfies

$$\langle v, c\rangle = \langle \varphi, Bc\rangle = 0. \quad \text{(KIRCHHOFF LOOP)}$$

More generally, for a connection cochain a and a supplied comparison face Σ,

$$\langle a, \partial\Sigma\rangle = \langle da, \Sigma\rangle. \quad \text{(CURVED LOOP)}$$

The zero-voltage loop is the exact or flat case; nonzero loop return is the curvature or changing interior carried by the spanning face.

At one lossless junction let a_j, b_j be incoming and outgoing amplitudes on branches with positive admittances Y_j. Continuity and oriented current balance are

$$u = a_j + b_j, \quad \sum_j Y_{j(b_j - a_j)} = 0.$$

Hence

$$u = 2\frac{\sum_j Y_j a_j}{\sum_j Y_j}, \quad b_j = 2\frac{\sum_l Y_l a_l}{\sum_l Y_l} - a_j. \quad \text{(JUNCTION RETURN)}$$

The resulting scattering map $S : a \mapsto b$ preserves the Y-weighted flux. With one incoming parent and daughter branches its reflected amplitude is

$$\frac{Y_{\text{parent}} - \sum_d Y_d}{Y_{\text{parent}} + \sum_d Y_d}. \quad \text{(BRANCH REFLECTION)}$$

It vanishes exactly at admittance matching.

Let an edge of exact length ℓ_e carry phase $P_{e(\kappa)} = e^{i\kappa\ell_e}$ and let $P(\kappa)$ be the diagonal propagation law. A standing current is precisely a fixed current of one complete propagation and junction return:

$$SP(\kappa)a = a. \quad \text{(STANDING RETURN)}$$

Along a closed path this is the phase-closure relation

$$\sum_e \kappa_e \ell_e + \sum_j \delta_j = n\Theta,$$

where Θ is one full turn and the δ_j are junction turns. In a uniform branch the least positive closure length is therefore

$$\lambda = \frac{\Theta}{|\kappa|}. \quad \text{(WAVELENGTH)}$$

Finally let $W \geq 0$ be an edge conductance or stiffness and $G \geq 0$ a storage operator. For $\operatorname{Re} z > 0$ define the dynamic incidence law

$$L(z) = d^* W d + zG. \quad \text{(DYNAMIC BODY)}$$

Split its variables into interior and boundary ports and write

$$L(z) = \begin{pmatrix} A(z) & B(z) \\ C(z) & D(z) \end{pmatrix}.$$

Whenever $A(z)$ is invertible, imposing zero interior injection returns the boundary response

$$S_{B(z)} = D(z) - C(z)A(z)^{-1}B(z). \quad \text{(DYNAMIC SHORT)}$$

It is positive real:

$$\operatorname{Re}\langle S_{B(z)} y, y\rangle \geq 0 \quad \text{for } \operatorname{Re} z > 0. \quad \text{(PASSIVE RETURN)}$$

> **Demonstration.** Pairing an additive current with the oriented boundary of one event gives ("EVENT BALANCE"). Summing those identities cancels every intermediate q_k by causal time parity and proves ("INTERVAL BALANCE").

The Kirchhoff identity is the adjoint relation $\langle d\varphi, c\rangle = \langle \varphi, Bc\rangle$. ("CURVED LOOP") is discrete Stokes. It also shows why a loop with changing enclosed flux has a returned interior term rather than violating conservation. At the junction, $b_j = u - a_j$. Substitution into current balance gives $u\sum_j Y_j = 2\sum_j Y_j a_j$, proving ("JUNCTION RETURN") and ("BRANCH REFLECTION"). In matrix form,

$$S = 2\mathbf{1}\frac{Y}{\mathbf{1}^* Y \mathbf{1}} - I,$$

so direct multiplication gives $S^* Y S = Y$. Propagation preserves the same flux when the phases have unit modulus. A standing current is therefore a unit-return eigenvector; multiplying its edge and junction phases around a closed path gives ("WAVELENGTH").

For the dynamic body,

$$\operatorname{Re}\langle L(z)x, x\rangle = \left\|W^{\frac{1}{2}} dx\right\|^2 + \operatorname{Re}(z)\left\|G^{\frac{1}{2}} x\right\|^2 \geq 0.$$

Given boundary value y, put $x = -A(z)^{-1}B(z)y$. Then $L(z)(x \oplus y) = 0 \oplus S_{B(z)}y$, so

$$\operatorname{Re}\langle S_{B(z)}y, y\rangle = \operatorname{Re}\langle L(z)(x \oplus y), x \oplus y\rangle \geq 0.$$

This proves ("PASSIVE RETURN").

Boundary. The oriented complex supplies legal composition and cancellation; it does not select A, W, G, Y, a metric, or a coefficient system. Those are parameters of the represented ecology. The theorem applies equally when the coefficients represent electrical charge, chemical storage, mechanical action, probability amplitude, or an analytic boundary field, provided the displayed hypotheses are actually supplied. A lossy junction gives a weighted contraction rather than a flux isometry. A globally nonexact connection may carry nontrivial loop holonomy even where its local curvature vanishes.

The missing source sign can now be stated as an ordinary passive world-tube law rather than as a scalar inequality detached from its causal interior. For an input u received up to logarithmic time τ, the completed source convolution carries the exact future-response state

$$x_{\tau(r)} = \int_0^\tau \psi_{\omega(\tau+r-s)} u(s)\, \mathrm{d}s, \quad r > 0. \quad \text{(CARRIED SOURCE STATE)}$$

This is not source history retained for its own sake. It is precisely the part of the past current which can still change a later output after the cut. When two time intervals are glued, x_τ is the same occurrence as the first interval's output state and the second interval's input state. Its two induced signs are causal time parity.

A source-derived proof of RH may therefore construct a nonnegative storage form $\mathcal{H}_\omega$ on this carried state and a nonnegative dissipation density $\mathcal{D}_\omega$ satisfying, for every finite interval $[\tau_1, \tau_2]$,

$$\mathcal{H}_{\omega\left(x_{\tau_2}\right)} - \mathcal{H}_{\omega\left(x_{\tau_1}\right)} + \int_{\tau_1}^{\tau_2} \left(|y(r)|^2 + \mathcal{D}_{\omega(r)}\right) \mathrm{d}r = \int_{\tau_1}^{\tau_2} |u(r)|^2\, \mathrm{d}r. \quad \text{(COMPLETED DISSIPATION IDENTITY)}$$

Here $y = \mathcal{V}_\omega u$ is the emitted completed response. Summing this identity over adjacent source cells cancels every intermediate storage term. With the rest state at the left boundary and nonnegative final storage, it gives

$$\left\|\mathcal{V}_{\omega,\tau} u\right\| \leq \|u\|$$

for every τ. The shared causal-defect theorem then gives both polarities $I \pm \mathcal{H}_{\omega,\tau} \geq 0$, and the established source equivalences give RH.

Conditional corollary 68 — A source-derived passive causal realization of the completed return proves RH Put

$$\Xi(z) = \xi\left(\frac{1}{2} + z\right), \quad F(z) = \frac{\Xi'(z)}{\Xi(z)}.$$

Suppose there is a directed family of finite source-derived occurrence complexes X_α with coboundaries d_α, nonnegative constitutive operators W_α, G_α, and dynamic bodies

$$L_{\alpha(z)} = d_\alpha^* W_\alpha d_\alpha + z G_\alpha, \quad \operatorname{Re} z > 0.$$

Let $S_{\alpha(z)}$ be their interior-shorted boundary responses from ("DYNAMIC SHORT") and let y_α be normalized scalar receiver ports. If

$$F_{\alpha(z)} = \langle S_{\alpha(z)} y_\alpha, y_\alpha \rangle \to F(z) \quad \text{(SOURCE RESPONSE LIMIT)}$$

locally uniformly on the right half-plane, then the Riemann Hypothesis holds.

Equivalently, it is sufficient to construct the same family directly as a causal storage/dissipation identity for the completed source convolution, or as one source-native Gram factorization of the Pick kernel

$$K_{F(p,q)} = \frac{F(p) + \overline{F(q)}}{p + \overline{q}}. \quad \text{(COMPLETED PICK RETURN)}$$

Through the completed theta bridge, this is exactly a factorization of the seam phase current $\mathcal{P}_0$ on the bilateral-exponential receiver span, not positivity of each isolated crossing layer.

Demonstration. The causal-parity Kirchhoff theorem makes every F_α positive real:

$$\operatorname{Re} F_{\alpha(z)} \geq 0 \quad \text{for } \operatorname{Re} z > 0.$$

Local uniform convergence preserves holomorphy and the nonnegative real part, so F is positive real. The completed-return-flux theorem makes this equivalent to RH.

A holomorphic function is positive real on the right half-plane exactly when its Pick kernel ("COMPLETED PICK RETURN") is positive semidefinite. The completed-theta phase-current theorem gives

$$\Xi(p)\overline{\Xi(q)} K_{F(p,q)} = 4 \int_{\mathbb{R}} \int_{\mathbb{R}} e^{pa} \mathcal{P}_0(a,b) e^{\overline{q}b} \, \mathrm{d}a \, \mathrm{d}b.$$

Multiplication by the nonzero analytic receiver factors preserves kernel sign away from zeros, and a right-half-plane zero is itself a failure of the positive-real law. Hence a source-native Gram or dissipative realization of this pullback supplies the same conclusion.

Boundary. Defining W_α, G_α, or a storage form by first assuming $\operatorname{Re} F \geq 0$ would be circular. The family must be constructed from the positive theta source together with the exact Euler, Gamma, endpoint, and causal-boundary relations already present before zero locations are known. Causal time parity proves that internal time faces cancel and that a local dissipation identity composes; it does not assign the required nonnegative constitutive operators by itself.

This identifies four previously separated proof faces:

- $F = \frac{\Xi'}{\Xi}$ is the completed boundary impedance;
- Θ_ω is its finite all-pass scattering face;
- $\mathcal{V}_\omega$ is its explicit one-sided causal source body; and
- $\mathcal{P}_0$ is its infinitesimal theta port-current kernel.

They are one realization problem. The source-native Gram map

$$4\langle g, \mathcal{P}_0 g \rangle = \|\mathcal{R} g\|^2$$

is the infinitesimal dissipation face of ("COMPLETED DISSIPATION IDENTITY"). The connected Hankel moments are its coefficient receiver, and the conditioned support short is its finite boundary receiver. No additional positivity mechanism is required.

Theorem 69 — The completed zeta source has an exact shift state and a local storage criterion Fix $0 < \omega < \frac{1}{2}$ and let ψ_ω be the completed causal source current from `theorem:completed-source-volterra-frame-defect`. On $\mathcal{X} = L^2(0, \infty)$ let

$$(T_\tau x)(r) = x(r + \tau), \quad Ax = \partial_r x, \quad B_\omega u = \psi_\omega u, \quad Cx = x(0)$$

on the natural smooth reachable core. For a smooth compactly supported input u, define

$$x_{\tau(r)} = \int_0^\tau \psi_{\omega(\tau + r - s)} u(s) \, \mathrm{d}s. \quad \text{(SOURCE SHIFT STATE)}$$

Then

$$\dot{x}_\tau = Ax_\tau + B_\omega u(\tau), \quad y(\tau) = Cx_\tau = (\psi_\omega * u)(\tau), \quad x_0 = 0. \quad \text{(SOURCE STATE LAW)}$$

Its transfer on $\operatorname{Re} p > 0$ is

$$C(pI - A)^{-1} B_\omega = \hat{\psi}_{\omega(p)} = \frac{\Xi(\omega - p)}{\Xi(\omega + p)}. \quad \text{(COMPLETED TRANSFER)}$$

Suppose there is a nonnegative self-adjoint storage operator P_ω whose form domain contains the reachable core and on which the block form

$$\begin{pmatrix} A^* P_\omega + P_\omega A + C^* C & P_\omega B_\omega \\ B_\omega^* P_\omega & -I \end{pmatrix} \le 0 \quad \text{(SOURCE STORAGE INEQUALITY)}$$

is well-defined. Then every finite causal cut is contractive:

$$\int_0^\tau |y(r)|^2 \, \mathrm{d}r \le \int_0^\tau |u(r)|^2 \, \mathrm{d}r \quad \text{for every } \tau > 0. \quad \text{(FINITE SOURCE PASSIVITY)}$$

Equivalently, the source Hankel cuts obey both polarities $I \pm \mathcal{H}_{\omega,a} \ge 0$.

If such a source-native P_ω is constructed for every $0 < \omega < \frac{1}{2}$, the Riemann Hypothesis follows.

Demonstration. The variation-of-constants formula for the left-shift semigroup is

$$x_\tau = \int_0^\tau T_{\tau - s} B_\omega u(s) \, \mathrm{d}s,$$

which is (“SOURCE SHIFT STATE”). Differentiating on the smooth core proves (“SOURCE STATE LAW”). Since $CT_t B_\omega = \psi_{\omega(t)}$, Laplace transformation gives (“COMPLETED TRANSFER”).
Put $\mathcal{H}_{\omega(x)} = \langle P_\omega x, x \rangle$. Along a reachable trajectory,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \mathcal{H}_{\omega(x_\tau)} + |y(\tau)|^2 - |u(\tau)|^2$$

is the quadratic form of (“SOURCE STORAGE INEQUALITY”) on $(x_\tau, u(\tau))$, and is therefore nonpositive. Integration from 0 to τ, together with $x_0 = 0$ and $P_\omega \ge 0$, proves (“FINITE SOURCE PASSIVITY”). The shared causal-defect factorization turns contraction of the Volterra source into positivity of both Hankel polarities. Their established equivalence for every $0 < \omega < \frac{1}{2}$ then gives RH.

Boundary. The theorem identifies a local source equation, not a way to define P_ω from the already-assumed norm of the transfer. A proof must construct its kernel or feature map directly from the completed theta–Euler–Gamma source and verify the form domains, integer-event seams, and limit in ω. The identity storage $P = I$ does not solve the problem: the source injection contributes an uncompensated positive rank-one term.

The theorem also removes the last abstraction from the word **storage**. On $L^2(0, \infty)$ the causal state is driven by the left-shift generator $A = \partial_r$, the explicit source injection $B_\omega u = \psi_\omega u$, and the boundary receiver $Cx = x(0)$. Therefore the local proof-bearing object is one positive operator P_ω satisfying the source-side inequality

$$\begin{pmatrix} A^* P_\omega + P_\omega A + C^* C & P_\omega B_\omega \\ B_\omega^* P_\omega & -I \end{pmatrix} \le 0. \quad \text{(LOCAL COMPLETED STORAGE)}$$

This is an operator equation in the known source coordinates. It can be attacked through a kernel or feature representation of P_ω, and its failure can be localized to a smooth interval or integer-event seam. Once proved, causal time parity glues the local inequalities and removes every intermediate storage cut automatically.

The proof construction has consequently become local in law without becoming finite in scope. It is enough to derive one symbolic theta–Euler–Gamma storage update which remains nonnegative when:

- a smooth logarithmic interval is extended;
- an integer source cell enters at $\tau = \log n$;

- the source cell shares prime axes with earlier cells; and
- a completed interior is shorted into its returned boundary response.

Causal time parity then performs the induction over all completed cells. What must not be done is define the storage by the already-assumed contractivity of $\mathcal{V}_\omega$ or positivity of F; the storage and dissipation features must be exposed directly by the source kernel.

XIII.8.10 An off-seam zero is a supercritical causal mode

The scalar residual also shows exactly how failure enters. Let

$$\Xi(q_0) = 0, \quad q_0 = \alpha + i\gamma, \quad \alpha > 0.$$

For any $0 < \omega < \alpha$ away from isolated numerator cancellations, Θ_ω has a pole at

$$z_0 = -\gamma + i(\alpha - \omega).$$

Moving the inverse Mellin contour from the source convergence region to the real boundary crosses this pole. Its residue contributes a nonzero multiple of

$$x^{-\frac{1}{2}-iz_0} = x^{\alpha-\omega-\frac{1}{2}+i\gamma}$$

to the integrated causal source; repeated order only multiplies this by a polynomial in $\log x$. The squared radial factor is

$$x^{2(\alpha-\omega)-1},$$

which is not integrable against $\mathrm{d}x$ because $\alpha > \omega$. An off-critical zero therefore does not destroy the boundary all-pass law. It creates a causal mode whose growth outruns the receiver's half-density. This is the direct source meaning of the upper-half-plane pole.

XIII.8.11 The first integer cell crosses, but does not close the tail

The aperture induction now has a rigorous beginning. For $1 < a^2 < 2$, only the $n = 1$ source cell is active. Put $L = \log a$. On the effective interval $\left(\frac{1}{a}, a\right)$, the unitary logarithmic lift carries its kernel to

$$\varphi_{\omega(t+s)} = e^{-\frac{t+s}{2}} g_{\omega(e^{-(t+s)})} \mathbf{1}_{t+s>0}, \quad -L < t, s < L.$$

Reflection turns this into a truncated convolution, so Young's inequality gives

$$\|\boldsymbol{H}_{\omega,a}\| \le \int_0^{2L} \left| e^{-\frac{u}{2}} g_{\omega(e^{-u})} \right| \mathrm{d}u.$$

The right side tends to zero like a symbolic constant times L^ω. Hence every fixed $\omega > 0$ has a nonempty aperture interval immediately above $a = 1$ on which both polarities are strictly positive. The rest cut really can be crossed without assuming RH.

This does not prove contraction on the whole first integer interval. More importantly, no finite first cell can decide whether $1 - J_\omega$ has finite energy over the complete logarithmic ray. The finite Schur law and the causal tail criterion are therefore complementary coordinates of the same remaining obligation: control both orientations at every successor shell strongly enough that no supercritical residue can accumulate at infinity.

XIII.8.12 Divisor transport must diffuse; max-flow alone does not prove the sign

Theorem 70 — A divisor-supported source transport must be diffuse and does not erase the global cut Fix $0 < \omega < \frac{1}{2}$. Write

$$G_{\omega(r)} = g_\omega^{\langle 1 \rangle}(r),$$

and let y_ω be its unique zero in $(0,1)$. Thus

$$G_{\omega(r)} < 0 \quad \text{for } 0 < r < y_\omega, \quad G_{\omega(r)} > 0 \quad \text{for } y_\omega < r < 1. \quad \text{(SOURCE SIGN SEAM)}$$

For a finite scale $x > \frac{1}{y_\omega}$, define

$$\mathcal{N}^-_{\omega,x} = \Big\{ n \in \mathbb{N} : n \le x, \frac{n}{x} < y_\omega \Big\},$$

$$\mathcal{N}^+_{\omega,x} = \Big\{ m \in \mathbb{N} : m \le x, \frac{m}{x} > y_\omega \Big\},$$

with demands and capacities

$$d_{\omega,x}(n) = c_{\omega(n)} \Big| G_{\omega(\frac{n}{x})} \Big|, \quad n \in \mathcal{N}^-_{\omega,x},$$

$$b_{\omega,x}(m) = c_{\omega(m)} G_{\omega(\frac{m}{x})}, \quad m \in \mathcal{N}^+_{\omega,x}. \quad \text{(SIGNED SOURCE MASSES)}$$

A divisor-supported transport is a nonnegative population $\Pi_{\omega,x}(n,m)$ which vanishes unless $n \mid m$ and satisfies

$$\sum_{m \in \mathcal{N}^+} \Pi_{\omega,x}(n,m) \ge d_{\omega,x}(n), \quad \text{(DEMAND)}$$

$$\sum_{n \in \mathcal{N}^-} \Pi_{\omega,x}(n,m) \le b_{\omega,x}(m). \quad \text{(CAPACITY)}$$

Such a transport exists if and only if every subset $S \subseteq \mathcal{N}^-_{\omega,x}$ satisfies

$$\sum_{n \in S} d_{\omega,x}(n) \le \sum_{m \in \Gamma_{x(S)}} b_{\omega,x}(m), \quad \Gamma_{x(S)} = \big\{ m \in \mathcal{N}^+_{\omega,x} : \text{ some } n \in S \text{ divides } m \big\}. \quad \text{(DIVISOR HALL CUT)}$$

The full Hall cut does not reduce the RH sign. Since $1 \in \mathcal{N}^-_{\omega,x}$ and 1 divides every positive cell,

$$\Gamma_{x(\mathcal{N}^-_{\omega,x})} = \mathcal{N}^+_{\omega,x}.$$

Hence its Hall inequality is exactly

$$\sum_{m \in \mathcal{N}^+} c_{\omega(m)} G_{\omega(\frac{m}{x})} \ge \sum_{n \in \mathcal{N}^-} c_{\omega(n)} \Big| G_{\omega(\frac{n}{x})} \Big|,$$

or equivalently

$$h^{\langle 1 \rangle}_\omega(x) \ge 0. \quad \text{(GLOBAL SOURCE CUT)}$$

A divisor-flow certificate is therefore a sufficient geometric realization of the source sign, but the abstract existence problem is a stronger condition which still contains the original RH-equivalent cut.

Moreover, no bounded-degree local transport can work. The exact small-r asymptotic is

$$G_{\omega(r)} = -A_\omega r^{\omega-1} + O\Big(r^{-\frac{1}{2}}\Big),$$

where

$$A_\omega = \frac{4\omega\pi^\omega}{(1-2\omega)\Gamma(\omega)} \mathrm{B}\Big(\frac{3-2\omega}{2}, \omega\Big) > 0. \quad \text{(OLD-CELL GROWTH)}$$

Put

$$M_\omega = \max_{y_\omega \le r \le 1} G_{\omega(r)} < \infty.$$

If a transport carries the demand of one fixed integer n, then

$$\operatorname{card}\{m \in \mathcal{N}^+_{\omega,x} : \Pi_{\omega,x}(n,m) > 0\}$$

is bounded below by

$$\left[\frac{A_\omega c_{\omega(n)} n^{\omega-1}}{M_\omega} + o(1)\right] x^{1-2\omega} \quad \text{as } x \to \infty. \quad \text{(DIFFUSE DEGREE)}$$

The required degree diverges because $1 - 2\omega > 0$.

This is not an individual reachability obstruction. Define the total positive descendant capacity

$$Z_{\omega,x}(n) = \sum_{m \in \mathcal{N}^+_{\omega,x};\, n|m} b_{\omega,x}(m). \quad \text{(DESCENDANT CAPACITY)}$$

For every fixed n, choose $y_\omega < \eta_0 < \eta_1 < 1$. Prime descendants $m = np$ with

$$\eta_0 \frac{x}{n} \le p \le \eta_1 \frac{x}{n}$$

give, by the prime number theorem,

$$Z_{\omega,x}(n) \ge C_{\omega,n,\eta_0,\eta_1} \frac{x^{1+\omega}}{n^{1+\omega} \log x}$$

for all sufficiently large x, with a positive constant C. Thus

$$\frac{d_{\omega,x}(n)}{Z_{\omega,x}(n)} = O_{\omega,n}\left(\frac{\log x}{x^{2\omega}}\right) \to 0. \quad \text{(INDIVIDUAL CAPACITY SURPLUS)}$$

Old cells have abundant descendants individually; the unresolved question is simultaneous sharing of those descendants.

One canonical diffuse candidate makes that collision explicit. When $Z_{\omega,x}(n) > 0$, put

$$\Pi^{\propto}_{\omega,x}(n,m) = \begin{cases} d_{\omega,x}(n) \frac{b_{\omega,x}(m)}{Z_{\omega,x}(n)} & n \mid m \\ 0 & \text{otherwise.} \end{cases} \quad \text{(PROPORTIONAL DESCENDANT FLOW)}$$

Every row then carries its exact demand. Its load at a positive cell m is

$$b_{\omega,x}(m) C_{\omega,x}(m),$$

where

$$C_{\omega,x}(m) = \sum_{n \in \mathcal{N}^-_{\omega,x};\, n|m} \frac{d_{\omega,x}(n)}{Z_{\omega,x}(n)}. \quad \text{(DIVISOR CONGESTION)}$$

Therefore this explicit transport succeeds exactly when

$$C_{\omega,x}(m) \le 1 \quad \text{for every } m \in \mathcal{N}^+_{\omega,x}. \quad \text{(CONGESTION LAW)}$$

> **Demonstration.** The sign seam and the asymptotic formula follow from Suzuki's explicit incomplete-Beta expression for G_ω. For $0 < \omega < \frac{1}{2}$, its $r^{\omega-1}$ term dominates the $r^{-\frac{1}{2}}$ term at zero; changing the sign of $2\omega - 1$ gives the positive constant A_ω.
> Attach a source to a sink by an edge exactly when the source divides the sink, add a supersource with edge capacity $d_{\omega,x}(n)$ to every negative cell, and add edges of capacity $b_{\omega,x}(m)$ from positive cells to a supersink. Infinite

capacities on divisor edges turn the finite max-flow/min-cut theorem into ("DIVISOR HALL CUT"). The cell 1 proves the full-neighborhood identity, and separating the positive and negative summands in the definition of $h_\omega^{\langle 1\rangle}$ proves ("GLOBAL SOURCE CUT").

For fixed n, ("OLD-CELL GROWTH") gives

$$d_{\omega,x}(n) = \left[A_\omega c_{\omega(n)} n^{\omega-1} + o(1)\right] x^{1-\omega}.$$

Every positive cell has

$$b_{\omega,x}(m) \le M_\omega c_{\omega(m)} \le M_\omega x^\omega.$$

Dividing the row demand by this maximum per-sink capacity proves ("DIFFUSE DEGREE").

If p is a sufficiently large prime not dividing fixed n, then

$$c_{\omega(np)} = c_{\omega(n)} p^{\omega(1-p^{-2\omega})}.$$

On the closed receiver band $[\eta_0, \eta_1]$, G_ω has a positive minimum. Summing the resulting capacities over primes in the displayed interval and applying the prime number theorem proves ("INDIVIDUAL CAPACITY SURPLUS").

Finally, summing ("PROPORTIONAL DESCENDANT FLOW") over m gives $d_{\omega,x}(n)$. Summing it over divisors n of a fixed m gives $b_{\omega,x}(m)C_{\omega,x}(m)$, proving the equivalence with ("CONGESTION LAW").

Boundary. This theorem invalidates a bounded local pairing of old cells with one or finitely many newer multiples. The required transport is genuinely diffuse, with degree at least of order $x^{1-2\omega}$ for every fixed old cell.

It does not prove RH and it does not prove that divisor-supported transport exists. The full Hall cut is exactly Suzuki's RH-equivalent source sign, so merely invoking max-flow would rename the obstruction. The material reduction supplied here is narrower: individual reachability is not the problem, and the proportional construction isolates all remaining failure in the explicit divisor congestion $C_{\omega,x}(m)$.

Proving ("CONGESTION LAW") uniformly would be a valid source-derived proof. A symbolic violation would refute this proportional flow but not every divisor flow or RH. Any continuation must therefore analyze the displayed congestion itself; it may not return to an unspecified "uniform tail-energy inequality."

The one-sign seam of $G_\omega = g_\omega^{\langle 1\rangle}$ permits an exact transport question. At scale x, old cells $\frac{n}{x} < y_\omega$ carry negative demand

$$d_{\omega,x}(n) = c_{\omega(n)} \left|G_{\omega(\frac{n}{x})}\right|,$$

while cells $\frac{m}{x} > y_\omega$ carry positive capacity

$$b_{\omega,x}(m) = c_{\omega(m)} G_{\omega(\frac{m}{x})}.$$

Joining n to m only when $n \mid m$ turns the proposal into a finite capacitated divisor graph.

The max-flow criterion is exact, but it exposes an important limitation. Because 1 is an old cell for large x and divides every positive cell, the Hall cut containing all negative cells is precisely

$$\sum_m b_{\omega,x}(m) \ge \sum_n d_{\omega,x}(n),$$

which is just $h_\omega^{\langle 1\rangle}(x) \ge 0$. Abstract flow existence therefore does not reduce the RH-equivalent sign; it strengthens it with additional subset cuts.

The local picture is nevertheless decided. For fixed n,

$$d_{\omega,x}(n) \sim A_\omega c_{\omega(n)} n^{\omega-1} x^{1-\omega},$$

whereas one positive descendant carries at most order x^ω. Any valid row must consequently use at least order

$$x^{1-2\omega}$$

different descendants. A one-to-one pairing, a bounded prime path, or any fixed-degree local routing is impossible.

Conversely, prime descendants in any closed sub-band of $(y_\omega x, x)$ give every fixed old cell much more total capacity than its own demand. The obstruction is therefore not reachability. It is collision: many old cells attempt to use the same positive descendants.

The proportional diffuse construction isolates that collision in one explicit divisor sum. If

$$Z_{\omega,x}(n) = \sum_{\frac{m}{x} > y_\omega;\, n|m} b_{\omega,x}(m),$$

distribute row n proportionally across those descendants. Its load at m is lawful exactly when

$$C_{\omega,x}(m) = \sum_{\frac{n}{x} < y_\omega;\, n|m} \frac{d_{\omega,x}(n)}{Z_{\omega,x}(n)} \le 1. \quad \text{(EXACT DIVISOR CONGESTION)}$$

This is the first nonredundant question produced by the transport idea. It is an explicit finite divisor-incidence inequality. Proving it uniformly would construct the source sign; disproving it would reject this particular proportional flow without rejecting RH.

XIII.9 Organize the successor's coherence as a formulation correspondence

Take as local generators the transformations already present in the completed relation:

- logarithmic lift and exponential return;
- Gaussian width u and Fourier reciprocal $u \to -u$;
- Poisson summation and Mellin transport;
- completed involution J;
- Euler-place admission $S \to S \cup \{p\}$; and
- support-aperture inclusion $R \to R'$.

For every generator, name its domain, target fiber, anchor into the classical test space, adjoint, response defect, and composition law. Then compute the curvature of every smallest available comparison cell. A closed cell establishes path independence for that local transport; an OPEN cell localizes the missing completion or boundary term.

This organization is secondary to the induction identity. Its decisive theorem is that the symbolic successor is independent of the order in which commuting prime places are charted, and that receiver recharts carry the same anchored form. It need not enumerate all formulations or replace the successor proof with compactness.

XIII.10 Express the aperture connection in the native semilocal chart

The formula

$$\kappa_{S,p}(f) = \mathrm{Tr}\Big(\Theta_{S(f)}(\Pi_p - P_S)\Theta^*_{S(f)}\Big)$$

is exact, but still expressed through the transported projection. The next derivation should seek a native kernel or prolate-coordinate expression for $A^{-1}B$, with declared Schatten and domain control. The purpose is not to force a sign; it is to make the coupling analyzable without an implicit coordinate reconstruction. This κ term is one component of the connection above, not the entire connection.

XIII.11 Join place growth to support growth

The place maps compose exactly, while the support apertures form an infinite-dimensional filtration. A useful theorem must be uniform in both:

Place axis	bounded Euler successor $J_{S,p}$, metric rebase, and endpoint coherence;

Support axis	convolution growth, continuously changing archimedean face, and discrete admissions $m \log p \leq R$;
Required bridge	a uniform completed-return comparison which remains valid as both axes grow and which does not define its positivity from Q_W itself.

XIII.12 Sixth: use exact finite probes to disprove bad conjectures early

The finite Guinand–Weil dictionary and the laboratory's directed-interval instruments can test bounded projections of any proposed connection law. A negative certified probe can refute an overstrong local monotonicity or domination claim. Passing finitely many probes cannot prove the universal theorem.

The useful experimental object is therefore a complete receipt:

$$\left(f, S, R, G_p, \Pi_p, W_p, \kappa, D_S, D_{S\cup\{p\}}\right)$$

with exact or certified interval arithmetic and no floating scalar standing in for the relation. This is a small analytical instrument, not an engine feature and not a scale campaign.

XIII.13 A proof receiver is a joint face, not a favorable view

Definition 71 — Co-present receiver atlas, joint face, and receiver grain Fix one actual event incidence e and its participating construction X_e. Let

$$q_i : X_e \to Y_i, \quad i \in I_e,$$

be its contemporary receiver maps. The **joint receiver face** is the actual image

$$\mathcal{J}_e = \operatorname{im}(q_e), \quad q_e = (q_i)_{i\in I_e} : X_e \to \prod_{i\in I_e} Y_i. \quad \text{(JOINT FACE)}$$

Thus $\mathcal{J}_e$ contains only tuples which arise from one occurrence of X_e; it is not completed to the Cartesian product of independently selectable receiver values. The family is **entangled at** e whenever this image does not factor as the product of the individual images.

When receiver overlaps carry partial transition maps

$$g_{ij} : Y_i|_{U_{ij}} \to Y_j|_{U_{ji}},$$

the compatible interior presented by the atlas is the fibered relation

$$\operatorname{Int}(\mathcal{J}_e) = \left\{(y_i) \in \mathcal{J}_e : g_{ij}(y_i) = y_j \ \text{ on every admitted overlap}\right\}. \quad \text{(COMPATIBLE INTERIOR)}$$

It is an available interior of the contemporary faces, not an ambient volume assumed before them.

A finite **receiver grain** $\mathcal{P}_i$ is a partition or cover of the distinctions in Y_i. Its center is a supplied pivot section c_i, not an absolute origin, and its aspect is a typed comparison of two receiver axes. Two occurrences are indistinguishable at this grain when their faces occupy one common aperture and retain one declared consequence signature. A difference is **tolerated** precisely while every realization in that unresolved fiber preserves the currently declared consequence and overlap relations.

One event successor is supplied jointly:

$$\mathcal{J}_e^+ = \Lambda_{e\left(\mathcal{J}_e;(q_i,\mathcal{P}_i,c_i)_{i\in I_e}\right)}. \quad \text{(JOINT SUCCESSOR)}$$

Receiver-local continuation may be evaluated independently after this co-present relation is formed. Evaluation order across the finite faces is not event chronology.

> Boundary. The index I_e contains receivers participating in this event, not every historical or potential receiver. Finite grain limits what one face can distinguish; it does not determine what exists. Pairwise overlap agreement need not supply a

compatible higher interior unless the transition domains and cocycle or wider descent conditions are also satisfied. A stable nonidentity return may be curvature rather than error.

Theorem 72 — Three marked members fix a projective pivot; the fourth carries the swing Let A, B, C, D be four distinct marked occurrences on one projective line $\mathbb{P}^1(K)$. There is a unique projectivity h sending

$$(A, B, C) \mapsto (\infty, 0, 1).$$

The remaining coordinate is

$$h(D) = \chi(A, B; C, D) = \frac{(C-A)(D-B)}{(C-B)(D-A)}. \quad \text{(SWING)}$$

For every projectivity g defined on the quadruple,

$$\chi(g(A), g(B); g(C), g(D)) = \chi(A, B; C, D). \quad \text{(RECEIVER COVARIANCE)}$$

Hence three marked members establish the local projective gauge and the fourth supplies its first nontrivial projective invariant.

The same statement applies to four lines in a projective pencil through one pivot in any higher-dimensional projective space. Overlapping cells

$$(A, B, C) \to D, \quad (B, C, D) \to E$$

therefore grow a receiver atlas by retaining a pivot triple while adjoining one new swing coordinate at a time.

Every nondegenerate field element x has the exact face

$$x = \chi(\infty, 0; 1, x).$$

If a receiver selects a nonzero quantity m as its unit, then its local coordinate becomes 1 while the transition to a receiver with unit u carries $\frac{m}{u}$. Normalization moves the value into receiver transport; it does not erase its construction or lineage.

Demonstration. The group $\mathrm{PGL}_2(K)$ acts sharply triply transitively on ordered triples of distinct points of $\mathbb{P}^1(K)$, which gives the unique h. Direct substitution into the fractional-linear action proves ("RECEIVER COVARIANCE"). A projective pencil is itself a projective line, so the higher-dimensional local statement is the same theorem. The unit statement follows from the typed rebase law $x_{au}(q) = a^{-1} x_{u(q)}$.

Boundary. Four is the canonical local population for one projective swing, not a universal maximum number of receivers. A receiver may expose several independent marks, while higher-rank or degenerate incidence may require additional overlapping cells. Cross-ratio equality does not identify complete causal interiors.

Theorem 73 — A jointly separating receiver refines a locally tolerated fiber Let $x, x' \in X_e$ be two actual occurrences. Suppose one receiver grain $\mathcal{P}_i$ identifies them, while another receiver grain $\mathcal{P}_j$ separates them:

$$\left[q_{i(x)}\right]_{\mathcal{P}_i} = \left[q_{i(x')}\right]_{\mathcal{P}_i}, \quad \left[q_{j(x)}\right]_{\mathcal{P}_j} \neq \left[q_{j(x')}\right]_{\mathcal{P}_j}.$$

Then the joint receiver face separates x and x'. More generally, refining any one receiver partition can only split, never merge, the fibers of the joint quotient.

Therefore a finite difference may remain **grain-dark** at some receivers while being consequential in the joint atlas. Coarsening is receiver-exact for a declared objective exactly when every pair newly identified by that coarsening retains the same admitted consequence and overlap signature.

Demonstration. Equality in the joint quotient requires equality in every component. The inequality in the j component therefore separates the two joint tuples. If $\mathcal{P}'_j$ refines $\mathcal{P}_j$, every $\mathcal{P}'_j$-class lies inside one $\mathcal{P}_j$-class, so replacing the latter component by the former can only split joint fibers. The final statement is the definition of receiver-exact factorization at the declared objective.

Boundary. More resolution is not automatically more useful. A refinement which exposes no consequential distinction may lawfully depart, while a coarse receiver may remain necessary because it carries a stable invariant or wider aperture unavailable to a fine receiver.

The receiver-atlas formulation does not weaken the inertia boundary above. If several co-present receivers carry one current, their proof-bearing object is the actual joint image of that current, not the Cartesian product of independently selected favorable faces. A negative anchored direction cannot be removed by assigning a coarser grain to the receiver which detects it. Conversely, one receiver may leave a finite change grain-dark while another receiver and the shared cross-ratio carry it exactly.

This gives the finite-core/tail construction a sharper form. A resolved receiver family may distribute finite modes unevenly, but every overlap must transport the same anchored completed response. The unresolved family may be factored only after a receiver-independent lower carrier is proved.

Lemma 74 — The seventh harmonic archimedean tail margin is strictly positive Put

$$\delta_7 = \frac{363}{140} - \gamma - \log(\pi \log 3) - \frac{\log 3}{4} - \frac{\log 2}{\sqrt{2}}.$$

Then

$$\delta_7 > \frac{1}{96} > 0. \quad \text{(SEVENTH-MODE MARGIN)}$$

Demonstration. Every comparison is made between rational endpoints. For $y > 1$, set $z = \frac{y-1}{y+1}$ and use

$$\log y = 2\sum_{k=0}^{N-1} \frac{z^{2k+1}}{2k+1} + R_N, \quad 0 < R_N < \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

This encloses $\log 2$ with $N = 32$ and $\log 3$ with $N = 40$. Machin's identity

$$\pi = 16 \arctan\left(\frac{1}{5}\right) - 4 \arctan\left(\frac{1}{239}\right)$$

is enclosed by the alternating arctangent series after respectively eighteen and six terms. Adjacent dyadic rationals with denominator 2^{96} enclose $\sqrt{2}$. At $n = 64$, Euler–Maclaurin gives

$$H_n - \log n - \frac{1}{2n} + \frac{1}{12n^2} - \frac{1}{120n^4} < \gamma$$

and

$$\gamma < H_n - \log n - \frac{1}{2n} + \frac{1}{12n^2} - \frac{1}{120n^4} + \frac{1}{252n^6}.$$

Applying the logarithm series for fifty-six terms to the rational enclosure of $\pi \log 3$, then substituting every upper endpoint into the subtracted terms, gives by exact integer cross-multiplication $\delta_7 > \frac{1}{96}$.

Boundary. This lemma certifies the already-derived uniform high-mode margin. It does not sign the finite low-mode Schur block, establish the symbolic successor law, or by itself prove RH.

The exact seventh-mode margin supplies such a carrier only for the already reduced high-mode region. It does not sign the remaining finite Schur block. Its methodological contribution is nevertheless decisive: receiver grain can vary without introducing floating-point coordinates, because every constant, projection aperture, and unresolved series tail can remain inside rational endpoint bounds.

XIII.14 Seventh: separate four possible closure routes

There are at least four honest proof presentations:

1. **Domination route.** Prove $Q_{W(f)} \geq \Sigma_{S_f}(f) \geq 0$ on the complete admissible family.
2. **Transport-atlas route.** Prove that every anchored admissible current lies in an orbit of an independently positive base cell and that every transition and loop preserves the completed response by positive congruence.

3. **Spectral route.** Construct a genuine self-adjoint or unitary generator whose complete trace identity reproduces the zeta zero spectrum and every arithmetic/archimedean term.
4. **Squeeze route.** Construct independently positive final-receiver approximants and a noncircular complete error bound tending to zero for every admissible returned current. The resolution theorem above shows that this is not a shortcut around domination: beyond already-controlled finite tails, it owes the same positive completed remainder.

The positive-covariant-monodromy theorem shows that routes 2 and 3 are not competing descriptions. A positive atlas supplies the metric; a returned spectral flow supplies its holonomy and trace. Their coincidence in one constructed body is a direct proof route. The Euler-successor geometry constrains that body but does not yet construct it.

XIV Record map

The integrated argument was reconstructed from the following current laboratory records:

Record family	Contribution used here
Affordance / arc	conductance/probability separation, critical line as unitary scale seam, and the complete Weil closure target;
Adjoint / unitary seam	Mellin involution, explicit-formula convention, triangular probe, and the noncircular positive-carrier obligation;
Support aperture / filtration	support algebra, prime-power admission seams, and infinite-dimensional aperture warning;
Prime / receiver rebase	rejection of the frozen predecessor block and requirement of successor covariance;
Euler difference / metric rebase	exact semilocal successor, metric, projection, adjoint, trace-class, coherence, and normal metric recurrence;
Euler metric / aperture turn	exact connection term, completed defect recurrence, finite no-sign control, and the first unsupported inequality;
Configuration / swing	configuration paths, projective differential, reparameterization invariance, simplicial connection, and the distinction between one-path continuation and curvature;
Counting / three-body frame	numerals as receiver glyphs of discrete local lineages, three independently ordered frames, actual co-presence incidence, triangular return, and independent-rechart covariance;
Causal formulation ecology and $\frac{\pi}{e}$	parametric complex powers, reciprocal Gaussian scale, formal exponential faces, formulation transitions, and receiver-exact compression boundaries;
Formula / theta / explicit formula	bounded embodiment of recurrence, tail transfer, changing support, critical-line wave transport, and exact limitations.
Remainder squeeze / centered prime character	exact boundary convergence, retained logarithmic path, circular phase, hyperbolic normal sheets, local flatness, persistent aperture deficits, the positive-remainder factorization, and the exact finite archimedean tail squeeze.
Coarea / half-Jacobian / carrier FTC	receiver-local direct-integral response, Cartesian–polar control, logarithmic Mellin half-density, Riemannian and Lorentzian hyperbolic separation, operator metric square root, and signed covariant seams.

Record family	Contribution used here
Knot / orbit / monodromy	prime powers as repeated primitive returns, Alexander return determinants, an exact reciprocal-but-hyperbolic counterexample, and the positive covariant metric required to force unitary holonomy.
Causal parity / Kirchhoff return	opposed incidence on shared temporal faces, interval rather than instantaneous balance, passive dynamic shorting, junction scattering, standing phase return, and the source-side storage identity required by the completed impedance.

Superseded static-block records were read as audit evidence but do not govern the argument. Historical visual projections and finite prime-pattern classifications remain useful examples of receiver dependence; they are not premises of the RH line.

XV Conclusion

The laboratory's RH research is no longer best described as a search for a hidden picture among prime points. Its strongest form is an operator geometry of three changing counting frames over an actual incidence atlas. Arithmetic place/valuation, archimedean Fourier–Poisson–Mellin transport, and the receiving current/aperture do not share a master clock. A one-parameter trajectory is a receiver section through their incidence, while the first closed relational datum is their triangular return up to conjugacy.

Prime powers arise twice from the same exact arithmetic structure: as repeated terms in the logarithm of the Euler product and as the Fourier modes of the normal derivative of the Euler successor metric. The centered Mellin involution fixes the critical line and turns returned currents into norm squares there. More invariantly, the critical line is one affine chart of the fixed seam of the completed involution. The parametric Gaussian g_u makes the $\frac{\pi}{e}$ relation operational: exponentiation creates multiplicative scale, Fourier transport sends u to $-u$, π fixes the self-dual normalization, and Mellin transport carries reciprocal scale into the completed reflection. Theta/Poisson transport supplies the archimedean completion. The explicit formula joins the complete prime-power-plus-infinite-place population to the completed zero population.

Those local faces now assemble into one scalar metric before any support split:

$$|\xi(s)|^2 = \frac{1}{4}|s(s-1)|^2\pi^{-\operatorname{Re}(s)}\left|\Gamma\left(\frac{s}{2}\right)\right|^2\prod_p |1-p^{-s}|^{-2}$$

in the Euler half-plane. Its outward normal logarithmic derivative is the complete endpoint–archimedean–prime current. Centered at the seam, that current is $2\operatorname{Re}\left(\frac{\Xi'}{\Xi}\right)$. The RH is exactly its positive-real continuation through the right half-plane, and the Weil form is its boundary Poisson energy. The support effective tensions below are consequently shorts of this same field, not independent sign mechanisms.

Causal time parity now gives that field an exact compositional interior. The log-time source convolution carries only the past response still capable of affecting a later cut. A nonnegative theta–Euler–Gamma storage and dissipation identity on that state would telescope across every smooth interval and every integer admission, making the completed boundary impedance positive real. The seam phase-current Gram, the causal-frame capacity, the connected Hankel laws, and the support effective tensions are therefore receiver faces of one passive realization problem.

The square-axis fold $H(w) = \frac{\Xi(\sqrt{w})}{\Xi(0)}$ gives an algebraic receiver for the same problem. Its impedance $\frac{H'}{H}$ must be Stieltjes. If $\frac{H'}{H} = \sum_{n(-1)}^{n} \mu_n w^n$, the complete proof obligation is the simultaneous positivity of the two symbolic Hankel families

$$[\mu_{i+j}] \geq 0, \quad [\mu_{i+j+1}] \geq 0.$$

The μ_n are obtained recursively from the positive Riemann theta moments and are the connected, logarithmic constituents of the return. This is the current non-enumerative construction target: derive one positive incidence or Gram factorization for both families from the theta/Euler/Gamma source.

The knot/monodromy refinement exposes the decisive distinction inside that join. Centered prime return has multiplier $\lambda_\ell(z) = e^{-\ell z}$, and the completed involution sends it to $\frac{1}{\lambda_\ell(z)}$. Reciprocal monodromy alone permits paired expansion and contraction: the figure-eight knot supplies an exact two-dimensional example. A positive metric transported by the same return does not. Its closed holonomy is unitary, so a generator $\Theta = \frac{1}{2}I + A$ whose independently established trace spectrum is the complete zero set would place every nontrivial zero on the fixed seam.

Adjoining a prime does not add a detached coordinate to an unchanged field. It transports the whole semilocal receiver. The pulled-back Euler metric turns the Sonin aperture, producing a connection term in the positive trace. The completed defect then obeys one exact recurrence:

$$D_{S\cup\{p\}} = D_S - W_p - \kappa_{S,p}.$$

The two signed terms in that recurrence are now independently reduced to endpoint amplitudes. The normalized Euler resolvent folds the complete prime-power series into

$$W_p = \log p \Big(\big\| \mathcal{R}_p L_R f \big\|^2 - \| L_R f \|^2 \Big),$$

while metric-standardizing the successor Sonin projection gives

$$\kappa_{S,p} = \Big\| \Theta_{S(f)} \tilde{P}_p \Big\|_{\mathrm{HS}}^2 - \Big\| \Theta_{S(f)} P_S \Big\|_{\mathrm{HS}}^2.$$

Hence the recurrence is one exact difference $D_{S\cup\{p\}}(f) = \|Xf\|^2 - \|Yf\|^2$. At fixed support its positive face is an independently constructed cross-channel contraction $\Gamma_{S,p,R} : Xf \mapsto Yf$.

The published Sonin theorem closes a different part which the earlier draft had left open. Prime admission is the invertible difference $\Delta_p = I - p^{-\frac{1}{2}} U_p$; its dual Euler population return is $\left(\Delta_p^*\right)^{-1}$; the pairing is exact; and both maps commute with nested support apertures. Consequently the dyadic periodization—with its nonzero kernel—cannot be a restriction of the successor. It is the lower-metric spectral defect face of that successor.

The proof architecture is therefore ordinary induction. Put all conditioned amplitudes supported in $\left[n^{-\frac{1}{2}}, n^{\frac{1}{2}}\right]$ in $\mathcal{T}_n$, and let $P(n)$ assert Weil positivity there. The archimedean theorem supplies $P(2)$. In $\mathcal{T}_{n+1} = \mathcal{T}_n \oplus J_n \mathcal{B}_n$, one source-derived identity

$$q_{n+1}(x + \tilde{J}_n y) = \|Z_n x + Y_n y\|^2 + \|R_n y\|^2$$

valid for symbolic n proves $P(n) \to P(n+1)$. Induction proves every $P(n)$, and compact support gives the complete Weil criterion.

Category theory governs coherence of this one successor law across commuting prime orders and receiver recharts; it does not require an exhaustive atlas as a separate proof method. Invertible perspective changes still cannot alter inertia.

For the first successor, the invariant residual now has a sharper name and meaning:

$$S_{2,3} = D_{2,3} - C_{2,3}^* A_2^{-1} C_{2,3}.$$

It is the effective boundary tension after the positive $P(2)$ interior has relaxed. The full form and $A_2 \oplus S_{2,3}$ have the same inertia. The earlier "global Gram" is only the componentwise coordinate expansion of the absorbed old-body response. Moreover strict positivity already continues to some $R_* > 2$ by continuity of the localized ground value. What is not known is a quantitative continuum bound carrying that tension through the complete first cell to $R = 3$.

The squeeze refinement locates the object still more sharply. Increasing support cannot wash out a negative compact witness, while increasing finite-rank resolution squeezes only the Hilbert–Schmidt tail of the positive Sonin return. If the structural defect has a positive carrier $K_{S,R} \geq 0$, then finite-rank Parseval tails provide the complete $E_{n(f)} \to 0$ immediately. Under one prime admission that carrier must obey

$$K_{S\cup\{p\},R} = K_{S,R} - C_{S,p}, \quad C_{S,p} = W_p + \kappa_{S,p},$$

and remain nonnegative. The archimedean integration tail already has the required positive order on a fixed certified Galerkin band; the semilocal remainder and the band/support completion do not.

The coarea refinement gives this carrier a constructive geometry. In an ordinary chart the square-root Jacobian transports amplitude so its squared response is invariant. The logarithmic chart makes that half-density $e^{\frac{u}{2}}$ and carries Fourier analysis to the Mellin line $\mathrm{Re}(s) = \frac{1}{2}$. In one Euler transition the corresponding operator is $G_p^{\frac{1}{2}}$; it standardizes the successor aperture and gives an exact positive contemporary complement. Along a changing receiver, however, positive endpoint norms have signed covariant derivatives and seam jumps. Accordingly $C_{S,p} = W_p + \kappa_{S,p}$ need not itself be positive.

The global carrier remains an amplitude field and measure:

$$D_{S(f)} = \int_{\Lambda_{S,R}} \left\|B_{S,R,\xi} f\right\|^2 \mathrm{d}\nu_{S,R}(\xi),$$

constructed from the completed arithmetic and archimedean transport and obeying the exact signed prime-seam recurrence. It must also define, after any controlled null quotient, the positive metric covariantly preserved by the prime-orbit return whose trace is the explicit formula. Neither polar geometry, hyperbolic geometry, knot reciprocity, nor the positive Euler complement supplies that identification by analogy.

This is the present proof problem, in dependency order:

1. for the first cell, estimate the already source-defined family $S_a = D_a - C_a^* A_2^{-1} C_a$ strongly enough to carry its nonnegative sign from $a = \frac{\log 2}{2}$ through $a = \frac{\log 3}{2}$;
2. derive the same effective-tension family uniformly for symbolic support successors, including dual Euler transport at prime steps and repeated-place crossings at prime-power steps;
3. prove nonnegativity before choosing a square root, then let the resulting $R_n = S_n^{\frac{1}{2}}$ supply the inductive factorization; and
4. retain the explicit trace identity and control any null radical in the Hilbert completion used by the induction.

The dyadic edge now has an exact interface and exact input/output amplitudes. Its symmetric C_2 character space is exactly the negative prime-defect space, and dyadic periodization constructs the minimum isolated positive filler $\frac{\log 2}{\sqrt{2}} P_+$. The first old-body/new-shell cross is now carried and its conditioned shell is source-defined. The missing content is the sign of their effective tension throughout the rest of the first prime-active interval. Domain and incidence declarations belong inside that estimate; writing them around an otherwise open triangle is not an additional proof step.

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